Queclink

# GV58LAU @Track Air Interface Protocol

## GSM/GPRS/WCDMA/LTE CAT4/GNSS Tracker

TRACGV58LAUAN009

Version: 9.00


Driving Smarter IoT

www.queclink.com

| | |
|---|---|
| **Document Title** | GV58LAU @Track Air Interface Protocol |
| **Version** | 9.00 |
| **Date** | 2025-06-25 |
| **Status** | Release |
| **Document Control ID** | TRACGV58LAUAN009 |

**General Notes**

Queclink offers this information as a service to its customers, to support application and engineering efforts that use the products designed by Queclink. The information provided is based upon requirements specifically provided to Queclink by the customers. Queclink has not undertaken any independent search for additional relevant information, including any information that may be in the customer's possession. Furthermore, system validation of this product designed by Queclink within a larger electronic system remains the responsibility of the customer or the customer's system integrator. All specifications supplied herein are subject to change.

# Contents

Contents ....0
1 Revision History ....8
2 Overview ....12
2.1 Scope of This Document ....12
2.2 Terms and Abbreviations ....12
2.3 System Architecture ....13
2.4 Frame Types ....13
3 Commands ....15
3.1 Network Settings ....15
3.1.1 BSI (Bearer Setting Information) ....15
3.1.2 SRI (Backend Server Registration Information) ....16
3.1.3 QSS (Quick Start Settings) ....20
3.1.4 MQT (MQTT Server Information) ....21
3.1.5 TLS (TLS Data Encryption) ....22
3.2 Device Configuration ....24
3.2.1 ASC (Axis Self-Calibration) ....24
3.2.2 CFG (Global Configuration) ....25
3.2.3 DOG (Protocol Watchdog) ....30
3.2.4 GAM (GNSS-Assisted Motion) ....32
3.2.5 HMC (Hour Meter Count) ....33
3.2.6 HRM (HEX Report Mask) ....34
3.2.7 OWH (Outside Working Hours) ....41
3.2.8 PDS (Preserving Device's States) ....43
3.2.9 PIN (Auto-unlock PIN) ....44
3.2.10 TMA (Time Adjustment) ....45
3.2.11 WLT (Allowlist) ....46
3.2.12 AVS (Accelerometer Virtual Ignition) ....47
3.2.13 VVS (Voltage Virtual Ignition) ....48
3.2.14 VMS (Virtual Ignition Mode Selection) ....49
3.2.15 CMD (Command String Storage) ....51
3.2.16 UDF (User Defined Function) ....52
3.2.17 UDS (User Defined Speed Event) ....57
3.2.18 GNA (GNSS Assistance) ....58
3.3 Bluetooth Settings ....59
3.3.1 BTS (Bluetooth Setting) ....59
3.3.2 BAS (Bluetooth Accessory) ....61
3.3.3 BID (Bluetooth Beacon ID) ....70
3.3.4 AEX (Analog Input Expansion) ....77
3.3.5 SVR (Stolen Vehicle Recovery) ....78
3.3.6 BFS (Bluetooth Fields Settings) ....80
3.4 Tracking Settings ....81

3.4.1 FRI (Fixed Report Information) ........ 81
3.4.2 FFC (Frequency Change of FRI) ........ 84
3.5 Alarm Settings ........ 86
3.5.1 CRA (Crash Detection) ........ 86
3.5.2 GEO (Geo-Fence) ........ 88
3.5.3 PEO (Polygon Geo-Fence) ........ 90
3.5.4 PEG (Polygon Geo-Fence Group) ........ 92
3.5.5 HBM (Harsh Behavior Monitoring) ........ 94
3.5.6 IDL (Excessive Idling Detection) ........ 98
3.5.7 JBS (Jamming Behavior Setting) ........ 99
3.5.8 JDC (Network Jamming Detection) ........ 102
3.5.9 RMD (Roaming Detection) ........ 103
3.5.10 SOS (SOS Setting) ........ 106
3.5.11 BZA (Buzzer Alarm) ........ 108
3.5.12 SPA (Overspeed Alarm) ........ 109
3.5.13 SPD (Speed Alarm) ........ 111
3.5.14 SSR (Start/Stop Report) ........ 112
3.5.15 TOW (Tow Alarm) ........ 114
3.6 IO Applications ........ 116
3.6.1 DIS (Digital Input Setting) ........ 116
3.6.2 DOS (Digital Output Setting) ........ 117
3.6.3 EPS (External Power Monitoring) ........ 121
3.6.4 GDO (Gradual Digital Output) ........ 124
3.6.5 IOB (Input/Output Binding) ........ 125
3.6.6 ROS (Restrict Output Setting) ........ 127
3.7 Accessories Applications ........ 128
3.7.1 DAT (Transparent Data) ........ 128
3.7.2 IDA (ID Authentication) ........ 129
3.7.3 IDS (ID Setting) ........ 131
3.7.4 IEX (Input Expansion) ........ 132
3.7.5 OEX (Output Expansion) ........ 134
3.7.6 CAN (CANBUS Setting) ........ 135
3.7.7 CLT (Canbus Alarm Setting) ........ 140
3.7.8 CFU (CAN100 FOTA Upgrade) ........ 147
3.8 OTA Update Service ........ 148
3.8.1 FVR (Configuration File Version Record) ........ 149
3.8.2 UPC (Update Configuration) ........ 150
3.8.3 UPD (Firmware Update) ........ 151
3.8.3.1 Start the Firmware Update ........ 152
3.8.3.2 Stop the Firmware Update ........ 153
3.8.3.3 Firmware Update Process ........ 154
3.8.3.3.1 Initiation of the Update Process ........ 154
3.8.3.3.2 Confirmation of the Update Process ........ 154
3.8.3.3.3 Download of the Update Package ........ 155

3.8.3.3.4 Update of the Firmware ....................................................155
3.8.3.3.5 An Example of Successful Update ....................................156
3.9 Real Time Operation ....................................................156
3.9.1 RTO (Real Time Operation) ....................................................156
3.9.1.1 Sub Command ....................................................157
3.9.1.2 Optional Parameters ....................................................162
3.9.2 SWL (SMS Allowlist)....................................................165
3.9.3 SMS Location Request....................................................166
3.9.4 RTP (Remote File Transfer)....................................................166
3.9.5 LTP (Local File Transfer) ....................................................167
4 Report(Ascii)....................................................169
4.1 ACK (Acknowledgement)....................................................169
4.2 Position Related Report ....................................................172
4.2.1 Generic Location Report ....................................................172
4.2.2 FRI(Fixed Report Information)....................................................176
4.2.3 ERI(Expand Fixed Report Information)....................................................180
4.2.4 EPS | AIS ....................................................187
4.2.5 LBC(Location By Call Alarm) ....................................................188
4.2.6 IDA(ID Authentication Alarm) ....................................................189
4.2.7 GES(Parking-Fence Information) ....................................................191
4.2.8 GIN|GOT(Geo-Fence Information)....................................................192
4.2.9 SPE(Speed Alarm inside the Geo-Fence)....................................................195
4.2.10 Google Maps Hyperlink....................................................196
4.2.11 GSM(Cells Information)....................................................197
4.3 Device Information Reports ....................................................200
4.3.1 INF(Device Information)....................................................200
4.4 RTO Reports ....................................................203
4.4.1 GPS(GNSS Information)....................................................203
4.4.2 ALM(Command Configuration Information)....................................................204
4.4.3 CID(ICCID Information)....................................................206
4.4.4 CSQ(Network Signal Information) ....................................................206
4.4.5 VER(Firmware Version Information) ....................................................207
4.4.6 BAT(Battery Information)....................................................208
4.4.7 IOS(IO Status Information) ....................................................209
4.4.8 TMZ(Time Zone Information)....................................................210
4.4.9 AIF(Basic Device Information) ....................................................210
4.4.10 GSV(GPS Satellite Information) ....................................................212
4.4.11 RSV(GLONASS Satellite Information) ....................................................213
4.4.12 BSV(Beidou Satellite Information) ....................................................214
4.4.13 CVN(CAN Version Information) ....................................................214
4.4.14 CSN(CAN Serial Number)....................................................215
4.4.15 ATI(Version Information) ....................................................216
4.4.16 CML(CAN100 Synchronization) ....................................................217
4.4.17 BTI(Bluetooth Peripheral Information) ....................................................218

4.4.18 SCS(Acceleration Calibration Information) ....220
4.4.19 ASV(Galileo Satellite Information) ....221
4.5 Event Reports ....222
4.5.1 Generic Event Report ....222
4.5.2 MPN | MPF | BTC ....223
4.5.3 JDR(Network Jamming Indication Notification) ....224
4.5.4 JDS(Network Jamming Indication Notification) ....226
4.5.5 STC(Battery Stops Charging) ....227
4.5.6 BPL(Backup Battery Low Alarm) ....228
4.5.7 STT(Device Status Notification) ....229
4.5.8 IGN(Ignition-on Report) ....230
4.5.9 VGN(Virtual Ignition-on Report) ....232
4.5.10 IGF(Ignition-off Report) ....233
4.5.11 VGF(Virtual Ignition-off Report) ....234
4.5.12 IDF(Exit Idle State) ....236
4.5.13 GSS(GNSS Signal Status) ....237
4.5.14 STR | STP | LSP | IDN ....238
4.5.15 DOS(Wave Shape 1 Output Status) ....240
4.5.16 RMD(Roaming State Report) ....241
4.5.17 CLT(CANBUS Information Alarm) ....242
4.5.18 CFU(CAN100 FOTA Upgrade Report) ....245
4.5.19 SVR(Stolen Vehicle Recovery) ....246
4.5.20 RTP(Remote File Transfer) ....248
4.5.21 PNR(Power-on Reason) ....249
4.5.22 PFR(Power-off Reason) ....250
4.5.23 CRA(Crash Alarm) ....251
4.5.24 BCS(Bluetooth Connection Report) ....252
4.5.25 BDS(Bluetooth Disconnection Report) ....253
4.5.26 ASC(Acceleration Calibration Alarm) ....255
4.5.27 HBE(Acceleration Information for HBM) ....256
4.5.28 DOM(Waveform Occurs) ....258
4.5.29 BAA(Bluetooth Accessory Alarm) ....260
4.5.30 BID(Bluetooth Beacon ID Report) ....264
4.5.31 BAR(Bluetooth Accessory Report) ....267
4.5.32 UDF(User Defined Function Report) ....269
4.6 Data Flow Reports ....271
4.6.1 CRG(Crash GNSS Data Packet) ....271
4.6.2 CRD(Crash Data Packet) ....273
4.7 Data Transmission ....275
4.7.1 DAT(Transparent Data Transmission) ....275
4.7.1.1 Short Format ....275
4.7.1.2 Long Format ....276
4.8 CANBUS Information Report ....277
4.8.1 CAN(CANBUS Device Information) ....277

4.9 Update Configuration....285
4.9.1 UPC(Configuration Update Notification)....285
4.9.2 UPD(Firmware Upgrade Report)....286
4.9.2.1 Update Confirmation....286
4.9.2.2 Package Download....287
4.9.2.3 Firmware Update....288
4.9.3 EUC(Extended Configuration Update Report)....289
4.9.4 EUD(Extended Firmware Update Report in ASCII Format)....290
4.9.4.1 Update Confirmation....290
4.9.4.2 Package Download....291
4.9.4.3 Firmware Update....292
4.10 SACK (Server Acknowledgement)....293
4.11 HBD (Heartbeat Data)....294
4.11.1 Heartbeat Report....294
4.11.2 Heartbeat Acknowledgement....295
4.12 Buffer Report....295
4.12.1 Overview....295
4.12.2 Description Information....296
5 Report(Hex)....297
5.1 ACK (Acknowledgement)....297
5.2 Position Related Report....300
5.2.1 Generic Location Report....301
5.2.2 FRI(Fixed Report Information)....307
5.2.3 LBC(Location By Call Alarm)....308
5.2.4 SOS(SOS Alarm Notification)....310
5.2.5 IDA(ID Authentication Alarm)....312
5.2.6 ERI(Expand Fixed Report Information)....314
5.2.7 GIN|GOT(Geo-Fence Information)....318
5.2.8 SPE(Speed Alarm inside the Geo-Fence)....320
5.3 Device Information Reports....322
5.3.1 INF(Device Information)....322
5.3.2 ATI(Version Information)....329
5.4 Event Reports....330
5.4.1 Generic Event Report....330
5.4.2 BPL(Backup Battery Low Alarm)....335
5.4.3 JDR(Jamming Indication Notification)....337
5.4.4 JDS(Jamming Indication Notification)....338
5.4.5 IGN | IGF....340
5.4.6 VGN | VGF....341
5.4.7 IDF(Exit Idle State)....343
5.4.8 GSS(GNSS Signal Status)....345
5.4.9 DOS(Wave Shape 1 Output Status)....346
5.4.10 GES(Parking-Fence Information)....348
5.4.11 RMD(Roaming State Report)....350

5.4.12 CLT(CANBUS Information Alarm) ....351
5.4.13 PNR(Power-on Reason) ....353
5.4.14 PFR(Power-off Reason) ....354
5.4.15 ASC(Acceleration Calibration Alarm) ....356
5.4.16 HBE(Acceleration Information for HBM) ....358
5.4.17 CRA(Crash Alarm) ....360
5.4.18 BCS(Bluetooth Connection Alarm) ....362
5.4.19 BDS(Bluetooth Disconnection Alarm) ....363
5.4.20 BAA(Bluetooth Accessory Alarm) ....365
5.4.21 BID(Bluetooth Beacon ID Report) ....371
5.4.22 BAR(Bluetooth Accessory Report) ....373
5.4.23 RTP(Remote File Transfer) ....376
5.4.24 SVR(Stolen Vehicle Recovery) ....379
5.4.25 CFU(CAN100 FOTA Upgrade Report) ....380
5.4.26 DOM(Waveform Occurs) ....382
5.4.27 UDF(User Defined Function Report) ....383
5.5 Data Flow Reports ....385
5.5.1 Generic Data Flow Report ....385
5.5.2 CRD(Crash Data Packet) ....387
5.6 Data Transmission ....389
5.6.1 DAT(Transparent Data Transmission) ....389
5.7 CANBUS Information Report ....391
5.7.1 CAN(CANBUS Device Information) ....391
5.8 Update Configuration ....399
5.8.1 UPC(Configuration Update Notification) ....399
5.8.2 UPD(Firmware Upgrade Report) ....401
5.8.3 EUC(Extended Configuration Update Report) ....402
5.9 HBD (Heartbeat Data) ....404
5.9.1 Heartbeat Report ....404
5.10 Buffer Report in Hex Format ....405
6 Appendices ....406
6.1 Two's Complement ....406
6.2 Accessory Index ....407
6.3 CRC-16 Calculation ....407

# 1 Revision History

| Version | Date | Author | Description of Change |
|---|---|---|---|
| 1.00 | 2022-03-09 | John Wang | 1. Initial. |
| 2.00 | 2023-03-10 | Allen Zhang | 1. Upgraded the protocol version. |
| 3.00 | 2023-07-07 | Archie Li | 1. Added model 5 to parameter <Accessory Model> in the command **AT+GTBAS**. |
| 3.01 | 2023-07-10 | Allen Zhang | 1. Added the command **AT+GTMQT**.<br>2. Added the command **AT+GTTLS**.<br>3. Added the command **AT+GTRTP**.<br>4. Added the command **AT+GTLTP**.<br>5. Added mode 9 to the parameter <Report Mode> in the command **AT+GTSRI**. |
| 3.02 | 2023-07-19 | Harper Kuang | 1. Added type 12 and 13 to parameter <Accessory Type> in the command **AT+GTBAS**.<br>2. Added model 4 to parameter <Beacon ID Model> in the command **AT+GTBID**. |
| 3.03 | 2023-07-19 | Todd Zheng | 1. Added model 4 to parameter <Accessory Model> in the command **AT+GTBAS**. |
| 3.04 | 2023-08-19 | Archie Li | 1. Added the command **AT+GTSVR**.<br>2. Added mode 2 to parameter <Mode> in the command **AT+GTSVR**. |
| 3.05 | 2023-08-19 | Vincent Cui | 1. Added the command **AT+GTIDA**.<br>2. Added the parameter <INF Expansion2 Mask> to the command **AT+GTHRM**.<br>3. Added the parameter <Network Type> to the message **+RESP:GTINF**.<br>4. Deleted the mode 1 from the parameter <Virtual Ignition Mode> in the command **AT+GTVMS**.<br>5. Added Bit 36 <Network jamming is detected> and Bit 37 <Network jamming is not detected> to <Input ID Mask> in **AT+GTUDF**. |
| 4.00 | 2023-10-09 | Allen Zhang | 1. Added the parameter <GNSS On Time> to the command **AT+GTCFG**.<br>2. Added mode 99 to the parameter <Mode> in the command **AT+GTFRI**. |

| Version | Date | Author | Description of Change |
|---|---|---|---|
| | | | 3. Added the command **AT+GTROS**.<br>4. Added mode 3/4/5 to the parameter <Mode> in the command **AT+GTEPS**.<br>5. Modified the range of parameter <Initial Hour Meter Count> in **AT+GTHMC** to 00000:00:00 - 1193000:00:00.<br>6. Modified the range of the parameters <Samples Before Crash> and <Samples After Crash> in **AT+GTCRA** from 1 - 1500 to 0 - 1500.<br>7. Added the command **AT+GTSWL**.<br>8. Modified the range of parameter <GEO ID> in **AT+GTPEO** from 20 to 200.<br>9. Modified the range of parameter <Min Threshold> in **AT+GTEPS** to 0 - 32000mV.<br>10. Added Bit 51 <The device fixes GNSS successfully> and Bit 52 <The device fails to fix GNSS> to <Input ID Mask> in **AT+GTUDF**.<br>11. Added the command **AT+GTPEG**.<br>12. Added the parameter <EOF> to the message **+RESP:GTALM**.<br>13. Deleted the parameters <Brake Speed Threshold>, <Delta Speed Threshold> and <Delta Heading Threshold> from the command **AT+GTASC**.<br>14. Added the parameter <Report Mode> to the command **AT+GTASC**.<br>15. Added the parameter <Format Version> to the command **AT+GTRTO**.<br>16. Added the parameter <Format Version> to the message **+RESP:GTALM**. |
| 4.01 | 2023-11-23 | Allen Zhang | 1. Added the parameter <Fast STP Mode> to the command **AT+GTSSR**.<br>2. Added the parameter <Special Wave Monitor Mask> to the command **AT+GTDOS**. |
| 4.02 | 2023-12-22 | Allen Zhang | 1. Added the parameter <+DAT Mask> to the command **AT+GTHRM**. |

| Version | Date | Author | Description of Change |
|---|---|---|---|
| 5.00 | 2024-02-20 | Todd Zheng | 1. Added parameter <No CAN Interval> to the command **AT+GTDOG**.<br>2. Added parameter <Random Reboot Time> to the command **AT+GTDOG**. |
| 5.01 | 2024-03-15 | Todd Zheng | 1. Added parameter <Report Mode> to **AT+GTUDF** command. |
| 5.02 | 2024-03-28 | Allen Zhang | 1. Added mode 10/12 to the parameter <GNSS Working Mode> in the command **AT+GTCFG**. |
| 5.03 | 2024-04-03 | Allen Zhang | 1. Added the sub-command 0xE to the command **AT+GTRTO**.<br>2. Added Bit14 and Bit0 to the parameter <INF Expansion Mask> in the command **AT+GTHRM**.<br>3. Added Bit2 and Bit5 to the parameter <INF Expansion2 Mask> in the command **AT+GTHRM**. |
| 6.00 | 2024-04-03 | Allen Zhang | 1. Added the mode 6 to <Network Mode> in the command **AT+GTBSI**.<br>2. Added the parameter <Input Improve Mask> to the command **AT+GTUDF**.<br>3. Added the command **AT+GTGNA**. |
| 7.00 | 2024-09-09 | Allen Zhang | 1. Added model 5 to parameter <Beacon ID Model> in the command **AT+GTBID**.<br>2. Added command **AT+GTBFS**. |
| 7.01 | 2024-09-21 | Bennett Cui | 1. Deleted the parameter <Read Interval> from the command **AT+GTBAS**. |
| | | Harper Kuang | 2. Added command **AT+GTIDS**, and added option 2 to parameter <ID List Type> in **AT+GTIDA**.<br>3. Added Bit 1 to parameter <Report Mode> in the command **AT+GTIDA**. |
| | 2024-09-23 | Todd Zheng | 4. Added bit3(GNSS accuracy factor) to parameter <Position Append Mask> in the command **AT+GTCFG**. |
| 7.02 | 2024-10-11 | Allen Zhang | 1. Added mode 6 to the parameter <CarBLE Detection Mode> in the command **AT+GTBID**. |
| 7.03 | 2024-10-17 | Harper Kuang | 1. Added the mode 2 to the command **AT+GTIDA**. |

| Version | Date | Author | Description of Change |
|---|---|---|---|
| 8.00 | 2024-11-25 | John Wang | 1. Modified the description of <SACK Mode> in the command **AT+GTSRI**.<br>2. Added Enhanced <Satellite in Use> to <+RSP Expansion Mask> and <+EVT Expansion Mask> in the **AT+GTHRM** command. |
| | | Claire Liu | 3. Added type 3 to parameter <ID Report Type> in the message **+RESP:GTIDA**. |
| | 2024-11-28 | Todd Zheng | 4. Added type 13 to the parameter <Update Type> in the **AT+GTUPD** command. |
| 8.01 | 2024-12-09 | Harper Kuang | 1. Added the parameter <IMS Mode> to the command **AT+GTBSI**. |
| 8.02 | 2024-12-16 | Harper Kuang | 1. Added Bit 57 <Output1 is activated>, Bit58 <Output1 is deactivated>, Bit 59 <Output2 is activated>, Bit 60 <Output2 is deactivated>, Bit 61 <Output3 is activated> and Bit 62 <Output3 is deactivated> to <Input ID Mask> in **AT+GTUDF**. |
| 8.03 | 2025-02-14 | Vincent Cui | 1. Added the command **AT+GTUDS**.<br>2. Added the parameters <Siren Alarm Duration (T7)> and <Preparing Alarm Timer (T8)> to the command **AT+GTJBS**.<br>3. Added parameter <Tow Distance> to the command **AT+GTTOW**.<br>4. Added the parameter <GNSS off Latency> to the command **AT+GTCFG**.<br>5. Added the parameter <Battery Percentage> to the message **Google Maps Hyperlink**. |
| 9.00 | 2025-04-21 | Vincent Cui | 1. Added the model 10 to the parameter <Beacon ID Model> in the command **AT+GTBID**. |

# 2 Overview

## 2.1 Scope of This Document

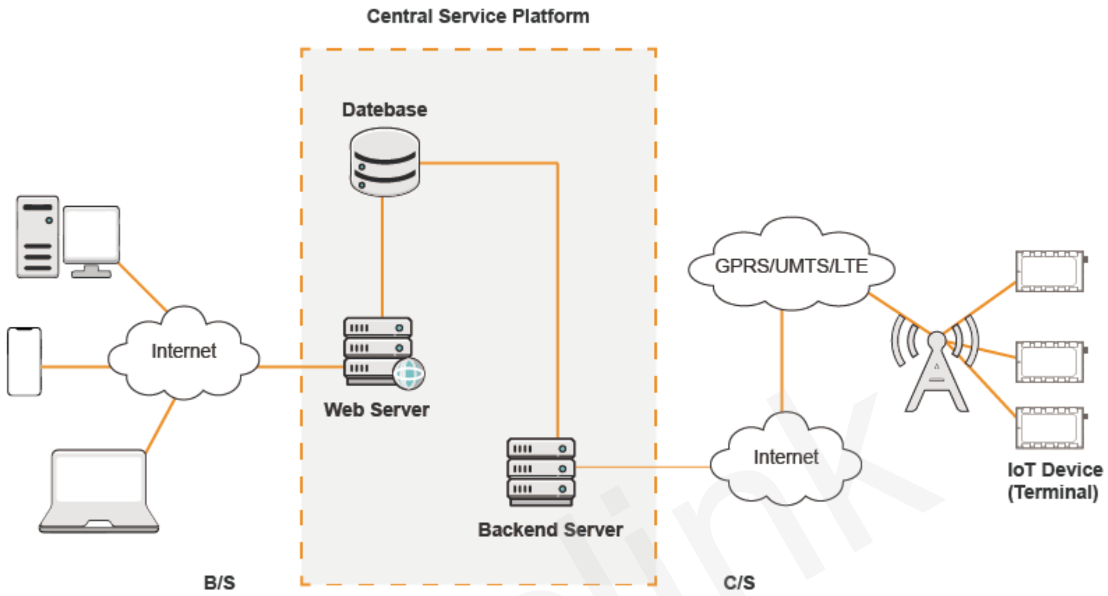
The @Track Air Interface Protocol is a digital communication interface between Queclink Trackers and the backend server. It is used for all communication between the backend server and the terminal via SMS or GPRS. The backend server sends a command to the terminal and then the terminal confirms the receipt with an acknowledgement message. If configured, the terminal also sends report messages to the backend server.

The purpose of this document is to describe how to build the backend server based on the @Track Air Interface Protocol.

## 2.2 Terms and Abbreviations

| Abbreviation | Description |
|---|---|
| APN | Access Point Network |
| ASCII | American National Standard Code for Information Interchange |
| GPRS | General Packet Radio Service |
| GSM | Global System for Mobile Communications |
| GNSS | Global Navigation Satellite System |
| HDOP | Horizontal Dilution of Precision |
| ICCID | Integrated Circuit Card identity |
| IP | Internet Protocol |
| SMS | Short Message Service |
| TCP | Transmission Control Protocol |
| UDP | User Datagram Protocol |
| UTC | Coordinated Universal Time |

## 2.3 System Architecture

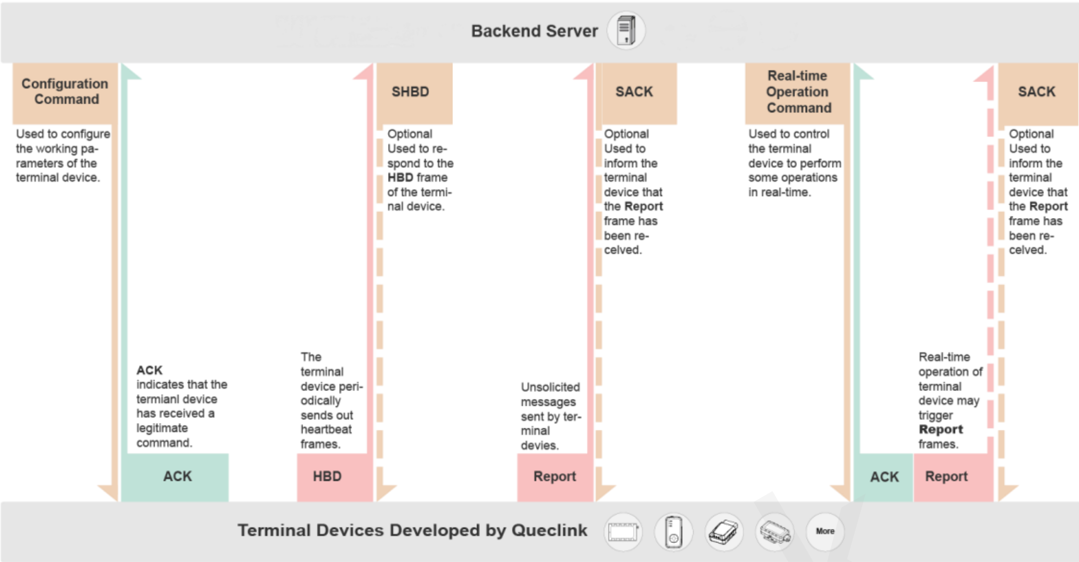



The backend server needs to be accessible by many terminals and should have the following abilities:

- The backend server should be able to access the Internet and listen for the connection requests originating from the terminal.
- The backend server should be able to support TCP or UDP connection with the terminal. It should be able to receive data from the terminal and send data to the terminal.
- The backend server should be able to receive and send SMS.

## 2.4 Frame Types

The @Track protocol contains the following types of frames, depending on their purpose:




*Frame Types and Encoding*

For each frame, the unit of Length is Byte. Please refer to Commands/ASCII/HEX for the detailed format of the above types of frames.

# 3 Commands

Commands are used to set the working parameters of the terminal or to cause the terminal to perform certain operations. Please refer to the details below.

## 3.1 Network Settings

This section describes the commands related to the connection between the terminal device and external units (such as the backend server). Please refer to the details below.

### 3.1.1 BSI (Bearer Setting Information)

The command **AT+GTBSI** is used to configure the parameters for GSM/GPRS/LTE Cat4/WCDMA data connection.

| Example: |
|---|
| AT+GTBSI=gv58lau,cmnet,,,3gnet,,,0,,,1,0000$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | BSI | BSI |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | APN | <=64 | | |
| | APN User Name | <=30 | | |
| | APN Password | <=30 | | |
| | Backup APN | <=64 | | |
| | Backup APN User Name | <=30 | | |
| | Backup APN Password | <=30 | | |
| | Network Mode | 1 | 0-6 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | IMS Mode | 1 | 0 \| 1 | 1 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *APN*

Access point name (APN).

✧ *APN User Name*

The user name of APN.

✧ *APN Password*

The password of APN.

✧ *Backup APN*

Backup access point name. If <APN> does not work, <Backup APN> will be used.

✧ *Backup APN User Name*

The user name of backup APN.

✧ *Backup APN Password*

The password of backup APN.

Note

Please set the parameter value to "%CLR%" if there is a need to erase the parameter <APN>, <APN User Name>, <APN Password> , <Backup APN>, <Backup APN User Name> or <Backup APN Password>. These parameter values will not be cleared if they are empty.

✧ *Network Mode*

- **0** - Auto (LTE & GSM/EGPRS & WCDMA)
- **1** - GSM/EGPRS Only
- **2** - WCDMA
- **3** - LTE Only
- **4** - GSM/EGPRS & WCDMA (GSM/EGPRS First)
- **5** - WCDMA & GSM/EGPRS (WCDMA First)
- **6** - WCDMA & LTE & GSM/EGPRS (WCDMA First)

✧ *IMS Mode*

Specify whether to enable IMS of the device.

- **0** - Disable
- **1** - Enable

✧ *Serial Number*

The serial number of the command. It will be included in the ACK message for the command.

✧ *Tail*

A character which indicates the end of the command. It must be "$".

### 3.1.2 SRI (Backend Server Registration Information)

The command **AT+GTSRI** is used to configure how to report all the messages, including the server information and the method of communication between the backend server and the terminal. If the terminal is configured correctly, it should be able to report data to the backend server.

Example:
AT+GTSRI=gv58lau,3,,1,60.174.225.173,20581,60.174.225.173,20581,13812341234,15,1,0,0,,30,
0.0.0.0,0.0.0.0,,,0001$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | SRI | SRI |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Report Mode | 1 | 0-7 \| 9 | 0 |
| | Reserved | 0 | | |
| | Buffer Mode | 1 | 0-2 | 1 |
| | Main Domain | <=60 | ASCII (not including '=' and ',') | |
| | Main Port | <=5 | 0 - 65535 | |
| | Backup Domain | <=60 | ASCII (not including '=' and ',') | |
| | Backup Port | <=5 | 0 - 65535 | |
| | SMS Gateway | <=20 | | |
| | Heartbeat Interval | <=3 | 0 \| 2 - 360(min) | 0 |
| | SACK Mode | 1 | 0-2 | 0 |
| | Protocol Format | 1 | 0 \| 1 | 0 |
| | SMS ACK Mode | 1 | 0 \| 1 | 0 |
| | Reserved | 0 | | |
| | Connection Life | <=3 | 0 \| 10 - 600(s) | 30 |
| | Primary DNS Server | <=15 | | 0.0.0.0 |
| | Secondary DNS Server | <=15 | | 0.0.0.0 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Report Mode*

This parameter defines the method of communication between the backend server and the terminal. Supported report modes are as follows:

- **0** - Stop mode.
- **1** - TCP short-connection preferred mode.

  The connection is based on TCP protocol. The terminal connects to the backend server every time it needs to send data and will shut down the connection when the terminal finishes sending data. If the terminal fails to establish TCP connection to the backend server (both Main Server and Backup Server), it will try to send data to the SMS gateway via SMS.
- **2** - TCP short-connection forced mode.

  The connection is based on TCP protocol. The terminal connects to the backend server every time it needs to send data and will shut down the connection when the terminal finishes sending data. If the terminal fails to establish TCP connection to the backend server (both Main Server and Backup Server), it will store the data in the memory buffer if the buffer report function is enabled. Otherwise, the data is discarded.

- **3** - TCP long-connection mode.
  The connection is based on TCP protocol. The terminal connects to the backend server and maintains the connection using heartbeat data. The backend server should respond to the heartbeat data from the terminals.
- **4** - UDP mode.
  The terminal will send data to the backend server through the UDP protocol. Receiving protocol commands via UDP is supported if the GPRS network allows it. It is recommended to enable heartbeat and **PDP** (ASCII)/(HEX) report when receiving commands via UDP.
- **5** - Forced SMS mode. Only SMS is used for data transmission.
  Note
  The messages **GSM** (ASCII)/(HEX) , **ALM** (ASCII) , and +DAT are sent via TCP short connection when the report mode is forced SMS mode.
- **6** - UDP with fixed local port mode.
  Like the UDP mode, the terminal will send data using UDP protocol. The difference is the terminal will use a fixed local port rather than a random port to communicate with the server in this mode. Thus the backend server could use the identical port to communicate with all terminals if the backend server and the terminals are all in the same VPN network. The port number the device uses is the same as the port number of the main server.
- **7** - Backup server supported TCP long-connection mode.
  The connection is based on TCP protocol. The terminal connects to the backend server and maintains the connection using the heartbeat data. The backend server should respond to the heartbeat data from the terminals. If the connection to the main server is lost, the terminal will try to connect to the backup server. If the connection to the backup server is also lost, the terminal will try to connect to the main server again.
- **9** - MQTT mode.
  MQTT is a Client-Server based on message Publish/Subscribe transport protocol. The protocol is built on the TCP protocol. This mode has a default username(**admin**) and password(**password**) as well as a subscription topic(**quec_msg**) and a publication topic(**quec_ctrl**).

✧ *Buffer Mode*

The working mode of the buffer report function.

If the buffer report function is enabled and the device goes into areas without network coverage, it will store all reports locally.

If the device goes into areas with network coverage again, it will then send all the buffered reports.

- **0** - Disable.
- **1** - Low priority. In this mode, the device will send the buffered messages after real-time messages.
- **2** - High priority. In this mode, the device will send all the buffered messages before real-time messages, except the messages **+RESP:GTSOS**, **+RESP:GTPFA**, **+RESP:GTPFR**,

**+RESP:GTPDP**, **+RESP:GTUPD**, **+RESP:GTIDN** , **+RESP:GTIDF**, **+RESP:GTASC**, **+RESP:GTRMD**, **+RESP:GTRTP**, **+RESP:GTSPD**, **+RESP:GTTOW**.

✧ *Main Domain*

The IP address or the domain name of the main server.

✧ *Main Port*

The port number of the main server.

✧ *Backup Domain*

The IP address or the domain name of the backup server.

✧ *Backup Port*

The port number of the backup server.

✧ *SMS Gateway*

It is a maximum of 20 characters including the optional national code starting with "+" for sending SMS messages. Short code (for example, 10086) is also supported.

✧ *Heartbeat Interval*

The interval for sending heartbeat messages (**+ACK:GTHBD**) when report mode is TCP long-connection mode or UDP mode. If it is set to 0, no heartbeat message will be sent.

✧ *SACK Mode*

Specify whether the terminal needs to wait for the SACK message from the backend server after successfully sending a message to the backend server. Please note that it requires the backend server to be able to reply to the terminal with the SACK message if <SACK Mode> is not 0.

- **0** - The terminal does not wait for the SACK message after sending a message to the backend server.
- **1** - The terminal will wait for the SACK message for 20 seconds after sending a message to the backend server. It will resend the message up to 4 times if the SACK message from backend server cannot be received correctly or the serial number of the SACK message does not match the last message sent.
- **2** - Similar to <SACK Mode> 1, the difference is that the terminal does not check the serial number of the SACK message in this mode.

Note

If the terminal receives **+SACK:GTHBD** from the backend server, the terminal must check the serial number of the SACK message **+SACK:GTHBD** regardless of the value of <SACK Mode>.

✧ *Protocol Format*

This parameter defines the format of the report messages sent from the device to the backend server.

- **0** - ASCII format.
- **1** - HEX format.

✧ *SMS ACK Mode*

This parameter defines whether to reply with the ACK confirmation via SMS when the command is sent via SMS.

- **0** - The device will send the ACK confirmation using the mode specified by <Report Mode>.
- **1** - The device will send the ACK confirmation via SMS to the phone number from which the command is sent via SMS.

✧ *Connection Life*

A numeral to indicate the time to maintain TCP connection for receiving commands from the server. If there is no data transmission within the time of <Connection Life>, the TCP connection will be closed. Unit: second.

✧ *Primary DNS Server*

The address of primary DNS server.

✧ *Secondary DNS Server*

The address of secondary DNS server.

Note

If both <Primary DNS Server> and <Secondary DNS Server> are 0.X.X.X, 127.X.X.X or 255.X.X.X, the default DNS server obtained from network will be used.

### 3.1.3 QSS (Quick Start Settings)

The command **AT+GTQSS** is used to configure GPRS and backend server parameters if the length of all the settings is less than 160 bytes. Otherwise, the two commands AT+GTBSI and AT+GTSRI are used to set those parameters.

Example:
AT+GTQSS=gv58lau,cmnet,,,3,,1,60.174.225.173,20581,60.174.225.173,20581,13812341234,15,1,0,0,0.0.0.0,0.0.0.0,,0002$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | QSS | QSS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | APN | <=64 | | |
| | APN User Name | <=30 | | |
| | APN Password | <=30 | | |
| | Report Mode | 1 | 0-7 \| 9 | 0 |
| | Reserved | 0 | | |
| | Buffer Mode | 1 | 0-2 | 1 |
| | Main Domain | <=60 | ASCII (not including '=' and ',') | |
| | Main Port | <=5 | 0 - 65535 | |
| | Backup Domain | <=60 | ASCII (not including '=' and ',') | |
| | Backup Port | <=5 | 0 - 65535 | |
| | SMS Gateway | <=20 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Heartbeat Interval | <=3 | 0 \| 2 - 360(min) | 0 |
| | SACK Mode | 1 | 0-2 | 0 |
| | Protocol Format | 1 | 0 \| 1 | 0 |
| | SMS ACK Mode | 1 | 0 \| 1 | 0 |
| | Primary DNS Server | <=15 | | 0.0.0.0 |
| | Secondary DNS Server | <=15 | | 0.0.0.0 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

### 3.1.4 MQT (MQTT Server Information)

The command **AT+GTMQT** is used to configure the username and password to connect to the MQTT server and to subscribe and publish topics. If the terminal is configured correctly, it should be able to report data to the MQTT server.

Example:
AT+GTMQT=gv58lau,0,60,admin,password,,quec_ctrl,,quec_msg,,#,,,,,FFFF$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | MQT | MQT |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Server ID | 1 | 0 | 0 |
| | Keep Alive | <=4 | 0 \| 5 - 1080(min) | 60 |
| | Username | <=64 | ASCII (not including ',') | admin |
| | MQTT Password | <=64 | ASCII String | password |
| | Reserved | 0 | | |
| | Subscribe Topic | <=64 | ASCII String | quec_ctrl |
| | Reserved | 0 | | |
| | Publish Topic | <=64 | ASCII String | quec_msg |
| | Reserved | 0 | | |
| | Client ID | <=64 | '0'-'9' 'a'-'z' 'A'-'Z' '#' | # |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Keep Alive*

It is used to set keep alive mechanism for MQTT. It is a time interval measured in minutes. In particular, 0 means the keep alive mechanism is disabled. Unit: minute.

✧ *Username*

If the MQTT server uses authentication, the username is specified here.

✧ *MQTT Password*

If the MQTT server uses authentication, the password is specified here.

✧ *Subscribe Topic*

The client subscribes to topic name.

Note

If the MQTT server supports wildcards, the rules for using the ('#' U+0023) and ('+' U+002B) wildcards in the <Subscribe Topic> are as follows:

The number sign ('#' U+0023) is a multi-level wildcard that can match any number of levels within a topic. When used, it must be specified either on its own or following a topic level separator ('/' U+002F). In either case, it must be the last character specified in the Topic Filter.

The plus sign ('+' U+002B) is a single-level wildcard that matches only one level within a topic. The single-level wildcard can be used at any level in the Topic Filter, including the first and the last levels. Where it is used, it must occupy the entire level of the filter. It can be used at more than one level in the Topic Filter and can also be used in conjunction with the multi-level wildcard.

✧ *Publish Topic*

The client publishes topic name.

✧ *Client ID*

Each client connected to the server has a unique client identifier (Client ID). Both the client and the server must use the Client ID to identify the state associated with the MQTT session between them. The Client ID can contain only uppercase letters, lowercase letters and numeric characters.

**For example:**

0123456789abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ

Note

In addition, a single character "#" is defined to indicate the use of the IMEI number that will be used internally as the Client ID.

### 3.1.5 TLS (TLS Data Encryption)

The command **AT+GTTLS** is used to configure TLS encryption parameters.

Example:
AT+GTTLS=gv58lau,0,1,1,,,FFFF$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | TLS | TLS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Server ID | 1 | 0 | 0 |
| | Mode | 1 | 0 \| 1 | 0 |
| | Verification Mode | 1 | 0-2 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *Server ID*

The index of the Server.

- **0** - The server is configured in the AT+GTSRI command.

✧ *Mode*

The working mode of the TLS function.

- **0** - Disable.
- **1** - Enable.

✧ *Verification Mode*

It specifies the certificate verification method for the terminal.

- **0** - Do not verify the certificates.
  In this method, no certificates need to be built into the terminal.
- **1** - Only server certification.
  In this method, at least the CA file needs to be built into the terminal.
- **2** - Two-way certification between server and client.
  In this method, at least the CA file, Client Certificate file, Client key file need to be built into the terminal.

Note

TLS encryption is only valid for TCP and MQTT connections.

## 3.2 Device Configuration

This section describes the commands related to the device generic configurations. Please refer to the details below.

### 3.2.1 ASC (Axis Self-Calibration)

The command **AT+GTASC** is used to define the condition for calibrating the directions of accelerometer. When the auto self-calibration factor is updated, the device will report the event message **ASC** (ASCII)/(HEX) containing the calibration result to the backend server. The pre-condition for the calibration is ignition on and movement.

Note
To avoid possible inaccuracies caused by historical calibration data, please clear the self-calibration status of the acceleration data via the sub command 25 in AT+GTRTO after the device is installed.

Example:
AT+GTASC=gv58lau,,,,,1,,,,,,,,FFFF$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | ASC | ASC |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Report Mode | 1 | 0 \| 1 | 1 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Report Mode*

Enable/disable the **+RESP:GTASC** report.

- **0** - Disable.
- **1** - Enable.

### 3.2.2 CFG (Global Configuration)

The **AT+GTCFG** command is used to configure global parameters.

Example:
AT+GTCFG=gv58lau,gv58lau,GV58LAU,0,0.0,,,003F,1,00,1BDEF,0,0,0,300,00,1,0,0,001F,0,4,150,,,,,300,FFFF$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | CFG | CFG |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | New Password | 4 - 20 | '0'-'9' 'a'-'z' 'A'-'Z' | |
| | Device Name | <=20 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' | GV58LAU |
| | ODO Mode | 1 | 0 \| 1 | 0 |
| | ODO Initial Mileage | <=9 | 0.0 - 4294967.0(km) | 0.0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Report Item Mask | <=4 | 0000 - FFFF | 003F |
| | Power Saving Mode | 1 | 0-2 | 1 |
| | Position Append Mask | 2 | 00 - FF | 00 |
| | Event Mask | <=8 | 00000000 - FFFFFFFF | 1BDEF |
| | Pin Mask | 1 | 0 - F | 0 |
| | LED Mode | 1 | 0-2 | 0 |
| | Device Information Report | 1 | 0 \| 1 | 0 |
| | Device Information Interval | <=5 | 30 - 86400(s) | 300 |
| | Location Request Mask | 2 | 00 - 23 | 00 |
| | Battery Working Mode | 1 | 0 \| 1 | 1 |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Power Mode | 1 | 0 \| 1 | 0 |
| | AGPS Mode | 1 | 0-2 | 0 |
| | Cell Report | 4 | 0000 - FFFF | 001F |
| | GNSS Lost Time | <=2 | 0 - 30(min) | 0 |
| | GNSS Working Mode | <=2 | 0 \| 1 \| 4 \| 10 \| 12 | 4 |
| | GNSS On Time | <=3 | 60 - 150 | 150 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | GNSS off Latency | <=3 | 0 - 300 (s) | 300 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *New Password*

It is set to change the current password.

✧ *Device Name*

An ASCII string which represents the name of the device.

✧ *ODO Mode*

Enable/disable the odograph function to calculate the total mileage. The current mileage is included in every position report message.

- **0** - Disable.
- **1** - Enable the odograph function to calculate the total mileage by GNSS.

✧ *ODO Initial Mileage*

The initial value for calculating the total mileage.

✧ *Report Item Mask*

Bitwise mask to configure the composition of report messages, especially the composition of GNSS information.

- **Bit 0** - for <Speed>
- **Bit 1** - for <Azimuth>
- **Bit 2** - for <Altitude>
- **Bit 3** - for Cell information, including <MCC>, <MNC>, <LAC>, <Cell ID>.
- **Bit 4** - for <Mileage>
- **Bit 5** - for <Send Time>, the time when the report message is generated.
- **Bit 6** - for <Device Name>

For each bit, set it to 1 to enable the corresponding component in the report, and set it to 0 to disable. This mask is valid for all report messages.

Note

The GSM tower data is not controlled by Bit 3 in the message **GSM** (ASCII)/(HEX).

- *Power Saving Mode*

The mode of the power saving function.

If <Power Saving Mode> is set to 0, the fixed report will follow <IGF Report Interval> when the engine is off.

If <Power Saving Mode> is set to 1, the fixed report **FRI**(ASCII)/(HEX) or **ERI** (ASCII)/(HEX), geo-fence **GIN/GOT**(ASCII)/(HEX) and speed alarm **SPD**(ASCII)/(HEX) report messages will be suspended when the device is stationary or the engine is off (Auto parking fence and manual parking fence will not be suspended in this case).

If <Power Saving Mode> is set to 2, it is mostly like Mode 1 and the difference is that the fixed report will not be suspended and the fix and send interval of it will be set to <IGF Report Interval> in AT+GTFRI when the engine is off.

- **0** - Disable the power saving function.
- **1** - GNSS deep saving mode.
- **2** - GNSS low saving mode.

- *Position Append Mask*

A bitwise numeral to control whether to include the corresponding fields in each position after <Cell ID>.

- **Bit 0** - The number of satellites in use for the current position (GNSS Satellite Number).
- **Bit 1** - Reserved for indicating <GNSS Trigger Type> in **FRI** (ASCII)/(HEX) and **ERI** (ASCII)/(HEX) report, which cannot be set here.
- **Bit 2** - For device status reporting in almost all the event messages (except **+RESP:GTPNA**, **+RESP:GTPFA**, **+RESP:GTPNR**, **+RESP:GTPFR**, **+RESP:GTUPC**, **+RESP:GTEUC**, **+RESP:GTCFU** and **+RESP:GTPDP**).
- **Bit 3** - GNSS accuracy factor for the current position (Horizontal GNSS Accuracy, Vertical GNSS Accuracy and 3D GNSS Accuracy).

Note

If <Wrap Corner Point> is set to 1, the bit 1 of <Position Append Mask> in **FRI** (ASCII)/(HEX) and **ERI** (ASCII)/(HEX) will be set to 1, but it will always be 0 in other messages.

- *Event Mask*

Bitwise mask to configure which event report should be sent to the backend server.

- **Bit 0** - for **PNA** (ASCII)/(HEX)
- **Bit 1** - for **PFA** (ASCII)/(HEX)
- **Bit 2** - for **MPN** (ASCII)/(HEX)
- **Bit 3** - for **MPF** (ASCII)/(HEX)
- **Bit 5** - for **BPL** (ASCII)/(HEX)
- **Bit 6** - for **BTC** (ASCII)/(HEX)
- **Bit 7** - for **STC** (ASCII)/(HEX)
- **Bit 8** - for **STT** (ASCII)/(HEX)
- **Bit 10** - for **PDP** (ASCII)/(HEX)
- **Bit 11** - for the power on **RTL** (ASCII)/(HEX)
- **Bit 12** - for the ignition report **IGN** (ASCII)/(HEX), **IGF** (ASCII)/(HEX), **VGN** (ASCII)/(HEX) and **VGF** (ASCII)/(HEX)
- **Bit 13** - for ignition on and off location report **IGL** (ASCII)/(HEX) and **VGL**(ASCII)/(HEX)

- **Bit 15** - for **PNR** (ASCII)/(HEX)
- **Bit 16** - for **PFR** (ASCII)/(HEX)

For each bit, set it to 1 to enable the corresponding event report, and set it to 0 to disable. If **+RESP:GTPNR** and **+RESP:GTPFR** events are enabled, **+RESP:GTPNA** and **+RESP:GTPFA** will not be reported even if they are enabled.

Note

The configuration in order to avoid generating a large amount of **STT** (ASCII)/(HEX) messages, 1A (Fake Tow) state is ignored.

✧ *Pin Mask*

It configures the working mode of PIN on the connector.

- **Bit 2** - for PIN6. Set it to 0 as digital input 1, set it to 1 as output3.

Note

The configuration of these PINs must be consistent with actual usage. Otherwise, these PINs may be corrupted.

✧ *LED Mode*

It configures the working mode of the LED lights.

- **0** - Each time the device is powered on, all the LEDs will work for 30 minutes and then turn off.
- **1** - All the LEDs turn on as configured.
- **2** - After the device is powered on, all the LEDs will work for 10 minutes and then turn off.

✧ *Device Information Report*

Enable/disable the device information report **INF**(ASCII)/(HEX).

The device information includes state of the device, ICCID, GSM signal strength, voltage of external power supply, battery voltage, charging status, working mode of LED lights, the last known time of GNSS fix, analog input voltage, status of all digital inputs and outputs, time zone information and daylight-saving setting.

- **0** - Disable.
- **1** - Enable.

✧ *Device Information Interval*

The interval for reporting the device information.

✧ *Location Request Mask*

Bitwise mask for SMS. 4 high bits for SMS request. Each bit represents one kind of report. Set it to 1 to enable the corresponding report, and set it to 0 to disable.

- 4 high bits for SMS request.
  - **0** - Ignore SMS location request.
  - **1** - Report the current location **LBC**(ASCII)/(HEX) to server.
  - **2** - Send current location with Google map link to Caller ID via SMS.
- 4 low bits for how to handle the incoming call .
  - **0** - Hang up only.
  - **1** - Hang up and report the current location **LBC**(ASCII)/(HEX) to server.
  - **2** - Hang up the call and send current location with Google map link to Caller ID via SMS.
  - **3** - Hang up and report the current location **LBC**(ASCII)/(HEX) to server, simultaneously send current location with Google map link to Caller ID via SMS.

✧ *Battery Working Mode*

It configures whether to enable backup battery. The backup battery will only be used when this parameter is set to 1 and the external power is not connected.

- **0** - Disable backup battery.
- **1** - Enable backup battery.

✧ *Power Mode*

It configures the power supply mode of the terminal.

- **0** - If the main power supply is connected, the backup battery is charged as needed.
- **1** - If the main power supply is connected, the backup battery is only charged when ignition is on. The charge process will begin 3 minutes after ignition on and stop when the ignition is off.

✧ *AGPS Mode*

A numeral which indicates whether to enable AGPS. AGPS helps increase the chances of getting GNSS position successfully and reduces the time needed to get GNSS position.

- **0** - Disable the AGPS function.
- **1** - Enable the AGPS function.
- **2** - Enable the AGPS online function.

✧ *Cell Report*

A hexadecimal numeral to indicate how to report cell information **GSM**(ASCII)/(HEX).

2 high bits represent the GSM working mode.

- **0b00** - Do not allow the cell information report.
- **0b01** - Allow the cell information report after failing to get GNSS position if cell information is available.
- **0b10** - Report the message **GSM**(ASCII)/(HEX) after getting GNSS position successfully every time if cell information is available.
- **0b11** - Report the message **GSM**(ASCII)/(HEX) no matter whether it is successful to get GNSS position if cell information is available.

Other bits control whether the following events will trigger the report **GSM**(ASCII)/(HEX).

- **Bit 0** - for **RTL** (ASCII)/(HEX)
- **Bit 1** - for **LBC** (ASCII)/(HEX)
- **Bit 2** - for **FRI**(ASCII)/(HEX) / **ERI** (ASCII)/(HEX)
- **Bit 3** - for **SOS** (ASCII)/(HEX)
- **Bit 4** - for **TOW** (ASCII)/(HEX)
- **Bit 5 - 13** - Reserved

For each bit, set it to 1 to enable the corresponding event report, and set it to 0 to disable.

✧ *GNSS Lost Time*

A time parameter to monitor the GNSS signal. If there is no GNSS signal or no successful GNSS fix for consecutive <GNSS Lost Time>, the device will send the event report **GSS**(ASCII)/(HEX) to indicate "GNSS signal lost". If the GNSS signal is recovered or a successful fix is obtained again, the device will send the event report **GSS**(ASCII)/(HEX) to indicate the recovery. 0 means "Disable this feature".

Note

If the device is rebooted, it will not report **GSS**(ASCII)/(HEX) to indicate GNSS signal recovery even if it has reported **GSS**(ASCII)/(HEX) to indicate "GNSS signal lost" before reboot.

✧ *GNSS Working Mode*

The working mode of GNSS chip. If the current GNSS chip doesn't support combination mode, the device will get position by GPS only.

- **0** - GPS and GLONASS positioning systems. In this mode, the device fixes positions with GPS and GLONASS systems.
- **1** - GPS positioning system. In this mode, the device fixes positions only with GPS system.
- **4** - GPS and Beidou positioning systems. In this mode, the device fixes positions with GPS and Beidou systems.
- **10** - GPS, GLONASS and Galileo positioning systems. In this mode, the device fixes positions with GPS, GLONASS and Galileo systems.
- **12** - GPS, Galileo and Beidou positioning systems. In this mode, the device fixes positions with GPS, Galileo and Beidou systems.

✧ *GNSS On Time*

A time parameter to set the time for GNSS on. When GNSS fix fails, GNSS will be on until the time period specified by this parameter is exceeded. The unit is second.

✧ *GNSS off Latency*

The time to turn off GNSS after engine off when <Power Saving Mode> is enabled.

### 3.2.3 DOG (Protocol Watchdog)

The **AT+GTDOG** command is used to reboot the device in a time-based manner or upon ignition to prevent the device from working improperly for a long time.

Besides these two automatic reboot methods, the device also supports triggering the reboot manually by digital input.

| |
|---|
| Example:<br>AT+GTDOG=gv58lau,1,60,30,0200,,1,0,,60,60,60,,,,0013$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | DOG | DOG |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-2 | 0 |
| | Ignition Frequency | <=3 | 10 - 120 (min) | 60 |
| | Interval | <=2 | 1 - 30 (day) | 30 |
| | Time | 4 | HHMM | 0200 |
| | Reserved | 0 | | |
| | Report Before Reboot | 1 | 0 \| 1 | 1 |
| | Input ID | 1 | 0 - 1 | 0 |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Reserved | 0 | | |
| | No Network Interval | <=4 | 0 \| 5 - 1440(min) | 60 |
| | No Activation Interval | <=4 | 0 \| 5 - 1440(min) | 60 |
| | Send Failure Timeout | <=4 | 0 \| 5 - 1440(min) | 60 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | No CAN Interval | 4 | 0 \| 5 - 1440(min) | 60 |
| | Random Reboot Time | <=2 | 0 - 60(min) | 0 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the watchdog function.

- **0** - Disable.
- **1** - Reboot periodically according to the <Interval> and <Time> settings.
- **2** - Reboot when the ignition is turned on.

✧ *Ignition Frequency*

If the interval between the current ignition-on and last ignition-on reboot is greater than the value specified by this parameter when <Mode> is 2, the device will automatically reboot upon ignition on. The device will reboot automatically upon the second ignition-on for the first time use whatever the time interval from the first ignition-on is.

✧ *Interval*

The interval for rebooting the device. It is measured in days. Rebooting the device for the first time will ignore this interval.

✧ *Time*

The time at which the reboot operation is performed when <Interval> condition is met.

✧ *Report Before Reboot*

Whether to report the **DOG**(ASCII)/(HEX) message before reboot. 0 means "Do not report the **DOG**(ASCII)/(HEX) message", and 1 means "Report the **DOG**(ASCII)/(HEX) message". If this parameter is enabled, the device will initiate a real-time fix before sending the message with the current location information.

✧ *Input ID*

The ID of the digital input port which is used to trigger the manual reboot. 0 means "Do not use manual reboot".

✧ *No Network Interval*

The interval for rebooting the device when there is no network signal. 0 means "Do not reboot the device".

✧ *No Activation Interval*

The interval for rebooting the device when the device is unable to be registered on the EGPRS/LTE network. 0 means "Do not reboot the device".

- *Send Failure Timeout*

  The length of time (in minutes) before rebooting the device when the device fails to send a message. 0 means "Do not reboot the device".

- *No CAN Interval*

  The time interval in minutes for rebooting the device when the device cannot receive CANBUS messages from CAN100. 0 means "Do not reboot the device".

- *Random Reboot Time*

  When the reboot periodically time is reached, the device will reboot at a random moment between <Time> and <Time> + <Random Reboot Time>. 0 means "Do not use <Random Reboot Time>".

### 3.2.4 GAM (GNSS-Assisted Motion)

The command **AT+GTGAM** is used for assisting in measuring motion with GNSS if the sensor detects stationary state while the vehicle is ignition on.

| Example: |
|---|
| AT+GTGAM=gv58lau,1,1,25,10,60,60,,,,,0006$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | GAM | GAM |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 \| 1 | 1 |
| | Speed Mode | 1 | 0 \| 1 | 1 |
| | Motion Speed Threshold | <=2 | 5 - 50(km/h) | 25 |
| | Motion Cumulative Time | <=3 | 10 - 100(s) | 10 |
| | Motionless Cumulative Time | <=3 | 10 - 250(s) | 60 |
| | GNSS Fix Failure Timeout | <=4 | 5 - 1800(s) | 60 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

- *Password*

  The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".
- *Mode*

  The working mode of the GNSS-assisted motion measurement function.
  - **0** - Disable this function.
  - **1** - Enable this function.
- *Speed Mode*

  Enable/disable the use of GNSS speed to assist with motion status measurement based on motion sensor state.
  - **0** - Disable.
  - **1** - Enable.
- *Motion Speed Threshold*

  The speed threshold which is combined with GNSS speed to measure the status of movement.
- *Motion Cumulative Time*

  If the average speed is higher than <Motion Speed Threshold> for <Motion Cumulative Time>, the device is considered to be in moving state.
- *Motionless Cumulative Time*

  If the average speed is lower than <Motion Speed Threshold> for <Motionless Cumulative Time>, the device is considered to be in stationary state.
- *GNSS Fix Failure Timeout*

  If the time of GNSS fix is more than <GNSS Fix Failure Timeout>, the device will update motion status by motion sensor.

### 3.2.5 HMC (Hour Meter Count)

The command **AT+GTHMC** is used to measure the accumulated use time of the device with each actuation of ignition on. When the device sends **FRI** (ASCII)/(HEX), **IGN** (ASCII)/(HEX) or **IGF** (ASCII)/(HEX) message, <Hour Meter Count> will be included in these reports.

```
Example:
AT+GTHMC=gv58lau,1,00000:00:00,,,,,,,,,0018$
```

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | HMC | HMC |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-2 | 0 |
| | Initial Hour Meter Count | 11-13 | 00000:00:00 - 1193000:00:00 | 00000:00:00 |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

Enable/disable the hour meter count function. If the hour meter count function is enabled, the hour meter count will be increased when the device detects the vehicle ignition is turned on.

- **0** - Disable the hour meter count function.
- **1** - Difference Calculation Mode.
  The current <Hour Meter Count> reported in **+RESP:GTFRI**, **+RESP:GTIGN** and **+RESP:GTIGF** is equal to (=) <Hour Meter Count> + current <Initial Hour Meter Count> - last <Initial Hour Meter Count>.
- **2** - Initial Value Mode.
  The current <Hour Meter Count> reported in **+RESP:GTFRI**, **+RESP:GTIGN** or **+RESP:GTIGF** is the same as the value of <Initial Hour Meter Count>.

✧ *Initial Hour Meter Count*

The initial value of <Hour Meter Count> ranges from 00000:00:00 to 1193000:00:00. It consists of three parts separated by ":", the first part is the hour digit and the length of it is between 5 to 7 bytes, the second part is the 2-byte minute digit, and the last part is the 2-byte second digit. When the ignition is turned on for the first time, the <Hour Meter Count> reported in **+RESP:GTFRI** , **+RESP:GTIGN** or **+RESP:GTIGF** will be increased based on this value.

### 3.2.6 HRM (HEX Report Mask)

The **AT+GTHRM** command consists of <+ACK Mask>, <+RSP Mask>, <+EVT Mask>, <+INF Mask>, <+HBD Mask>, <+CRD Mask> , <+DAT Mask>, <+CAN Mask>which control the composition of the corresponding HEX report message. In each HEX report message, the corresponding mask for the report indicates which part is reported.

| |
|---|
| Example:<br>AT+GTHRM=gv58lau,,,6F,FE1FBF,FE1FBF,FF7D,EF,7F,7D,7FF,,0018$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | HRM | HRM |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Reserved | 0 | | |
| | Reserved | 0 | | |
| | +ACK Mask | <=2 | 0 - FF | 6F |
| | +RSP Mask | <=16 | 0 - FFFFFFFFFFFFFFFF | FE1FBF |
| | +EVT Mask | <=16 | 0 - FFFFFFFFFFFFFFFF | FE1FBF |
| | +INF Mask | <=16 | 0 - FFFFFFFFFFFFFFFF | FF7D |
| | +HBD Mask | <=2 | 0 - FF | EF |
| | +DAT Mask | <=8 | 0 - FFFFFFFF | 7F |
| | +CRD Mask | <=4 | 0 - FFFF | 7D |
| | +CAN Mask | <=8 | 0 - FFFFFFFF | 7FF |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

- ✧ *Password*

  The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

- ✧ *+ACK Mask*

  Component mask of the acknowledgement received.

| Mask Bit | Item |
|---|---|
| Bit 7 | Reserved |
| Bit 6 | <Count Number> |
| Bit 5 | <Send Time> |
| Bit 4 | <Device Name> |
| Bit 3 | <Firmware Version> |
| Bit 2 | <Protocol Version> |
| Bit 1 | <Device Type> |
| Bit 0 | <Length> |

- ✧ *+RSP Mask*

  Component mask of the location report message.

| Mask Bit | Item |
|---|---|
| Bit 31 | Extend Mask Flag |
| Bit 30 | Reserved |
| ... | Reserved |

| Mask Bit | Item |
|---|---|
| Bit 27 | Reserved |
| Bit 26 | Reserved |
| Bit 25 | <CAN Data> |
| Bit 24 | Reserved |
| Bit 23 | <Total Hour Meter Count> |
| Bit 22 | <Current Hour Meter Count> |
| Bit 21 | <Total Mileage> |
| Bit 20 | <Current Mileage> |
| Bit 19 | <Satellite in Use> |
| Bit 18 | <Motion Status> |
| Bit 17 | <Digital IO Status> |
| Bit 16 | Reserved |
| Bit 15 | Reserved |
| Bit 14 | Reserved |
| Bit 13 | Reserved |
| Bit 12 | <External Power Voltage> |
| Bit 11 | <Battery Level> |
| Bit 10 | <Firmware Version> |
| Bit 9 | <Protocol Version> |
| Bit 8 | <Device Type> |
| Bit 7 | <Length> |
| Bit 6 | <Device Name> |
| Bit 5 | <Count Number> |
| Bit 4 | <Send Time> |
| Bit 3 | <MCC / MNC / LAC / Cell ID / Reserved> |
| Bit 2 | <Altitude> |
| Bit 1 | <Azimuth> |
| Bit 0 | <Speed> |

✧ *+RSP Expansion Mask*

Bit 32 - Bit 63 of <+RSP Mask>, component mask of the expand information report. Each bit indicates which expand information is included when the device reports the message **+RSP** (HEX).

| Mask Bit | Item |
|---|---|
| Bit 2 - Bit 31 | Reserved |
| Bit 1 | Enhanced <Satellite in Use> |
| Bit 0 | Reserved |

Note

If Bit 1 of <+RSP Expansion Mask> is set to 1, Bit 19 of <+RSP Mask> will be forced to 0.

✧ *+EVT Mask*

Component mask of the event report message.

| Mask Bit | Item |
|---|---|
| Bit 31 | Extend Mask Flag |
| Bit 30 | Reserved |
| Bit 29 | Reserved |
| Bit 28 | Reserved |
| Bit 27 | Reserved |
| Bit 26 | Reserved |
| Bit 25 | <CAN Data> |
| Bit 24 | Reserved |
| Bit 23 | <Total Hour Meter Count> |
| Bit 22 | <Current Hour Meter Count> |
| Bit 21 | <Total Mileage> |
| Bit 20 | <Current Mileage> |
| Bit 19 | <Satellite in Use> |
| Bit 18 | <Motion Status> |
| Bit 17 | <Digital IO Status> |
| Bit 16 | Reserved |
| Bit 15 | Reserved |
| Bit 14 | Reserved |
| Bit 13 | Reserved |
| Bit 12 | <External Power Voltage> |
| Bit 11 | <Battery Level> |
| Bit 10 | <Firmware Version> |
| Bit 9 | <Protocol Version> |
| Bit 8 | <Device Type> |
| Bit 7 | <Length> |
| Bit 6 | <Device Name> |
| Bit 5 | <Count Number> |
| Bit 4 | <Send Time> |
| Bit 3 | <MCC / MNC / LAC / Cell ID / Reserved> |
| Bit 2 | <Altitude> |
| Bit 1 | <Azimuth> |
| Bit 0 | <Speed> |

✧ *+EVT Expansion Mask*

Bit 32 - Bit 63 of <+EVT Mask>, component mask of the expand event report. Each bit indicates which expand information is included when the device reports the message **+EVT** (HEX).

| Mask Bit | Item |
|---|---|
| Bit 2 - Bit 31 | Reserved |
| Bit 1 | Enhanced <Satellite in Use> |
| Bit 0 | Reserved |

Note

If Bit 1 of <+EVT Expansion Mask> is set to 1, Bit 19 of <+EVT Mask> will be forced to 0.

✧ *+INF Mask*

Component mask of the information report message. Bit 8 - Bit 15 indicate which groups of information items are included when the device reports the message **INF** (HEX).

| Mask Bit | Item |
|---|---|
| Bit 15 | +RESP:GTGSM |
| Bit 14 | +RESP:GTTMZ |
| Bit 13 | +RESP:GTCSQ |
| Bit 12 | +RESP:GTCID |
| Bit 11 | +RESP:GTBAT |
| Bit 10 | +RESP:GTGPS |
| Bit 9 | +RESP:GTIOS |
| Bit 8 | +RESP:GTVER |
| Bit 7 | <INF Expansion Mask> |
| Bit 6 | <Count Number> |
| Bit 5 | <Send Time> |
| Bit 4 | <Firmware Version> |
| Bit 3 | <Protocol Version> |
| Bit 2 | <Device Type> |
| Bit 1 | <Device Name> |
| Bit 0 | <Length> |

✧ *INF Expansion Mask*

Bit 16 - Bit 31 of <+INF Mask>, component mask of the expand information report. Each bit indicates which expand information is included when the device reports the message **+RESP:GTINF**.

| Mask Bit | Item |
|---|---|
| Bit 15 | <INF Expansion2 Mask> |
| Bit 14 | +RESP:GTRSV |
| Bit 13 | Reserved |
| Bit 12 | +RESP:GTSCS |
| ... | Reserved |
| Bit 9 | +RESP:GTBTI |

| Mask Bit | Item |
|---|---|
| Bit 8 | Reserved |
| Bit 7 | Reserved |
| Bit 6 | +RESP:GTCML |
| Bit 5 | Reserved |
| Bit 4 | Reserved |
| Bit 3 | +RESP:GTCSN |
| Bit 2 | +RESP:GTCVN |
| Bit 1 | Reserved |
| Bit 0 | +RESP:GTGSV |

✧ *INF Expansion2 Mask*

Bit 32 - Bit 47 of <+INF Mask>, component mask of the expand information report. Each bit indicates which expand information is included when the device reports the message **+RESP:GTINF**.

| Mask Bit | Item |
|---|---|
| Bit 7 - 15 | Reserved |
| Bit 6 | Reserved |
| Bit 5 | +RESP:GTASV |
| Bit 4 | <Network Type> |
| Bit 3 | Reserved |
| Bit 2 | +RESP:GTBSV |
| Bit 1 | Reserved |
| Bit 0 | Reserved |

✧ *+HBD Mask*

Component mask of the heartbeat data.

| Mask Bit | Item |
|---|---|
| Bit 7 | <UID> |
| Bit 6 | <Count Number> |
| Bit 5 | <Send Time> |
| Bit 4 | <Device Name> |
| Bit 3 | <Firmware Version> |
| Bit 2 | <Protocol Version> |
| Bit 1 | <Device Type> |
| Bit 0 | <Length> |

✧ *+DAT Mask*

Component mask of the data report message.

| Mask Bit | Item |
|---|---|
| Bit 7 - 31 | Reserved |

| Mask Bit | Item |
|---|---|
| Bit 6 | <Count Number> |
| Bit 5 | <Send Time> |
| Bit 4 | <Device Name> |
| Bit 3 | <Firmware Version> |
| Bit 2 | <Protocol Version> |
| Bit 1 | <Device Type> |
| Bit 0 | <Length> |

✧ *+CRD Mask*

Component mask of the crash data packet.

| Mask Bit | Item |
|---|---|
| Bit 7 - 15 | Reserved |
| Bit 6 | <Count Number> |
| Bit 5 | <Send Time> |
| Bit 4 | <Firmware Version> |
| Bit 3 | <Protocol Version> |
| Bit 2 | <Device Type> |
| Bit 1 | <Device Name> |
| Bit 0 | <Length> |

✧ *+CAN Mask*

Component mask of the CANBUS Information packet in HEX format.

| Mask Bit | Item |
|---|---|
| Bit 11-31 | Reserved |
| Bit 10 | <Firmware Version> |
| Bit 9 | <Protocol Version> |
| Bit 8 | <Device Type> |
| Bit 7 | <Length> |
| Bit 6 | <Device Name> |
| Bit 5 | <Count Number> |
| Bit 4 | <Send Time> |
| Bit 3 | <MCC / MNC / LAC / Cell ID / Reserved> |
| Bit 2 | <Altitude> |
| Bit 1 | <Azimuth> |
| Bit 0 | <Speed> |

### 3.2.7 OWH (Outside Working Hours)

To protect the privacy of the driver when he is off duty, the device could be configured to report empty location information outside working hours.

The command **AT+GTOWH** is used to define the working hours and the working mode to protect the privacy. If this function is enabled in non-working hours, in all ASCII format reports except **SOS** (ASCII)/(HEX), **JDR** (ASCII)/(HEX) and **JDS** (ASCII)/(HEX), the fields Latitude, Longitude, MCC, MNC, LAC, Cell ID will be empty, <Position Append Mask> will be 00 and the optional parameters next to it will not exist. Meanwhile, in HEX format reports where location should be hidden, the fields Latitude and Longitude will be filled with 0x054C5638, and the fields MCC, MNC, LAC Cell ID will be filled with 0.

Example:
AT+GTOWH=gv58lau,3,1F,0900,1200,1300,1800,,,0,0,0,0,0,,,,,0012$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | OWH | OWH |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-3 | 0 |
| | Day of Work | <=2 | 0 - 7F | 1F |
| | Working Hours Start1 | 4 | HHMM | 0900 |
| | Working Hours End1 | 4 | HHMM | 1200 |
| | Working Hours Start2 | 4 | HHMM | 1300 |
| | Working Hours End2 | 4 | HHMM | 1800 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Digital Input ID | 1 | 0 - 1 | 0 |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

A numeral to indicate the working mode of this function.

- **0** - Disable this function.
- **1** - Manual start mode.
  By connecting an external unit to a specified digital input of the device, the driver manually enables time check. If it is outside of working hours, the device will hide the location information in the report messages. Otherwise, the location information will be reported normally.
- **2** - Full manual mode.
  By connecting an external unit to a specified digital input of the device, the driver has full control of the privacy protection. The device will not check the time against the working hours range automatically. It hides location information if the input is activated and reports location information normally if the input is deactivated.
- **3** - Automatic mode.
  In this mode, the device will ignore the status of the digital input. It will automatically check the current time against the working hours range. If it is outside of working hours, location information will be hidden. Otherwise, location information will be reported normally.

✧ *Day of Work*

It specifies the working days in a week in a bitwise manner.

- **Bit 0** - for Monday
- **Bit 1** - for Tuesday
- **Bit 2** - for Wednesday
- **Bit 3** - for Thursday
- **Bit 4** - for Friday
- **Bit 5** - for Saturday
- **Bit 6** - for Sunday

For each bit, 0 means "Off Day", and 1 means "Working Day".

✧ *Working Hours Start1, Working Hours End1*

The first period of working hours in a day.

✧ *Working Hours Start2, Working Hours End2*

The second period of working hours in a day.

✧ *Digital Input ID*

The input ID used to trigger this function when <Mode> is 1 .

The working parameters of the specified input must be set by AT+GTDIS first. If interruptible digital input is used, please connect slide switch instead of tact button to the input for this function.

✧ *Output ID, Output Status, Duration and Toggle Times*

If this function is enabled and it is outside of working hours, the specified wave will be output on the specified output.

### 3.2.8 PDS (Preserving Device's States)

The command **AT+GTPDS** is used to preserve specified logic state for the device. The function works according to the <Mode> setting, and the logic state to be saved are selected according to the value of <Mask>.

| |
|---|
| Example:<br>AT+GTPDS=gv58lau,1,11,,,,,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | PDS | PDS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-2 | 0 |
| | Mask | <=4 | 0 - FFFF | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of preserving special logic state of the device.

- **0** - Disable this function.
- **1** - Preserve specified logic state of the device according to the value of <Mask>.
- **2** - Reset all the specified logic state listed in <Mask> after receiving the command, and then preserve the specified logic state according to the value of <Mask>.

✧ *Mask*

Bitwise mask to configure the device state to be preserved.

Each bit represents a state.

- **Bit 0** - State of GEO
- **Bit 1** - Reserved
- **Bit 2** - Reserved
- **Bit 3** - Information of last known position

- **Bit 4** - State of ignition
- **Bit 5** - State of wave shape 1
- **Bit 6** - State of digital input
- **Bit 7** - State of SPD
- **Bit 8** - State of SSR
- **Bit 9** - State of main power
- **Bit 10** - State of PEO
- **Bit 11** - Reserved
- **Bit 12** - Reserved

### 3.2.9 PIN (Auto-unlock PIN)

The command **AT+GTPIN** is used to configure the auto-unlock PIN function of the device. Some operators offer SIM card with PIN code protection by default. To make the device work with the PIN-protected SIM card, this command is used to configure the device to auto-unlock the SIM PIN with the preset PIN code.

| Example: |
|---|
| AT+GTPIN=gv58lau,1,1234,,,,,,0014$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | PIN | PIN |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Enable Auto-unlock PIN | 1 | 0 \| 1 | 0 |
| | PIN | 4 - 8 | '0'–'9' | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Enable Auto-unlock PIN*

Set it to 1 to enable the auto-unlock PIN function, and set it to 0 to disable this function.

✧ *PIN*

The code used to unlock the SIM PIN.

### 3.2.10 TMA (Time Adjustment)

The command **AT+GTTMA** is used to remotely adjust the local time of the device. Upon receiving this command, the device will set the time zone and daylight saving accordingly. Then it will use the given UTC time to adjust for the local time based on the time zone and daylight-saving setting. This command will also be a trigger for the device to start GNSS. After a successful GNSS fix, the device will update the local time with the GNSS UTC time again.

Example:
AT+GTTMA=gv58lau,+,3,52,0,20090917203500,1,,,,FFFF$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | TMA | TMA |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Sign | 1 | + \| - | + |
| | Hour Offset | <=2 | 0 - 12 | 0 |
| | Minute Offset | <=2 | 0 - 59 | 0 |
| | Daylight Saving Mode | 1 | 0 \| 1 | 0 |
| | UTC Time | 14 | YYYYMMDDHHMMSS | |
| | Network Time Checking | 1 | 0 \| 1 | 1 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Sign*

It indicates the positive or negative offset of the local time from UTC time.

✧ *Hour Offset*

UTC offset in hours.

✧ *Minute Offset*

UTC offset in minutes.

✧ *Daylight Saving Mode*

Enable/disable daylight saving time.

- **0** - Disable.
- **1** - Enable.

✧ *UTC Time*
UTC time used to adjust for the local time.

✧ *Network Time Checking*
A numeral to indicate whether to check GNSS UTC time with network time.
- **0** - Only correct device time when the device can't get GNSS UTC time.
- **1** - Always correct device time with network time.

## 3.2.11 WLT (Allowlist)

The command **AT+GTWLT** is used to configure a list of authorized phone numbers which are allowed to perform the location by call and SMS functions.

| Example: |
|---|
| AT+GTWLT=gv58lau,0,1,2,13813888888,13913999999,,,,,0018$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | WLT | WLT |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Call Filter | 1 | 0 - 1 \| 4 - 5 | 0 |
| | Start Index | <=2 | 1 - 10 | |
| | End Index | <=2 | 1 - 10 | |
| | Phone Number List | <=20*10 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*
The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Call Filter*
It configures the working mode of this function. If a bit is set to 1, only the phone numbers in allowlist will be valid for the corresponding feature. If a bit is set to 0, the corresponding allowlist will be ignored.
- **Bit 0** - Allowlist for location by call.
- **Bit 1** - Reserved.

- **Bit 2** - Allowlist for SMS. The <SMS Gateway> (refer to AT+GTSRI) is always available even if it is not in the allowlist.

✧ *Start Index, End Index*

The index range of the allowlist to which the phone numbers are to be updated. For example, if <Start Index> is set to 1 and <End Index> is set to 2, then the first two phone numbers in the allowlist will be updated by the numbers provided in the parameter <Phone Number List>. <Start Index> and <End Index> determine the number of phone numbers that will be updated. If either one is empty, there should be no <Phone Number List>.

✧ *Phone Number List*

A list of comma-separated phone numbers to be updated to the allowlist. The total number of the phone numbers is determined by <Start Index> and <End Index>. The format of the phone numbers includes area code and phone number. The area code is optional.

Note

If more phone numbers are needed, please adjust <Start Index> and <End Index> for appropriate setup. If some phone numbers in <Phone Number List> are empty, then the corresponding phone numbers will be deleted.
For example, to delete the 4th, 5th and 6th numbers of the <Phone Number List>, please set <Start Index> to 4 and set <End Index> to 6 and keep those three phone numbers of <Phone Number List> empty.

### 3.2.12 AVS (Accelerometer Virtual Ignition)

The command **AT+GTAVS** is used to configure parameters for detecting virtual ignition status based on motion status. It works when Accelerometer (motion status) virtual ignition mode is enabled by AT+GTVMS.

Note
Please make sure hard-wired ignition line is not connected.

| |
|---|
| Example:<br>AT+GTAVS=gv58lau,30,60,,,,000B$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | AVS | AVS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Rest Validity | <=3 | 1 - 255(s) | 30 |
| | Movement Validity | <=3 | 1 - 255(s) | 60 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

- *Password*

  The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".
- *Rest Validity*

  A time parameter to determine whether the device enters stationary state. The device will be considered in stationary state after the motion sensor detects stationary state and the stationary state is maintained for the period specified by the parameter <Rest Validity>.
- *Movement Validity*

  A time parameter to determine whether the device enters moving state. The device will be considered in moving state after the motion sensor detects movement and the moving state is maintained for the period specified by the parameter <Movement Validity>.

### 3.2.13 VVS (Voltage Virtual Ignition)

The command **AT+GTVVS** is used to configure parameters for detecting virtual ignition state by voltage. It works when Voltage Virtual Ignition mode is enabled by AT+GTVMS.

Note
Please make sure hard-wired ignition line is not connected.

Example:
AT+GTVVS=gv58lau,13500,600,10,1,10,000B$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | VVS | VVS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Ignition On Voltage | <=5 | 250 - 28000(mV) | 13500 |
| | Voltage Offset | <=4 | 200 - 2000(mV) | 600 |
| | Ignition On Debounce | <=3 | 5 - 255(x2s) | 10 |
| | Smart Voltage Adjustment | 1 | 0 \| 1 | 1 |
| | Ignition Off Debounce | <=3 | 5 - 255(x2s) | 10 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

- *Password*
  The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".
- *Ignition On Voltage*
  The external power voltage in ignition on state. Different vehicles have different voltage in ignition on state. This parameter should be set very close to the original external power so that the device can detect ignition event more accurately.
- *Voltage Offset*
  The offset from <Ignition On Voltage> used to determine ignition on or ignition off state.
  If the voltage of the external power is higher than **<Ignition On Voltage> - <Voltage Offset>** and is maintained for <Ignition On Debounce> seconds, the device will consider it as virtual ignition on state.
  If the voltage of the external power is lower than **<Ignition On Voltage> - <Voltage Offset>** and is maintained for <Ignition Off Debounce> seconds, the device will consider it as virtual ignition off state.

Note

If necessary, the values of <Ignition On Voltage> and <Voltage Offset> are automatically adjusted based on the measured external supply voltage data for more accurate ignition judgment.

- *Ignition On Debounce*
  The debounce time before updating virtual ignition on state according to the external power voltage. Unit: second.
- *Smart Voltage Adjustment*
  Enable/disable smart voltage adjustment algorithm.
  - **0** - Disable. The values of <Ignition On Voltage> and <Voltage Offset> will remain unchanged.
  - **1** - Enable. The values of <Ignition On Voltage> and <Voltage Offset> will dynamically change according to the actual ignition on and off voltage.
- *Ignition Off Debounce*
  The debounce time before updating the virtual ignition off state according to the external power voltage.

### 3.2.14 VMS (Virtual Ignition Mode Selection)

The command **AT+GTVMS** is used to configure the mode of virtual ignition state detection.

| Example: |
|---|
| AT+GTVMS=gv58lau,7,03,03,1,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | VMS | VMS |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Virtual Ignition Mode | 1 | 0 \| 2 \| 4 \| 7 | 0 |
| | Virtual Ignition On Mask | 2 | 00 - 03 | 03 |
| | Virtual Ignition Off Mask | 2 | 00 - 03 | 03 |
| | Virtual Ignition On Logic | 1 | 0 \| 1 | 1 |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Virtual Ignition Mode*

A numeral to define the working mode of virtual ignition state detection.

- **0** - Disable the virtual ignition detection function.
- **2** - Voltage virtual ignition detection mode.
  The ignition state is related to the voltage of the external power supply. Please use the command AT+GTVVS to configure the parameters.
- **4** - Accelerometer virtual ignition detection mode.
  Ignition state can be indicated by the motion state determined by <Rest Validity> and <Movement Validity> defined in the AT+GTAVS command. Stationary state indicates ignition off and moving state indicates ignition on.
- **7** - Combined detection mode.
  In this mode, ignition on/off trigger conditions can be selected using parameters <Virtual Ignition On Mask> and <Virtual Ignition Off Mask>.
  Note
  <Virtual Ignition off Mask> must contain <Virtual Ignition On Mask> to prevent logic errors.

✧ *Virtual Ignition On Mask*

Bitwise mask to detect the ignition on event. The logic of each bit is controlled by the parameter <Virtual Ignition On Logic>.

- **Bit0 (01)** - Voltage virtual ignition detection
- **Bit1 (02)** - Motion status virtual ignition detection

✧ *Virtual Ignition Off Mask*

Bitwise mask to detect ignition off event. All bits matched are considered as ignition off event.

- **Bit0 (01)** - Voltage virtual ignition detection
- **Bit1 (02)** - Motion status virtual ignition detection

For example:

Bit (00000003): Voltage virtual ignition detection and motion status virtual ignition detection combined mode. Only when ignition off is detected by both Mode 2 and Mode 4 will the device be considered in ignition off state.

✧ *Virtual Ignition On Logic*

The logic of each bit in <Virtual Ignition On Mask>.

- **0** - AND logic. All bits matched are considered as ignition on event.
- **1** - OR logic. Any one bit matched is considered as ignition on event.

Note

Records of hard-wired ignition are forcibly saved when PDS mode is enabled. When the hard wired is recorded, virtual ignition detection will be ignored.

### 3.2.15 CMD (Command String Storage)

The **AT+GTCMD** command is used to store the commands to be used by the command AT+GTUDF.

| |
|---|
| Example:<br>AT+GTCMD=gv58lau,1,0,AT+GTRTO=gv58lau,0,,,,,,000B$,,,,,0005$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | CMD | CMD |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 \| 1 | 0 |
| | Command ID | <=2 | 0 - 31 | |
| | Command String | <=200 | AT command | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of storing command string.

- **0** - Delete the stored command.
- **1** - Add the stored command.

✧ *Command ID*

A numeral to identify the stored command.

✧ *Command String*

The whole content of the stored command.

### 3.2.16 UDF (User Defined Function)

The **AT+GTUDF** command is used to bind input events and stored commands. The input events will trigger the corresponding stored commands.

| |
|---|
| Example:<br>AT+GTUDF=gv58lau,0,0,FF,0,0,0,0,1,0,0,,,0005$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | UDF | UDF |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-2 | 0 |
| | Group ID | <=2 | 0 - 31 | |
| | Input ID Mask | <=16 | 0 - FFFFFFFFFFFFFFFF | 0 |
| | Debounce Time | <=5 | 0 - 86400(s) | 0 |
| | Inzizo Mask | <=5 | 00000 - FFFFF | 0 |
| | Outzizo Mask | <=5 | 00000 - FFFFF | 0 |
| | Command ID Mask | <=8 | 0 - FFFFFFFF | 0 |
| | Command Ack Mode | 1 | 0 \| 1 | 0 |
| | Inpeo Mask | <=5 | 00000 - FFFFF | 0 |
| | Outpeo Mask | <=5 | 00000 - FFFFF | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Input Improve Mask | <=16 | 0 - FFFFFFFFFFFFFFFF | 0 |
| | Report Mode | 1 | 0 \| 1 | 0 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the user defined function.

- **0** - Disable the group.
- **1** - Enable the group.
- **2** - Delete the group.

✧ *Group ID*

A numeral to identify the group of input events and stored commands to be executed.

✧ *Input ID Mask*

The bitwise mask to indicate the input events included in the group.

- **Bit 0 (00000001)** - Select ID1
- **Bit 1 (00000002)** - Select ID2
- **Bit 2 (00000004)** - Select ID3
- **Bit 3 (00000008)** - Select ID4

For example:

- **Bit (00000003)** - SelectID1 and ID2
- **Bit (00000017)** - Select ID1, ID2, ID3, and ID5

| ID | Bit | Item |
|---|---|---|
| 1 | Bit 0 | Power on finished |
| 2 | Bit 1 | Ignition on |
| 3 | Bit 2 | Ignition off |
| 4 | Bit 3 | Attached to the GPRS network |
| 5 | Bit 4 | Not attached to the GPRS network |
| 6 | Bit 5 | Registered on the GSM network |
| 7 | Bit 6 | Not registered on the GSM network |
| 8 | Bit 7 | Network roaming |
| 9 | Bit 8 | Network non-roaming |
| 11 | Bit 10 | GNSS is on |
| 12 | Bit 11 | GNSS is off |
| 13 | Bit 12 | The device is stationary |
| 14 | Bit 13 | The device is moving |
| 15 | Bit 14 | External charge inserted |
| 16 | Bit 15 | No external charge |
| 17 | Bit 16 | The device is charging |
| 18 | Bit 17 | The device is not charging |
| 21 | Bit 20 | Digital input 1 is activated |
| 22 | Bit 21 | Digital input 1 is deactivated |
| 23 | Bit 22 | SIM card is inserted |
| 24 | Bit 23 | SIM card is not inserted |
| 27 | Bit 26 | Inside the speed range |
| 28 | Bit 27 | Outside the speed range |
| 29 | Bit 28 | Messages need to be sent |
| 30 | Bit 29 | No messages need to be sent |
| 37 | Bit 36 | Network jamming is detected |
| 38 | Bit 37 | Network jamming is not detected |

| ID | Bit | Item |
|---|---|---|
| 52 | Bit 51 | The device fixes GNSS successfully |
| 53 | Bit 52 | The device fails to fix GNSS |
| 58 | Bit 57 | Output1 is activated |
| 59 | Bit 58 | Output1 is deactivated |
| 60 | Bit 59 | Output2 is activated |
| 61 | Bit 60 | Output2 is deactivated |
| 62 | Bit 61 | Output3 is activated |
| 63 | Bit 62 | Output3 is deactivated |

- *Debounce Time*

  The debounce time for input events before the specified stored commands are executed.
- *Inzizo Mask*

  The bitwise mask to indicate the input events within the circular GEO-Fence.

| ID | Bit | Item |
|---|---|---|
| 1 | Bit 0 | Inside the GEO 0 |
| 2 | Bit 1 | Inside the GEO 1 |
| 3 | Bit 2 | Inside the GEO 2 |
| 4 | Bit 3 | Inside the GEO 3 |
| 5 | Bit 4 | Inside the GEO 4 |
| 6 | Bit 5 | Inside the GEO 5 |
| 7 | Bit 6 | Inside the GEO 6 |
| 8 | Bit 7 | Inside the GEO 7 |
| 9 | Bit 8 | Inside the GEO 8 |
| 10 | Bit 9 | Inside the GEO 9 |
| 11 | Bit 10 | Inside the GEO 10 |
| 12 | Bit 11 | Inside the GEO 11 |
| 13 | Bit 12 | Inside the GEO 12 |
| 14 | Bit 13 | Inside the GEO 13 |
| 15 | Bit 14 | Inside the GEO 14 |
| 16 | Bit 15 | Inside the GEO 15 |
| 17 | Bit 16 | Inside the GEO 16 |
| 18 | Bit 17 | Inside the GEO 17 |
| 19 | Bit 18 | Inside the GEO 18 |
| 20 | Bit 19 | Inside the GEO 19 |

- *Outzizo Mask*

  The bitwise mask to indicate the input events outside the circular GEO-Fence.

| ID | Bit | Item |
|---|---|---|
| 1 | Bit 0 | Outside the GEO 0 |

| ID | Bit | Item |
|---|---|---|
| 2 | Bit 1 | Outside the GEO 1 |
| 3 | Bit 2 | Outside the GEO 2 |
| 4 | Bit 3 | Outside the GEO 3 |
| 5 | Bit 4 | Outside the GEO 4 |
| 6 | Bit 5 | Outside the GEO 5 |
| 7 | Bit 6 | Outside the GEO 6 |
| 8 | Bit 7 | Outside the GEO 7 |
| 9 | Bit 8 | Outside the GEO 8 |
| 10 | Bit 9 | Outside the GEO 9 |
| 11 | Bit 10 | Outside the GEO 10 |
| 12 | Bit 11 | Outside the GEO 11 |
| 13 | Bit 12 | Outside the GEO 12 |
| 14 | Bit 13 | Outside the GEO 13 |
| 15 | Bit 14 | Outside the GEO 14 |
| 16 | Bit 15 | Outside the GEO 15 |
| 17 | Bit 16 | Outside the GEO 16 |
| 18 | Bit 17 | Outside the GEO 17 |
| 19 | Bit 18 | Outside the GEO 18 |
| 20 | Bit 19 | Outside the GEO 19 |

✧ *Command ID Mask*

The bitwise mask of the stored commands to be executed after the state of the group becomes TRUE (i.e. all the included input events occur).

✧ *Command Ack Mode*

A numeral to indicate whether to return an acknowledgement message after a stored command is executed.

- **0** - Do not send an acknowledgement message when a stored command is executed.
- **1** - Send an acknowledgement message when a stored command is executed.

✧ *Inpeo Mask*

The bitwise mask to indicate the input events within the polygon GEO-Fence.

| ID | Bit | Item |
|---|---|---|
| 1 | Bit 0 | Inside the PEO 0 |
| 2 | Bit 1 | Inside the PEO 1 |
| 3 | Bit 2 | Inside the PEO 2 |
| 4 | Bit 3 | Inside the PEO 3 |
| 5 | Bit 4 | Inside the PEO 4 |
| 6 | Bit 5 | Inside the PEO 5 |
| 7 | Bit 6 | Inside the PEO 6 |

| ID | Bit | Item |
|---|---|---|
| 8 | Bit 7 | Inside the PEO 7 |
| 9 | Bit 8 | Inside the PEO 8 |
| 10 | Bit 9 | Inside the PEO 9 |
| 11 | Bit 10 | Inside the PEO 10 |
| 12 | Bit 11 | Inside the PEO 11 |
| 13 | Bit 12 | Inside the PEO 12 |
| 14 | Bit 13 | Inside the PEO 13 |
| 15 | Bit 14 | Inside the PEO 14 |
| 16 | Bit 15 | Inside the PEO 15 |
| 17 | Bit 16 | Inside the PEO 16 |
| 18 | Bit 17 | Inside the PEO 17 |
| 19 | Bit 18 | Inside the PEO 18 |
| 20 | Bit 19 | Inside the PEO 19 |

- *Outpeo Mask*

  The bitwise mask to indicate the input events outside the polygon GEO-Fence.

| ID | Bit | Item |
|---|---|---|
| 1 | Bit 0 | Outside the PEO 0 |
| 2 | Bit 1 | Outside the PEO 1 |
| 3 | Bit 2 | Outside the PEO 2 |
| 4 | Bit 3 | Outside the PEO 3 |
| 5 | Bit 4 | Outside the PEO 4 |
| 6 | Bit 5 | Outside the PEO 5 |
| 7 | Bit 6 | Outside the PEO 6 |
| 8 | Bit 7 | Outside the PEO 7 |
| 9 | Bit 8 | Outside the PEO 8 |
| 10 | Bit 9 | Outside the PEO 9 |
| 11 | Bit 10 | Outside the PEO 10 |
| 12 | Bit 11 | Outside the PEO 11 |
| 13 | Bit 12 | Outside the PEO 12 |
| 14 | Bit 13 | Outside the PEO 13 |
| 15 | Bit 14 | Outside the PEO 14 |
| 16 | Bit 15 | Outside the PEO 15 |
| 17 | Bit 16 | Outside the PEO 16 |
| 18 | Bit 17 | Outside the PEO 17 |
| 19 | Bit 18 | Outside the PEO 18 |
| 20 | Bit 19 | Outside the PEO 19 |

✧ *Input Improve Mask*

This parameter is used to improve the logic of <Input ID Mask>. After enabling <Input Improve Mask>, the event in <Input ID Mask> is triggered only if the device state changes compared to the state before the reboot.

| ID | Bit | Item |
|---|---|---|
| 1 - 18 | Bit 0 - 17 | Reserved |
| 21 | Bit 20 | Digital input 1 is active |
| 22 | Bit 21 | Digital input 1 is inactive |
| 23 - 30 | Bit 22 - 29 | Reserved |

✧ *Report Mode*

When triggering the corresponding storage command, whether to report **UDF** (ASCII)/(HEX) to obtain the corresponding <Group ID>.

- **0** - Do not report **+RESP:GTUDF**
- **1** - Report **+RESP:GTUDF**

Note

The maximum number of the stored commands to be executed in a group is five.

## 3.2.17 UDS (User Defined Speed Event)

The command **AT+GTUDS** is used to configure speed thresholds, it will trigger the corresponding <Group ID> in **AT+GTUDF** when the conditions are met. To trigger the command **AT+GTUDF**, the speed threshold needs to be met simultaneously with the Mask selected in **AT+GTUDF**.

Example:
AT+GTUDS=gv58lau,0,0,0,,60,,,,000C$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | UDS | UDS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Group ID | <=2 | 0 - 31 | 0 |
| | Mode | 1 | 0 \| 1 | 0 |
| | Speed Threshold | <=3 | 0 - 400(km/h) | 0 |
| | Reserved | 0 | | |
| | Validity | <=4 | 0 - 3600(s) | 60 |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *Group ID*

A number to indicate how many sets of speed thresholds. Please refer to the parameter <Group ID> in **AT+GTUDF**.

✧ *Mode*

- **0** - Disable.
- **1** - Enable.

  In this mode, the corresponding Group ID in **AT+GTUDF** will be triggered when the speed trigger condition is met.

✧ *Speed Threshold*

The minimum value that triggers **AT+GTUDF** commands.

✧ *Validity*

If the speed meets the alarm condition and is maintained longer than <Validity>, the speed alarm will be triggered.

### 3.2.18 GNA (GNSS Assistance)

The command **AT+GTGNA** is used to configure GNSS assistance data download and update.

Example:
AT+GTGNA=gv58lau,1,,http://offline-live1.services.u-blox.com/,,,,,,,,FFFF$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | GNA | GNA |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 \| 1 | 0 |
| | Reserved | 0 | | |
| | Offline URL | <= 50 | '0'-'9' 'a'-'z' 'A'-'Z' ':' '/' '.' | http://offline-live1.services.u-blox.com/ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

A numeral which indicates whether to enable AGPS. AGPS helps increase the chances of getting GNSS position successfully and reduces the time needed to get GNSS position. This parameter is the same as <AGPS Mode> in the AT+GTCFG command.

- **0** - Disable the AGPS function.
- **1** - Enable the AGPS function.

✧ *Offline URL*

It specifies the URL to download GNSS assistance offline data.

Note

The parameters <Mode> and <Offline URL> will not be reset by the AT+GTRTO-4 command.

## 3.3 Bluetooth Settings

This section describes the commands related to the Bluetooth setting and Bluetooth accessories configuration. Please refer to the details below.

### 3.3.1 BTS (Bluetooth Setting)

The command **AT+GTBTS** is used to configure Bluetooth settings for the device to report certain events.

| |
|---|
| Example:<br>AT+GTBTS=gv58lau,1,,GV58LAU%IMEI,,3,0,1D03,0003,0,123456,,,,,,,,,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Command Word | 3 | BTS | BTS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 \| 1 | 1 |
| | Reserved | 0 | | |
| | Bluetooth Name | <=18 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' '%' | GV58LAU%IMEI |
| | Reserved | 0 | | |
| | Discoverable Mode | 1 | 0-3 | 3 |
| | Discoverable Time | <=4 | 0 - 1440(min) | 0 |
| | Bluetooth Report Mask | 4 | 0000 - FFFF | 1D03 |
| | Bluetooth Event Mask | 4 | 0000 - FFFF | 0003 |
| | PIN Code Mode | 1 | 0 \| 1 | 0 |
| | PIN Code | 4\|6 | '0'–'9' | 123456 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the Bluetooth.

- **0** - Disable Bluetooth.
- **1** - Enable Bluetooth.

✧ *Bluetooth Name*

The name of the device for Bluetooth identification. The "%IMEI" in the Bluetooth broadcast name is replaced with "_" and IMEI number.

✧ *Discoverable Mode*

The mode to configure the Bluetooth to be non-discoverable or discoverable for the period according to <Discoverable Time>.

- **0** - Non-discoverable mode.
- **1** - General discoverable mode: The device will remain discoverable for <Discoverable Time> minutes after the ignition is turned on.
- **2** - General discoverable mode: The device will remain discoverable for <Discoverable Time> minutes after the ignition is turned off.
- **3** - General discoverable mode: The device will remain discoverable for <Discoverable Time> minutes after it is powered on.

✧ *Discoverable Time*

The time period for the device to remain discoverable. If it is set to 0, the device will always be discoverable when a specific condition as described in <Discoverable Mode> is met.

✧ *Bluetooth Report Mask*

Bitwise mask to configure the composition of Bluetooth information in report messages.

- **Bit 0** - for <Bluetooth Name>
- **Bit 1** - for <Bluetooth MAC Address>
- **Bit 2 ... Bit 7** - Reserved
- **Bit 8** - for <Peer Role>
- **Bit 9** - Reserved
- **Bit 10** - <Peer Address Type>
- **Bit 11** - <Peer MAC Address>
- **Bit 12** - <Peer Name>
- **Bit 13 ... Bit 15** - Reserved

For each bit, set it to 1 to enable the corresponding component in the report, and set it to 0 to disable.

This mask is valid for **+RESP:GTBCS**, **+RESP:GTBDS**, **+RESP:GTBAR** messages.

✧ *Bluetooth Event Mask*

Bitwise mask to configure which event report should be sent to the backend server.

- **Bit 0** - for **BCS** (ASCII)/(HEX)
- **Bit 1** - for **BDS** (ASCII)/(HEX)

✧ *PIN Code Mode*

It defines whether a PIN code for pairing is needed or not.

- **0** - No PIN code is needed.
- **1** - PIN code is needed for pairing.

✧ *PIN Code*

PIN code for pairing if needed.

### 3.3.2 BAS (Bluetooth Accessory)

The command **AT+GTBAS** is used for device scanning or connecting Bluetooth accessories to obtain data such as humidity and temperature. To use this command, the parameter <Mode> in the command AT+GTBTS must be enabled.

Example:
AT+GTBAS=gv58lau,0,6,5,WTH301,7805412CF340,7FBF,30,2400,,0,0,10,2,300,0,20,30,2,300,,0,0,0,0,,,FFFF$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | BAS | BAS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Index | 1 | 0 - 9 | 0 |
| | Accessory Type | <=2 | 0-2 \| 4 \| 6-8 \| 10-13 | 0 |
| | Accessory Model | 1 | 0-5 | 0 |
| | Accessory Name | <=20 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' | |
| | Accessory MAC | 12 | 000000000000 - FFFFFFFFFFFF | FFFFFFFFFFFF |
| | Accessory Append Mask | <=4 | 0 - FFFF | 7FBF |
| | Reserved | 0 | | |
| | Low Voltage Threshold | <=4 | 0 - 5000(mV) | 2400 |
| | Reserved | 0 | | |
| | Accessory Parameters (Optional) | | | |
| | Reserved | 0 | | |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Index*

The index of the Bluetooth accessory.

✧ *Accessory Type*

The type of the Bluetooth accessory which is defined in the <Index>. The following is supported now:

- **0** - No Bluetooth accessory.

- **1** - Escort Bluetooth accessory.
- **2** - Beacon temperature sensor.
- **4** - CAN accessory.
- **6** - Multi-functional beacon sensor.
- **7** - Technoton accessory.
- **8** - External Input Output Bluetooth accessory.
- **10** - Mechatronics Bluetooth accessory.
- **11** - Magnet Bluetooth accessory.
- **12** - BLE TPMS sensor.
- **13** - Relay Bluetooth accessory.

✧ *Accessory Model*

The model of the Bluetooth accessory which is defined in <Accessory Type>.

The following is supported now:

- The model of Escort Bluetooth accessory (<Accessory Type> is 1):
  - **0** - TD_BLE fuel sensor
  - **3** - Angle sensor
- The model of beacon temperature sensor (<Accessory Type> is 2):
  - **0** - WTS300 (Temperature sensor)
  - **1** - Temperature ELA
- The model of CAN accessory (<Accessory Type> is 4):
  - **0** - BLE CAN100
- The model of multi-functional beacon sensor (<Accessory Type> is 6):
  - **2** - WTH300 (Temperature and humidity sensor)
  - **3** - RHT ELA (Temperature and humidity sensor)
  - **4** - WMS301 (Door sensor with embedded temperature and humidity sensor)
  - **5** - WTH301 (Temperature and humidity sensor)
- The model of Technoton accessory (<Accessory Type> is 7):
  - **0** - DUT-E S7 (Fuel level sensor)
  - **1** - DFM 100 S7 (Fuel flowmeter sensor)
  - **2** - DFM 250DS7 (Fuel flowmeter sensor)
  - **3** - GNOM DDE S7 (Axle load sensor)
  - **4** - GNOM DP S7 (Axle load sensor)
- The External Input Output Bluetooth accessory (<Accessory Type> is 8):
  - **0** - WBC300 (External input output sensor)

  Note

  The current version supports only one WBC300 configuration.
- The model of Mechatronics accessory (<Accessory Type> is 10):
  - **0** - Fuel sensor
  - **1** - Angle sensor
- The model of Magnet sensor (<Accessory Type> is 11):
  - **0** - MAG ELA (Used to check whether the door is open or closed)
- The model of TPMS sensor accessory (<Accessory Type> is 12):
  - **0** - MLD BLE TPMS (ATP100/ ATP102)
- The model of Relay sensor (<Accessory Type> is 13):

– **0** - WRL300 sensor

✧ *Accessory Name*

The name of the Bluetooth accessory. For details about whether the accessory can be connected by name, please refer to Appendix.

✧ *Accessory MAC*

The MAC address of the Bluetooth accessory. If <Accessory MAC> of the Bluetooth accessory is valid and the MAC address of the Bluetooth accessory is unique, the device will use the MAC address to scan or connect Bluetooth accessories. If <Accessory MAC> is the default value, the device will search for the Bluetooth accessory by accessory name.

✧ *Accessory Append Mask*

If the device is connected with the Bluetooth accessory, and Bit 8 (for <Bluetooth Accessory Data>) of <ERI Mask> is set to 1, the device will report Bluetooth accessory data via **ERI** (ASCII)/(HEX) instead of **FRI** (ASCII)/(HEX).

This mask is used to configure the accessory data fields to be reported in the **ERI** (ASCII)/(HEX) and **BAA** (ASCII)/(HEX). messages. To obtain the <Accessory Append Mask> supported by the accessory, refer to the BLE Accessory Appendix.

If Bit14 is set to 1, the device will report <Relay state> but not report <Relay Config Result> in the message +RESP:GTERI.

| Mask Bit | Item | Description |
|---|---|---|
| Bit 0 | <Accessory Name> | Accessory name |
| Bit 1 | <Accessory MAC> | Accessory MAC |
| Bit 2 | <Accessory Status> | Accessory Bluetooth connection status |
| Bit 3 | <Accessory Battery Voltage> | Accessory battery voltage |
| Bit 4 | <Accessory Temperature> | Accessory temperature |
| Bit 5 | <Accessory Humidity> | Accessory humidity |
| Bit 6 | Reserved | Reserved |
| Bit 7 | <Accessory Input Output Data> | Including <Accessory Output status>, <Accessory Digital Input status>, <Accessory Analog Input voltage> |
| Bit 8 | <Accessory Event Notification Data> | Including <Accessory Mode>, <Accessory Event> |
| Bit 9 | <Tire pressure> | Tire pressure |
| Bit 10 | <Timestamp> | Timestamp |
| Bit 11 | <Enhanced Temperature> | Enhanced temperature |
| Bit 12 | <Magnet Data> | Including <Magnet ID>, <MAG Event Counter>, <Magnet state> |
| Bit 13 | <Accessory Battery Percentage> | Accessory battery percentage |

| Mask Bit | Item | Description |
|---|---|---|
| Bit 14 | <Relay Data> | Including <Relay Config Result>, <Relay State> |

✧ *Low Voltage Threshold*

It specifies the lower voltage limit. When the voltage of Bluetooth accessory is below this value, the device will report message **BAA** (ASCII)/(HEX) to the backend server. 0 means "Disable low voltage detection".

Note

The two functions of **AT+GTBAS** cannot be used at the same time.

✧ *Accessory Parameters (Optional)*

Some parameters for Bluetooth accessories. For different Accessory Types, there are different definitions.

If the Accessory Type is **1** (Escort Bluetooth Accessory), <Accessory Parameters (Optional)> will be unfolded as follows.

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Event Notification | 1 | 0 \| 1 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

- *Event Notification*

  It configures whether to enable event notification function.

  – **0** - Disable.

  – **1** - Enable. If a new event occurs on the accessory, the device will report the message **BAA** (ASCII)/(HEX).

If the Accessory Type is **2** (Beacon temperature sensor), <Accessory Parameters (Optional)> will be unfolded as follows.

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Mode | 1 | 0-3 | 0 |
| Low Temperature | <=3 | -40 - 80 ( ℃ ) | 0 |
| High Temperature | <=3 | -40 - 80 ( ℃ ) | 10 |
| Validity | <=2 | 1 - 10(s) | 2 |
| Send Interval | <=5 | 30 - 43200(s) | 300 |

The device will report the message **BAA** (ASCII)/(HEX). to the backend server when the temperature outside or inside the range is detected.

- *Mode*

  The working mode of the temperature alarm.

  – **0** - Disable temperature alarm.

  – **1** - Report temperature alarm if the current temperature is within the temperature range defined by <Low Temperature> and <High Temperature>.

– **2** - Report temperature alarm if the current temperature is outside the temperature range defined by <Low Temperature> and <High Temperature>.
– **3** - Report temperature alarm only once if the current temperature enters or exits the temperature range defined by <Low Temperature> and <High Temperature>. In this mode, <Send Interval> will be ignored.

- *Low Temperature*
  It specifies the lower temperature limit in centigrade.
- *High Temperature*
  It specifies the upper temperature limit in centigrade.
- *Validity*
  If the sensor detects that the ambient temperature meets the alarm condition, it will continuously check the temperature. If the temperature keeps meeting the alarm condition for <Validity> time, the temperature alarm will be triggered.

If the Accessory Type is **4** (CAN accessory), <Accessory Parameters (Optional)> will be unfolded as follows.

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| PIN Code | 4\|6 | 0000 - 9999 \| 000000 - 999999 | 0000 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

- *PIN Code*
  The code needs to be input when it is paired with accessories.

If the Accessory Type is **6** (Multi-functional beacon sensor), <Accessory Parameters (Optional)> will be unfolded as follows.

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Temperature Mode | 1 | 0 - 3 | 0 |
| Low Temperature | <=3 | -40 - 80 ( ℃ ) | 0 |
| High Temperature | <=3 | -40 - 80 ( ℃ ) | 10 |
| Temperature Validity | <=2 | 1 - 10(s) | 2 |
| Temperature Send Interval | <=5 | 30 - 43200(s) | 300 |
| Humidity Mode | 1 | 0 - 3 | 0 |
| Low Humidity | <=3 | 0 - 100 (rh) | 20 |
| High Humidity | <=3 | 0 - 100 (rh) | 30 |
| Humidity Validity | <=2 | 1 - 10(s) | 2 |
| Humidity Send Interval | <=5 | 30 - 43200(s) | 300 |

The device will report the **BAA** (ASCII)/(HEX) message to the backend server when the temperature and humidity meet alarm conditions.

- *Temperature Mode*

The working mode of the temperature alarm.

- **0** - Disable temperature alarm.
- **1** - Report temperature alarm if the current temperature is within the temperature range defined by <Low Temperature> and <High Temperature>.
- **2** - Report temperature alarm if the current temperature is outside the temperature range defined by <Low Temperature> and <High Temperature>.
- **3** - Report temperature alarm only once if the current temperature enters or exits the temperature range defined by <Low Temperature> and <High Temperature>. In this mode, <Temperature Send Interval> will be ignored.

- *Low Temperature*

  It specifies the lower temperature limit in centigrade.
- *High Temperature*

  It specifies the upper temperature limit in centigrade.
- *Temperature Validity*

  If the sensor detects that the ambient temperature meets the alarm condition, it will continuously check the temperature. If the temperature keeps meeting the alarm condition for <Temperature Validity> times, the temperature alarm will be triggered.
- *Humidity Mode*

  The working mode of the humidity alarm.
  - **0** - Disable humidity alarm.
  - **1** - Report humidity alarm if the current humidity is within the humidity range defined by <Low Humidity> and <High Humidity>.
  - **2** - Report humidity alarm if the current humidity is outside the humidity range defined by <Low Humidity> and <High Humidity>.
  - **3** - Report humidity alarm only once if the current humidity enters or exits the humidity range defined by <Low Humidity> and <High Humidity>. In this mode, <Humidity Send Interval> will be ignored.
- *Low Humidity*

  It specifies the lower humidity limit.
- *High Humidity*

  It specifies the upper humidity limit.
- *Humidity Validity*

  If the sensor detects that the ambient humidity meets the alarm condition, it will continuously check the humidity. If the humidity keeps reaching to the alarm condition for <Humidity Validity> time, the humidity alarm will be triggered.

If the Accessory Type is **7** (Technoton accessory), <Accessory Parameters (Optional)> will be unfolded as follows.

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| IGN Send Interval | <=5 | 0 \| 10 - 86400(s) | 30 |
| IGF Send Interval | <=5 | 0 \| 10 - 86400(s) | 60 |
| Actual Sensor Length | <=5 | 0 - 6000(mm) | 0 |
| Empty Frequency | <=7 | 0 - 4294967(Hz) | 0 |
| Full Frequency | <=7 | 0 - 4294967(Hz) | 0 |

- *IGN Send Interval*

  The interval for sending the report message **BAR** (ASCII)/(HEX) to the backend server when the ignition is on. 0 means "Do not report the message **+RESP:GTBAR**".
- *IGF Send Interval*

  The interval for sending the report message **+RESP:GTBAR** to the backend server when the ignition is off. 0 means "Do not report the message **+RESP:GTBAR**".
- *Actual Sensor Length*

  The actual length of sensor after cutting.
- *Empty Frequency*

  The frequency of empty sensor (not immersed in fuel).
- *Full Frequency*

  The frequency of sensor fully immersed in fuel.

Note

The parameters <Actual Sensor Length>, <Empty Frequency> and <Full Frequency> only supported by DUT-E fuel sensor, and if they are set to non-default values, the value of fuel level will send the message **ERI** (ASCII)/(HEX) instead of **+RESP:GTBAR**.

If the Accessory Type is **8** (External Input Output Bluetooth Accessory), <Accessory Parameters (Optional)> will be unfolded as follows.

Note

The input/output configuration of the accessory can be found in the commands AT+GTIEX, AT+GTAEX and AT+GTOEX.

If the Accessory Type is **10** (Mechatronics Bluetooth accessory), <Accessory Parameters (Optional)> will be unfolded as follows.

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Fuel Level Format | 1 | 0 \| 1 | 0 |
| Reserved | | | |
| Reserved | | | |
| Reserved | | | |
| Reserved | | | |

- *Fuel Level Format*

  The format of fuel level in the message.
  - **0** - The original value of fuel level.
  - **1** - The percentage of fuel level.

If the Accessory Type is **11** (Magnet Bluetooth accessory), <Accessory Parameters (Optional)> will be unfolded as follows.

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| MAG Event Notification | 1 | 0 \| 1 | 0 |
| Reserved | | | |
| Reserved | | | |
| Reserved | | | |
| Reserved | | | |

- *MAG Event Notification*

It configures whether to enable magnet event notification function.

– **0** - Disable.
– **1** - Enable. If a new event occurs on the accessory, the device will report the message **BAA** (ASCII)/(HEX).

If the Accessory Type is **12** (BLE TPMS sensor), this parameter will work as follows.

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | | | |
| Tire Pressure Alarm Mode | 1 | 0 - 3 | 0 |
| Low Tire Pressure | <=3 | 0 - 500(kPa) | 150 |
| High Tire Pressure | <=3 | 0 - 500(kPa) | 300 |
| Validity | <=2 | 1 - 10 | 2 |
| Alarm Send Interval | <=5 | 0 - 86400(s) | 60 |
| Available Validity Time | <=5 | 60 - 86400(s) | 60 |

The device will report the **BAA** (ASCII)/(HEX) message to the backend server when the tire pressure keeps meeting alarm conditions.

- *Tire Pressure Alarm Mode*

  The working mode of the tire pressure alarm.

  – **0** - Disable tire pressure alarm.
  – **1** - Report tire pressure alarm if the current tire pressure is within the tire pressure range defined by <Low Tire Pressure> and <High Tire Pressure>.
  – **2** - Report tire pressure alarm if the current tire pressure is outside the tire pressure range defined by <Low Tire Pressure> and <High Tire Pressure>.
  – **3** - Report tire pressure alarm only once if the current tire pressure enters/exits the tire pressure range defined by <Low Tire Pressure> and <High Tire Pressure>. In this mode, <Alarm Send Interval> will be ignored.

- *Low Tire Pressure*

  The lower tire pressure limit.

- *High Tire Pressure*

  The upper tire pressure limit.

- *Validity*

  If the tire pressure keeps meeting the alarm condition for <Validity> times, the tire pressure alarm will be triggered.

- *Alarm Send Interval*

  After the <Validity> checking, the device will report tire pressure alarm every <Alarm Send Interval> times of tire pressure reading based on reading timer of the tire pressure sensor. If <Alarm Send Interval> is set to 0, the device will only report the tire pressure alarm once.

- *Available Validity Time*

  If the device does not detect the tire pressure sensor in the <Available Validity Time>, the <Accessory Status> in the message **+RESP:GTERI** report messages will be 0.

If the Accessory Type is **13** (Relay Bluetooth accessory), this parameter will work as follows.

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Relay Event Notification | 1 | 0 \| 1 | 0 |

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Password | <=6 | '0'-'9' 'a'-'z' 'A'-'Z' | 123456 |
| New Password | <=6 | '0'-'9' 'a'-'z' 'A'-'Z' | 123456 |
| Reserved | | | |
| Reserved | | | |

- *Relay Event Notification*

  It configures whether to enable relay event notification function.
  - **0** - Disable.
  - **1** - Enable. If a new event occurs on the accessory, the device will report the **BAA** (ASCII)/(HEX) message.
- *Password*

  It is the current password for the accessory device.
- *New Password*

  It is set to change the current password.

Note

If <New Password> is set successfully, <Password> will be changed to <New Password>.

✧ *Output ID*

The ID of the output port to output the specified wave shape when the **BAA** (ASCII)/(HEX) event is detected.

Note

If <Accessory Type> is 4 or <Accessory Type> is 7, the parameters <Accessory Append Mask>, <Low Voltage Threshold>, <Output ID>, <Output Status>, <Duration> and <Toggle Times> are invalid. And the data would not be reported via the message **ERI** (ASCII)/(HEX).

### 3.3.3 BID (Bluetooth Beacon ID)

The command **AT+GTBID** is used for the device to scan Bluetooth beacon accessories and report the scanning results via **BID** (ASCII)/(HEX) messages. To use this function, the parameter <Mode> in the command AT+GTBTS must be 1.

The MAC address of some Bluetooth accessories can be used as authorization ID. For detailed information, please refer to AT+GTIDA.

| |
|---|
| Example:<br>AT+GTBID=gv58lau,0,1,1,000A,2400,,1,1,,,0,30,,,AC233F,0,0,0,0,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | BID | BID |
| | Leading Symbol | 1 | = | = |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| | Index | 1 | 0 - 2 | 0 |
| | Mode | 1 | 0 \| 1 \| 4 | 0 |
| | Beacon ID Model | <=2 | 0-2 \| 4 \| 5 \| 10 | 0 |
| | Accessory Append Mask | <=8 | 0000 - FFFF \| 00000000 - FFFFFFFF | 000A |
| | Low Voltage Threshold | <=4 | 0 - 5000(mV) | 2400 |
| | Reserved | 0 | | |
| | Start Index | <=3 | 1 - 300 | |
| | End Index | <=3 | 1 - 300 | |
| Body | MAC List | <=12*75 | | |
| | Accessory Parameters(Optional) | | | |
| | OUI | 6 | 000000 - FFFFFF | |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Index*

The index of the beacon Bluetooth accessory.

✧ *Mode*

The working mode of this function.

- **0** - Disable
- **1** - MAC or OUI
- **4** - MAC or NAME

Note

Mode 4 is valid only when <Beacon ID Model> is CarBLE accessory and it is necessary to configure the corresponding NAME using the AT+GTBFS command.

✧ *Beacon ID Model*

The model of the beacon Bluetooth accessory. The following is supported now:

- **0** - WKF300
- **1** - iBeacon E6

- **2** - ID ELA
- **4** - WID310
- **5** - CarBLE
- **10** - WID330

✧ *Accessory Append Mask*

Bitwise mask to configure the composition of Bluetooth accessory information in messages **BAA** (ASCII)/(HEX), **BID** (ASCII)/(HEX) .

- **Bit 0** - Reserved
- **Bit 1** - <Accessory MAC>
- **Bit 2** - Reserved
- **Bit 3** - <Accessory Battery Level> , iBeacon E6 is not supported.
- **Bit 4** - Reserved
- **Bit 5** - Reserved
- **Bit 6** - <Accessory Signal Strength>
- **Bit 7** - <Beacon Type> and <Beacon Data>
- **Bit 15** - Expanded mask for Bit 16 - Bit 31
- **Bit 22** - <CarBLE Information>

✧ *Low Voltage Threshold*

It specifies the lower voltage limit. When the voltage of the Bluetooth accessory falls below this value, the device will report the message **+RESP:GTBAA** to the backend server. 0 means "Disable low voltage detection".

✧ *Start Index, End Index*

The index range of the MAC list to which the MAC addresses are to be updated.

For example, if <Start Index> is set to 1 and <End Index> is set to 2, then the first two MAC addresses in the MAC list will be updated by the MAC addresses provided in the parameter <MAC List>. <Start Index> and <End Index> determine the total number of MAC addresses that will be updated. If either one is empty, there should be no <MAC List> following the empty value. A maximum of 75 MAC addresses can be updated each time.

✧ *MAC List*

A list of comma-separated MAC addresses to be updated to the MAC list. The number of the MAC addresses is determined by <Start Index> and <End Index>.

Note

If more accessories are needed, please adjust <Start Index> and <End Index> for appropriate setup. If some MAC addresses in <MAC List> are empty, then the corresponding MAC addresses will be deleted.

For example, to delete the 4th, 5th and 6th MAC addresses of the <MAC List>, please set <Start Index> to 4 and set <End Index> to 6 and keep those three MAC addresses of <MAC List> empty. The maximum number of MAC addresses for all indexes is 300.

✧ *Accessory Parameters(Optional)*

Some parameters for Bluetooth Beacon. For different Beacons, there are different definitions.

If Beacon ID Model is **0** (WKF300), the following parameters are used:

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Push Button Event | 1 | 0 \| 1 | 0 |

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Keyfob Detection Mode | 1 | 0-2 | 0 |
| Keyfob Detection Interval | <=3 | 30 - 600(s) | 30 |
| Reserved | 0 | | |
| Reserved | 0 | | |

- *Push Button Event*

  If this parameter is set to 1 and the button on WKF300 is pushed, the device will report the message **BAA** (ASCII)/(HEX) to the backend server.

- *Keyfob Detection Mode*

  It specifies the mode of detecting Keyfob.

  – **0** - Disable Keyfob detection.

  – **1** - Scan only once.

    After entering ignition on and moving state, the device will scan Keyfob once for the time period specified by <Keyfob Detection Interval> and then send the message **BID** (ASCII)/(HEX) to report information of Keyfob. If more than 3 Keyfobs are detected, the message **+RESP:GTBID** contains information of the top 3 Keyfobs with the strongest signal.

  – **2** - Conditional Continuous Scan1.

    After entering ignition on and moving state, the device will keep scanning Keyfob(s) continuously. If the device detects Keyfob or change of available Keyfob, it will send the message **+RESP:GTBID** to report information of Keyfob. If more than 3 Keyfobs are detected, the message **+RESP:GTBID** contains information of the top 3 Keyfobs with the strongest signal.

- *Keyfob Detection Interval*

  The device scans Keyfob for the time period specified by this parameter.

If Beacon ID Model is **1** (iBeacon E6), the following parameters are used:

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| E6 Detection Mode | 1 | 0 - 2 | 0 |
| E6 Detection Interval | <=3 | 30 - 600(s) | 30 |
| Reserved | 0 | | |
| Reserved | 0 | | |

- *E6 Detection Mode*

  It specifies the mode of detecting Beacon E6.

  – **0** - Disable detection.

  – **1** - Scan only once.

    After entering ignition on and moving state, the device will scan E6 one time for the time period specified by <E6 Detect Interval> and then it will send the message **+RESP:GTBID** to report information of E6. If more than 15 iBeacon E6 are detected, the message **+RESP:GTBID** contains the information about the top 15 iBeacon E6.

  – **2** - Conditional Continuous Scan1.

    After entering ignition on and moving state, the device will keep scanning E6 continuously. If the device detects E6 or change of available E6, it will send the message **+RESP:GTBID** to report information of E6. If more than 15 iBeacon E6 are detected, the message **+RESP:GTBID** contains the information of the top 15 iBeacon E6.

- *E6 Detection Interval*

  The device scans E6 for the period specified by this parameter.

If Beacon ID Model is **2** (ID ELA), the following parameters are used:

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Alarm Mode | 1 | 0 - 3 | 0 |
| IDELA Detection Mode | 1 | 0-3 | 0 |
| IDELA Detection Interval | <=3 | 30 - 600(s) | 30 |
| Reserved | 0 | | |
| Reserved | 0 | | |

- *Alarm Mode*

  It specifies the mode of alarm report.
  - **0** - Disable alarm.
  - **1** - Alarm when receiving Bluetooth broadcast.
  - **2** - Alarm when Bluetooth broadcast is not received.
  - **3** - Alarm when receiving Bluetooth broadcast or exiting Bluetooth broadcast.
- *IDELA Detection Mode*

  It specifies the mode of detecting ID ELA.
  - **0** - Disable detection.
  - **1** - Scan only once.

    After entering ignition on and moving state, the device will scan IDELA one time for the time period specified by <IDELA Detect Interval> and then it will send the message **+RESP:GTBID** to report information of IDELA. If more than 15 IDELA are detected, the message **+RESP:GTBID** contains the information of the top 15 IDELA.
  - **2** - Conditional Continuous Scan1.

    After entering ignition on and moving state, the device will keep scanning IDELA continuously. If the device detects IDELA or change of available IDELA, it will send the message **+RESP:GTBID** to report information of IDELA. If more than 15 IDELA are detected, the message **+RESP:GTBID** contains the information of the top 15 IDELA.
  - **3** - Continuous Scan.

    In this mode, the device will keep scanning continuously. If the device detects IDELA or change of available IDELA, it will send the message **+RESP:GTBID** to report information of IDELA. If more than 15 IDELA are detected, the message **+RESP:GTBID** contains the information of the top 15 IDELA.
- *IDELA Detection Interval*

  The device scans IDELA for the time period specified by this parameter.

If Beacon ID Model is **4** (WID310), the following parameters are used:

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Detection Mode | 1 | 0 \| 2 \| 3 | 0 |
| Detection Interval | <=3 | 30 - 600(s) | 30 |
| IGN Scan & Report Interval | <=5 | 0 \| 10 - 86400(s) | 60 |
| IGF Scan & Report Interval | <=5 | 0 \| 10 - 86400(s) | 60 |

- *Detection Mode*
  The parameter which specifies the mode of detecting WID310.
  - **0** - Disable detection.
  - **2** - Conditional Continuous Scan1.
    After entering ignition on and moving state, the device will keep scanning WID310 continuously. If the device detects WID310 or change of available WID310, it will send the **+RESP:GTBID** messages to report information of WID310. If more than 15 WID310 are detected, the message **+RESP:GTBID** contains the information of the top 15 WID310.
  - **3** - Continuous Scan.
    In this mode, the device will keep scanning continuously. The **+RESP:GTBID** message is sent to the backend server periodically according to the parameter <IGN Scan & Report Interval>/<IGF Scan & Report Interval>. If more than 15 WID310 are detected, the message **+RESP:GTBID** contains the information of the top 15 WID310.
- *Detection Interval*
  The device scans WID310 for the time specified by this parameter when <Detection Mode> is 2.
- *IGN Scan & Report Interval*
  The time interval for sending **+RESP:GTBID** messages when the engine is on and <Detection Mode> is 3.
- *IGF Scan & Report Interval*
  The time interval for sending **+RESP:GTBID** messages when the engine is off and <Detection Mode> is 3.

If Beacon ID Model is **5** (CarBLE), the following parameters are used:

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Alarm Mode | 1 | 0 - 3 | 0 |
| CarBLE Detection Mode | 1 | 0-3 \| 6 | 0 |
| CarBLE Detection Interval | <=3 | 15 - 600(s) | 30 |
| Reserved | 0 | | |
| Reserved | 0 | | |

- *Alarm Mode*
  It specifies the mode of alarm report.
  - **0** - Disable alarm.
  - **1** - Alarm when receiving Bluetooth broadcast.
  - **2** - Alarm when Bluetooth broadcast is not received.

- **3** - Alarm when receiving Bluetooth broadcast or exiting Bluetooth broadcast.

- *CarBLE Detection Mode*

  It specifies the mode of detecting CarBLE.

  - **0** - Disable detection.
  - **1** - Scan only once.

    After entering ignition on and moving state, the device will scan CarBLE one time for the time period specified by <CarBLE Detect Interval> and then it will send the message **+RESP:GTBID** to report information of CarBLE. If more than 15 CarBLE are detected, the message **+RESP:GTBID** contains the information of the top 15 CarBLE.
  - **2** - Conditional Continuous Scan1

    After entering ignition on and moving state, the device will keep scanning CarBLE continuously. If the device detects CarBLE or change of available CarBLE, it will send the message **+RESP:GTBID** to report information of CarBLE. If more than 15 CarBLE are detected, the message **+RESP:GTBID** contains the information of the top 15 CarBLE.
  - **3** - Continuous Scan1

    In this mode, the device will keep scanning continuously. If the device detects CarBLE or change of available CarBLE, it will send the message **+RESP:GTBID** to report information of CarBLE. If more than 15 CarBLE are detected, the message **+RESP:GTBID** contains the information of the top 15 CarBLE.
  - **6** - Continuous Scan2

    It will periodically send the message **+RESP:GTBID** to report information of CarBLE. If more than 15 CarBLE are detected, the message **+RESP:GTBID** contains the information of the top 15 CarBLE.

- *CarBLE Detection Interval*

  The device scans CarBLE for the time period specified by this parameter.

If Beacon ID Model is **10** (WID330), the following parameters are used:

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Detection Mode | 1 | 0 \| 2 \| 3 | 0 |
| Detection Interval | <=3 | 30 - 600(s) | 30 |
| IGN Scan & Report Interval | <=5 | 0 \| 10 - 86400(s) | 60 |
| IGF Scan & Report Interval | <=5 | 0 \| 10 - 86400(s) | 60 |

- *Detection Mode*

  The parameter which specifies the mode of detecting WID330.

  - **0** - Disable detection.
  - **2** - Conditional Continuous Scan1.

    After entering ignition on and moving state, the device will keep scanning WID330 continuously. If the device detects WID330 or change of available WID330, it will send the **+RESP:GTBID** messages to report information of WID330. If more than 15 WID330 are detected, the message **+RESP:GTBID** contains the information of the top 15 WID330.

- **3** - Continuous Scan.
  In this mode, the device will keep scanning continuously. The **+RESP:GTBID** message is sent to the backend server periodically according to the parameter <IGN Scan & Report Interval>/<IGF Scan & Report Interval>. If more than 15 WID330 are detected, the message **+RESP:GTBID** contains the information of the top 15 WID330.

- *Detection Interval*
  The device scans WID330 for the time specified by this parameter when <Detection Mode> is 2.
- *IGN Scan & Report Interval*
  The time interval for sending **+RESP:GTBID** messages when the engine is on and <Detection Mode> is 3.
- *IGF Scan & Report Interval*
  The time interval for sending **+RESP:GTBID** messages when the engine is off and <Detection Mode> is 3.

✧ *OUI*

It is the first three bytes of Bluetooth address, which is composed of NAP and UAP. Only one Organization Unique Identifier (OUI) is allowed for each type of Bluetooth accessory.
For example, "AC233F" represents the Bluetooth iBeacon E6. The "AC23" is NAP, the "3F" is UAP. If the device detects this OUI, the message **+RESP:GTBID** will be reported. If the value is empty, it means "Disable this function".

### 3.3.4 AEX (Analog Input Expansion)

The command **AT+GTAEX** is used to configure the parameters of External Bluetooth Analog Input accessories. All these inputs are customizable.

| Example: |
|---|
| AT+GTAEX=gv58lau,0,0,1,5,0,8000,32000,,,,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | AEX | AEX |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Analog Input Expansion Type | 1 | 0 | 0 |
| | Bind BAS Index | 1 | 0 - 9 | 0 |
| | Analog Input Number | 1 | 1 | |
| | Analog Input ID | 1 | 5 | |
| | Mode | 1 | 0-2 | 0 |
| | Min Threshold | <=5 | 8000 - 32000 (mV) | 8000 |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Max Threshold | <=5 | 8000 - 32000 (mV) | 32000 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Analog Input Expansion Type*

It indicates which extended device the following parameters are used for.

- **0** - Bluetooth Analog Input accessory setting.

✧ *Bind BAS Index*

It is used to bind the specific configuration in AT+GTBAS.

✧ *Analog Input Number*

The total number of configured Bluetooth Analog Input Setting. In one configuration, <Analog Input ID>, <Mode>, <Min Threshold> and <Max Threshold> are included and others are reserved.

✧ *Analog Input ID*

The ID of 5 is Bluetooth Analog Input ID, the parameters <Accessory Type> and <Accessory Model> should be set in AT+GTBAS.

✧ *Mode*

The working mode of the analog input alarm **AIS**(ASCII)/(HEX).

- **0** - Disable analog input alarm.
- **1** - Enable analog input alarm: If the current input voltage is within the range of (<Min. Threshold>, <Max. Threshold>), the alarm will be triggered.
- **2** - Enable analog input alarm: If the current input voltage is outside the range of (<Min. Threshold>, <Max. Threshold>), the alarm will be triggered.

✧ *Min Threshold*

This parameter specifies the lower voltage limit for the analog input port to trigger the alarm when the <Mode> is set to 1 or 2.

✧ *Max Threshold*

This parameter specifies the upper voltage limit for the analog input port to trigger the alarm when the <Mode> is set to 1 or 2.

### 3.3.5 SVR (Stolen Vehicle Recovery)

The command **AT+GTSVR** is used to configure the Bluetooth settings for the device. If the Ghost /Primary device cannot connect with the Primary/Ghost device within the time defined by

<Connect Interval> * <Connect Fail Count>, it will report the message **SVR** (ASCII) / (HEX) to the backend server. When the Ghost/Primary device connects to the Primary/Ghost device for the first time, and meanwhile the Ghost/Primary matches the Primary/Ghost's MAC address, IMEI and UTC time successfully, the device will report the message **+RESP:GTSVR** to the backend server, and the CELL LED will be ON (2 seconds) and OFF (2 seconds) periodically which lasts 40 seconds.

| Example: |
|---|
| AT+GTSVR=gv58lau,0,78054125E8D8,10,1,863574046024485,0,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | SVR | SVR |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-2 | 0 |
| | Ghost/Primary MAC Address | 12 | 000000000000 - FFFFFFFFFFFF | |
| | Connect Interval | <=4 | 10 \| 20 \| 30 \| 60 \| 120 \| 180 \| 240 \| 360 \| 480 \| 720 \| 1440 | 10 |
| | Connect Fail Count | <=2 | 1 - 10 | 1 |
| | Match Connected IMEI | 15 | '0'–'9' | |
| | BTI Report Interval | <=4 | 0 - 1440 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the Primary device's stolen vehicle recovery function.

- **0** - Disable.
- **1** - Enable this function as Primary device.
- **2** - Enable this function as Ghost device.

Note

If <Mode> in **AT+GTSVR** is set to 1, the broadcast service for AT+GTBTS will not work. If <Mode> is set to 2, the fixed report, geo-fence, speed alarm report functions are suspended and the RF, Bluetooth, GNSS will be turned off when the Ghost device is connected with the Primary device.

✧ *Ghost/Primary MAC Address*

The MAC address of the Ghost/Primary device which is used to pair with the Primary/Ghost device.

✧ *Connect Interval*

The maximum amount of time allowed for the Ghost device to establish a connection with the Primary device. Unit: minute.

Note

The parameter <Connect Interval> for the Ghost device should be set to a value less than or equal to the <Connect Interval> set for the Primary device.

✧ *Connect Fail Count*

The maximum number of connection failures between the Primary device and the Ghost device.

✧ *Match Connected IMEI*

The IMEI of the Ghost device which is used to match with the Primary device.

✧ *BTI Report Interval*

It specifies the interval for reporting the **BTI** (ASCII)/(HEX) message regarding the status and Bluetooth information of the Ghost device.

0 means "Disable the **+RESP:GTBTI** report". Unit: minute.

Note

If the connection between the Primary device and the Ghost device fails, the message **+RESP:GTBTI** will not be reported.

### 3.3.6 BFS (Bluetooth Fields Settings)

The command **AT+GTBFS** is specifically used to configure Bluetooth fields scanned with the AT+GTBID command.

| |
|---|
| Example:<br>AT+GTBFS=gv58lau,0,,,0,0,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | BFS | BFS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Index | 1 | 0-2 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Name Numbers | 1 | 0 - 1 | 0 |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Name 1(Optional) | <=16 | '0' - '9', 'a' - 'z', 'A' - 'Z', '-', '_', ' ' | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

- *Password*

  The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".
- *Index*

  The index of the beacon Bluetooth accessory.
- *NAME Numbers*

  The number of <NAME>.
- *NAME*

  When the parameter <Mode> of **AT+GTBID** is set to 4, the device detects Bluetooth accessories by NAME.

Note

If <NAME> is empty, the corresponding value will be deleted.

## 3.4 Tracking Settings

This section describes the commands related to the location messages. Please refer to the details below.

### 3.4.1 FRI (Fixed Report Information)

The command **AT+GTFRI** is used to configure the parameters for fixed report **FRI** (ASCII)/(HEX) or **ERI** (ASCII)/(HEX).

| |
|---|
| Example:<br>AT+GTFRI=gv58lau,0,1,,1,0000,0000,0,30,1000,1000,,0,600,00000000,0,,0,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | FRI | FRI |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | <=2 | 0-5 \| 99 | 0 |
| | Discard No Fix | 1 | 0 \| 1 | 1 |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Period Mode | 1 | 0 \| 1 | 1 |
| | Start Time | 4 | HHMM | 0000 |
| | End Time | 4 | HHMM | 0000 |
| | Check Interval | <=5 | 0 - 86400(s) | 0 |
| | Send Interval | <=5 | 1 - 86400(s) | 30 |
| | Distance | <=5 | 50 - 65535(m) | 1000 |
| | Mileage | <=5 | 50 - 65535(m) | 1000 |
| | Reserved | 0 | | |
| | Corner Value | <=3 | 0 - 180 | 0 |
| | IGF Report Interval | <=5 | 0 \| 5 - 86400(s) | 600 |
| | ERI Mask | 8 | 00000000 - FFFFFFFF | 00000000 |
| | Continue Time | <=2 | 0 - 10(min) | 0 |
| | Reserved | 0 | | |
| | Wrap Corner Point | 1 | 0 \| 1 | 0 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the fixed report function.

- **0** - Disable this function.
- **1** - Fixed Time Report.
  The position report message is sent to the backend server periodically according to the parameter <Send Interval>.
- **2** - Fixed Distance Report.
  The position report message is sent to the backend server when the straight-line distance between the current GNSS position and the last sent GNSS position is greater than or equal to the distance specified by the parameter <Distance>.
- **3** - Fixed Mileage Report.
  The position report message is sent to the backend server when the path length between the current GNSS position and the last sent GNSS position is greater than or equal to the mileage specified by the parameter <Mileage>.
- **4** - Optimum Report.
  The device simultaneously checks both time interval and path length between two adjacent position reports. Device position will be reported if the calculated time interval between the current time and the time of last report is greater than <Send Interval>, and the length of the path between the current position and the last position is greater than <Mileage>.
- **5** - Fixed Time or Mileage Report.

It checks either time interval or path length between two adjacent reports. Device position will be reported if the calculated time interval between current time and time of last report is greater than <Send Interval>, or the length of path between the current position and the last position is greater than <Mileage>.

- **99** - Real Time Report.
  The message **+RESP:GTFRI** or **+RESP:GTERI** is immediately sent to the backend server, and the value of <Report ID / Report Type> in **+RESP:GTFRI** or **+RESP:GTERI** is 00.

Note

1. For mode 2/3/4/5, the vehicle ignition signal should be connected to the specified digital input port of the device or enabling virtual ignition detection.
2. If the engine is off, the position report message will be sent to the backend server periodically according to the parameter <IGF Report Interval>.

✧ *Discard No Fix*
Enable/disable report when there is no GNSS fix.
- **0** - Enable.
- **1** - Disable.

✧ *Period Mode*
Enable/disable the time range specified by <Start Time> and <End Time>. If the time range is enabled, the position reporting will be limited within the time range.

✧ *Start Time*
The start time of the fixed report. The valid format is "HHMM". The value range of "HH" is "00"-"23". The value range of "MM" is "00"-"59".

✧ *End Time*
The end time of the fixed report. The valid format and range are the same as those of <Start Time>.

✧ *Check Interval*
The interval for updating GNSS position. If the parameter value is 0, the device will update GNSS position according to the value of <Send Interval>. Make sure <Check Interval> is not greater than <Send Interval> so that position data is ready before sending time arrives.

✧ *Send Interval*
The interval for sending the position information. If <Report Mode> in AT+GTSRI is set to forced SMS mode, this parameter should not be less than 15 seconds, otherwise the device will send the position information via TCP short connection.

Note

Due to the limit of the maximum report message length, make sure the <Send Interval>/<Check Interval> ratio is less than or equal to 15.

✧ *Distance*
The specified distance for sending the position information when <Mode> is 2. Unit: meter.

✧ *Mileage*
The specified length for sending the position information when <Mode> is 3/4/5. Unit: meter.

✧ *Corner Value*
A numeral indicating whether to report **+RESP:GTFRI** message based on change in the

direction of device movement.

- **0** - Disable.
- **1 - 180** - The angle used to determine whether the device turns around a corner. If the change in the direction of device movement is greater than the specified value, the device is considered turning around a corner. Unit: degree.

Note

If FRI multi-point report occurs at the same time with corner report, the corner point will be included in multi-point report message, and the <Report Type> of the **+RESP:GTFRI** message will be 0.

✧ *IGF Report Interval*

The interval for acquiring and sending the position information when <Mode> is not 0 and <Power Saving Mode> in AT+GTCFG is set to 0 or 2 with engine off. If <IGF Report Interval> is less than 60 seconds, the GNSS will be always on.

✧ *ERI Mask*

When the device is connected with a peripheral, and the bit for the peripheral is set to 1, the device will report **+RESP:GTERI** instead of **+RESP:GTFRI**. This mask is used to configure whether to report the data from peripherals via **+RESP:GTERI**.

- **Bit 2** - For the <CAN Data> field in **+RESP:GTERI**. This mask only works in ASCII-formatted **+RESP:GTERI** message.
- **Bit 5** - Reserved.
- **Bit 8** - For the <Bluetooth Accessory Data> field in **+RESP:GTERI**.
- **Bit 9** - Reserved.

✧ *Continue Time*

After the ignition is turned off, the message **+RESP:GTFRI/+RESP:GTERI** will continue to be reported according to <Send Interval> for an extra period of time specified by <Continue Time> and then at <IGF Report Interval>.

✧ *Wrap Corner Point*

A numeral to indicate whether to wrap corner point with other fixed GNSS points and wait to send **+RESP:GTFRI** or **+RESP:GTERI** message according to the value of <Mode>.

- **0** - Do not wrap corner point and send the corner point immediately when it is found.
- **1** - Wrap corner point and wait to send **+RESP:GTFRI** or **+RESP:GTERI** according to the value of <Mode>.

Note

If <Wrap Corner point> is set to 1, all the wrap corner points and normal points will be influenced by AT+GTOWH if it is used.

### 3.4.2 FFC (Frequency Change of FRI)

The command **AT+GTFFC** is used to change the parameters of fixed report when a certain event occurs, so that different report interval requirements can be met. When the event disappears, the device will resume its previous settings.

The device supports up to 5 sets of parameters for different events. Priority is assigned among these events. Only the parameters for the highest priority event are applied if more than one event occurs at the same time.

| Example: |
|---|
| AT+GTFFC=gv58lau,0,1,0,30,500,500,300,0,0,,,0000$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | FFC | FFC |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Priority | 1 | 0 - 4 | 0 |
| | Mode | 1 | 0-3 | 0 |
| | FRI Mode | 1 | 0-5 | 0 |
| | FRI IGN Report Interval | <=5 | 5 - 86400(s) | 30 |
| | FRI Report Distance | <=5 | 50 - 65535(m) | 500 |
| | FRI Report Mileage | <=5 | 50 - 65535(m) | 500 |
| | FRI IGF Report Interval | <=5 | 0 \| 5 - 86400(s) | 300 |
| | FRI IGN Check Interval | <=5 | 0 - 86400(s) | 0 |
| | Corner Value | <=3 | 0 - 180 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Priority*

The priority of the event which triggers the change of fixed report parameters. 0 means "The highest priority".

✧ *Mode*

It specifies the trigger event for the fixed report parameter change.

- **0** - Disable the parameters of the specified priority.
- **1** - Change the fixed report parameters when the device enters any of the defined Geo-Fence.
- **2** - Change the fixed report parameters when the device enters known roaming state. Please refer to the AT+GTRMD command for more information.
- **3** - Change the fixed report parameters when the device enters unknown roaming state.

✧ *FRI Mode*

If the specified event occurs, the working mode of the fixed report will be changed according to this parameter.

- **0** - Do not change the working mode.
- **1** - Change the working mode to "Fixed Time Report".
- **2** - Change the working mode to "Fixed Distance Report".
- **3** - Change the working mode to "Fixed Mileage Report".
- **4** - Change the working mode to "Optimum Report".
- **5** - Change the working mode to "Fixed Time or Mileage Report".

✧ *FRI IGN Report Interval*

The interval for sending the position information when the ignition is on. Unit: second.

✧ *FRI Report Distance*

The specified distance for sending the position information when the mode is fixed distance report. Unit: meter.

✧ *FRI Report Mileage*

The specified path length for sending the position information when the mode is fixed mileage report or optimum report. Unit: meter.

✧ *FRI IGF Report Interval*

The interval for fixing and sending the position information when the ignition is off and <Power Saving Mode> in AT+GTCFG is set to 0 or 2. Unit: second.

✧ *FRI IGN Check Interval*

The interval for GNSS fix. Unit: second.

✧ *Corner Value*

A numeral indicating whether to report **+RESP:GTFRI** message based on change in the direction of device movement.

- **0** - Disable.
- **1 - 180** - The angle used to determine whether the device turns around a corner. If the change in the direction of device movement is greater than the specified value, the device is considered turning around a corner. Unit: degree.

## 3.5 Alarm Settings

This section describes the commands related to the alarm settings. Please refer to the details below.

### 3.5.1 CRA (Crash Detection)

The command **AT+GTCRA** is used to configure the parameters for crash detection. When the current acceleration in a certain direction exceeds the configured threshold, the device will report the **CRA** (ASCII)/(HEX) event message and data packets **CRD**(ASCII)/(HEX) to the backend server.

| |
|---|
| Example:<br>AT+GTCRA=gv58lau,1,50,50,50,0,500,500,0,0,0,0,,0,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | CRA | CRA |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-2 | 0 |
| | Threshold_X | <=3 | 0 - 160 | 50 |
| | Threshold_Y | <=3 | 0 - 160 | 50 |
| | Threshold_Z | <=3 | 0 - 160 | 50 |
| | Sampling Start Mode | 1 | 0 \| 1 | 0 |
| | Samples Before Crash | <=4 | 0 - 1500 | 500 |
| | Samples After Crash | <=4 | 0 - 1500 | 500 |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Report GNSS Mode | 1 | 0 \| 1 | 0 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the crash detection function.

- **0** - Disable.
- **1** - Enable. In this mode, no three-axis self-calibration is required.
- **2** - Enable. In this mode, the acceleration sensor data will be converted in accordance with three-axis self-calibration.
  In the new coordinate system, the positive X-axis points in the direction the vehicle travels; the positive Y-axis is perpendicular to X-axis, and the direction conforms to the right-hand rule; and the positive Z-axis is the opposite direction of gravity.

Note

The device will keep monitoring crash event based on the original three-axis data from sensor until it detects the first **ASC** (ASCII)/(HEX) event.

✧ *Threshold_X*

The acceleration threshold for crash in X direction. The smaller the parameter, the easier it is to detect crash events. If <Threshold_X> is 0, the device will not monitor crash event in X-axis. The unit is 0.1g.

✧ *Threshold_Y*

The acceleration threshold for crash in Y direction. The smaller the parameter, the easier it is to detect crash events. If <Threshold_Y> is 0, the device will not monitor crash event in Y-axis. The unit is 0.1g.

- *Threshold_Z*

  The acceleration threshold for crash in Z direction. The smaller the parameter, the easier it is to detect crash events. If <Threshold_Z> is 0, the device will not monitor crash event in Z-axis. The unit is 0.1g.

- *Sampling Start Mode*

  A numeral to indicate when to start sampling acceleration data.
  - **0** - Start acceleration sampling after the device is powered on.
  - **1** - Start acceleration sampling when ignition is on.

- *Samples Before Crash*

  The number of recorded XYZ-axis acceleration samples before crash.

- *Samples After Crash*

  The number of recorded XYZ-axis acceleration samples after crash.

- *Report GNSS Mode*

  A numeral which indicates whether to report the GNSS information of 10 seconds before/after crash to the backend server.
  - **0** - Disable the GNSS information report.
  - **1** - Enable the GNSS information report. The device will report the GNSS information of 10s before crash and 10 seconds after crash to the backend server in the message **CRG** (ASCII)/(HEX).

### 3.5.2 GEO (Geo-Fence)

The command **AT+GTGEO** is used to configure the parameters of circular Geo-Fence. Circular Geo-Fence is a virtual perimeter around a geographic area using a location-based service. When the terminal with geo-fencing enters or exits the area, a notification containing information about the terminal's location is generated that can be sent to the backend server.

| |
|---|
| Example:<br>AT+GTGEO=gv58lau,0,3,112.129273,32.839031,50,0,0,0,0,0,0,0,0000,0000,0,000A$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | GEO | GEO |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | ID | <=2 | 0 - 99 | |
| | Mode | 1 | 0-3 | 0 |
| | Longitude | <=11 | (-)xxx.xxxxxx | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Latitude | <=10 | (-)xx.xxxxxx | |
| | Radius | <=7 | 50 - 6000000(m) | 50 |
| | Check Interval | <=5 | 0 \| 5 - 86400(s) | 0 |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Trigger Mode | <=2 | 0 \| 21 \| 22 | 0 |
| | Trigger Report | 1 | 0 \| 1 | 0 |
| | Start Time | 4 | HHMM | 0000 |
| | End Time | 4 | HHMM | 0000 |
| | State Mode | 1 | 0 \| 1 | 0 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *ID*

The ID of the circular Geo-Fence. A total of 20 zones (0 - 99) are supported.

✧ *Mode*

The working mode for the device to report the Geo-Fence message **GIN**(ASCII)/(HEX) or **GOT**(ASCII)/(HEX) to the backend server.

- **0** - Disable the zone's circular Geo-Fence function.
- **1** - Enter the zone. The device will send the message **+RESP:GTGIN** to the server when the vehicle enters the Geo-Fence.
- **2** - Exit the zone. The device will send the message **+RESP:GTGOT** to the server when the vehicle exits the Geo-Fence.
- **3** - Report **+RESP:GTGIN** and **+RESP:GTGOT** when entering and exiting the Geo-Fence respectively.

✧ *Longitude*

The longitude of a point which is defined as the center of the circular Geo-Fence. The unit is degree, and accuracy is 6 decimal places. West longitude is defined as negative starting with the sign "-" and east longitude is defined as positive without "+".

✧ *Latitude*

The latitude of a point which is defined as the center of the circular Geo-Fence. The unit is degree, and accuracy is 6 decimal places. South latitude is defined as negative starting with the minus sign "-" and north latitude is defined as positive without "+".

✧ *Radius*

The radius of the circular Geo-Fence. Unit: meter.

✧ *Check Interval*

The interval for GNSS to check location information based on Geo-Fence alarms.

- *Trigger Mode*

  A numeral to indicate the trigger mode of the Geo-Fence function.

  - **0** - Disable automatic triggering.
  - **21** - Automatic parking fence.

    Automatically set up Geo-Fence after ignition off. In this mode, the device will automatically set up a Geo-Fence with the current location as the center point of the Geo-Fence when the ignition is off. It will only send the alarm report when the device exits the Geo-Fence, and the Geo-Fence will be cancelled after the device exits the zone.
  - **22** - Manual parking fence.

    Manually enable Geo-Fence after ignition off. In this mode, the device will automatically set up a Geo-Fence with the current location as the center point of the Geo-Fence when the ignition is off. It will only send the alarm report when the device exits the Geo-Fence. After the device exits the Geo-Fence, it will cancel the Geo-Fence and disable the trigger mode at the same time. If the driver wants to use this trigger mode again, he has to manually set the trigger mode again.

- *Trigger Report*

  Whether to report the **GES** (ASCII)/(HEX) message when the specified trigger mode is activated or the Geo-Fence is cancelled.

  - **0** - Disable the **+RESP:GTGES** report.
  - **1** - Enable the **+RESP:GTGES** report.

- *Start Time*

  The time to start monitoring when the device enters or exits the Geo-Fence. The valid format is "HHMM". The value range of "HH" is "00"-"23". The value range of "MM" is "00"-"59".

- *End Time*

  The time to end monitoring when the device enters or exits the Geo-Fence. The valid format and range are the same as those of <Start Time>.

- *State Mode*

  The mode for reporting the device's GEO state.

  - **0** - Report when getting the GEO state for the first time.
  - **1** - Do not report until the GEO state changes.

### 3.5.3 PEO (Polygon Geo-Fence)

The command **AT+GTPEO** is used to configure the parameters of polygon Geo-Fence. Polygon Geo-Fence is a virtual perimeter around a geographic area using a location-based service. When the terminal with geo-fencing enters or exits the area, a notification containing information about the terminal's location is generated that can be sent to the backend server.

Note

This command can be used to set less than ten sets of longitude and latitude coordinates each time.

Example:
AT+GTPEO=gv58lau,0,0,1,3,121.412240,31.187801,121.412248,31.187891,121.412258,31.187991,0,0,0,0,0,1,,,,000B$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | PEO | PEO |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | ID | <=3 | 0 - 199 | 0 |
| | Mode | 1 | 0-3 | 0 |
| | Start Point | <=2 | 1 - 10 | 1 |
| | End Point | <=2 | 3 - 10 | 3 |
| | Longitude | <=11 | (-)xxx.xxxxxx | |
| | Latitude | <=10 | (-)xx.xxxxxx | |
| | Check Interval | <=5 | 0 \| 5 - 86400(s) | 0 |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | State Mode | 1 | 0 \| 1 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *ID*

The ID of the polygon Geo-Fence. A total of 200 zones (0 - 199) are supported.

✧ *Mode*

The working mode for the device to report the polygon Geo-Fence message to the backend server.

- **0** - Disable the zone's polygon Geo-Fence function.
- **1** - Enter the zone. The **GIN**(ASCII)/(HEX) message will be generated only when the terminal enters the Geo-Fence.
- **2** - Exit the zone. The **GOT** (ASCII)/(HEX) message will be generated only when the terminal exits the Geo-Fence.

- **3** - Report the **+RESP:GTGIN** and **+RESP:GTGOT** when entering and exiting the Geo-Fence zone respectively.

✧ *Start Point*

The start point of the polygon GEO-Fence formed by a set of points.

✧ *End Point*

The end point of the polygon GEO-Fence formed by a set of points.

✧ *Longitude*

The longitude of a point which is defined as the endpoint of the polygon Geo-Fence region. The unit is degree, and accuracy is 6 decimal places. West longitude is defined as negative starting with the minus sign "-" and east longitude is defined as positive without "+".

✧ *Latitude*

The latitude of a point which is defined as the endpoint of the polygon Geo-Fence region. The unit is degree, and accuracy is 6 decimal places. South latitude is defined as negative starting with the minus sign "-" and north latitude is defined as positive without "+".

Note

If more sets of <Longitude> and <Latitude> are needed, please adjust <Start Point> and <End Point> for appropriate setup. If some sets of <Longitude> and <Latitude> are empty, then the corresponding vertices will be deleted. For example, to delete the 4th, 5th, and 6th vertices of a polygon Geo-Fence, please set <Start Point> to 4 and set <End Point> to 6 and keep the three sets of <Longitude> and <Latitude> empty.

✧ *Check Interval*

The interval for GNSS to check location information based on polygon Geo-Fence alarms.

✧ *State Mode*

The mode for reporting the device's PEO state.

- **0** - Report when getting the PEO state for the first time.
- **1** - Do not report until the PEO state changes.

### 3.5.4 PEG (Polygon Geo-Fence Group)

The command **AT+GTPEG** is used to configure the groups of Polygon Geo-Fence.

| |
|---|
| Example:<br>AT+GTPEG=gv58lau,0,0,0,19,,131,,60,,0,1,0,0,,,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | PEG | PEG |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Group ID | 1 | 0 - 9 | 0 |
| | Mode | 1 | 0 \| 1 | 0 |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Start ID | <=3 | 0 - 199 | 0 |
| | End ID | <=3 | 0 - 199 | 19 |
| | Reserved | 0 | | |
| | Speed Threshold | <=3 | 0 - 400(km/h) | 0 |
| | Reserved | 0 | | |
| | Validity | <=4 | 0 - 3600(s) | 60 |
| | Reserved | 0 | | |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0-2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Group ID*

The ID of a GEO group. The smaller the ID, the higher the priority of a group. If the device enters Polygon GEO-Fence within a higher priority Polygon Geo-Fence group, it will not check the GEO status of those Polygon Geo-Fence groups of relatively low priority.

✧ *Mode*

Enable/disable the group's Polygon Geo-Fence function. If this parameter is set to 0 for all groups, the **AT+GTPEG** command does not take effect, and the device works according to configuration of AT+GTPEO.

- **0** - Disable.
- **1** - Enable.

✧ *Start ID, End ID*

The range of the GEO IDs. It indicates which IDs are currently in the group. A GEO ID cannot be set in two groups. A group supports a maximum of 32 GEO IDs.

✧ *Speed Threshold*

The minimum value that triggers the speed alarm when the device is inside the Geo Fence within this group. It will report **+RESP:GTSPE** for the alarm. 0 means the speed alarm function is disabled.

✧ *Validity*

If the speed is higher than the speed threshold for period longer than the time specified by <Validity>, the speed alarm will be triggered.

✧ *Output ID, Output Status, Duration, Toggle Times*

When the speed alarm is triggered, the specified wave will be output at the specified output.

## 3.5.5 HBM (Harsh Behavior Monitoring)

The command **AT+GTHBM** is used to monitor the harsh driving behavior based on GNSS or motion sensor. The same harsh behavior within 30 seconds will only be reported once if only GNSS is used to measure harsh driving behavior.

| |
|---|
| Example:<br>AT+GTHBM=gv58lau,1,3,1,100,5,5,,60,10,10,,,30,30,,0,0,0,0,25,50,20,50,,,25,50,0,0010$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | HBM | HBM |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-5 | 0 |
| | Behavior Duration | 1 | 3 - 5 | 3 |
| | Discard Unknown Event | 1 | 0 \| 1 | 1 |
| | High Speed | <=3 | 100 - 400(km/h) | 100 |
| | ΔVhb | <=3 | 0 - 100(km/h) | 0 |
| | ΔVha | <=3 | 0 - 100(km/h) | 0 |
| | Reserved | 0 | | |
| | Medium Speed | <=3 | 60 - 100(km/h) | 60 |
| | ΔVmb | <=3 | 0 - 100(km/h) | 0 |
| | ΔVma | <=3 | 0 - 100(km/h) | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | ΔVlb | <=3 | 0 - 100(km/h) | 0 |
| | ΔVla | <=3 | 0 - 100(km/h) | 0 |
| | Reserved | 0 | | |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Cornering Threshold | <=3 | 10 - 100 | 25 |
| | Cornering Duration | <=3 | 10 - 250 (x10ms) | 50 |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Acceleration Threshold | <=3 | 5 - 100 | 20 |
| | Acceleration Duration | <=3 | 10 - 250 (x10ms) | 50 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Brake Threshold | <=3 | 10 - 100 | 25 |
| | Brake Duration | <=3 | 10 - 250(x10ms) | 50 |
| | Report Mode | 1 | 0 \| 1 | 0 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the harsh behavior monitoring function.

- **0** - Disable this function.
- **1** - Enable this function: Detected by GNSS only.
  In this mode, 2 harsh behaviors are monitored, namely harsh braking and harsh acceleration. According to the speed read from GNSS, 3 levels of speed are defined including high speed, medium speed and low speed. For each speed level, 2 thresholds of speed change are defined to determine harsh braking and harsh acceleration. If the change of speed within <Behavior Duration> seconds is greater than the corresponding threshold, the device will report the **HBM** (ASCII)/(HEX) message to the backend server to indicate the harsh behavior.
- **2** - Enable this function: Detected by motion sensor only.
  In this mode, three types of harsh behavior can be detected, namely harsh braking, harsh acceleration and harsh cornering. The device needs GNSS information to get the harsh behavior direction, so it is necessary to keep GNSS always on to collect all the information needed.
- **3** - Enable this function: Detected by motion sensor or GNSS.
- **4** - Enable this function: Detected by motion sensor and GNSS.
- **5** - Enable this function: Detected by XYZ-axis acceleration data only.
  In this mode, the XYZ-axis data will be converted in accordance with 3-axis self-calibration. If the value change of the positive X-axis (vehicle driving direction) is greater than <Acceleration Threshold> for the time of <Acceleration Duration>, the device will report the message **+RESP:GTHBM** to the backend server to indicate harsh acceleration.
  If the value change of opposite direction of positive X-axis is greater than <Brake Threshold> for the time of <Brake Duration>, the device will report the message **+RESP:GTHBM** to indicate harsh braking behavior.
  If the value change of the Y-axis (left and right direction of vehicle moving direction) is greater than <Cornering Threshold> for the time of <Cornering Duration>, the device will report the message **+RESP:GTHBM** to indicate harsh cornering behavior, and the message

**HBE** (ASCII)/(HEX) will be reported when the harsh behavior is over.

Note

The device will not detect harsh behavior until it has already detected the first **ASC** (ASCII)/(HEX) event .

✧ *Behavior Duration*

The speed change within <Behavior Duration> is measured.

✧ *Discard Unknown Event*

It configures whether to discard the **+RESP:GTHBM** message that indicates unknown harsh behavior.

- **0** - Do not discard unknown harsh behavior message.
- **1** - Discard unknown harsh behavior message.

✧ *High Speed, Medium Speed*

If the last known speed of the device read from GNSS is greater than or equal to <High Speed>, the vehicle that the device is attached to is considered to be at high speed.

If the last known speed is less than <High Speed> but greater than or equal to <Medium Speed>, the vehicle is considered to be at medium speed.

If the last known speed is less than <Medium Speed>, the vehicle is considered to be at low speed.

✧ *ΔVhb*

The threshold for harsh braking at high speed level.

If the current speed is less than the last known speed and the change of the speed is greater than or equal to this value within <Behavior Duration> seconds, harsh braking is detected at high speed level.

If it is set to 0, it means "Do not monitor harsh braking behavior at high speed level".

✧ *ΔVha*

The threshold for harsh acceleration at high speed level.

If the current speed is greater than the last known speed and the change of the speed is greater than or equal to this value within <Behavior Duration> seconds, harsh acceleration is detected at high speed level.

If it is set to 0, it means "Do not monitor harsh acceleration behavior at high speed level".

✧ *ΔVmb*

The threshold for harsh braking at medium speed level.

If the current speed is less than the last known speed and the change of the speed is greater than or equal to this value within <Behavior Duration> seconds, harsh braking is detected at medium speed level.

If it is set to 0, it means "Do not monitor harsh braking behavior at medium speed level".

✧ *ΔVma*

The threshold for harsh acceleration at medium speed level.

If the current speed is greater than the last known speed and the change of the speed is greater than or equal to this value within <Behavior Duration> seconds, harsh acceleration is detected at medium speed level.

If it is set to 0, it means "Do not monitor harsh acceleration behavior at medium speed level".

✧ *ΔVlb*

The threshold for harsh braking at low speed level.

If the current speed is less than the last known speed and the change of the speed is greater than or equal to this value within <Behavior Duration> seconds, harsh braking is detected at low speed level.

If it is set to 0, it means "Do not monitor harsh braking behavior at low speed level".

- ✧ *ΔVla*

  The threshold for harsh acceleration at low speed level.

  If the current speed is greater than the last known speed and the change of the speed is greater than or equal to this value within <Behavior Duration> seconds, harsh acceleration is detected at low speed level.

  If it is set to 0, it means "Do not monitor harsh acceleration behavior at low speed level".

- ✧ *Output ID*

  It specifies the ID of the output port to output specified wave shape when the harsh behavior is detected. If it is set to 0, there will be no output wave.

- ✧ *Cornering Threshold*

  The threshold for the motion sensor to measure whether the device is in harsh cornering.

- ✧ *Cornering Duration*

  A time parameter to measure whether the device enters harsh cornering. If the driving behavior is maintained for a period of time longer than <Cornering Duration>, the harsh cornering or harsh braking event will be triggered.

- ✧ *Acceleration Threshold*

  The threshold for the motion sensor to measure whether the device is in harsh acceleration state.

- ✧ *Acceleration Duration*

  A time parameter to measure whether the device enters harsh acceleration state. If the driving behavior is maintained for a period of time longer than <Acceleration Duration>, the harsh acceleration event will be triggered.

- ✧ *Brake Threshold*

  The threshold for the motion sensor to measure whether the device is in harsh braking state.

- ✧ *Brake Duration*

  A time parameter to measure whether the device enters harsh braking state. If the driving behavior is maintained for a period of time longer than <Brake Duration>, the harsh braking event will be triggered.

- ✧ *Report Mode*

  Whether to report **+RESP:GTHBE** / **+RESP:GTHBM** to the backend server.

  - **0** - Do not report **+RESP:GTHBE**, only report **+RESP:GTHBM**.
  - **1** - Report **+RESP:GTHBE** and **+RESP:GTHBM**.

Note

The threshold for harsh driving should be smaller than the threshold for crash in the AT+GTCRA command.

## 3.5.6 IDL (Excessive Idling Detection)

The command **AT+GTIDL** is used to detect the engine excessive idling (vehicle stays stationary while ignition is on). To use this command, the device must be in the ignition state. If the vehicle enters the idle state, the device will report the event message **IDN** (ASCII)/(HEX) to the backend server. If the vehicle exits the idle state, the device will report the event message **IDF** (ASCII)/(HEX) to the backend server.

Example:
AT+GTIDL=gv58lau,0,2,1,0,,,,0,0,0,0,,,,,000F$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | IDL | IDL |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 \| 1 | 0 |
| | Time to Idling | <=2 | 1 - 30(min) | 2 |
| | Time to Movement | 1 | 1 - 5(min) | 1 |
| | Debounce Distance | <=4 | 0 \| 25 - 9999(m) | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of this function.

- **0** - Disable.

- **1** - Enable.

✧ *Time to Idling*

If the vehicle is detected to be stationary with ignition on within the period specified by this parameter, it is considered to be idling.

✧ *Time to Movement*

After entering the idle state, if the vehicle moves again and the moving state is maintained for the time specified by this parameter or ignition off is detected, it is considered to exit the idle state.

If ignition off is detected, the vehicle is considered to leave idling status regardless of the <Time to Movement> setting.

✧ *Debounce Distance*

After entering the idle state, if the vehicle moves more than <Debounce Distance>, it is considered to exit the idle state.

✧ *Output ID*

It specifies the ID of the output port to output specified wave shape when the vehicle enters the idle state. If it is set to 0, there will be no output wave.

### 3.5.7 JBS (Jamming Behavior Setting)

The command **AT+GTJBS** is used for the Jamming Behavior Setting function. There are two modes of Jamming Behavior Setting, namely Jamming Behavior Setting Configure Mode and Jamming Behavior Setting Reset Mode. The output1 is used for "fuel cut-off" and the output2 is used for "siren".

✧ **Jamming Behavior Setting Configure Mode**

| |
|---|
| Example:<br>AT+GTJBS=gv58lau,0,,10,10,1800,1,30,0,0,5,1,0,0,0,60,30,,001A$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | JBS | JBS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 \| 1 | 0 |
| | Reserved | 0 | | |
| | Siren On Timer (T1) | <=3 | 1 - 600 (x100ms) | 10 |
| | Siren Off Timer (T2) | <=3 | 1 - 600 (x100ms) | 10 |
| | Ready Fuel Release Timer (T3) | <=5 | 1 - 65535 (s) | 1800 |
| | Check Speed | 1 | 0 \| 1 | 1 |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Speed Limit | <=3 | 0 - 999(km/h) | 30 |
| | Output 1 Init State | 1 | 0 \| 1 | 0 |
| | Motion Sensor | 1 | 0 \| 1 | 0 |
| | GNSS Fix Failure Timeout Timer (T4) | <=3 | 1 - 100 (min) | 5 |
| | Enable Siren | 1 | 0 \| 1 | 1 |
| | Release Fuel Cut-off Timer (T5) | <=4 | 0 - 1000 (min) | 0 |
| | Check Jamming in T3 | 1 | 0 \| 1 | 0 |
| | Waiting Release Fuel Timer (T6) | <=5 | 0 - 65535 (s) | 0 |
| | Siren Alarm Duration (T7) | <=5 | 1 - 65535 (s) | 60 |
| | Preparing Alarm Timer (T8) | <=5 | 1 - 65535 (s) | 30 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

A numeral to indicate the working mode of the Jamming Behavior Setting function.

- **0** - Disable the JBS function.
- **1** - Jamming Behavior Setting Configure Mode.

✧ *Siren On Timer (T1)*

It specifies the length of time the siren is on.

✧ *Siren Off Timer (T2)*

It specifies the length of time the siren is off.

✧ *Ready Fuel Release Timer (T3)*

It indicates the length of time when the fuel is cut off and whether jamming state is checked every 2 seconds is determined according to the <Check Jamming in T3>.

✧ *Check Speed*

Whether to check speed when the device enters jamming state.

- **0** - Disable speed check.
- **1** - Enable speed check.

✧ *Speed Limit*

The speed limit to cut off fuel.

✧ *Output 1 Init State*

It is used to set the initial state of output 1.

✧ *Motion Sensor*

Whether the motion sensor needs to measure the motion state to cut off fuel when the GNSS fix failure timeout expires. If <Motion Sensor> is set to 0, the state machine will always measure the GNSS fix state.

- **0** - Disable motion sensor.
- **1** - Enable motion sensor.

✧ *GNSS Fix Failure Timeout Timer (T4)*

It indicates the time length of GNSS timeout.

✧ *Enable Siren*

It defines whether to control siren with digital output 2 in the current JBS state.

✧ *Release Fuel Cut-off Timer (T5)*

If the device enters JBS and then cuts off fuel, it will check the current jamming state when the <Fuel Cut-off Timer (T3)> condition is met. If the device doesn't quit the jamming state and the value of <Release Fuel Cut-off Timer> is greater than 0, the device will release fuel cut-off and <Release Fuel Cut-off Timer> will start to work.

When the <Release Fuel Cut-off Timer> condition is met, the device will check the current jamming state. If the device doesn't quit the jamming state, it will check the condition and decides whether to cut off fuel again. If the device doesn't quit the jamming state and the value of <Release Fuel Cut-off Timer> is 0, the device will keep fuel cut-off status until it quits the jamming state.

✧ *Check Jamming in T3*

It indicates whether the JBS state machine starts T3 timer to check jamming state and starts T6 timer if the device quits jamming state.

- **0** - Do not check jamming state (compatible with old JBS state machines).
- **1** - Check jamming state and start T6 timer.

✧ *Waiting Release Fuel Timer (T6)*

It indicates the length of time to be waited before releasing fuel and quitting JBS state machine.

✧ *Siren Alarm Duration (T7)*

It indicates the length of time the siren alarm sounds.

✧ *Preparing Alarm Timer (T8)*

It indicates the length of time for alarm preparation.

✧ **Jamming Behavior Setting Reset Mode**

| |
|---|
| Example:<br>AT+GTJBS=gv58lau,2,,,,,001A$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | JBS | JBS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 2 | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

- *Mode*

  A numeral to indicate the working mode of the JBS function.

  - **2** - Jamming Behavior Setting Reset Mode.

### 3.5.8 JDC (Network Jamming Detection)

The command **AT+GTJDC** is used to configure the parameters for jamming detection. When the detection condition is met, the device will report the event message **JDR** (ASCII)/(HEX) or **JDS** (ASCII)/(HEX) to the backend server according to the <Mode> setting.

| |
|---|
| Example:<br>AT+GTJDC=gv58lau,2,15,70,,70,2,60,10,0,1,0,0,,001A$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | JDC | JDC |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-2 | 0 |
| | LTE Threshold1 | <=2 | 0 - 20 | 15 |
| | LTE Threshold2 | <=3 | 0 - 130 | 70 |
| | Reserved | 0 | | |
| | 2G Threshold1 | <=2 | 0 - 99 | 70 |
| | 2G Threshold2 | 1 | 0 - 6 | 2 |
| | Enter Jamming Duration | <=4 | 0 - 3600(s) | 60 |
| | Exit Jamming Duration | <=4 | 0 - 3600(s) | 10 |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the jamming detection function.

- **0** - Disable jamming detection function.
- **1** - Enable jamming detection function.
  If jamming is detected, the device will report the **+RESP:GTJDR** message when it enters the jamming state.
- **2** - Enable jamming detection function.
  If jamming is detected, the device will report the **+RESP:GTJDS** message when it enters or exits the jamming state.

✧ *LTE Threshold1, LTE Threshold2*

The built-in LTE jamming detection algorithm uses these two parameters to measure whether the device is currently being jammed. The smaller the parameter value, the more sensitive the detection.

✧ *2G Threshold1, 2G Threshold2*

The built-in 2G jamming detection algorithm uses these two parameters to measure whether the device is currently being jammed. The smaller the parameter value, the more sensitive the detection.

✧ *Enter Jamming Duration*

When the device detects jamming, it will trigger the "enter jamming" event based on the <Enter Jamming Duration>.

✧ *Exit Jamming Duration*

When the device quits jamming, it will trigger the "quit jamming" event based on the <Exit Jamming Duration>.

Note

The device starts to judge jamming state only after it is disconnected from the network.

## 3.5.9 RMD (Roaming Detection)

The command **AT+GTRMD** is used to configure the parameters for roaming detection.

Example:
AT+GTRMD=gv58lau,0,,,,,1,1,46003,,,2,2,46007,,,3,3,46001,,,3DEF,,,3DEF,,,,,0,0,0,0,,,0001$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | RMD | RMD |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Body | Mode | 1 | 0 \| 1 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Home Operator Start | <=2 | 1 - 10 | |
| | Home Operator End | <=2 | 1 - 10 | |
| | Home Operator List | <=6*10 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Roaming Operator Start | <=3 | 1 - 100 | |
| | Roaming Operator End | <=3 | 1 - 100 | |
| | Roaming Operator List | <=6*100 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Blocklist Operator Start | <=2 | 1 - 20 | |
| | Blocklist Operator End | <=2 | 1 - 20 | |
| | Blocklist Operator | <=6*20 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Known Roaming Event Mask | <=6 | 000000 - FFFFFF | 3DEF |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Unknown Roaming Event Mask | <=6 | 000000 - FFFFFF | 3DEF |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the roaming detection function.

- **0** - Disable this function.
- **1** - Enable this function.

✧ *Operator Start*

A numeral which indicates the first index of the allowlist operator numbers to be input.

For example, if the value is 1, the device will update the allowlist of operators from the first one. If the parameter is empty, there should be no allowlist number.

✧ *Operator End*

A numeral which indicates the last index of the allowlist operator numbers to be input.

For example, if the value is 2, the device will update the allowlist of operators until the second one. If the parameter is empty, there should be no allowlist number.

✧ *Home Operator List*

An allowlist of PLMN operator numbers. The numbers are composed of MCC and MNC, both of which consist of 3 digits. And the last digit of MNC can be omitted, for example, both "46001F" and "46001" are the PLMN of CHINA UNICOM.

The operators in this list will be considered as in "Home" state. Two adjacent operator numbers are separated with ",". The number of the operators in the list is determined by the parameters <Operator Start> and <Operator End>. For example, if <Operator Start> is 1 and <Operator End> is 2, the operator list should include 2 operator numbers (empty value acceptable) and the two numbers are separated with ",".

"MCCFF" type code is used to identify operators across a whole country, for example, "460FF" covers the mobile network operators across China.

✧ *Roaming Operator List*

It is mostly like <Home Operator List>, and the difference is that the operators in this list will be considered to be in "Known Roaming" state.

✧ *Blocklist Operator*

It is mostly like <Home Operator List>, and the difference is that the operators in this list will be considered to be in "Blocking Report" state. In this state, the device works properly but all reports will be buffered instead of being sent.

Note

Operators that are not in <Home Operator List>, <Roaming Operator List> or <Blocklist Operator> will be considered to be in "Unknown Roaming" state.

✧ *Known Roaming Event Mask*

Bitwise mask to configure which event report should be sent to the backend server when roaming state is detected. If the roaming state is "Known Roaming", <Known Roaming Event Mask> will be valid; if the roaming state is "Unknown Roaming", <Unknown Roaming Event

Mask> will be valid.

- **Bit 0** - for **PNA** (ASCII)/(HEX)
- **Bit 1** - for **PFA** (ASCI)/(HEX)
- **Bit 2** - for **MPN** (ASCII)/(HEX)
- **Bit 3** - for **MPF** (ASCII)/(HEX)
- **Bit 5** - for **BPL** (ASCII)/(HEX)
- **Bit 6** - for **BTC** (ASCII)/(HEX)
- **Bit 7** - for **STC** (ASCII)/(HEX)
- **Bit 8** - for **STT** (ASCII)/(HEX)
- **Bit 10** - for **PDP** (ASCII)/(HEX)
- **Bit 11** - for the power on **RTL** (ASCII)/(HEX)
- **Bit 12** - for the ignition report **IGN** (ASCII)/(HEX), **IGF** (ASCII)/(HEX), **VGN** (ASCII)/(HEX) and **VGF** (ASCII)/(HEX)
- **Bit 13** - for the ignition on/off location reports **IGL**(ASCII)/(HEX) and **VGL** (ASCII)/ (HEX)
- **Bit 15** - for **PNR** (ASCII)/(HEX)
- **Bit 16** - for **PFR** (ASCII)/(HEX)

For each bit, set it to 1 to enable the corresponding event report, and set it to 0 to disable the corresponding event report.

✧ *Unknown Roaming Event Mask*

It is mostly like <Known Roaming Event Mask>.

✧ *Output ID, Output Status, Duration, Toggle Times*

If this function is enabled and roaming state is detected, the specified wave will be output on the specified output.

Note

If more operators are needed, please adjust <Operator Start> and <Operator End> for appropriate setup. If some operators in <Operator List> are empty, then the corresponding operators will be deleted. For example, to delete the 4th, 5th and 6th operators of the <Operator List>, please set <Operator Start> to 4 and set <Operator End> to 6 and keep those three operators of <Operator List> empty.

### 3.5.10 SOS (SOS Setting)

This command is used to configure a specified input port for emergency.

When an emergency occurs, the end user can use the specified input port to trigger the SOS function and the device will report the position message **SOS** (ASCII)/(HEX) to the backend server. A specified wave shape can be configured to output on a specified output port.

| |
|---|
| Example: |
| AT+GTSOS=gv58lau,0,0,,0,0,0,0,,,1,,000D$ |

| Parts | Fields | Length | Range/Format | Default |
| --- | --- | --- | --- | --- |
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | SOS | SOS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 \| 2 \| 4 | 0 |
| | Digital Input ID | 1 | 0 - 1 | 0 |
| | Reserved | 0 | | |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | GNSS Position Type | 1 | 1-3 | 1 |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the SOS function.

- **0** - Disable SOS function.
- **2** - Send the current position to the backend server only.
- **4** - Send the current position to the SMS gateway via SMS.

✧ *Digital Input ID*

The ID of the digital input port which triggers the SOS function. 0 means "Disable the SOS function".

The digital input port should first be configured by the command AT+GTDIS for the SOS function. If a digital input port is configured to trigger the SOS function, there is no **DIS** (ASCII)/(HEX) report message for the specified digital input port.

✧ *GNSS Position Type*

A numeral to indicate the type of the GNSS position (last GNSS position or current GNSS position) to be reported to the backend server.

- **1** - Try to report the current GNSS location when the SOS event is triggered.
- **2** - Only report the last GNSS location immediately when the SOS event is triggered.
- **3** - Report the last GNSS location immediately when the SOS event is triggered and then the device tries to get the current GNSS location to be reported.

Note

If <GNSS Position Type> is set to 2, the **GSM** (ASCII)/(HEX) report will not be triggered.

### 3.5.11 BZA (Buzzer Alarm)

This command is used to set buzzer alarm. There are four kinds of alarms, and each alarm outputs a different buzzer sound. Before using these alarms, the output ID connected to the buzzer should be configured and enabled.

The overspeed alarm event (refer to the command AT+GTSPA) can trigger the buzzer alarm defined by this command.

| |
|---|
| Example:<br>AT+GTBZA=gv58lau,2,,,,1,0,0,,,0,10,1,,,2,0,0,,,1,2,10,,,,,,,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | BZA | BZA |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Output ID | 1 | 0 \| 2 - 3 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Alarm 1 Output Status | 1 | 0-2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Alarm 2 Output Status | 1 | 0-2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Alarm 3 Output Status | 1 | 0-2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Alarm 4 Output Status | 1 | 0-2 | 0 |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Output ID*

The output port connected to the buzzer.

### 3.5.12 SPA (Overspeed Alarm)

This command is used to set the speed thresholds and bind an alarm type to each speed threshold for the buzzer alarm. If the current speed meets one of the thresholds, the buzzer will emit a sound corresponding to the alarm type.

Example:
AT+GTSPA=gv58lau,1,50,,60,0,,,70,,60,0,,,90,,60,0,,,110,,60,0,,,,,,,,,000C$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | SPA | SPA |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-2 | 0 |
| | Speed Threshold 1 | <=3 | 0 - 400(km/h) | 50 |
| | Reserved | 0 | | |
| | Validity | <=4 | 0 - 3600(s) | 60 |
| | Alarm Type | 1 | 0 \| 1 - 4 | 0 |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Speed Threshold 2 | <=3 | 0 - 400(km/h) | 70 |
| | Reserved | 0 | | |
| | Validity | <=4 | 0 - 3600(s) | 60 |
| | Alarm Type | 1 | 0 \| 1 - 4 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Speed Threshold 3 | <=3 | 0 - 400(km/h) | 90 |
| | Reserved | 0 | | |
| | Validity | <=4 | 0 - 3600(s) | 60 |
| | Alarm Type | 1 | 0 \| 1 - 4 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Speed Threshold 4 | <=3 | 0 - 400(km/h) | 110 |
| | Reserved | 0 | | |
| | Validity | <=4 | 0 - 3600(s) | 60 |
| | Alarm Type | 1 | 0 \| 1 - 4 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the overspeed alarm function.

- **0** - Disable this function.
- **1** - Strict standard mode. In this mode, the device will check the speed and trigger the buzzer alarm when accelerating or decelerating.
- **2** - Warning mode. In this mode, the device will only check the speed and trigger the buzzer alarm when accelerating.

- *Speed Threshold*
  The minimum speed to trigger the buzzer alarm.
- *Validity*
  If the speed meets the alarm condition and is maintained for a period longer than <Validity>, the buzzer alarm will be triggered.
- *Alarm Type*
  The alarm type for each speed threshold. 0 means "No buzzer alarm".

## 3.5.13 SPD (Speed Alarm)

The command **AT+GTSPD** is used to set a speed range for the speed alarm function of the terminal. According to the working mode, the device will report the message **SPD** (ASCII)/(HEX) to the backend server when its moving speed is outside or within the range.

| Example: |
|---|
| AT+GTSPD=gv58lau,1,0,0,60,300,0,0,0,0,,,,,,,,,,,,000C$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | SPD | SPD |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-3 | 0 |
| | Min Speed | <=3 | 0 - 400(km/h) | 0 |
| | Max Speed | <=3 | 0 - 400(km/h) | 0 |
| | Validity | <=4 | 0 - 3600(s) | 60 |
| | Send Interval | <=4 | 30 - 3600(s) | 300 |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the speed alarm function.

- **0** - Disable speed alarm.
- **1** - Report speed alarm if the current speed is within the speed range specified by <Min Speed> and <Max Speed>.
- **2** - Report speed alarm if the current speed is outside the speed range specified by <Min Speed> and <Max Speed>.
- **3** - Report speed alarm only once if the current speed is within or outside the speed range specified by <Min Speed> and <Max Speed>. In this mode, <Send Interval> will be ignored.

✧ *Min Speed*

The lower speed limit.

✧ *Max Speed*

The upper speed limit.

✧ *Validity*

If the speed meets the alarm condition and is maintained longer than <Validity>, the speed alarm will be triggered.

✧ *Send Interval*

The interval for sending the speed alarm message.

### 3.5.14 SSR (Start/Stop Report)

The command **AT+GTSSR** is used to detect the state of vehicle. When the vehicle enters Start state, the device will report the event message **STR** (ASCII)/(HEX) to the backend server. When the vehicle exits the Start state and then enters Stop state, the device will report the event message **STP** (ASCII)/(HEX) to the backend server.

```
Example:
AT+GTSSR=gv58lau,1,2,1,5,30,0,1,,000F$
```

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | SSR | SSR |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 \| 1 | 0 |
| | Time to Stop | <=4 | 0 - 30(min)\|0 - 1800(s) | 2(min) |
| | Time to Start | <=3 | 0 - 5(min)\|0 - 300(s) | 1(min) |
| | Start Speed | <=2 | 1 - 10(km/h) | 5 |
| | Long Stop | <=5 | 0 - 43200(min) | 0 |
| | Time Unit | 1 | 0 \| 1 | 0 |
| | Fast STP Mode | 1 | 0 \| 1 | 1 |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the Start/Stop report function.

- **0** - Disable this function.
- **1** - Enable this function.

✧ *Time to Stop*

After entering the Start state, if the vehicle is stationary again and remains stationary for the time specified in this parameter, it is considered to exit the Start state.

✧ *Time to Start*

If the vehicle is moving with ignition on for the period specified by this parameter, it is considered to be in Start state.

✧ *Start Speed*

The start speed threshold to determine whether the vehicle is started or not. When it is detected by the built-in motion sensor that the vehicle is moving with ignition on, the device will start to check the speed from GNSS.

If the device speed is maintained at a level higher than <Start Speed> for the time longer than <Time to Start>, the vehicle is considered to be in Start state, and the event message **+RESP:GTSTR** will be reported.

If the device speed stays at a level lower than or equal to <Start Speed> for the time longer than <Time to Stop>, the vehicle is considered to quit Start state and the event message **+RESP:GTSTP** will be reported.

If abnormal GNSS fix lasts more than 1 minute, the built-in motion sensor is used to detect the Start/Stop state only without checking the speed.

✧ *Long Stop*

After the vehicle enters Stop state and stays in Stop state for the time specified by this parameter, the **LSP** (ASCII)/(HEX) message will be reported. 0 means "Disable this parameter".

✧ *Time Unit*

It controls the time unit of <Time to Stop> and <Time to Start>.

- **0** - Unit: minute.
- **1** - Unit: second.

✧ *Fast STP Mode*

Whether to report **+RESP:GTSTP** message immediately when the engine is turned off.

- **0** - The event message **+RESP:GTSTP** will be reported when the engine is turned off and the time meets the <Time to Stop>.
- **1** - The event message **+RESP:GTSTP** will be reported immediately when the engine is turned off, but it only works when <Mode> is set to 1.

### 3.5.15 TOW (Tow Alarm)

The **AT+GTTOW** command is used to configure sensitivity setting of the motion sensor and the tow alarm parameters.

| Example: |
|---|
| AT+GTTOW=gv58lau,1,10,1,300,0,0,0,0,2,3,2,600,,,,,,,,000B$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | TOW | TOW |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 \| 1 | 0 |
| | Engine Off to Tow | <=2 | 5 - 15(min) | 10 |
| | Fake Tow Delay | <=2 | 0 - 10(min) | 1 |
| | Tow Interval | <=5 | 30 - 86400(s) | 300 |
| | Tow Output ID | 1 | 0 - 3 | 0 |
| | Tow Output Status | 1 | 0 - 2 | 0 |
| | Tow Output Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Tow Output Toggle Times | <=3 | 0 - 255 | 0 |
| | Rest Duration | <=3 | 1 - 255(x15s) | 2 |
| | Motion Duration | <=2 | 1 - 10(x100ms) | 3 |
| | Motion Threshold | 1 | 1 - 9 | 2 |
| | Tow Distance | <=5 | 0 \| 50 - 65535(m) | 600 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

Enable/disable the tow alarm function.

- **0** - Disable.
- **1** - Enable.

✧ *Engine Off to Tow*

A time parameter used to measure whether the device is considered to be towed after the engine is turned off. If the motion sensor does not detect non-movement within the specified time after the engine is off, the device is being towed.

✧ *Fake Tow Delay*

If the motion sensor detects movement after engine off and non-movement are detected previously, the device enters a state called fake tow. If the device stays in fake tow longer than the time specified by <Fake Tow Delay>, it is considered to be towed.

✧ *Tow Interval*

The interval for sending the tow alarm message. If <Tow Interval> is less than 60 seconds, the GNSS will keep working.

✧ *Tow Output ID*

The ID of the output port to output the specified wave shape when the tow event is detected.

✧ *Tow Output Status*

Please refer to the parameter <Output Status> in AT+GTDOS.

✧ *Tow Output Duration*

Please refer to the parameter <Duration> in AT+GTDOS.

✧ *Tow Output Toggle Times*

Please refer to the parameter <Toggle Times> in AT+GTDOS.

✧ *Rest Duration*

A time parameter to measure whether the device enters stationary state. The state of the device will be changed to rest if the motion sensor detects stationary and the stationary state is maintained for a period longer than the time specified by the parameter <Rest Duration>.

✧ *Motion Duration*

A time parameter to measure whether the device enters moving state. The state of the device will be changed to motion if the motion sensor detects motion and the moving state is maintained for a period specified by the parameter <Motion Duration>.

✧ *Motion Threshold*

The threshold for the motion sensor to measure whether the device is moving. The smaller the value, the more sensitive the motion detection. Please note that the default value is the ideal setting for motion and static detection. Therefore, exercise caution when modifying threshold parameters; otherwise, motion and static detection may be affected.

- *Tow Distance*

  A distance parameter to determine whether the device is considered to be towed after the engine is turned off. If the vehicle moves a distance greater than or equal to <Tow Distance> after engine is off, the device is considered to be towed and the message **+RESP:GTTOW** will be reported.

## 3.6 IO Applications

This section describes the commands related to Input and Output port settings.

### 3.6.1 DIS (Digital Input Setting)

The command **AT+GTDIS** is used to configure the parameters of the digital input ports. The input <Ignition Detection> is dedicated to ignition detection, others can be customized. If the logic status is changed on one of the customized digital input ports, the device will report the message **DIS** (ASCII)/(HEX) to the backend server.

| |
|---|
| Example:<br>AT+GTDIS=gv58lau,0,,5,0,1,0,2,0,,,,,,,,,,0,,,0005$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | DIS | DIS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Ignition Detection | 1 | 0 | 0 |
| | Reserved | 0 | | |
| | Debounce Time | <=2 | 0 - 20(x10ms) | 5 |
| | Validity Time | <=2 | 0 \| 1 - 12(x2s) | 0 |
| | Input ID 1 | 1 | 1 | 1 |
| | Mode1 | 1 | 0 \| 1 | 0 |
| | Debounce Time1 | <=2 | 0 - 20(x10ms) | 2 |
| | Validity Time1 | <=2 | 0 \| 1 - 12(x2s) | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Last Position | 1 | 0 \| 1 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

- *Password*

  The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".
- *Ignition Detection*

  The ID of the ignition detection port.
- *Input ID*

  The digital input port ID.
- *Mode*

  Enable/disable the interrupt input.
  - **0** - Disable.
  - **1** - Enable.
- *Debounce Time*

  The time for interruptible input port debouncing.
- *Validity Time*

  The validity time of the input port. 0 means "Do not check the validity time".
- *Last Position*

  A numeral to indicate whether to include real time position or last known position in the report **+RESP:GTDIS**.
  - **0** - Include real time position in the report **+RESP:GTDIS**. It takes time to get GNSS position before the report.
  - **1** - Include the last known position in the report **+RESP:GTDIS**. The report can be sent to the backend server immediately after digital input state changes.

### 3.6.2 DOS (Digital Output Setting)

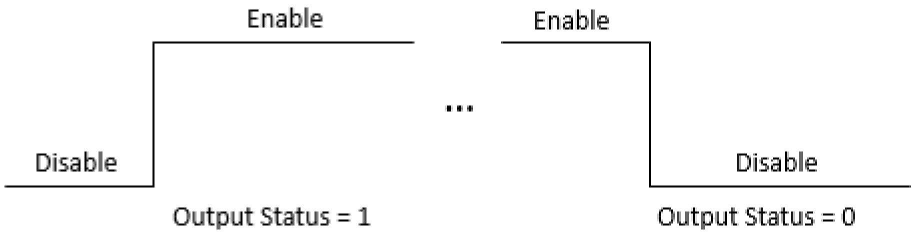
The **AT+GTDOS** command is used to output specified wave shape from the digital output ports. The supported wave shapes are shown below. If it is set to wave shape 1, the device will maintain this wave shape at the specified output port after power reset.

The digital output 1 is a latched output. The final status of the output will be latched during

power off.
If a specified output port is set to wave shape 4, then the port will output square wave. When the main power is off, the port will stop outputting the wave; if the main power is turned on again, the port will start to output the wave again. If the device is rebooted, the port will still output the wave.

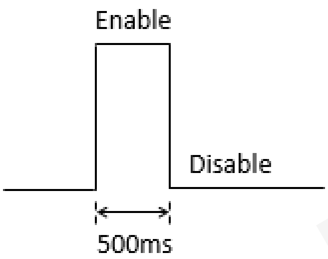
**Wave Shape 1:**

✓ *<Duration>* = 0ms, *<Toggle Times>* = 0



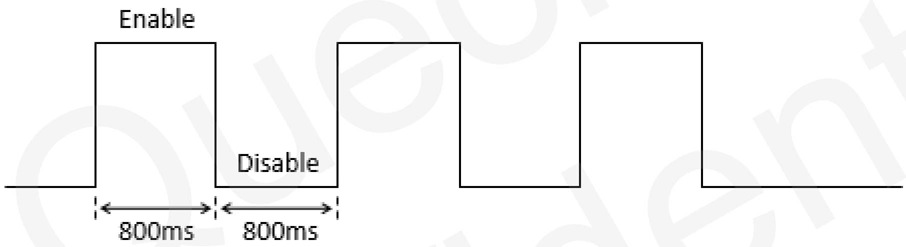

**Wave Shape 2:**

✓ *<Duration>* = 500ms, *<Toggle Times>* = 1



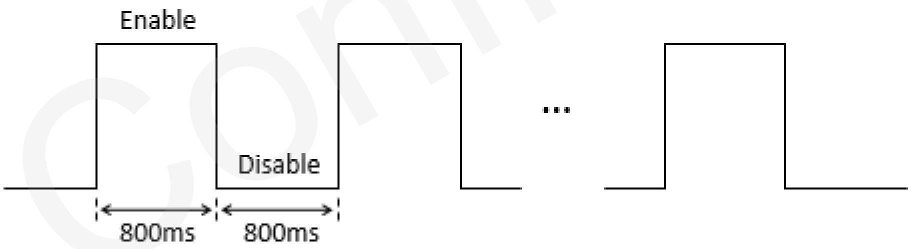

**Wave Shape 3:**

✓ *<Duration>* = 800ms, *<Toggle Times>* = 3



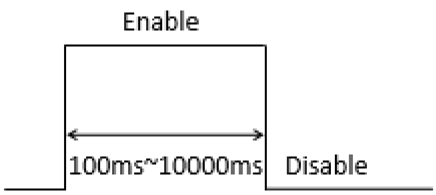

**Wave Shape 4:**

✓ *<Duration>* = 800ms, *<Toggle Times>* = 0




**Wave Shape 5:**

✓ *<Duration>* = [100,10000]ms, *<Toggle Times>* = 1



Example:
AT+GTDOS=gv58lau,0,1,3,2,,190,0,,0,0,5,,,,FFFF$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | DOS | DOS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Command Frame Version | 1 | 0 | 0 |
| | Number of Output Ports | 1 | 1 - 3 | 3 |
| | Output ID | 1 | 1 - 3 | |
| | Output Status | 1 | 0-2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | |
| | Toggle Times | <=3 | 0 - 255 | |
| | Long Operation | <=3 | 0 - 120(min) | |
| | Reserved | 0 | | |
| | DOS Report | 1 | 0 - 7 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Special Wave Monitor Mask | <=2 | 00 - FF | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

- *Password*

  The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".
- *Command Frame Version*

  It indicates the version of this command. Its value will be changed if this command needs to be extended.
- *Number of Output Ports*

  The number of output ports to be set. The six parameters <Output ID>, <Output Status>, <Duration>, <Toggle Times>, <Long Operation> and <Reserved> following this parameter will repeat <Number of Output Ports> times.
- *Output ID*

  It indicates the ID of output ports.
- *Output Status*

  It is used to set the final status of the output port and valid only for wave shape 1.

- **0** - Inactive status.
- **1** - Active status.
- **2** - Gradual-progressive-wave-shape active status. For detailed information, please refer to AT+GTGDO.

✧ *Duration*

Please refer to the wave shapes shown above. Unit: 100ms.

✧ *Toggle Times*

Please refer to the wave shapes shown above. When <Duration> is set to 0, <Toggle Times> must be set to 0; otherwise, the command is invalid.

✧ *DOS Report*

Whether to report **DOS** (ASCII)/(HEX) when the status of wave shape 1 output changes.

- **Bit 0** - For output 1 (not) to report **+RESP:GTDOS**
- **Bit 1** - For output 2 (not) to report **+RESP:GTDOS**
- **Bit 2** - For output 3 (not) to report **+RESP:GTDOS**

For each bit, set it to 1 to enable the report, and set it to 0 to disable the report.

Note

This parameter is also valid for the Gradual-progressive-wave-shape function.

✧ *Long Operation*

The long operation time for output and it is valid only when the output wave shape is 1 or 4. The output wave will be stopped on the special output port after the length of time specified by the parameters.

✧ *Special Wave Monitor Mask*

A mask bit that indicates exactly which waveforms need to be monitored, and the **DOM** (ASCII)/(HEX) message will be reported when that waveform occurs.

Special Wave Monitor Mask Table:

| Mask Bit | Item |
|---|---|
| Bit 7 | Reserved For Extend |
| Bit 6 | Reserved |
| Bit 5 | Reserved |
| Bit 4 | Reserved |
| Bit 3 | Reserved |
| Bit 2 | Reserved |
| Bit 1 | Reserved |
| Bit 0 | Wave5 |

### 3.6.3 EPS (External Power Monitoring)

The command **AT+GTEPS** is used to configure the parameters for external power supply monitoring. The device will measure and monitor the voltage of the external power supply. If the voltage of the external power supply meets the predefined alarm condition, the device will report an alarm message **EPS** (ASCII)/(HEX) to the backend server to notify the status of the external power supply.

To ensure that this function works properly, switch on the internal battery in case the voltage of the external power supply drops to a very low level.

| |
|---|
| Example:<br>AT+GTEPS=gv58lau,0,250,12000,0,0,0,0,0,0,0,0,0,0,0007$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | EPS | EPS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-5 | 0 |
| | Min Threshold | <=5 | 0 - 32000 (mV) | 0 |
| | Max Threshold | <=5 | 0 - 32000 (mV) | 0 |
| | Sample Period | <=2 | 0 \| 1 - 12(x2s) | 0 |
| | Debounce Time | 1 | 0 - 5 (x1s) | 0 |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Sync with FRI | 1 | 0 \| 1 | 0 |
| | Voltage Margin Error | <=3 | 0 - 100(x10mV) | 0 |
| | Debounce Voltage Threshold | <=3 | 0 - 100(x100mV) | 0 |
| | MPN / MPF Validity Time | 1 | 0 - 5(x1s) | 0 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the external power supply monitoring function.

- **0** - Disable.
- **1** - Enable. If the current voltage is within the range of (<Min Threshold>, <Max Threshold>), the **+RESP:GTEPS** alarm will be triggered.
- **2** - Enable. If the current voltage is outside the range of (<Min Threshold>, <Max Threshold>), the **+RESP:GTEPS** alarm will be triggered.
- **3** - Enable. If the current voltage is within or outside the range of (<Min Threshold>, <Max Threshold>), the **+RESP:GTEPS** alarm will be triggered.

- **4** - Enable. If the current voltage is higher than the <Max Threshold>, the **+RESP:GTEPS** alarm will be triggered.
- **5** - Enable. If the current voltage is lower than the <Min Threshold>, the **+RESP:GTEPS** alarm will be triggered.

✧ *Min Threshold*

The lower voltage limit of the external power supply to trigger the alarm.

✧ *Max Threshold*

The upper voltage limit of the external power supply to trigger the alarm.

✧ *Sample Period*

The time for sampling the external power supply.

✧ *Debounce Time*

The time for debouncing external power voltage to avoid false report caused by excessive voltage drop in a short period of time.

✧ *Output ID*

It specifies the ID of the output port to output specified wave shape when the **+RESP:GTEPS** alarm is triggered. If it is set to 0, there will be no output wave.

✧ *Sync with FRI*

Besides the **+RESP:GTEPS** alarm report, the device can also send the voltage of external power supply periodically along with the fixed report message.

- **0** - Do not report external power supply voltage with fixed report message.
- **1** - Report external power supply voltage with fixed report message.

✧ *Voltage Margin Error*

This parameter is used together with parameters <Min Threshold> and <Max Threshold>. It indicates the voltage margin error of <Min Threshold> and <Max Threshold>.

If the voltage value detected is within the <Voltage Margin Error> of <Min Threshold> or <Max Threshold>, it will not trigger the **+RESP:GTEPS** alarm report.

For example, if <Min Threshold> is set to 6000mV, <Max Threshold> is set to 12000mV, and <Voltage Margin Error> is set to ±100mV, the current voltage will not trigger **+RESP:GTEPS** alarm report when it meets the condition (5900mV < current voltage < 6100mV) or (11900mV < current voltage < 12100mV). This parameter improves the performance of **+RESP:GTEPS** alarm.

✧ *Debounce Voltage Threshold*

This parameter is used together with <Debounce Time>. If the voltage drops or bursts dramatically more than <Debounce Voltage Threshold>, the device will start to debounce voltage for the time specified by <Debounce Time>.

✧ *MPN / MPF Validity Time*

The validity time for detecting the device connected or disconnected from the main power supply. 0 means "Do not check the validity time".

If <MPN / MPF Validity Time> is not 0, and the device remains connected or disconnected from the main power supply for the time specified by this parameter, the device will report **MPN** (ASCII)/(HEX) or **MPF** (ASCII)/(HEX) to the backend server.

If <MPN / MPF Validity Time> is 0, the device will immediately report **MPN** (ASCII)/(HEX) or **MPF** (ASCII)/(HEX) to the backend server.

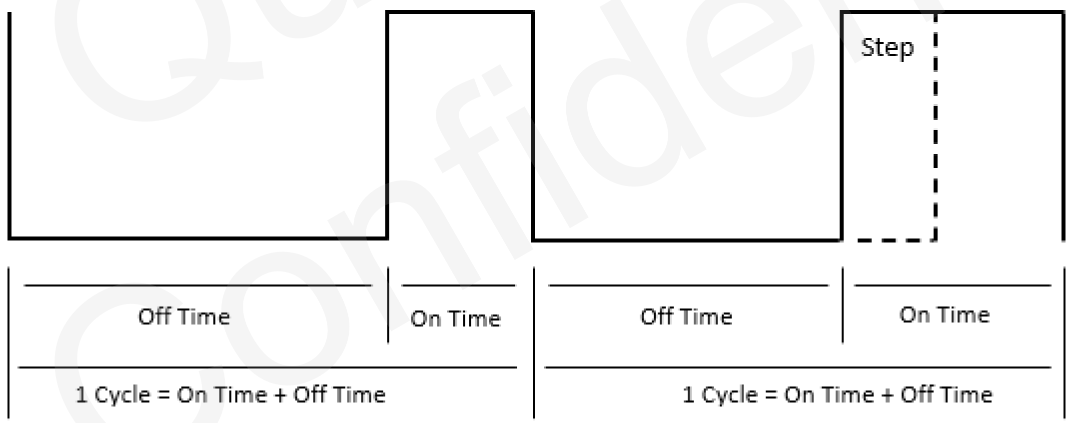
### 3.6.4 GDO (Gradual Digital Output)

The **AT+GTGDO** command is used to configure specified gradual progressive wave shape from the digital output ports.

For progressive output, an increment step is added to the ON Time until the ON Time (including the time increment) is equal to or greater than the Cycle time. This phase is defined as progressive state.
After the condition **On Time + Incremental Step >= Cycle Time** is reached, the output becomes steady until it is deactivated by the command AT+GTDOS. This phase in which the output is steady is defined as constant state. If the device reboots during constant state and the AT+GTPDS settings are configured and enabled, the device will restore the previous output state.
The next time the progressive output is activated, the cycle described above starts over regardless of the previous progressive state.
The figure below shows the components of an output cycle. Here are some notes:

- The time for one complete cycle is equal to OFF time plus ON time.
- For constant output, the ON time and the Cycle time should be the same.
- For progressive output, when the <Output Status> is set to 2, an incremental step will be added to the ON time at the end of a cycle before the start of the next.

Gradual Progressive Wave Shape




| Example: |
| --- |
| AT+GTGDO=gv58lau,0,30,1,,,,,,0004$ |

| Parts | Fields | Length | Range/Format | Default |
| --- | --- | --- | --- | --- |
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | GDO | GDO |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Body | First ON Time | <=3 | 0 - 100 | 0 |
| | Cycle Time | <=3 | 0 - 100 | 30 |
| | Incremental Step | <=3 | 0 - 100 | 1 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *First ON Time*

The time that the output is in active state during the first cycle. Unit: 100ms.

✧ *Cycle Time*

The time that forms a complete cycle. Unit: 100ms.

✧ *Incremental Step*

The time that shall be added to the ON/OFF time before starting the next cycle. If this parameter value is 0, the cycles are equal. Unit: 100ms.

Note

The value of <First ON Time> cannot be greater than <Cycle Time>.

### 3.6.5 IOB (Input/Output Binding)

This command is used to configure the user defined output port actions triggered by input ports. If the I/O binding is configured and the corresponding condition is met, the device will output specified wave shape on the specified output port. Otherwise, the device will restore the initial state of the specified output port. The device will report the message **IOB** (ASCII)/(HEX) to the backend server when the logic state of the bound input ports changes.

Example:
AT+GTIOB=gv58lau,3,3,3,0,0,0,0,0,,,,,0006$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | IOB | IOB |
| | Leading Symbol | 1 | = | = |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | ID | 1 | 0 - 3 | |
| | Input Mask | 1 | 0 - 3 | 0 |
| | Trigger Mask | 1 | 0 - 3 | 0 |
| | Input Sample Period | <=2 | 0 \| 1 - 12 (x2s) | 0 |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *ID*

The ID of the user defined IO binding.

✧ *Input Mask*

Bitwise mask for input ports composition. Each bit represents one digital input port. Set it to 1 to enable corresponding input port and set it to 0 to disable.

- **Bit 0** - Ignition detection
- **Bit 1** - Digital input 1

✧ *Trigger Mask*

Bitwise mask for trigger condition composition of the corresponding input ports. Each bit represents the logic state of the corresponding input port to trigger the IOB event. Set it to 1 to use "Active state" as the trigger condition and 0 to use "Inactive state" as the trigger condition. Only when the logic state of all the input ports in one IO binding meets the trigger condition will the IOB event be triggered.

- **Bit 0** - Ignition detection
- **Bit 1** - Digital input 1

✧ *Input Sample Period*

The interval for checking the state of all the digital input ports in one IO binding. **AT+GTIOB** and AT+GTDIS use separate sample periods to check the input port state even for the same input port.

✧ *Output ID*

The ID of the output port to output specified wave when the trigger condition is met. 0 means "No wave will be output".

### 3.6.6 ROS (Restrict Output Setting)

The command **AT+GTROS** is used to restrict the OUTPUT when the corresponding event is triggered.

| |
|---|
| Example:<br>AT+GTROS=gv58lau,0,0,0,,,0,,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | ROS | ROS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Reserved | 0 | | |
| | Output ID | 1 | 0 - 1 | 0 |
| | Output Status | 1 | 0 - 1 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Event MASK | <=4 | 0 - FFFF | 0 |
| | Event Parameter (Optional) | | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Event MASK*

A mask bit used to indicate additional events for the output.

- **Bit 0** - Over Speed.

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Over Speed Value | <=3 | 0 - 400(km/h) | 0 |
| Debounce Count | <=2 | 0 - 20 | 0 |

- *Over Speed Value*

  If the current speed is less than or equal to this value, output will be activated. If the current speed is greater than the value, output will not be activated until the speed is less than or equal to this value.

- *Debounce Count*

  The debounce count for getting GNSS speed. If the GNSS fix is successful, **1 Debounce Count=1s**. If <Debounce Count> is set to 3, it means that the current speed is considered to be less than the configured speed only if there are three successful GNSS fix points and the speed is less than the configured speed. If fix fails during the period, the previously successful location point count will not be reset.
- **Bit 1** - Ignition Off.

| Fields | Length | Range/Format | Default |
|---|---|---|---|
| Check IGF | 1 | 0 \| 1 | 0 |

- *Check IGF*

  A numeral to indicate whether to check ignition state before activating output. It is recommended to enable the AT+GTPDS to avoid problems when the device reboots.
  - **0** - Do not check ignition state, activate output immediately.
  - **1** - Check ignition state first. If ignition is off, activate output immediately. If ignition is on, output is not activated unless the ignition is turned off.

## 3.7 Accessories Applications

This section describes the commands related to the configuration accessories which is connected by uart port. Please refer to the details below.

### 3.7.1 DAT (Transparent Data)

The command **AT+GTDAT** is used to transmit data between the backend server and the equipment connected to the Bluetooth of the device. Data to the backend server is wrapped into the message **DAT** (ASCII)/(HEX) and sent to the backend server while data to the equipment is directly output to the

Bluetooth unrestricted by the @Track protocol. And all data is transparent to the device. Before using this command, the parameters of the Bluetooth should be set in the command AT+GTBTS.

| |
|---|
| Example:<br>AT+GTDAT=gv58lau,0,,wferyhtrhtrh,0,,,,0017$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | DAT | DAT |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Command Type | 1 | 0 \| 2 \| 4 \| 5 | |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Data | <= (200\|1280) | ASCII String | |
| | Need ACK | 1 | 0 \| 1 | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *Command Type*

The command type which indicates how to send data.

- **0** - Send message to the backend server with **DAT (Short Format)** (ASCII)/(HEX).
- **2** - Send message to the backend server with **DAT (Long Format)** (ASCII)/(HEX).
- **4** - Send the raw data directly to the Bluetooth.
- **5** - Send the raw data directly to the Bluetooth without CRLF.

✧ *Data*

Data to be transferred between the backend server and the equipment connected to the secondary serial port or Bluetooth of the device. <Data> sent to the backend server cannot contain the character "$" when the value of <SACK Mode> in AT+GTSRI and AT+GTQSS is 1.

Note

The maximum length of the <Data> is 200 when the remote device sends data to the terminal device through Bluetooth.

✧ *Need ACK*

Whether to report **+ACK:GTDAT**.

- **0** - Do not need the ACK report.
- **1** - Need the ACK report.

## 3.7.2 IDA (ID Authentication)

The command **AT+GTIDA** is used to protect against unauthorized use. This is achieved by obtaining IDs from different accessories for driver identification, and connecting an external relay (normally closed relay is recommended) to cut off the starter or the fuel pump.

To use this command, the external relay must be connected to the device. When the device reads an ID, it will report the event message **IDA** (ASCII)/(HEX) to the backend server. If the ID is in the allowlist of the ID numbers, it will be authorized until next time the ignition is turned off. After the ignition is turned off again, the authentication will last for a settable short period of time, during which the driver can turn on the engine again without the need to re-identify himself.

| Example: |
|---|
| AT+GTIDA=gv58lau,1,,,,30,1,30,,,,0,0,0,0,,1,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | IDA | IDA |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0-2 | 0 |
| | Reserved | | | |
| | Reserved | | | |
| | Reserved | | | |
| | Timeout after Ignition Off | <=3 | 0 \| 15 - 600(s) | 30 |
| | Report Mode | 1 | 0 - 7 | 0 |
| | ID Validity Time | <=3 | 15 - 600(s) | 30 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | ID List Type | 1 | 1 \| 2 | 1 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the ID authentication function.

- **0** - Disable this function.
- **1** - Authorized ID Unlock. Only authorized ID cards can unlock the vehicle.
- **2** - Any ID Unlock. Any ID card can unlock the vehicle.

- *Timeout after Ignition Off*
  If the ignition is turned off, it will still be authorized for a short time. In this period, re-authentication is not needed. 0 means "Lock the vehicle when the ignition is turned off".
- *Report Mode*
  The mode of reporting ID.
  - **Bit 0** - Report the authorized ID.
  - **Bit 1** - Report the unauthorized ID.
  - **Bit 2** - Report the ID which has logged out. If authorized ID meets the trigger conditions <ID Validity Time> and <Timeout after Ignition Off>, then the **+RESP:GTIDA** report will be sent to indicate the logout event.

  For each bit, set it to 1 to enable the report, and set it to 0 to disable. If <Report Mode> is 0, no **+RESP:GTIDA** message will be reported.
- *ID Validity Time*
  It will remain authorized for this period of time when the ID is valid.
- *Output ID*
  It specifies the ID of the output port to output specified wave shape when it is authorized.
- *ID List Type*
  This parameter indicates which ID List to use.
  - **1** - Use <MAC List> and <Organization Unique Identifier> in the command AT+GTBID. Please make sure the function AT+GTBID is configured properly.
    To enable this feature, the following conditions must be met:
    – WKF300: Set <Push Button Event> to 1 when the <Keyfob Detection Mode> is set to mode 1, 2 . And there is at least one MAC address in the <MAC List>.
    – ID ELA: Enable <IDELA Detection Mode>, and there is at least one MAC address in the <MAC List>.
    – WID310 / WID330 : Enable <WID310 Detection Mode> , <WID330 Detection Mode> , and there is at least one MAC address in the <MAC List>.
  - **2** - Use <ID Number List> in the command AT+GTIDS.

### 3.7.3 IDS (ID Setting)

The command **AT+GTIDS** is used to set ID Numbers for driver authentication.

| Example: |
|---|
| AT+GTIDS=gv58lau,1,0,8,D2C4FBC58765432122222222,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | IDS | IDS |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 \| 1 | 0 |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | ID Type | 1 | 0 | 0 |
| | Length of ID Number | <=2 | 8 - 28 | |
| | ID Number List | <=700 | '0'-'9' 'a'-'f' 'A'-'F' | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the ID authentication function.

- **0** - Delete ID Number. If <ID Number List> is empty, all ID Number will be deleted. Otherwise, the given ID Numbers in <ID Number List> will be deleted.
- **1** - Append ID Number. In this mode, all given ID Numbers will be appended to the saved ID Numbers.

✧ *ID Type*

A numeral to indicate the ID type.

✧ *Length of ID Number*

A numeral to indicate the length of ID Numbers in <ID Number List>.

✧ *ID Number List*

A list of ID numbers. It is necessary to set <Length of ID Number> so that the quantity of the ID Numbers can be calculated.

Note

The ID numbers with different length cannot be issued together, and only a maximum of (1000/<Length of ID Number>) IDs can be stored at a time.This command can be issued multiple times, but it can only store a maximum of 700 IDs and the configured IDs are valid only if <ID List Type> is set to 2 in the AT+GTIDA command .

### 3.7.4 IEX (Input Expansion)

The command **AT+GTIEX** is used to configure the parameters of expanded inputs. If the logic status is changed on one of the expanded input ports, the device will report the message **DIS** (ASCII)/(HEX) to the backend server.

| |
|---|
| Example:<br>AT+GTIEX=gv58lau,1,0,1,8,,,,,,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | IEX | IEX |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Input Expansion Type | 1 | 1 | 1 |
| | Bind BAS Index | 1 | 0 - 9 | 0 |
| | Input Number | 1 | 1 | |
| | Input ID | 1 | 8 | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Input Expansion Type*

It is used to indicate which extended device the following parameters are used for.

- **1** - Bluetooth Accessory Input Setting. In this mode, it's necessary to enable external Input/Output Bluetooth Accessory.

Note

Due to the limitation of the detection mechanism, changing the input state in a short time may not be detected.

✧ *Bind BAS Index*

It is used to bind the specific configuration in AT+GTBAS when <Input Expansion Type> is set to 1. The value is the same as the index in **AT+GTBAS**.

✧ *Input Number*

The total number of the configured inputs.

If <Input Expansion Type> is set to 1, it indicates the number of configured Bluetooth Input Setting. In one configuration, <Input ID> is included and others are reserved.

✧ *Input ID*

The ID of the configured inputs.

The ID of 8 is Bluetooth accessory input and the parameters <Accessory Type> and <Accessory Model> should be set in AT+GTBAS.

### 3.7.5 OEX (Output Expansion)

The **AT+GTOEX** command is used to output wave shape 1 on expanded output.

Example:
AT+GTOEX=gv58lau,1,0,2,7,0,0,0,8,0,0,0,,,0,,,,,FFFF$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | OEX | OEX |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Output Expansion Type | 1 | 1 | 1 |
| | Bind BAS Index | 1 | 0 - 9 | 0 |
| | Output Number | 1 | 1 - 2 | |
| | Output ID | 1 | 7 - 8 | |
| | Status | 1 | 0 \| 1 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | DOS Report | 1 | 0 - 3 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Output Expansion Type*

It's used to indicate which extended device the following parameters are used for.

- **1** - Bluetooth Output Accessory Setting. In this mode, it's necessary to enable external Input/Output Bluetooth Accessory.

✧ *Output Number*

It is used to indicate the number of the following special parameters.

If <Output Expansion Type> is set to 1, it indicates the number of configured Bluetooth Output Setting. In one configuration, <Bind BAS Index>, <Output ID>, <Status>, <Duration>, <Toggle

Times>, <Reserved>, <Reserved> and <DOS Report> are included and others are reserved.

✧ *Output ID*

The ID of the configured outputs.

The ID ranges from 7 to 8 is Bluetooth accessory output and the parameters <Accessory Type> and <Accessory Model> should be set in AT+GTBAS.

✧ *DOS Report*

Whether to report **DOS** (ASCII)/(HEX) when the state of output wave shape 1 changes. 1 means "Report **+RESP:GTDOS**", and 0 means "Do not report **+RESP:GTDOS**".

If <Output Expansion Type> is set to 1:

- **Bit 0** - For output 7 (not) to report **+RESP:GTDOS**.
- **Bit 1** - For output 8 (not) to report **+RESP:GTDOS**.

### 3.7.6 CAN (CANBUS Setting)

The command **AT+GTCAN** is used to set the CANBUS device configuration for reporting CANBUS device information in **CAN** (ASCII)/(HEX), which includes VIN, vehicle speed, engine speed, engine coolant temperature and other information.

| |
|---|
| Example:<br>AT+GTCAN=gv58lau,0,30,60,C00FFFFF,0,,1FFFFF,0,,,,,,,158,FFFFFFFF,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | CAN | CAN |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 \| 1 | 0 |
| | CAN Report Interval | <=5 | 0 \| 5 - 86400(s) | 0 |
| | CAN Report Interval IGF | <=5 | 0 \| 5 - 86400(s) | 0 |
| | CAN Report Mask | <=8 | 00000000 - FFFFFFFF | C00FFFFF |
| | Additional Event Mask | <=2 | 00 - FF | 0 |
| | Reserved | 0 | | |
| | CAN Report Expansion Mask | <=8 | 00000000 - FFFFFFFF | 1FFFFF |
| | GNSS Assisted Mode | 1 | 0 \| 1 | 0 |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of this function.

- **0** - Disable this function.
- **1** - Enable the CAN chip. In this mode, only CANBUS device information will be reported.

✧ *CAN Report Interval*

The interval for sending the message **+RESP:GTCAN** to the backend server when the ignition is on. 0 means "Do not report the message **+RESP:GTCAN**".

✧ *CAN Report Interval IGF*

The interval for sending the message **+RESP:GTCAN** to the backend server when the ignition is off. 0 means "Do not report the message **+RESP:GTCAN**".

✧ *Additional Event Mask*

A bitwise numeral to control whether to send the message **+RESP:GTCAN** by additional event mask. 0 means "Ignore all additional events".

- **Bit 0** - By ignition on/off event.

✧ *CAN Report Mask*

Bitwise mask to configure the composition of CAN report message.

Note

Bit 31 (for <GSM Information>) and Bit 30 (for <GNSS Information>) of the <CAN Report Mask> only control the composition of **+RESP:GTCAN** in ASCII format.

<+CAN Mask> in AT+GTHRM controls the GSM and GNSS information in HEX format **+RESP:GTCAN**.

Bit 22 of this parameter only controls <Total Distance Impulses> in HEX format **+RESP:GTCAN**.

| Bit | Item | Description |
|---|---|---|
| Bit 31 | <GSM Information> | Including <MCC>, <MNC>, <LAC>, <Cell ID>. |
| Bit 30 | <GNSS Information> | Including <GNSS Accuracy>, <Speed>, <Azimuth>, <Altitude>, <Longitude>, <Latitude>, <GNSS UTC Time>. |
| Bit 29 | <CAN Report Expansion Mask> | If this bit is set to 1, the parameter <CAN Report Expansion Mask> in **AT+GTCAN** is valid.<br>If this bit is set to 0, the parameter <CAN Report Expansion Mask> in **AT+GTCAN** is invalid. |
| Bit 28 | Reserved | |
| Bit 27 | Reserved | |
| Bit 26 | Reserved | |
| Bit 25 | Reserved | |
| Bit 24 | Reserved | |
| Bit 23 | Reserved | |
| Bit 22 | <Total Distance Impulses> | Vehicle total distance measured in Impulses if distance from dashboard is not available. |

| Bit | Item | Description |
|---|---|---|
| Bit 21 | <Total Vehicle Engine Overspeed Time> | The total time when the vehicle engine speed is greater than the limit defined in CAN100 configuration. |
| Bit 20 | <Total Vehicle Overspeed Time> | The total time when the vehicle speed is greater than the limit defined in CAN100 configuration. |
| Bit 19 | <Doors> | An 8-bit hexadecimal number. Each bit contains information of one door. |
| Bit 18 | <Lights> | An 8-bit hexadecimal number. Each bit contains information of one light. |
| Bit 17 | <Detailed Information/Indicators> | A hexadecimal number. Each bit indicates information of one indicator. |
| Bit 16 | <Tachograph Information> | Two bytes. The higher byte describes driver 2 (the one whose card is inserted in tachograph slot 2), and the lower byte describes driver 1. |
| Bit 15 | <Axle Weight 2nd> | Weight of vehicle's second axle. |
| Bit 14 | <Total Idle Fuel Used> | Number of liters of fuel used since vehicle manufacture or device installation. Total idle fuel or energy used - with vehicle speed 0 km/h. |
| Bit 13 | <Total Engine Idle Time> | Time of engine running during idle state since vehicle manufacture or device installation. |
| Bit 12 | <Total Driving Time> | Time of engine running during driving (non-zero speed) since vehicle manufacture or device installation. |
| Bit 11 | <Total Engine Hours> | Time of engine running since vehicle manufacture or device installation. |
| Bit 10 | <Accelerator Pedal Pressure> | The pressure applied on acceleration pedal. |
| Bit 9 | <Range> | The number of kilometers to drive on remaining fuel. |
| Bit 8 | <Fuel Level> | The level of fuel in vehicle's tank (in liter or percentage). |
| Bit 7 | <Fuel Consumption> | The fuel consumption of the engine. |
| Bit 6 | <Engine Coolant Temperature> | The temperature of the engine coolant. |
| Bit 5 | <Engine RPM> | Revolutions per minute of the engine. |
| Bit 4 | <Vehicle Speed> | The speed of vehicle. |

| Bit | Item | Description |
|---|---|---|
| Bit 3 | <Total Fuel Used> | The number of liters of fuel used since vehicle manufacture or device installation. Total fuel or energy used - read from vehicle. |
| Bit 2 | <Total Distance> | Vehicle total distance. |
| Bit 1 | <Ignition Key> | Ignition status. |
| Bit 0 | <VIN> | Vehicle identification number. |

✧ *CAN Report Expansion Mask*

Bitwise mask to configure the composition of expanded CANBUS information of the **+RESP:GTCAN** message.

| Bit | Item | Description |
|---|---|---|
| Bit 31 | Reserved | |
| Bit 30 | Reserved | |
| Bit 29 | Reserved | |
| Bit 28 | Reserved | |
| Bit 27 | Reserved | |
| Bit 26 | Reserved | |
| Bit 25 | Reserved | |
| Bit 24 | Reserved | |
| Bit 23 | <Engine Torque> | The engine torque. Unit: Percentage. |
| Bit 22 | <Rapid Accelerations> | The number of total rapid accelerations since installation and it is calculated based on CAN100 settings of speed increase time and value. |
| Bit 21 | <Rapid Brakings> | The number of total rapid brakings since installation and it is calculated based on CAN100 settings of speed decrease time and value. |
| Bit 20 | <Expansion Information> | A hexadecimal number. Each bit represents the information of one indicator. |
| Bit 19 | <Registration Number> | The vehicle registration number |
| Bit 18 | <Tachograph Driver 2 Name> | The name of tachograph driver 2 |
| Bit 17 | <Tachograph Driver 1 Name> | The name of tachograph driver 1 |
| Bit 16 | <Tachograph Driver 2 Card Number> | The card number of tachograph driver 2 |
| Bit 15 | <Tachograph Driver 1 Card Number> | The card number of tachograph driver 1 |

| Bit | Item | Description |
|---|---|---|
| Bit 14 | <Total Brake Applications> | Count of applying brake pedal (braking process initiated by brake pedal) |
| Bit 13 | <Total Accelerator Kick-down Time> | Total time when accelerator pedal is pressed over 90% |
| Bit 12 | <Total Cruise Control Time> | Total time when the vehicle speed is controlled by cruise-control module |
| Bit 11 | <Total Effective Engine Speed Time> | Total time when the vehicle engine speed is effective |
| Bit 10 | <Total Accelerator Kick-downs> | Count of accelerator pedal kick-downs (with the pedal pressed over 90%) |
| Bit 9 | <Pedal Braking Factor> | It measures how often the driver brakes with brake pedal or engine and stores both counts, which are always increasing. Decreasing speed with brake pedal pressed causes increase of pedal braking factor. |
| Bit 8 | <Engine Braking Factor> | It measures how often the driver brakes with brake pedal or engine and stores both counts, which are always increasing. Decreasing speed with no pedal pressed causes increase of engine braking factor. |
| Bit 7 | <Analog Input Value> | Analog input value |
| Bit 6 | <Tachograph Driving Direction> | Vehicle driving direction from tachograph |
| Bit 5 | <Tachograph Vehicle Motion Signal> | Vehicle motion signal from tachograph |
| Bit 4 | <Tachograph Overspeed Signal> | Tachograph overspeed signal for the vehicle |
| Bit 3 | <Axle Weight 4th> | Weight of vehicle's fourth axle |
| Bit 2 | <Axle Weight 3rd> | Weight of vehicle's third axle |
| Bit 1 | <Axle Weight 1st> | Weight of vehicle's first axle |
| Bit 0 | <Ad-Blue Level> | The level of Ad-Blue |

✧ *GNSS Assisted Mode*

It specifies whether to use GNSS to calculate total distance when the communication between the device and CAN100 is abnormal.

- **0** - Disable.
- **1** - Enable. When the communication between the device and CAN100 is abnormal, the device will use GNSS to calculate total distance and report <Total Distance> in **+RESP:GTCAN**. This parameter works only when the unit of total distance is hectometer.

### 3.7.7 CLT (Canbus Alarm Setting)

The command **AT+GTCLT** is used to set alarm threshold of CANBUS data. It supports a maximum of 20 CANBUS alarm groups. Each CAN alarm trigger condition consists of <Alarm Mask 1>, <Alarm Mask 2> and <Alarm Mask 3>.

For the CAN alarm trigger event information, please refer to <Detailed Information / Indicators>, <Lights>, <Doors> and <Engine RPM> of the message **CAN** (ASCII)/(HEX). If <Alarm Mask 1>, <Alarm Mask 2> and <Alarm Mask 3> meet each trigger condition at the same time, and the trigger event duration time is longer than <Debounce Time>, the alarm message **CLT** (ASCII)/(HEX) will be sent.
Note
The **AT+GTCLT** and AT+GTCAN commands are used together. Only when all of <Alarm Mask 1>, <Alarm Mask 2> and <Alarm Mask 3> meet trigger condition and the trigger event duration time is longer than <Debounce Time> will the **+RESP:GTCLT** alarm message be sent.

Example:
AT+GTCLT=gv58lau,0,0,0,000FFFFF,0,0,0,30,8,001FFFFF,,60,15,0,0,0,0,,0006$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | CLT | CLT |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Group ID | <=2 | 0 - 19 | 0 |
| | Mode | 1 | 0 \| 1 | 0 |
| | Debounce Time | <=3 | 0 - 255(x1s) | 0 |
| | CAN Data Mask | 8 | 00000000 - FFFFFFFF | 000FFFFF |
| | Alarm Mask 1 | <=8 | 00000000 - FFFFFFFF | 0 |
| | Alarm Mask 2 | <=8 | 00000000 - FFFFFFFF | 0 |
| | Alarm Mask 3 | <=8 | 00000000 - FFFFFFFF | 0 |
| | High RPM Threshold | <=3 | 1 - 100(x100 rpm) | 30 |
| | Low RPM Threshold | <=3 | 0 - 99(x100 rpm) | 8 |
| | CAN Report Expansion Mask | <=8 | 00000000 - FFFFFFFF | 001FFFFF |
| | Reserved | 0 | | |
| | Vehicle Speed High Threshold | <=4 | 1 - 6553(km/h) | 60 |
| | Vehicle Speed Low Threshold | <=4 | 0 - 6552(km/h) | 15 |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Output ID | 1 | 0 - 3 | 0 |
| | Output Status | 1 | 0 - 2 | 0 |
| | Duration | <=3 | 0 - 255(x100ms) | 0 |
| | Toggle Times | <=3 | 0 - 255 | 0 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Group ID*

The ID of the CANBUS alarm group. A total of 20 groups are supported.

✧ *Mode*

The CANBUS alarm working mode for each group.

- **0** - Disable the CAN alarm function.
- **1** - Enable the CAN alarm function.

✧ *Debounce Time*

The time for CANBUS alarm trigger event debouncing.

✧ *CAN Data Mask*

Bitwise mask to configure the CAN data composition of the **+RESP:GTCLT** message. <CAN Data Mask> only works in **CLT** ASCII message.

| Bit | Item | Description |
|---|---|---|
| Bit 31 | Reserved | |
| ... | Reserved | |
| Bit 29 | <CAN Report Expansion Mask> | If this bit is set to 1, the parameter <CAN Report Expansion Mask> in **AT+GTCLT** is valid.<br>If this bit is set to 0, the parameter <CAN Report Expansion Mask> in **AT+GTCLT** is invalid. |
| Bit 22 | <Total Distance Impulses> | Vehicle total distance measured in impulses if distance from dashboard is not available. |
| Bit 21 | <Total Vehicle Engine Overspeed Time> | The total time when the vehicle engine speed is greater than the limit defined in CAN100 configuration. |
| Bit 20 | <Total Vehicle Overspeed Time> | The total time when the vehicle speed is greater than the limit defined in CAN100 configuration. |
| Bit 19 | <Doors> | An 8-bit hexadecimal number. Each bit contains information of one door. |

| Bit | Item | Description |
| --- | --- | --- |
| Bit 18 | <Lights> | An 8-bit hexadecimal number. Each bit contains information of a light. |
| Bit 17 | <Detailed Information/Indicators> | A hexadecimal number. Each bit contains information of one indicator. |
| Bit 16 | <Tachograph Information> | Two bytes. The higher byte describes driver 2 (the one whose card is inserted in tachograph slot 2), and the lower byte describes driver 1. |
| Bit 15 | <Axle Weight 2nd> | Weight of vehicle's second axle |
| Bit 14 | <Total Idle Fuel Used> | Number of liters of fuel used since vehicle manufacture or device installation |
| Bit 13 | <Total Engine Idle Time> | Time of engine running during idling status (vehicle at a standstill) since vehicle manufacture or device installation |
| Bit 12 | <Total Driving Time> | Time of engine running during driving (non-zero speed) since vehicle manufacture or device installation |
| Bit 11 | <Total Engine Hours> | Time of engine running since vehicle manufacture or device installation |
| Bit 10 | <Accelerator Pedal Pressure> | The pressure applied on acceleration pedal |
| Bit 9 | <Range> | The number of kilometers to drive on remaining fuel |
| Bit 8 | <Fuel Level> | The level of fuel in vehicle's tank in liters or percentage. |
| Bit 7 | <Fuel Consumption> | The fuel consumption of the engine |
| Bit 6 | <Engine Coolant Temperature> | The temperature of the engine coolant |
| Bit 5 | <Engine RPM> | Revolutions per minute of the engine |
| Bit 4 | <Vehicle Speed> | Vehicle road speed |
| Bit 3 | <Total Fuel Used> | The number of liters of fuel used since vehicle manufacture or device installation |
| Bit 2 | <Total Distance> | Vehicle total distance |

| Bit | Item | Description |
|---|---|---|
| Bit 1 | <Ignition Key> | Ignition state |
| Bit 0 | <VIN> | Vehicle identification number |

✧ *Alarm Mask 1*

Bitwise setting of the alarm mask. The alarm mask information is based on <Detailed Information / Indicators> and <Expansion Information> of the **+RESP:GTCAN** message.

Note

In the CAN100 firmware versions 2.0.xx and 2.1.xx, the parameter is 16-bit long (Bit 0 - Bit 15), which has been extended to 32 bits since the CAN100 version 2.2.0.

| Bit | Alarm Mask 1 |
|---|---|
| Bit 31 | Reserved |
| Bit 30 | Reserved |
| Bit 29 | Reserved |
| Bit 28 | OLL - oil level low indicator (1 - on, 0 - off or not available) |
| Bit 27 | SC - service call indicator (1 - on, 0 - off or not available) |
| Bit 26 | AIR - airbags indicator (1 - on, 0 - off or not available) |
| Bit 25 | CHK - "check engine" indicator (1 - on, 0 - off or not available) |
| Bit 24 | Reserved |
| Bit 23 | ABS - ABS failure indicator (1 - on, 0 - off or not available) |
| Bit 22 | EH - engine hot indicator (1 - on, 0 - off or not available) |
| Bit 21 | OP - oil pressure indicator (1 - on, 0 - off or not available) |
| Bit 20 | BF - brake system failure indicator (1 - on, 0 - off or not available) |
| Bit 19 | BAT - battery indicator (1 - on, 0 - off or not available) |
| Bit 18 | CLL - coolant level low indicator (1 - on, 0 - off or not available) |
| Bit 17 | BFL - brake fluid low indicator (1 - on, 0 - off or not available) |
| Bit 16 | W - webcast (1 - on, 0 - off or not available) |
| Bit 15 | T - trunk (1 - open, 0 - closed) |
| Bit 14 | D - doors (1 - any door open, 0 - all doors closed) |
| Bit 13 | FFL - front fog lights (1 - on, 0 - off) |
| Bit 12 | RFL - rear fog lights (1 - on, 0 - off) |

| Bit | Alarm Mask 1 |
|---|---|
| Bit 11 | HB - high beams (1 - on, 0 - off) |
| Bit 10 | LB - low beams (1 - on, 0 - off) |
| Bit 9 | RL - running lights (1 - on, 0 - off) |
| Bit 8 | R - reverse gear (1 - on, 0 - off) |
| Bit 7 | CL - central lock (1 - locked, 0 - unlocked) |
| Bit 6 | H - handbrake (1 - pulled-up, 0 - released) |
| Bit 5 | C - clutch pedal (1 - pressed; 0 - released) |
| Bit 4 | B - brake pedal (1 - pressed; 0 - released) |
| Bit 3 | CC - cruise control (1 - active, 0 - inactive) |
| Bit 2 | AC - air conditioning (1 - on, 0 - off) |
| Bit 1 | DS - driver seatbelt indicator (1 - indicator on, 0 - off). |
| Bit 0 | FL - fuel low indicator (1 - indicator on, 0 - off). |

- *Alarm Mask 2*

  Bitwise setting of the alarm mask. The alarm mask information is based on <Lights> and <Doors> of the **+RESP:GTCAN** message.

| Bit | Alarm Mask 2 |
|---|---|
| Bit 31 | Reserved |
| … | Reserved |
| Bit 21 | Hood (1 - open, 0 - closed) |
| Bit 20 | Trunk (1 - open, 0 - closed) |
| Bit 19 | Rear Right Door (1 - open, 0 - closed) |
| Bit 18 | Rear Left Door (1 - open, 0 - closed) |
| Bit 17 | Passenger Door (1 - open, 0 - closed) |
| Bit 16 | Driver Door (1 - open, 0 - closed) |
| … | Reserved |
| Bit 5 | Hazard Lights (1 - on, 0 - off) |
| Bit 4 | Rear Fog Light (1 - on, 0 - off) |
| Bit 3 | Front Fog Light (1 - on, 0 - off) |
| Bit 2 | High Beam (1 - on, 0 - off) |
| Bit 1 | Low Beam (1 - on, 0 - off) |
| Bit 0 | Running Lights (1 - on, 0 - off) |

- *Alarm Mask 3*

  Bitwise setting of the alarm mask. The alarm mask information is based on <Engine RPM> of the **+RESP:GTCAN** message.

| Bit | Alarm Mask 3 |
|---|---|
| Bit 31 | Reserved |
| … | … |

| Bit | Alarm Mask 3 |
|---|---|
| Bit 7 | Over Vehicle Speed High Threshold Event (1 - Triggered, 0 - not triggered). |
| Bit 6 | Under Vehicle Speed High Threshold Event (1 - Triggered, 0 - not triggered). |
| Bit 5 | Over Vehicle Speed Low Threshold Event (1 - Triggered, 0 - not triggered). |
| Bit 4 | Under Vehicle Speed Low Threshold Event (1 - Triggered, 0 - not triggered). |
| Bit 3 | Over High RPM Event (1 - Triggered, 0 - not triggered). |
| Bit 2 | Under High RPM Event (1 - Triggered, 0 - not triggered). |
| Bit 1 | Over Low RPM Event (1 - Triggered, 0 - not triggered). |
| Bit 0 | Under Low RPM Event (1 - Triggered, 0 - not triggered). |

- *High RPM Threshold*
  This is the threshold of the high engine RPM. If the current engine RPM is greater than <High RPM Threshold>, it will trigger over high RPM event.
- *Low RPM Threshold*
  This is the threshold of the low engine RPM. If the current engine RPM is less than <Low RPM Threshold>, it will trigger under low RPM event.
- *CAN Report Expansion Mask*
  Bitwise mask to configure the composition of CANBUS expansion information of the **+RESP:GTCLT** message.

| Bit | Item | Description |
|---|---|---|
| Bit 31 | Reserved | |
| ... | Reserved | |
| Bit 23 | <Engine Torque> | The engine torque. Unit: Percentage. |
| Bit 22 | <Rapid Accelerations> | The number of total rapid accelerations since installation and it is calculated based on CAN100 settings of speed increase time and value. |
| Bit 21 | <Rapid Brakings> | The number of total rapid brakings since installation and it is calculated based on CAN100 settings of speed decrease time and value. |
| Bit 20 | <Expansion Information> | A hexadecimal number. Each bit represents information of one indicator. |
| Bit 19 | <Registration Number> | The vehicle registration number |

| Bit | Item | Description |
|---|---|---|
| Bit 18 | <Tachograph Driver 2 Name> | The name of tachograph driver 2 |
| Bit 17 | <Tachograph Driver 1 Name> | The name of tachograph driver 1 |
| Bit 16 | <Tachograph Driver 2 Card Number> | The card number of tachograph driver 2 |
| Bit 15 | <Tachograph Driver 1 Card Number> | The card number of tachograph driver 1 |
| Bit 14 | <Total Brake Applications> | Count of applying brake pedal (braking process initiated by brake pedal) |
| Bit 13 | <Total Accelerator Kick-down Time> | Total time when accelerator pedal is pressed over 90% |
| Bit 12 | <Total Cruise Control Time> | Total time when the vehicle speed is controlled by cruise-control module |
| Bit 11 | <Total Effective Engine Speed Time> | Total time when the vehicle engine speed is effective |
| Bit 10 | <Total Accelerator Kick-downs> | Count of accelerator pedal kick-downs (with the pedal pressed over 90%) |
| Bit 9 | <Pedal Braking Factor> | It measures how often the driver brakes with brake pedal or with engine and stores both counts (which are always increasing).<br>Decreasing speed with brake pedal pressed causes increase of pedal braking factor. |
| Bit 8 | <Engine Braking Factor> | It measures how often the driver brakes with brake pedal or with engine and stores both counts (which are always increasing).<br>Decreasing speed with no pedal pressed causes increase of engine braking factor. |
| Bit 7 | <Analog Input Value> | Analog input value |
| Bit 6 | <Tachograph Driving Direction> | Vehicle driving direction from tachograph |

| Bit | Item | Description |
|---|---|---|
| Bit 5 | <Tachograph Vehicle Motion Signal> | Vehicle motion signal from tachograph |
| Bit 4 | <Tachograph Overspeed Signal> | Tachograph overspeed signal for the vehicle |
| Bit 3 | <Axle Weight 4th> | Weight of vehicle's fourth axle |
| Bit 2 | <Axle Weight 3rd> | Weight of vehicle's third axle |
| Bit 1 | <Axle Weight 1st> | Weight of vehicle's first axle |
| Bit 0 | <Ad-Blue Level> | The level of Ad-Blue |

- *Vehicle Speed High Threshold*

  This parameter is for high threshold of CANBUS speed alarm. If the current CANBUS speed is higher than or equal to the value of <Vehicle Speed High Threshold> and lasts for <Debounce Time>, it will trigger Over Vehicle Speed High Threshold Event. On the contrary, if the current CANBUS speed is lower than <Vehicle Speed High Threshold> and lasts for <Debounce Time>, it will trigger Under Vehicle Speed High Threshold Event.
- *Vehicle Speed Low Threshold*

  This parameter is for low threshold of CANBUS speed alarm. If the current CANBUS speed is higher than or equal to the value of <Vehicle Speed Low Threshold> and lasts for <Debounce Time>, it will trigger Over Vehicle Speed Low Threshold Event. On the contrary, if the current CANBUS speed is lower than <Vehicle Speed Low Threshold> and lasts for <Debounce Time>, it will trigger Under Vehicle Speed Low Threshold Event.

### 3.7.8 CFU (CAN100 FOTA Upgrade)

The command **AT+GTCFU** is used to upgrade the firmware of CAN100 over the air.

| |
|---|
| Example:<br>AT+GTCFU=gv58lau,0,10,0,name,pw,http://220.178.67.210:8208/GV58LAU/deltabin/csb_des_07_build1116.bin,0,0,,,0001$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | CFU | CFU |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Retry Times | 1 | 0 - 3 | 0 |
| | Timeout | 2 | 10 - 30(min) | 10 |
| | Protocol Type | 1 | 0 | 0 |
| | Server User Name | <=6 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Server Password | <=6 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' | |
| | Server URL | 100 | Legal URL | |
| | Update Type | 1 | 0 \| 1 | 0 |
| | Mode | <=1 | 0 \| 1 | 0 |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Retry Times*

It specifies the maximum number of retries to download the update package upon download failure.

✧ *Timeout*

It specifies the expiration timeout of one download. If the download expires, it is considered to be a failure.

✧ *Protocol Type*

The protocol used to download the package.

- **0** - HTTP. Only HTTP is supported now.

✧ *Server User Name*

If the file server uses authentication, the user name is specified here.

✧ *Server Password*

If the file server uses authentication, the password is specified here.

✧ *Server URL*

It specifies the URL to download the package.

✧ *Mode*

It indicates whether to cancel the CAN100 upgrade process.

- **0 or empty** - Start the CAN100 upgrade process.
- **1** - Cancel CAN100 upgrade process.

✧ *Update Type*

It specifies the type of CAN100 update over the air.

- **0** - CAN100 firmware update.
- **1** - CAN100 configuration update.

## 3.8 OTA Update Service

This section describes the commands related to the OTA update service. Please refer to the details below.

## 3.8.1 FVR (Configuration File Version Record)

The command **AT+GTFVR** is used to record information of the configuration file generated by Manage Tool for AT+GTUPC.

| |
|---|
| Example:<br>AT+GTFVR=gv58lau,name,0000,,,,,00000000000000000000000000000000,,,,,20190415172349,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | FVR | FVR |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Configuration Name | <=40 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' | |
| | Configuration Version | 4 | 0000 - 9999 | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Digital Signature | 32 | '0'-'9' 'a'-'z' 'A'-'Z' | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Generation Time | 14 | YYYYMMDDHHMMSS | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *Configuration Name*

The name of the configuration file.

✧ *Configuration Version*

The version number of the configuration file. The first two characters indicate the major version number, and the last two characters indicate the minor version number.

✧ *Digital Signature*

It is used to confirm the validity of subsequent commands.

✧ *Generation Time*

The time when the configuration file is generated.

Note

The **AT+GTFVR** command must be the first command in the configuration file.

### 3.8.2 UPC (Update Configuration)

The command **AT+GTUPC** is used to download configuration file over the air for the update of the local configuration.

```
Example:
AT+GTUPC=gv58lau,0,10,0,1,0,http://www.queclink.com/configure.ini,1,,0,00000000,,3,0001$
```

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | UPC | UPC |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Max Download Retry | 1 | 0 - 3 | 0 |
| | Download Timeout | <=2 | 5 - 30(min) | 10 |
| | Download Protocol | 1 | 0 | 0 |
| | Report Mode | 1 | 0 \| 1 | 0 |
| | Update Interval | <=4 | 0 - 8760(h) | 0 |
| | Download URL | <=100 | Legal URL | |
| | Mode | 1 | 0 \| 1 | 0 |
| | Reserved | 0 | | |
| | Extended Status Report | 1 | 0 \| 1 | 0 |
| | Identifier Number | 8 | 00000000 - FFFFFFFF | 0 |
| | Reserved | 0 | | |
| | Update Status Mask | 1 | 0 - F | 3 |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *Max Download Retry*

It specifies the maximum number of retries to download the configuration file upon download failure.

✧ *Download Timeout*

It specifies the expiration timeout of one download. If the download expires, it is considered to be failure.

✧ *Download Protocol*

The protocol used to download the file.

- **0** - HTTP

✧ *Report Mode*

A numeral to indicate whether to report the message **UPC** (ASCII)/(HEX) or **EUC** (ASCII)/(HEX) when the configuration is updated over the air.

- **0** - Do not report the message **+RESP:GTUPC** or **+RESP:GTEUC**.
- **1** - Report the message **+RESP:GTUPC** or **+RESP:GTEUC**.

✧ *Update Interval*

The interval for updating the configuration over the air.

✧ *Download URL*

It specifies the URL to download the configuration file. If <Download URL> ends with "/" which means the URL is just a path without file name, the <IMEI>.ini will be added as the default file name to complete the URL. If it is greater than 100 bytes in length, error will be returned.

✧ *Mode*

A numeral that indicates the working mode of downloading configuration over the air.

- **0** - Disable this function.
- **1** - Enable this function.

✧ *Extended Status Report*

A numeral to indicate the message to be reported for the configuration update status when <Report Mode> is 1.

- **0** - Report the message **UPC** (ASCII)/(HEX).
- **1** - Report the message **EUC** (ASCII)/(HEX) to include more information. If the <Protocol Format> in AT+GTSRI is set to HEX format, it is strongly recommended to enable this mode to avoid overflow if the value of parameter <Command ID> in **+RESP:GTUPC** is bigger than 255.

✧ *Identifier Number*

A numeral to identify the configuration update request. It will be included in the message **+RESP:GTEUC** to indicate the request it is related to.

✧ *Update Status Mask*

Bitwise mask to configure the status in which the device can update the configuration.

- **Bit 0** - for ignition off
- **Bit 1** - for ignition on

Note

✧ **1** - Make sure there is only one command per line in the configuration file and there is a "\r\n" between each two commands.

✧ **2** - There should be no space before each command.

✧ **3** - The configuration file should be a plain text file.

### 3.8.3 UPD (Firmware Update)

The command **AT+GTUPD** is used to start and stop the firmware update remotely.

### 3.8.3.1 Start the Firmware Update

To start the firmware update, the backend server sends the **AT+GTUPD** (sub:0) command to the device. Upon receiving this command, the device is informed of where to download the update package and how to download the package.

Example:
AT+GTUPD=gv58lau,0,0,20,0,,,http://60.174.225.173:10050/GV58LAU/deltabin/GV58LAU_00307_00307.bin,,0,0,,0001$

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | UPD | UPD |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Sub Command | 1 | 0 | |
| | Max Download Retry | 1 | 0 - 3 | 0 |
| | Download Timeout | 2 | 10 - 30(min) | 20 |
| | Download Protocol | 1 | 0 | 0 |
| | Download Username | <=6 | '0'-'9' 'a'-'z' 'A'-'Z' | |
| | Download Password | <=6 | '0'-'9' 'a'-'z' 'A'-'Z' | |
| | Download URL | <=100 | legal URL | |
| | Reserved | 0 | | |
| | Update Type | 1 | 0 \| 4 | 0 |
| | Extended Status Report | 1 | 0 \| 1 | 0 |
| | Identifier Number | 8 | 00000000 - FFFFFFFF | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z' and 'A' - 'Z'. The default value is "gv58lau".

✧ *Sub Command*

The sub command of **AT+GTUPD**. 0 means "Start the firmware update".

✧ *Max Download Retry*

It specifies the maximum number of retries to download the update package upon download failure.

✧ *Download Timeout*

It specifies the expiration timeout of one download. If the download expires, it is considered to be failure.

✧ *Download Protocol*

The protocol used to download the package.

- **0** - HTTP

✧ *Download Username*

If the file server uses authentication, the user name is specified here.

✧ *Download Password*

If the file server uses authentication, the password is specified here.

✧ *Download URL*

It specifies the URL to download the package.

✧ *Reserved*

Reserved for future extension.

✧ *Update Type*

It specifies the firmware type to be updated.

- **0** - BB firmware update
- **4** - BLE firmware update

✧ *Extended Status Report*

A numeral to indicate the message type to be reported for firmware update status.

- **0** - Report the **UPD** (ASCII)/(HEX) message to indicate the firmware update status.
- **1** - Report the **EUD** (ASCII) message to indicate the firmware update status.

✧ *Identifier Number*

A numeral to identify the firmware update request. It will be included in the **+RESP:GTEUD** message to indicate the request it is related to.

✧ *Serial Number*

The exact serial number will be sent back to the platform in ACK. It is in hexadecimal format. It should begin from 0000 and increases by 1 every time. It rolls back after "FFFF".

Note

**+RESP:GTEUD** in hex format has the same message format as **+RESP:GTUPD**.

### 3.8.3.2 Stop the Firmware Update

Before the device finishes downloading the update package, the backend server could use the **AT+GTUPD** (sub:1) command to cancel the current firmware update. If the package is downloaded successfully, this command is ignored.

| Example: |
|---|
| AT+GTUPD=gv58lau,1,,,,,0001$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | UPD | UPD |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Body | Sub Command | 1 | 1 | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Sub Command*

The Sub Command of **AT+GTUPD**. 1 means "Cancel the current firmware update process".

### 3.8.3.3 Firmware Update Process

#### 3.8.3.3.1 Initiation of the Update Process

The backend server sends the **AT+GTUPD** (sub: 0) command to the device to initiate the update process. Along with this command, the backend server sends necessary information for the device to start the update process.

It is the backend server's duty to decide when and how to initiate the firmware update process to all the devices it controls. As the response message collector and the controller, the backend server has all the information it needs to start an update process including the current firmware versions of the devices it controls (retrieved with the AT+GTRTO command), the version of the latest available firmware and the location of the proper update packages.

#### 3.8.3.3.2 Confirmation of the Update Process

Upon receiving the **AT+GTUPD** (sub: 0) command, the device will first check the current battery capacity. If the battery capacity cannot support the update process, it will report **+RESP:GTUPD** (code: 103) to notify the backend server that the update process is to be aborted because of low battery. If the battery capacity is ample, the device will send **+RESP:GTUPD** with confirmation information to the backend server. Then the update process proceeds to the next step.

If the update command is confirmed, the device will go into non-interactive mode. That is, the end user can no longer make phone call, and all incoming calls are rejected automatically until the update process finishes. In the meantime, the device will ignore all the commands received from the backend server if they are not related to the update process. Also the device will stop all the reports that are not related to the update process.

#### 3.8.3.3.3 Download of the Update Package

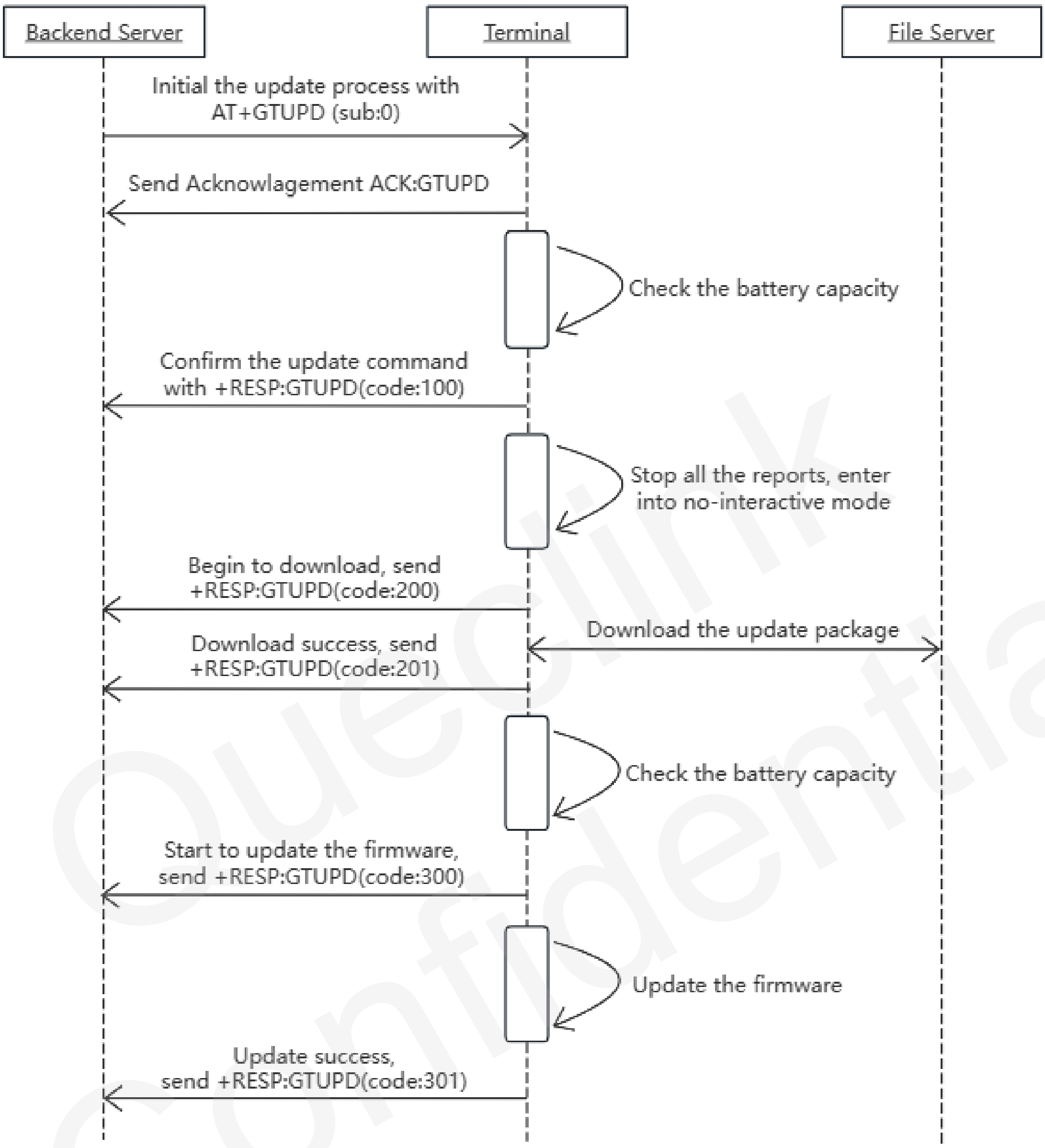
If the update command is confirmed, the device will use the information sent by the backend server to download the update package. If the download fails, the device will retry the specified times. If all attempts fail, the update process is aborted and the device will automatically reboot to go back to the normal working mode. If the download succeeds, the update process proceeds to the next step. Either way, the device will send **+RESP:GTUPD** with download information to the backend server.

Before the package is downloaded, the backend server could send the **AT+GTUPD** (sub: 1) command to cancel the current update process. This is the only chance to abort during the update process.

#### 3.8.3.3.4 Update of the Firmware

After downloading the package successfully, the device will check the battery capacity again. If the battery cannot support the update process, the device will report **+RESP:GTUPD** (code: 303) to notify the backend server that the update process is to be aborted because of low battery. If the battery capacity is ample, the device will send **+RESP:GTUPD** (code: 300) to the backend server to indicate the start of the update. Then it uses the update package to update the firmware. After the update, whether it succeeds or fails, the device will reboot automatically. After the device boots up, it sends **+RESP:GTUPD** with update information to the backend server and works as usual.

#### 3.8.3.3.5 An Example of Successful Update




## 3.9 Real Time Operation

This section describes the commands related to the runtime operations. Please refer to the details below.

### 3.9.1 RTO (Real Time Operation)

The command **AT+GTRTO** is used to retrieve information from the terminal or control the terminal when it executes certain actions.

| |
|---|
| Example:<br>AT+GTRTO=gv58lau,3,,,,,,0015$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | RTO | RTO |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Sub Command | <=2 | 0-E \| 10 \| 12-14 \| 1C \| 22 \| 25 \| 31 \| 3A | |
| | Parameter 1 | | | |
| | Parameter 2 | | | |
| | Parameter 3 | | | |
| | Parameter 4 | | | |
| | Parameter 5 | | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Ending Flag | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Sub Command*

A HEX value which indicates the sub command of **AT+GTRTO**.

✧ *Parameter 1-5*

Refer to the following description.

#### 3.9.1.1 Sub Command

✧ **0** - GNSS. Get the GNSS related information via the message **GPS**(ASCII)/(HEX).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **1** - RTL. Request the terminal to report its current position immediately via the message **RTL**(ASCII)/(HEX) .

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **2** - READ. Get the current configuration of the terminal via the message **ALM** (ASCII).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| AT Command | 3 - 79 | SRI | |
| Reserved | 0 | | |
| Format Version | <=3 | 0 - 999 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **3** - REBOOT. Reboot the terminal.

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **4** - RESET. Reset all parameters to factory settings.

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| AT Command\|Reset Level | 3 - 79\|1 | SRI\|0 - 1 | |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **5** - PWROFF. Power off the device.

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **6** - CID. Get the ICCID of the SIM card which is being used by the terminal via the message **CID**(ASCII)/(HEX).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

- ✧ **7** - CSQ. Get the current GSM signal level of the terminal via the message **CSQ**(ASCII)/(HEX) .

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

- ✧ **8** - VER. Get the version information of the device via the message **VER**(ASCII)/(HEX).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

- ✧ **9** - BAT. Get the power supply and adapter status of the terminal via the message **BAT**(ASCII)/(HEX).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

- ✧ **A** - IOS. Get status of all the IO ports via the message **IOS**(ASCII)/(HEX).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

- ✧ **B** - TMZ. Get the time zone settings via the message **TMZ**(ASCII)/(HEX).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **C** - GIR. Get cell information via the message **GSM**(ASCII)/(HEX).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **D** - AIF. Get APN, ICCID, base station ID, RSSI, cell ID, IP and DNS server via the message **AIF**(ASCII).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **E** - GSV/BSV/RSV/ASV. Request the device to report the GNSS satellites information via the message **GSV**(ASCII)/(HEX)/**RSV**(ASCII)/(HEX)/**BSV**(ASCII)/(HEX)/**ASV**(ASCII)/(HEX).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Satellite Information Mask | 2 | 00 - FF | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **10** - CAN. Get CAN information via the message **CAN**(ASCII)/(HEX).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **12** - CVN. Get the version number information of CAN100 via the message **CVN**(ASCII)/(HEX).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **13** - CSN. Get the serial number information of CAN100 via the message **CSN**(ASCII)/(HEX). It works only when CAN is connected and the configuration is correct.

If CAN is connected by BLE, set the <Accessory Type> of AT+GTBAS to 4 and the <Accessory Model> to 0. The <Mode> of AT+GTCAN also needs to be set to 1.

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **14** - BLE. Commands for Bluetooth.

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Bluetooth Command | 3 | BTI \| WUM | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **1C** - ATI. Get basic device information via the message **ATI**(ASCII).

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| ATI Mask | 8 | 00000000 - FFFFFFFF | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

✧ **22** - CAN100. Set car model for CAN100 or read car model from CAN100.

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| CAN100 Operation Mode | 1 | 0-2 | |
| Output Direction\|CAN100 Car Model ID | 1 \| 1-5 | 0 \| 1 \| 3 \| 1 - 65535 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

- **25** - SCS. Get the self-calibration status of the acceleration data via the message **SCS**(ASCII)/(HEX) or clear the self-calibration status. It is used together with <SCS Action> below.

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| SCS Action | 1 | 0 - 1 | |
| Output Direction | 1 | 0 \| 1 \| 3 | 0 |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

- **31** - DELBUF. Delete all the buffered reports.

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

- **3A** - RLY. Set the state of the WRL300.

| Parameter | Length | Range/Format | Default |
|---|---|---|---|
| RLY Operation Mode | 1 | 0 \| 1 | |
| Bind BAS Index | 1 | 0 - 9 | |
| Reserved | 0 | | |
| Reserved | 0 | | |
| Reserved | 0 | | |

### 3.9.1.2 Optional Parameters

Detailed explanation of the optional parameters.

- *AT Command*

- To get AT command configuration when <Sub Command> is set to **2**, please follow the format in the following example.
  For instance, to get the configuration of AT+GTFRI, set **AT+GTRTO=gv58lau,2,FRI,,,,,0015$**, and get it via **ALM** (ASCII). To get more than one AT command configuration, the neighbouring commands are separated by the ASCII character ":".
  For example, to get configuration of the commands AT+GTFRI, AT+GTOWH, AT+GTSRI, set **AT+GTRTO=gv58lau,2,FRI:OWH:SRI,,,,,0015$**, and get it via **ALM**(ASCII). Supports up to 20 commands to query at the same time.
- Set <Sub Command> to **4** to specify the configuration to be reset. To specify a configuration, use the last three letters of the protocol command.
  For example, to reset configuration of the **AT+GTFRI** command, send the command **AT+GTRTO=gv58lau,4,FRI,,,,,000F$**. To reset more than one AT command configuration, the neighbouring commands are separated by the ASCII character ":".
  For example, to reset configuration of the commands **AT+GTFRI**, **AT+GTOWH**, **AT+GTTOW**, send the command **AT+GTRTO=gv58lau,4,FRI:OWH:TOW,,,,,0015$**. The buffered messages saved can be deleted with the command **AT+GTRTO=gv58lau,4,BUF,,,,,000F$**.
  The mileage value saved during <GNSS Assisted Mode> enabling can be reset with the command **AT+GTRTO=gv58lau,4,DST,,,,,000F$**.
  Configuration of the commands AT+GTBSI, AT+GTSRI, AT+GTQSS, AT+GTMQT, AT+GTCFG, AT+GTTMA and AT+GTPIN cannot be reset by this command.

✧ *Reset Level*

If <Sub Command> is **4**, this parameter works as follows.

- **0** - Light Reset. Reset all configuration parameters, except parameters configured by AT+GTBSI, AT+GTSRI, AT+GTMQT, AT+GTCFG, AT+GTPIN and AT+GTTMA .
- **1** - Heavy Reset. Reset all configuration parameters, except the <New Password> of the AT+GTCFG and the command AT+GTPIN.

✧ *Satellite Information Mask*

If <Sub Command> is set to **E**, please get the satellite information message according to the following bitwise mask. The satellite information mask must be 2 bytes.

If it is less than 2 bytes, add 0 to the high bytes of the satellite information mask. If this field is reserved, the device will report **GSV**(ASCII)/(HEX) / **RSV**(ASCII)/(HEX)/ **BSV**(ASCII)/(HEX) / **ASV** (ASCII)/ (HEX).

| Bit | Message Name | Description |
|---|---|---|
| Bit 3 | +RESP:GTASV | Report Galileo satellite information |
| Bit 2 | +RESP:GTBSV | Report Beidou satellite information |
| Bit 1 | +RESP:GTRSV | Report GLONASS satellite information |
| Bit 0 | +RESP:GTGSV | Report GPS satellite information |

✧ *Bluetooth Command*

It specifies the Bluetooth command to be executed when <Sub Command> is set to **14**.

- **BTI** - Request the device to report the Bluetooth information and list of paired devices via the message **BTI** (ASCII)/(HEX).
- **WUM** - This command is used to wake up Ghost.

Note

The Primary device cannot wake up Ghost device immediately, but it can wake up Ghost device only when Ghost device connects to the Primary device every time via Bluetooth.

✧ *ATI Mask*

If <Sub Command> is set to **1C**, the basic information will be reported via the message **ATI** (ASCII)/(HEX) according to chosen <ATI Mask>.

- **ATI Mask Table**

| Mask Bit | Item |
|---|---|
| Bit 0 | Firmware Version |
| Bit 1 - Bit 3 | Reserved for SW |
| Bit 4 | BT Firmware Version |
| Bit 5 | BT Boot Firmware Version |
| Bit 6 - Bit 7 | Reserved for SW |
| Bit 8 | Reserved |
| Bit 9 | Reserved |
| Bit 10 - Bit 11 | Reserved for MCU |
| Bit 12 | Hardware Version |
| Bit 13 - Bit 15 | Reserved for HW |
| Bit 16 | Reserved |
| Bit 17 | Reserved |
| Bit 18 | Flash ID |
| Bit 19 | Sensor ID |
| Bit 20 | Modem IMEI |
| Bit 21 - Bit 31 | Reserved |

✧ *CAN100 Operation Mode*

If the sub command is **22**, this parameter will work as follows.

- **0** - Read the current car model and report it via the message **CML** (ASCII)/(HEX).
- **1** - Set car model. Please use the <CAN100 Car Model ID> parameter to set car model.
- **2** - Start CAN100 automatic synchronization. The synchronization result is reported via the **+RESP:GTCML** message.

Note

The entire synchronization takes about 10-40s, and CAN100 will restart immediately after the end of the synchronization regardless of the result. If automatic sync is enabled, please wait for the synchronization to finish before reading the current car model. If automatic synchronization has not ended, the subsequent synchronization command will be ignored.

✧ *CAN100 Car Model ID*

It works only when the sub command is **22** and the <CAN100 Operation Mode> is set to **1**. This parameter value should be car model ID described in supported car model list.

✧ *SCS Action*

If <Sub Command> is set to **25**, Read or Clear action is controlled by this parameter.

- **0** - Read self-calibration status.
- **1** - Clear self-calibration status.

✧ *RLY Operation Mode*

If the sub command is **3A**, this parameter will work as follows.

- **0** - Disable relay of WRL300, switch relay pin from NC to COM.
- **1** - Enable relay of WRL300, switch relay pin from NO to COM.

✧ *Bind BAS Index*

It is used to bind the specific configuration in AT+GTBAS when the <Sub Command> is set to **3A**. The value is the same as the index in **AT+GTBAS**.

✧ *Output Direction*

It determines the destination that the response message of the RTO command will be reported to and is invalid for <Sub Command> 2(READ), 3(REBOOT), 4(RESET), 5(PWROFF), 31(DELBUF), 3A(RLY).

- **0** - The message will be output to the backend server.
- **1** - The message will be output to the main serial port.
- **3** - If the command is received via SMS, the message will be output to the original SMS number; otherwise the message will be output to the backend server.

✧ *Format Version*

A numeral to indicate the format of the **+RESP:GTALM** message.

- **0** - Do not add cutoff characters to the message **+RESP:GTALM**.
- **1** - Add cutoff characters to the message **+RESP:GTALM**.
- **2-999** - Reserved.

### 3.9.2 SWL (SMS Allowlist)

The command **AT+GTSWL** is used to configure a list of authorized phone numbers that are allowed to configure the device.

| |
|---|
| Example:<br>AT+GTSWL=gv58lau,0,1,2,13813888888,13913999999,,,,,0018$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | SWL | SWL |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Number Filter | 1 | 0 \| 1 | 0 |
| | Start Index | <=2 | 1 - 10 | |
| | End Index | <=2 | 1 - 10 | |
| | Phone Number List | <=20*10 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau"

✧ *Number Filter*

The working mode of this function.

- **0** - Disable. Allow any phone number to configure the device.
- **1** - Enable. Only phone numbers saved in the allowlist could configure the device.

✧ *Start Index, End Index*

The index range of the allowlist to which the phone numbers are to be updated. The <Start Index> and <End Index> defines the total amount of phone numbers that will be updated. If either one is empty, there should be no <Phone Number List> followed. For example, if <Start Index> is set to 1 and <End Index> is set to 2, the first two phone numbers in the allowlist will be updated by the numbers provided in the parameter <Phone Number List>.

✧ *Phone Number List*

A comma-separated list of phone numbers to update to the allowlist. The amount of the phone numbers is defined by <Start Index> and <End Index>.

### 3.9.3 SMS Location Request

This command can only be sent via SMS. The device will send SMS report with Google maps hyperlink and the current location to corresponding phone number or report **LBC** (ASCII)/(HEX) to the backend server immediately. For more detail please refer to the parameter <Location Request Mask> of the command AT+GTCFG.

| Item | Format |
|---|---|
| Command | Get Position |

✧ *Command*

The detailed command string received via SMS.

"Get Position": This command is used to query the current location, Case is ignored.

### 3.9.4 RTP (Remote File Transfer)

The command **AT+GTRTP** is used to obtain files from the backend server.

| |
|---|
| Example:<br>AT+GTRTP=gv58lau,0,0,0,http://60.174.225.173:20581/GV58LAU/deltabin/server2.crt,,,,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | RTP | RTP |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Mode | 1 | 0 | |
| | Protocol Type | 1 | 0 | |
| | File Type | 1 | 0-2 | |
| | URL | <=100 | ASCII (not including '=') | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of file transfer.

- **0** - Download file.

✧ *Protocol Type*

The type of communication protocol used to obtain data from the backend server.

- **0** - HTTP.

✧ *File Type*

It defines the type of file to download from the server.

- **0** - CA certificate.
- **1** - Client certificate.
- **2** - Client key.

✧ *URL*

It specifies the URL to download the configuration file.

### 3.9.5 LTP (Local File Transfer)

The command **AT+GTLTP** is used to write the file to the device by subcontracting.

| |
|---|
| Example:<br>AT+GTLTP=gv58lau,,1,1,0,0,***Data***,***CRC***,,,,,FFFF$ |

| Parts | Fields | Length | Range/Format | Default |
|---|---|---|---|---|
| Head | Header | 5 | AT+GT | AT+GT |
| | Command Word | 3 | LTP | LTP |
| | Leading Symbol | 1 | = | = |
| | Password | 4-20 | '0' - '9' 'a' - 'z' 'A' - 'Z' | gv58lau |
| Body | Reserved | 0 | | |
| | Mode | 1 | 0 \| 1 | |
| | File Type | 1 | 0-2 | |
| | Current Number | <=3 | 0 - 100 | |
| | Total Number | <=3 | 0 - 100 | |
| | Data | <=512 | HEX | |
| | CRC | <=4 | 0 - FFFF | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| | Reserved | 0 | | |
| Tail | Serial Number | 4 | 0000-FFFF ('0'-'9', 'A'-'F') | |
| | Tail | 1 | $ | $ |

✧ *Password*

The valid characters for the password include '0' - '9', 'a' - 'z', and 'A' - 'Z'. The default value is "gv58lau".

✧ *Mode*

The working mode of the file.

- **0** - Delete file.
- **1** - Write file.

✧ *File Type*

It defines the type of file to download from the server.

- **0** - CA certificate
- **1** - Client certificate
- **2** - Client key

✧ *Current Number*

The location where the current data is written to the file.

✧ *Total Number*

Total serial number of write file data.

✧ *Data*

Data written to file.

✧ *CRC*

CRC verification data of <Data> is used to determine whether the data is correct.

# 4 Report(Ascii)

This section defines the ASCII formats of the report messages. Due to the size limit of an SMS message (160 bytes), it is recommended to carefully set <Report Item Mask> in AT+GTCFG to limit the length of the report which contains GNSS position information in the case of SMS transmission. Otherwise the report will be truncated to fit the length of SMS message.

## 4.1 ACK (Acknowledgement)

The frame format of ACK is as follows:

Example:

The Option field is not included.

+ACK:GTXXX,8020040900,864292043426376,,0000,20200613114927,FFFF$

The Option field is included.

+ACK:GTGEO,8020040803,866314060835525,GV58LAU,0,FFFF,20250313031651,145E$

+ACK:GTPEO,8020040803,866314060835525,GV58LAU,0,FFFF,20250313031710,145F$

+ACK:GTPEG,8020040803,866314060179445,GV58LAU,9,0070,20231208081922,00D5$

+ACK:GTIOB,8020040803,866314060179445,GV58LAU,0,0077,20231208082047,00D6$

+ACK:GTCLT,8020040803,866314060179445,GV58LAU,0,0080,20231208082541,00DE$

+ACK:GTCMD,8020040803,866314060179445,GV58LAU,31,FFFF,20231208082154,00D7$

+ACK:GTUDF,8020040803,866314060179445,GV58LAU,0,FFFF,20231208082218,00D8$

+ACK:GTROS,8020040803,866314060179445,GV58LAU,1,0079,20231208082423,00DA$

+ACK:GTBAS,8020040803,866314060179445,GV58LAU,7,FFFF,20231208082306,00D9$

+ACK:GTRTO,8020040803,866314060179445,GV58LAU,PWROFF,0089,20231208084647,00FE$

+ACK:GTIDS,8020040803,866314060835525,GV58LAU,2,1,FFFF,20250313031059,145A$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +ACK |
| | Leading Symbol | 1 | : |
| | Command Word | 5 | 'A'-'Z', '0'-'9' |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Option | N | Please see below. |
| Tail | Serial Number | 4 | 0000-FFFF |
| | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000-FFFF |
| | Tail | 1 | $ |

✧ *Command Word*

The "Command Word" in the configuration command. For example, it is **GTXXX**.

✧ *Full Protocol Version*

The protocol version that the device conforms to. It is separated into 3 parts. The first two or four characters represent the device type. As shown in the example above, 802004 means GV58LAU. The second part consists of two characters that represent the major version number of the protocol, and the last part consists of two characters that indicate the minor version number of the protocol. Both version numbers are hex digits. For example, 0A01 means version 10.01.

✧ *Unique ID*

The International Mobile Equipment Identity for the terminal device.

✧ *Device Name*

The name of terminal device.

✧ *Option*

The Acknowledgement of commands that contain the Option field are as follows:

- **AT+GTGEO** - This field represents the GEO command ID.

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | GEO ID | <=2 | 0 - 99 |

- **AT+GTPEO** - This field represents the PEO command ID.

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | GEO ID | <=3 | 0 - 199 |

- **AT+GTPEG** - This field represents the PEG command ID.

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Group ID | <=2 | 0 - 9 |

- **AT+GTIOB** - This field represents the IOB command ID.

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | IOB ID | 1 | 0 - 3 |

- **AT+GTCLT** - This field represents the CLT group ID.

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Group ID | <=2 | 0 - 19 |

- **AT+GTCMD** - This field represents the stored command ID.

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Stored Command ID | <=2 | 0 - 31 |

- **AT+GTUDF** - This field represents the UDF group ID.

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Group ID | <=2 | 0 - 31 |

- **AT+GTROS** - This field represents the ROS output ID.

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Output ID | 1 | 0 - 1 |

- **AT+GTBAS** - This field represents the BAS index.

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Index | 1 | 0 - 9 |

- **AT+GTRTO** - This field represents the RTO sub-command ID.

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Sub Command | <=6 | Sub Command String |

- **AT+GTIDS** - This field represents the IDS group ID.

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Result Code | <=2 | 0 - 2 |
| | Additional Number(optional) | <=4 | 0 - 5000 |

- *Result Code*

  A numeral to indicate the result of issuing AT+GTIDS.
  - **0** - Successful. <Additional Number> indicates the number of IDs that are configured.
  - **1** - ID Number list is full. <Additional Number> indicates the number of IDs that are discarded.
  - **2** - Many duplicate IDs are discarded. <Additional Number> indicates the number of duplicated IDs.
- For other commands, it is not included.

✧ *Serial Number*

The serial number in the configuration command.

✧ *Send Time*

The local time when the frame is generated in the format **YYYYMMDDHHMMSS**.
For example, "20191120135807" means 13:58:07 on November 20, 2019.

✧ *Count Number*

A self-increasing count number in each acknowledgement message. It begins from "0000" and increases by 1 for each acknowledgement message. And it rolls back after "FFFF".

Note

1. The ACK frame only indicates that the terminal device has received the command, but does not mean that the command has been successfully executed; the terminal device will send a report to inform the backend server of the execution result of the command when necessary.
2. Only after both the commands AT+GTBSI and AT+GTSRI are properly set can the ACK messages and other report messages be received by the backend server.

## 4.2 Position Related Report

This section describes the format of positioning related messages. Please refer to the details below.

### 4.2.1 Generic Location Report

- **TOW** (Tow Alarm Information)
  If the tow alarm is enabled by the command AT+GTTOW, the device will send the message **+RESP:GTTOW** to the backend server when the motion sensor detects tow.
- **DIS** (Digital Input Alarm)
  If the status change of digital inputs is detected, the device will send the message **+RESP:GTDIS** to the backend server.
- **IOB** (IO Ports Bind Alarm)
  If the IO combination is set and the corresponding condition is met, the device will report the message **+RESP:GTIOB** to the backend server.
- **SPD** (Speed Alarm Information)
  If the speed alarm is enabled, the device will send the message **+RESP:GTSPD** to the backend server when the speed of the device within the alarm range is detected .
- **SOS** (Digital Input Port Triggers SOS)
  If the SOS function is enabled, the device will send the message **+RESP:GTSOS** to the backend server when the corresponding digital input port triggers SOS.
- **RTL** (Realtime Location Information)
  After the device receives the command AT+GTRTO, it will start GNSS to get the current position and then send the message **+RESP:GTRTL** to the backend server.
- **DOG** (Watchdog Reboot Alarm)
  The protocol watchdog reboot message **+RESP:GTDOG**.
- **IGL** (Ignition Location Information)
  The location message **+RESP:GTIGL** for ignition on/off.
- **VGL** (Virtual Ignition Location Information)
  The location message **+RESP:GTVGL** for virtual ignition on/off.
- **HBM** (Harsh Behavior Alarm)
  If harsh behavior is detected, the message **+RESP:GTHBM** will be sent to the backend server.

Example:
+RESP:GTTOW,8020040803,866314060835525,GV58LAU,,00,1,1,0.0,0,95.7,117.129343,31.83918
3,20250312054144,0460,0000,550B,0E9E30A5,09,12,0.79,1.28,1.50,0.0,20250312054149,005C$

+RESP:GTDIS,8020040803,866314060835525,GV58LAU,,11,1,1,0.0,0,95.7,117.129343,31.83918
3,20250312054239,0460,0000,550B,0E9E30A5,09,12,0.86,1.45,1.69,0.0,20250312054240,005F$

+RESP:GTIOB,8020040803,866314060835525,GV58LAU,,01,1,1,0.0,0,95.7,117.129343,31.83918
3,20250312054327,0460,0000,550B,0E9E30A5,09,12,0.86,1.44,1.68,0.0,20250312054327,006A$

+RESP:GTSPD,8020040803,866314060835525,GV58LAU,,01,1,1,0.0,0,95.7,117.129343,31.83918
3,20250312054356,0460,0000,550B,0E9E30A5,09,12,0.82,1.29,1.53,0.0,20250312054357,006C$

+RESP:GTSOS,8020040803,866314060835525,GV58LAU,,10,1,1,0.0,0,95.7,117.129343,31.83918
3,20250312054431,0460,0000,550B,0E9E30A5,09,12,0.81,1.29,1.52,0.0,20250312054431,0071$

+RESP:GTRTL,8020040803,866314060835525,GV58LAU,,00,1,1,0.0,0,95.7,117.129343,31.83918
3,20250312054450,0460,0000,550B,0E9E30A5,09,12,0.80,1.41,1.62,0.0,20250312054450,0073$

+RESP:GTDOG,8020040803,866314060835525,GV58LAU,,13,1,1,0.0,0,95.7,117.129343,31.83918
3,20250312054544,0460,0000,550B,0E9E30A5,09,12,0.79,1.28,1.51,0.0,20250312054545,0079$

+RESP:GTIGL,8020040803,866314060835525,GV58LAU,,00,1,1,0.0,0,95.7,117.129343,31.83918
3,20250312054327,0460,0000,550B,0E9E30A5,09,12,0.86,1.44,1.68,0.0,20250312054327,0069$

+RESP:GTVGL,8020040803,866314060835525,GV58LAU,,20,1,1,0.0,0,42.8,117.129265,31.83939
9,20250312054816,0460,0000,550B,0E9E30A5,09,12,0.79,1.38,1.59,0.0,20250312054817,008F$

+RESP:GTHBM,8020040803,866314060268081,GV58LAU,,11,1,2,17.4,325,56.2,117.128108,31.8
52642,20250312072747,0460,0000,550B,0E9E30A5,09,6,2.26,4.45,4.99,6.4,20250312163748,04
58$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | TOW \| DIS \| IOB \| SPD \| SOS \| RTL \| DOG \| IGL \| VGL \| HBM |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 0 | |
| | Report ID / Report Type | 2 | X(0 - 4 \| 7)Y(0 - A) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Number | 1 | 1 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Report ID / Report Type*

It is a one-byte hexadecimal value represented by two ASCII bytes. The first byte (4 higher bits of the hexadecimal value) indicates Report ID and the second byte (4 lower bits of the hexadecimal value) indicates Report Type.

**Report ID** has different meanings in different messages as follows:

- The ID of digital input port which triggers the report message **+RESP:GTDIS** and **+RESP:GTSOS**.
- The ID of the bound IO which triggers the report message **+RESP:GTIOB**.
- The ID of the digital input port which triggers the reboot message **+RESP:GTDOG**.
- The speed level at which the harsh behavior is detected in the message **+RESP:GTHBM**. "3" indicates high speed, "2" indicates medium speed and "1" indicates low speed. If <Mode> in AT+GTHBM is set to 2, the value will always be 0 which indicates unknown speed.
- The value of <Virtual Ignition Mode> which indicates the trigger source of the message **+RESP:GTVGL**.
- For other messages, it will always be 0.

**Report Type** has different meanings in different messages as follows:

- In the **+RESP:GTDIS** report message generated by the digital input
  - 0 - The current logic status of the input port is "Inactive status".
  - 1 - The current logic status of the input is "Active status".
- In the **+RESP:GTIOB** report message generated by bound IO
  - 0 - The current logic status of the bound IO does not meet the alarm condition.
  - 1 - The current logic status of the bound IO meets the alarm condition.
- In the speed alarm message **+RESP:GTSPD**
  - 0 - Outside the predefined speed range
  - 1 - Within the predefined speed range
- In the protocol watchdog reboot message **+RESP:GTDOG**
  - 1 - Reboot device message for time based working mode
  - 2 - Reboot device message for ignition on working mode
  - 3 - Reboot device message for input triggered reboot
  - 4 - Reboot device message for network watchdog reboot
  - 5 - Reboot device message for WCDMA, EGPRS/GSM and LTE watchdog reboot
  - 6 - Reboot device message for send failure watchdog
  - A - The time interval in minutes for rebooting the device when the device cannot receive CANBUS messages from CAN100. 0 means “Do not reboot the device”.
- In the (virtual) ignition on/off message **+RESP:GTIGL/+RESP:GTVGL**
  - 0 - Ignition on
  - 1 - Ignition off
- In the harsh behavior monitoring message **+RESP:GTHBM**
  - 0 - Harsh braking behavior
  - 1 - Harsh acceleration behavior
  - 2 - Harsh cornering behavior
  - 3 - Harsh braking and cornering behavior
  - 4 - Harsh acceleration and cornering behavior
  - 5 - Unknown harsh driving behavior
- For other messages, it will always be 0.

✧ *Number*

The number of the GNSS positions included in the report message. Generally, it is 1.

✧ *GNSS Accuracy*

A numeral to indicate the GNSS fix status and HDOP of the GNSS position. 0 means the current GNSS fix fails and the last known GNSS position is used. A non-zero value (1 - 50) means the current GNSS fix is successful and represents the HDOP of the current GNSS position.

✧ *Speed*

The current speed. Unit: km/h.

✧ *Azimuth*

The azimuth of the GNSS fix.

✧ *Altitude*

The height above the sea level.

✧ *Longitude*

The longitude of the current position.

- *Latitude*
  The latitude of the current position.
- *GNSS UTC Time*
  The UTC time obtained from the GNSS chip.
- *MCC*
  Mobile country code. It is 3 digits in length and the range is from 000 to 999.
- *MNC*
  Mobile network code. It is 3 digits in length and the range is from 000 to 999.
- *LAC*
  Location area code in hex format.
- *Cell ID*
  The cell ID in hex format.
- *Position Append Mask*
  A bitwise numeral to control whether to include the corresponding fields in each position after <Cell ID>.
- *Satellites in Use*
  If bit 0 of <Position Append Mask> is enabled, this part will be displayed with the number of satellites in use for the current position.
- *Horizontal GNSS Accuracy*
  A numeral to indicate the GNSS fix status and HDOP of the GNSS position. 0 means the current GNSS fix fails and the last known GNSS position is used. A non-zero value means that the current GNSS fix is successful and it represents the HDOP of the current GNSS position. The smaller the value, the higher the positioning accuracy.
- *Vertical GNSS Accuracy*
  A numeral to indicate the GNSS fix status and VDOP of the GNSS position. 0 means the current GNSS fix fails and the last known GNSS position is used. A non-zero value means that the current GNSS fix is successful and it represents the VDOP of the current GNSS position. The smaller the value, the higher the positioning accuracy.
- *3D GNSS Accuracy*
  A numeral to indicate the GNSS fix status and PDOP of the GNSS position. 0 means the current GNSS fix fails and the last known GNSS position is used. A non-zero value means that the current GNSS fix is successful and it represents the PDOP of the current GNSS position. The smaller the value, the higher the positioning accuracy.
- *Mileage*
  The current total mileage.

### 4.2.2 FRI(Fixed Report Information)

If fixed report is enabled, the device will send the message **+RESP:GTFRI** to the backend server according to the working mode.

Example:
+RESP:GTFRI,8020040803,866314060835525,GV58LAU,12051,10,1,1,0.0,0,109.7,117.129305,31.

839371,20250312061316,0460,0000,550B,0E9E30A5,09,12,0.81,1.67,1.86,0.0,,,,,0,220000,,,,202
50312061318,00DF$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | FRI |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | External Power Voltage | <=5 | 0 - 32000 (mV) |
| | Report ID / Report Type | 2 | X(0-5)Y(0-6) |
| | Number | <=2 | 1 - 15 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | GNSS Trigger Type(Optional) | 1 | 0-4 |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| | Hour Meter Count | 13 | 0000000:00:00 - 1193000:00:00 |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Backup Battery Percentage | <=3 | 0 - 100 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Device Status | 6 | 000000 - FFFFFF |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *External Power Voltage*

The voltage of the external power supply. If the command AT+GTEPS is configured for the device to report the external power supply voltage periodically with fixed report, the device will send the current voltage along with the **+RESP:GTFRI** message to the backend server. If the **AT+GTEPS** command is not configured, this field will be empty.

✧ *Report ID / Report Type*

It is a one-byte hexadecimal value represented by two ASCII bytes. The first byte (4 higher bits of the hexadecimal value) indicates Report ID and the second byte (4 lower bits of the hexadecimal value) indicates Report Type.

Report ID has several meanings as follows.

- **0** - Real-time report
- **1** - Fixed time report
- **2** - Fixed distance report
- **3** - Fixed mileage report
- **4** - Fixed time and mileage report
- **5** - Fixed time or mileage report

Report Type has the following meanings.

- **0** - Normal fixed report.
- **1** - Normal corner report. If <Wrap Corner Point> is 0, this message indicates a turning point. If <Wrap Corner Point> is 1, this message indicates that the number of collected GNSS points reaches 15 or GNSS points collection stops.
- **2** - FFC fixed report. This message indicates that the device enters the state pre-configured by <Mode> in AT+GTFFC, such as Geo-Fence state, roaming state, etc.
- **3** - FFC corner report. If <Wrap Corner Point> is 0, this message means a turning point in condition of FRI report frequency changes by AT+GTFFC. If <Wrap Corner Point> is 1, it indicates the number of collected GNSS points reaches 15 or GNSS points collection stops.
- **4** - Mileage report when <Mode> in AT+GTFRI is set to 5.
- **5** - Reserved.
- **6** - Mileage report when <Mode> in AT+GTFRI is set to 5 and FFC works.

✧ *Number*

The number of the GNSS positions included in the report message. In the message **+RESP:GTFRI**, there may be one or several GNSS positions. If there are multiple positions in one **+RESP:GTFRI** message, the parameter between <Number> and <Mileage> will be repeated.

- *Position Append Mask*

  A bitwise numeral to control whether to include the corresponding fields in each position after <Cell ID>.

- *Satellites in Use*

  If bit 0 of <Position Append Mask> is enabled, this part will be displayed with the number of satellites in use for the current position.

- *GNSS Trigger Type*

  The trigger type of GNSS point has the following meanings.

  - **0** - Time point
  - **1** - Corner point
  - **2** - Distance point
  - **3** - Mileage point
  - **4** - Optimum point (time and mileage)

- *Horizontal GNSS Accuracy*

  A numeral to indicate the GNSS fix status and HDOP of the GNSS position. 0 means the current GNSS fix fails and the last known GNSS position is used. A non-zero value means that the current GNSS fix is successful and it represents the HDOP of the current GNSS position. The smaller the value, the higher the positioning accuracy.

- *Vertical GNSS Accuracy*

  A numeral to indicate the GNSS fix status and VDOP of the GNSS position. 0 means the current GNSS fix fails and the last known GNSS position is used. A non-zero value means that the current GNSS fix is successful and it represents the VDOP of the current GNSS position. The smaller the value, the higher the positioning accuracy.

- *3D GNSS Accuracy*

  A numeral to indicate the GNSS fix status and PDOP of the GNSS position. 0 means the current GNSS fix fails and the last known GNSS position is used. A non-zero value means that the current GNSS fix is successful and it represents the PDOP of the current GNSS position. The smaller the value, the higher the positioning accuracy.

- *Hour Meter Count*

  If the hour meter count function is enabled by the command AT+GTHMC, total hours the meter counts when the engine is on will be reported in this field. It consists of three parts separated by ":", the first part is the hour digit and the length of it is between 5 to 7 bytes, the second part is the 2-byte minute digit, and the last part is the 2-byte second digit. And it ranges from 00000:00:00 to 1193000:00:00. If the function is disabled, this field will be empty.

- *Backup Battery Percentage*

  The current level of the backup battery in percentage.

- *Device Status*

  The status of the device.

  From left to right, the first two bytes indicate the current motion status of the device, the middle two bytes indicate the input status, and the last two bytes indicate the output status.

  The current motion status of the device:

  - **16 (Tow)** - The device attached vehicle is ignition off and it is towed.
  - **1A (Fake Tow)** - The device attached vehicle is ignition off and it might be towed.
  - **11 (Ignition Off Rest)** - The device attached vehicle is ignition off and it is motionless.

- **12 (Ignition Off Motion)** - The device attached vehicle is ignition off and it is moving before it is considered as being towed.
- **21 (Ignition On Rest)** - The device attached vehicle is ignition on and it is motionless.
- **22 (Ignition On Motion)** - The device attached vehicle is ignition on and it is moving.
- **41 (Sensor Rest)** - The device attached vehicle is motionless without ignition signal detected.
- **42 (Sensor Motion)** - The device attached vehicle is moving without ignition signal detected.

The input status: A bitwise hex integer to represent the logic status of digital input. The low bit represents the ignition detection input and the high bit represents the digital input.
The output status: A bitwise hex integer to represent the logic status of digital output. The low bit represents digital output.

| Mask Bit | Item |
|---|---|
| ...(Optional) | Reserved |
| Bit 16-23 | Motion status of the device |
| Bit 9 | Digital input 1 |
| Bit 8 | Ignition detection |
| … | Reserved |
| Bit 2 | Digital output 3 |
| Bit 1 | Digital output 2 |
| Bit 0 | Digital output 1 |

### 4.2.3 ERI(Expand Fixed Report Information)

If **+RESP:GTERI** is enabled, the device will send the message **+RESP:GTERI** to the backend server instead of **+RESP:GTFRI**.

Example:
+RESP:GTERI,8020040803,866314060268081,GV58LAU,00000100,,10,1,2,14.2,71,35.6,117.243103,31.854079,20250320015848,0460,0000,550B,0E9E30A5,09,9,2.01,1.47,2.49,0.0,,,,,0,220100,,2,00,1,0,00000001,001F,TD_100109,FD6D3DE6D704,1,3600,18,01,1,3,0000006D,011F,DU_100361,F022A2143F36,1,3500,0,9,0,20250320015850,00D0$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | ERI |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | ERI Mask | 8 | 00000000 - FFFFFFFF |
| | External Power Supply | <=5 | 0 - 32000(mV) |
| | Report ID / Report Type | 2 | X(0-5)Y(0-6) |
| | Number | <=2 | 1 - 15 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | GNSS Trigger Type(Optional) | 1 | 0-4 |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| | Hour Meter Count | 13 | 0000000:00:00 - 1193000:00:00 |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Backup Battery Percentage | <=3 | 0 - 100 |
| | Device Status | 6 | 000000 - FFFFFF |
| | Reserved | 0 | |
| | CAN Data (Optional) | <=1000 | |
| | Bluetooth Accessory Data (Optional) | | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Position Append Mask*

A bitwise numeral to control whether to include the corresponding fields in each position after <Cell ID>.

✧ *Satellites in Use*

If bit 0 of <Position Append Mask> is enabled, this part will be displayed with the number of satellites in use for the current position.

✧ *Horizontal GNSS Accuracy*

A numeral to indicate the GNSS fix status and HDOP of the GNSS position. 0 means the current GNSS fix fails and the last known GNSS position is used. A non-zero value means that the current GNSS fix is successful and it represents the HDOP of the current GNSS position. The smaller the value, the higher the positioning accuracy.

✧ *Vertical GNSS Accuracy*

A numeral to indicate the GNSS fix status and VDOP of the GNSS position. 0 means the current GNSS fix fails and the last known GNSS position is used. A non-zero value means that the current GNSS fix is successful and it represents the VDOP of the current GNSS position. The smaller the value, the higher the positioning accuracy.

✧ *3D GNSS Accuracy*

A numeral to indicate the GNSS fix status and PDOP of the GNSS position. 0 means the current GNSS fix fails and the last known GNSS position is used. A non-zero value means that the current GNSS fix is successful and it represents the PDOP of the current GNSS position. The smaller the value, the higher the positioning accuracy.

✧ *CAN Data*

If Bit 2 of <ERI Mask> in AT+GTFRI is set to 1, the data got from CAN device will be displayed.

✧ *Bluetooth Accessory Data(Optional)*

| Fields | Length | Range/Format |
|---|---|---|
| Bluetooth Accessory Number | <=2 | 0 - 10 |
| Index | 2 | 0 - 9 |
| Accessory Type | <=2 | 0-2 \| 6-8 \| 10-13 |
| Accessory Model | 1 | 0-5 |
| Raw Data(Optional) | <=18 | |
| Accessory Append Mask | <=4 | 0 - FFFF |
| Accessory Name(Optional) | <=20 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| Accessory MAC(Optional) | 12 | 000000000000 - FFFFFFFFFFFF |
| Accessory Status(Optional) | 1 | 0 - 1 |
| Accessory Battery Voltage(Optional) | <=4 | 0 - 5000(mV) |
| Accessory Temperature(Optional) | <=3 | -40 - 80( ℃ ) |
| Accessory Humidity(Optional) | <=3 | 0 - 100%(rh) |
| Accessory Output status(Optional) | 2 | 00 - 03 |

| Fields | Length | Range/Format |
|---|---|---|
| Accessory Digital Input status(Optional) | 2 | 00 - 01 |
| Accessory Analog Input voltage(Optional) | <=5 | 0 - 32000(mV) |
| Accessory Mode(Optional) | <=2 | 0 - 10 |
| Accessory Event(Optional) | 1 | 0 - 2 |
| Tire pressure(Optional) | <=3 | 0 - 500(kPa) |
| Timestamp(Optional) | 14 | YYYYMMDDHHMMSS |
| Enhanced Temperature(Optional) | <=5 | -40.00 - 80.00 ( ℃ ) |
| Magnet ID(Optional) | 2 | 00 - FF |
| MAG Event Counter(Optional) | <=5 | 0 - 32767 |
| Magnet State(Optional) | 1 | 0 - 1 |
| Accessory Battery Percentage(Optional) | <=3 | 0 - 100(%) |
| Relay State(Optional) | 1 | 0 - 1 |

- *Bluetooth Accessory Number*

  It indicates the number of Bluetooth accessories connected with the device.
- *Index*

  The Index of the Bluetooth accessory.
- *Accessory Type*

  The type of the Bluetooth accessory.
- *Accessory Model*

  The model of the Bluetooth accessory.
- *Raw Data*

  The raw data is read from Bluetooth accessory. It varies depending on <Accessory Type> and <Accessory Model>.
  - **WTH300**

    It is a four-byte hexadecimal value. The higher 2 bytes of the hexadecimal value indicate temperature, the lower 2 bytes indicate humidity. The high byte is the integer part and the low byte is the fractional part.

    Temperature is equal to the low byte divided by 256 plus the high byte, in degree Centigrade. Humidity is equal to the low byte divided by 256 plus the high byte, and the unit is rh.
  - **WRL300**

    It is a four-byte hexadecimal value. It indicates the current state of the relay.
  - **WMS301/WTH301**

    It is a four-byte hexadecimal value. The lower 2 bytes of the hexadecimal value indicate temperature, the higher 2 bytes indicate humidity.

    Humidity is equal to the higher 2 bytes divided by 100, and the unit is rh. Temperature is equal to the lower 2 bytes divided by 100, in degree Centigrade.

- **Fuel sensor**

  It is a decimal value which indicates the fuel level.

  For Mechatronics fuel sensor, if the value starts with the character "F" ("FXX"), it indicates the fuel percentage, otherwise it indicates the original fuel level value.

  Note

  For more details, please refer to parameter <Fuel Level Format> in AT+GTBAS.

- **WTS300**

  It is a four-byte hexadecimal value. The higher 2 bytes of the hexadecimal value indicate battery voltage in millivolt, the lower 2 bytes indicate temperature. The high byte is the integer part and the low byte is the fractional part.

  Temperature is equal to the low byte divided by 256 plus the high byte, in degree Centigrade.

- **TEMP ELA**

  It is a two-byte hexadecimal Temperature value.

- **Escort Angle Sensor**

  It is a four-byte hexadecimal value.

  The first byte of the higher 2 bytes in the hexadecimal value is reserved, which is 00.

  The second byte of the higher 2 bytes indicates Event Notification of Angle sensor.

  And the lower 2 bytes of the hexadecimal value indicate Tilt Angle of sensor.

| Fields | Length | Range/Format |
|---|---|---|
| Reserved | 1 | 00 |
| Event Notification | 1 | 00 - FF |
| Tilt Angle | 2 | 0000 - FFFF |

- **Mechatronics Angle Sensor**

  It is an eight-byte hexadecimal value.

| Fields | Length | Range/Format |
|---|---|---|
| Reserved | 1 | 00 |
| X-plane angle | 1 | -90 - 90 |
| Y-plane angle | 1 | -90 - 90 |
| Z-plane angle | 1 | -90 - 90 |
| Sensor status | 1 | 0 - FF |
| Single events | 2 | 0 - FFFF |
| Complex events | 2 | 0 - FFFF |

- **RHT ELA**

  It is a four-byte hexadecimal value. The lower 2 bytes of the hexadecimal value indicate temperature, the higher 2 bytes indicate humidity.

- **ATP100/ATP102**

  It is a four-byte hexadecimal value.

  From left to right, the first byte is reserved(0x00).

  The second byte includes pressure value. Tire pressure is equal to the second byte multiply by 2.5.

  The third byte includes temperature and low power alarm. Temperature is equal to the third byte subtract by 40.

The higher 4 bits of last byte includes the product model, and the lower 4 bits of last byte includes the firmware version.

- **MAG ELA**

  It is a four-byte hexadecimal value. The lower 2 bytes of the hexadecimal value indicate MAG data, the higher 2 bytes indicate MAG ID.

✧ *Accessory Append Mask*

This parameter indicates which Bluetooth accessory data needs to be reported in the message. If Bit 15 is set to 0, only Bit 0 - Bit 14 are valid.

| Mask Bit | Item | Description |
|---|---|---|
| Bit 14 | <Relay Data> | Including <Relay Config Result>, <Relay state> |
| Bit 13 | <Accessory Battery Percentage> | Accessory Battery Percentage |
| Bit 12 | <Magnet Data> | Including <Magnet ID>, <MAG Event Counter>, <Magnet State> |
| Bit 11 | <Enhanced Temperature> | Enhanced temperature |
| Bit 10 | <Timestamp> | Timestamp |
| Bit 9 | <Tire pressure> | Tire pressure |
| Bit 8 | <Accessory Event Notification Data> | Including <Accessory Mode>, <Accessory Event> |
| Bit 7 | <Accessory Input Output Data> | Including <Accessory Output status>, <Accessory Digital Input status>, <Accessory Analog Input voltage> |
| Bit 6 | Reserved | Reserved |
| Bit 5 | <Accessory Humidity> | Accessory Humidity |
| Bit 4 | <Accessory Temperature> | Accessory Temperature |
| Bit 3 | <Accessory Battery Voltage> | Accessory Battery Voltage |
| Bit 2 | <Accessory Status> | Accessory Bluetooth Connection Status |
| Bit 1 | <Accessory MAC> | Accessory MAC |
| Bit 0 | <Accessory Name> | Accessory Name |

✧ *Accessory Name*

The name of the Bluetooth accessory.

✧ *Accessory MAC*

The MAC address of the Bluetooth accessory.

- *Accessory Status*

  It indicates the connection status of the Bluetooth accessory.
  - **0** - Disconnected.
  - **1** - Connected.
- *Accessory Battery Voltage*

  It indicates the voltage of the battery in Bluetooth accessory.
- *Accessory Temperature*

  It indicates the temperature measured by Bluetooth accessory.
- *Accessory Humidity*

  It indicates the humidity measured by the Bluetooth accessory.
- *Accessory Output status*

  The status of Bluetooth accessory output. Each bit indicates the status of one output. For example, 0X01 indicates that Output ID 7 is enabled and Output ID 8 is disabled.
- *Accessory Digital Input status*

  The status of Bluetooth accessory digital input.
- *Accessory Analog Input voltage*

  The status of Bluetooth accessory analog input voltage.
- *Accessory Mode*

  The working mode of Angle sensor.
- *Accessory Event*

  The event is generated by the Angle sensor.
- *Timestamp*

  Timestamp of the tire pressure value collection.
- *Enhanced Temperature*

  It instructs Bluetooth accessories to measure temperature with high precision.

  Note

  Temperature alarm uses integer value.
- *Magnet ID*

  ID corresponding to different magnet sensors.
- *MAG Event Counter*

  The total number of times detected by the magnet sensor.
- *Magnet State*

  The state of the two parts of the magnet sensor.
  - **0** - Separate
  - **1** - Closed
- *Accessory Battery Percentage*

  Percentage of Bluetooth accessory's battery power.
- *Relay state*

  The current state of the relay sensor.

Note

The word "Optional" means the item is controlled by the parameter <ERI Mask> of the AT+GTFRI command.

## 4.2.4 EPS | AIS

✧ **+RESP:GTEPS**

If the external power supply monitoring is enabled by the command AT+GTEPS, the device will send the message **+RESP:GTEPS** to the backend server when the voltage of the external power supply enters the alarm range.

✧ **+RESP:GTAIS**

The device will send the message **+RESP:GTAIS** to the backend server when analog input voltage enters the alarm range.

Example:
+RESP:GTEPS,8020040803,866314060835525,GV58LAU,12041,01,1,1,0.0,0,38.6,117.129217,31.839273,20250313055404,0460,0000,550B,0E9E30A5,09,12,0.73,1.20,1.40,0.0,20250313055404,14D0$

+RESP:GTAIS,8020040803,135790246811220,GV58LAU,1980,11,1,1,4.3,92,70.0,121.354335,31.222073,20090214013254,0460,0000,550B,6141,01,15,2000.0,20090214093254,11F0$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | EPS \| AIS |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | External Power Voltage/Analog Input VCC | <=5 | 0 - 32000(mV)\|0 - 16000(mV) |
| | Report ID / Report Type | 2 | X(0 \| 5 )Y(0 \| 1 ) |
| | Number | 1 | 1 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *External Power Voltage / Analog Input VCC*

The value of the external power supply voltage or the analog input voltage.

When the voltage meets the alarm condition as set by the command AT+GTEPS , the device will send the current voltage with the **+RESP:GTEPS** or **+RESP:GTAIS** message to the backend server.

✧ *Report ID / Report Type*

It is a one-byte hexadecimal value represented by two ASCII bytes. The first byte (4 higher bits of the hexadecimal value) indicates Report ID and the second byte (4 lower bits of the hexadecimal value) indicates Report Type.

- Report ID has different meanings as follows:
  - The value of Report ID for the report message **+RESP:GTEPS** is 0.
  - The ID of the analog input port which triggers the report message **+RESP:GTAIS**, the range is 5.
- Report type has the following meanings:
  - **0** - Outside the predefined range.
  - **1** - Within the predefined range.

✧ *Number*

The number of the GNSS positions included in the report message. Generally, it is 1.

### 4.2.5 LBC(Location By Call Alarm)

If the parameter <Location Request Mask> is enabled by the command AT+GTCFG, the device will get and send the current position to the backend server via the message **+RESP:GTLBC** when there is an SMS.

Example:
+RESP:GTLBC,8020040803,866314060835525,GV58LAU,17855503680,1,0.0,0,109.7,117.129305,

31.839371,20250312062641,0460,0000,550B,0E9E30A5,09,12,0.82,1.42,1.64,20250312062642,0
0F5$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | LBC |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Call Number | <=20 | phone number |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Call Number*

The phone number of the incoming call which triggers the report message.

### 4.2.6 IDA(ID Authentication Alarm)

If <Mode> in AT+GTIDA is set to 1 or 2, **+RESP:GTIDA** will be reported according to the setting.

If <Mode> in AT+GTIDA is set to 0, **+RESP:GTIDA** will always be reported without checking the status of ID authorization.

Example:
+RESP:GTIDA,8020040803,866314060835525,GV58LAU,,11111,1,1,1,0.0,0,142.6,117.129265,31.839839,20250312070032,0460,0000,550B,0E9E30A5,09,12,0.78,1.19,1.43,0.0,,,,,20250312070033,0122$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | IDA |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 0 | |
| | ID | <=28 | ASCII (not including '=' and ',') |
| | ID Report Type | 1 | 0-3 |
| | Number | 1 | 1 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *ID*

The ID that is read currently. If the driver does not place ibutton at the end of the alarm, the ID will be replaced by FFFFFFFFFFFFFFF.

✧ *ID Report Type*

A numeral to indicate the type of reported ID.

- **0** - IDA function is disabled.
- **1** - The ID is authorized.
- **2** - The ID has logged out.
- **3** - The ID is unauthorized.

### 4.2.7 GES(Parking-Fence Information)

The **+RESP:GTGES** message is reported according to <Trigger Mode> and <Trigger Report> in AT+GTGEO after the ignition is turned off.

Example:
+RESP:GTGES,8020040803,866314060268081,GV58LAU,,01,21,500,5,1,1,0.0,1,51.3,117.128739,31.846549,20250320020241,0460,0000,550B,0E9E30A5,09,8,1.59,2.21,2.72,0.0,20250320020241,00E2$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | GES |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 0 | |
| | Report ID / Report Type | <=3 | X(0 - 63)Y(0 - 1) |
| | Trigger Mode | <=2 | 0 \| 21 \| 22 |
| | Radius | <=7 | 50 - 6000000(m) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Check Interval | <=5 | 0\|5 - 86400(s) |
| | Number | 1 | 1 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Mileage | <=9 | 0.0 - 4294967.0(km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Report ID/Report Type*

It is a hexadecimal value represented by three ASCII bytes. The first two bytes indicates Report ID and the last byte indicates Report Type.

- Report ID

  The ID of Geo Fence in HEX format.

- Report Type
  - **0** - The current Parking-Fence is inactive.
  - **1** - The current Parking-Fence is active.

## 4.2.8 GIN|GOT(Geo-Fence Information)

✧ **+RESP:GTGIN**

If Geo-Fence is configured and enabled, the device will send the message **+RESP:GTGIN** to the backend server according to settings when the device enters the Geo-Fence.

### ✧ +RESP:GTGOT

If Geo-Fence is configured and enabled, the device will send the message **+RESP:GTGOT** to the backend server according to settings when the device leaves the Geo-Fence.

```
Example:
+RESP:GTGIN,8020040803,866314060268081,GV58LAU,,1,0,00000001,0,0,19,,1,1,10.2,90,80.4,1
17.131445,31.852366,20250320020524,0460,0000,550B,0E9E30A5,09,8,1.59,2.23,2.74,0.0,2025
0320020524,00EE$

+RESP:GTGOT,8020040803,866314060268081,GV58LAU,,1,0,00000001,0,0,19,,1,1,12.0,86,58.0,1
17.135599,31.852691,20250320020615,0460,0000,550B,0E9E30A5,09,7,1.59,2.23,2.74,0.0,2025
0320020615,00F4$
```

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | GIN \| GOT |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 0 | |
| | ID Report Format | 1 | 0 \| 1 |
| | Parameter 1 | | |
| | Parameter 2 | | |
| | Parameter 3 | | |
| | Parameter 4 | | |
| | Parameter 5 | | |
| | Reserved | 0 | |
| | Number | 1 | 1 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *ID Report Format*

A numerical value to indicate the format of GEO ID. The following five parameters have different definitions for each format.

- **0:** Mask Format.

  Each Geo ID will be indicated as a bit of <Area Mask>.

| Fields | Length | Range/Format |
|---|---|---|
| Area Type | 1 | 1 |
| Area Mask 1 | 8 | 00000000 - 000FFFFF |
| Area Mask 2 | 8 | HEX |
| Area Mask 3 | 8 | HEX |
| Area Mask 4 | 8 | HEX |

- **1:** Group Format.

  Using <Group ID>, <Start GEO ID> and <End GEO ID> to indicate reported Geo Fence.

| Fields | Length | Range/Format |
|---|---|---|
| Area Type | 1 | 0 |
| Area Mask | 8 | 00000000 - 000FFFFF |
| Group ID | 1 | 0 - 9 |
| Start GEO ID | <=3 | 0 - 199 |
| End GEO ID | <=3 | 0 - 199 |

✧ *Area Type*

This message is for polygon or circular area. 0 means "Polygon", and 1 means "Circular".

✧ *Area Mask*

For polygon area, it indicates the report message is for a single polygon or multiple polygons overlapping.

- **Bit 0** - For Polygon ID 0
- **Bit 1** - For Polygon ID 1

- ...
- **Bit 19** - For Polygon ID 19

For circular area, it indicates the report message is for a single circle or multiple circles overlapping.

- **Bit 0** - For Circular ID 0
- **Bit 1** - For Circular ID 1
- ...
- **Bit 99** - For Circular ID 99

For example, if the Area Mask is 03, it indicates entering or exiting events of GEO-ID0/PEO-ID0 and GEO-ID1/PEO-ID1 occur at the same time.

### 4.2.9 SPE(Speed Alarm inside the Geo-Fence)

If the <Speed Threshold> is greater than 0, the message **+RESP:GTSPE** will be sent to the backend server when the device speed is higher than <Speed Threshold> if it is inside the Geo-Fence.

| |
|---|
| Example:<br>+RESP:GTSPE,8020040803,866314060268081,GV58LAU,0,,,01,1,1,58.6,88,46.8,117.153051,31.853068,20250320020801,0460,0000,550B,0E9E30A5,09,8,1.41,1.38,1.97,0.0,20250320020801,00FC$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | SPE |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Group ID | 1 | 0 - 9 |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Report ID / Report Type | 2 | X(0 ) \| Y(0 \| 1 ) |
| | Number | 1 | 1 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Group ID*

A numeral to indicate in which group the speed alarm is triggered.

✧ *Report ID / Report Type*

It is a one-byte hexadecimal value represented by two ASCII bytes. The first byte (4 higher bits of the hexadecimal value) indicates Report ID and the second byte (4 lower bits of the hexadecimal value) indicates Report Type.

- *Report Type*

  Report type in the speed alarm message **+RESP:GTSPE** has the following meanings:

  – **0** - The value of the current speed is below the <Speed Threshold> or the device is outside the current Polygon Geo-Fence.

  – **1** - The value of the current speed exceeds the <Speed Threshold> when the device is inside the Polygon Geo-Fence.

## 4.2.10 Google Maps Hyperlink

If the Google Maps hyperlink reporting feature in parameter <Location Request Mask> of the command AT+GTCFG is enabled, the device will send its current position to the mobile phone which makes the call request or SMS request via SMS with a Google Maps hyperlink.

```
Example:
LBC:  GV58LAU
http://maps.Google.com/maps?q=31.222073,121.354335
F1  D2009/01/01  T00:00:00  B0  I1  V0.0
```

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Type Name | <=10 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Device Name | <=20 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Google Maps Hyperlink Header | 30 | http://maps.google.com/maps?q= |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | GNSS Fix | <=3 | F[0 - 50] |
| | GNSS UTC Time | 20 | DYYYY/MM/DDTHH:MM:SS |
| | Battery Percentage | <=4 | B[0 - 100] |
| | Ignition Status | 2 | I[0 \| 1 ] |
| | Speed | <=6 | V0.0 - V999.9(km/h) |

✧ *Type Name*

A string that includes the device name and GNSS positioning type. ("SOS", "IN GEO-i", "OUT GEO-i", "LBC")

✧ *Google Maps Hyperlink*

A string of a Google Maps hyperlink.

✧ *GNSS Fix*

The accuracy of the location information. F0 means "No GNSS fix".

✧ *Battery Percentage*

The percentage of the backup battery.

✧ *Ignition Status*

The status of ignition. 0 means "Ignition off", and 1 means "Ignition on".

✧ *Speed*

The current speed. Unit: km/h.

## 4.2.11 GSM(Cells Information)

The report for the information of the serving cell and the neighboring cells.

Example:
+RESP:GTGSM,8020040803,866314060268081,TOW,0460,0000,550B,085BE2AA,48,,0460,0000,550B,0E9E30AD,37,,0460,0000,550B,085BE2AE,25,,0460,0000,550B,085BE2AC,25,,,,,,,,,,,,,,0460,0000,550B,0E9E30A5,71,0D,8,160001,1.39,1.39,1.97,20250320021635,011B$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | GSM |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Fix Type | 3 | SOS \| RTL \| LBC \| FRI \| GIR \| ERI \| TOW \| |
| | MCC1 | 4 | 0XXX |
| | MNC1 | 4 | 0XXX |
| | LAC1 | 4 | XXXX |
| | Cell ID1 | 4\|8 | XXXX/XXXXXXXX |
| | RX Level1 | <=3 | 0 - 31 \| 99 \| 0 - 97 \| 255 |
| | Reserved | 0 | |
| | MCC2 | 4 | 0XXX |
| | MNC2 | 4 | 0XXX |
| | LAC2 | 4 | XXXX |
| | Cell ID2 | 4\|8 | XXXX/XXXXXXXX |
| | RX Level2 | <=3 | 0 - 31 \| 99 \| 0 - 97 \| 255 |
| | Reserved | 0 | |
| | MCC3 | 4 | 0XXX |
| | MNC3 | 4 | 0XXX |
| | LAC3 | 4 | XXXX |
| | Cell ID3 | 4\|8 | XXXX/XXXXXXXX |
| | RX Level3 | <=3 | 0 - 31 \| 99 \| 0 - 97 \| 255 |
| | Reserved | 0 | |
| | MCC4 | 4 | 0XXX |
| | MNC4 | 4 | 0XXX |
| | LAC4 | 4 | XXXX |
| | Cell ID4 | 4\|8 | XXXX/XXXXXXXX |
| | RX Level4 | <=3 | 0 - 31 \| 99 \| 0 - 97 \| 255 |
| | Reserved | 0 | |
| | MCC5 | 4 | 0XXX |
| | MNC5 | 4 | 0XXX |
| | LAC5 | 4 | XXXX |
| | Cell ID5 | 4\|8 | XXXX/XXXXXXXX |
| | RX Level5 | <=3 | 0 - 31 \| 99 \| 0 - 97 \| 255 |
| | Reserved | 0 | |
| | MCC6 | 4 | 0XXX |
| | MNC6 | 4 | 0XXX |
| | LAC6 | 4 | XXXX |
| | Cell ID6 | 4\|8 | XXXX/XXXXXXXX |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | RX Level6 | <=3 | 0 - 31 \| 99 \| 0 - 97 \| 255 |
| | Reserved | 0 | |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX/XXXXXXXX |
| | RX Level | <=3 | 0 - 31 \| 99 \| 0 - 97 \| 255 |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | HEX |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Fix Type*

A string to indicate the type of GNSS fix for which this cell information is used.

- **SOS** - This cell information is for SOS request.
- **RTL** - This cell information is for RTL request.
- **LBC** - This cell information is for LBC request.
- **TOW** - This cell information is for TOW request.
- **FRI** - This cell information is for FRI request.
- **GIR** - This cell information is for sub command "**C**" in the AT+GTRTO command.
- **ERI** - This cell information is for ERI request.

✧ *MCC(i)*

MCC of the neighboring cell i (i is the index of the neighboring cell).

✧ *MNC(i)*

MNC of the neighboring cell i.

✧ *LAC(i)*

LAC in hex format of the neighboring cell i.

✧ *Cell ID(i)*

Cell ID in hex format of the neighboring cell i.

✧ *RX Level(i)*

The signal strength of the neighboring cell i. This parameter is a 6-bit value coded in 1 dB steps:

For 2G/3G network:

| CSQ RSSI | Signal Strength (dBm) |
|---|---|
| 0 | <-113 |
| 1 | -111 |
| 2 - 30 | -109 - -53 |
| 31 | >-51 |
| 99 | Unknown |

For 4G network:

| CSQ RSRP | Signal Strength (dBm) |
|---|---|
| 0 | <-140 |
| 1 | -140 |
| 2 - 96 | -139 - -44 |
| 97 | >= -44 |
| 255 | Unknown |

- *MCC*
  MCC of the serving cell.
- *MNC*
  MNC of the serving cell.
- *LAC*
  LAC in hex format of the serving cell.
- *Cell ID*
  Cell ID in hex format of the serving cell.
- *RX Level*
  The signal strength of the serving cell.

Note

- **1** - It may include information of several neighboring cells or even no neighboring cell information. If no neighboring cell is found, all the fields of the neighboring cell will be empty.
- **2** - The "ffff" in the fields of <LAC(i)> and <Cell ID(i)> means the terminal does not know the value.
- **3** - This message cannot be sent via SMS.

## 4.3 Device Information Reports

This section describes the message related to device generic information. Please refer to the details below.

### 4.3.1 INF(Device Information)

If the device information report function is enabled by the command AT+GTCFG, the device will send the device information via the message **+RESP:GTINF** to the backend server periodically.

Example:
+RESP:GTINF,8020040803,866314060873492,GV58LAU,22,89860015123286906624,67,0,1,11916,3,0.00,0,1,,,20250314083427,4,,,,01,00,+0000,0,20250314083428,0086$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | INF |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Motion Status | 2 | 11 \| 12 \| 21 \| 22 \| 41 \| 42 \| 1A \| 16 |
| | ICCID | 20 | |
| | CSQ RSSI\|RSRP | <=3 | 0 - 31 \| 99 \| 0 - 97 \| 255 |
| | CSQ BER | <=2 | 0 - 7 \| 99 |
| | External Power Supply | 1 | 0 \| 1 |
| | External Power Voltage | <=5 | 0 - 32000(mV) |
| | Network Type | 1 | 0-3 |
| | Backup Battery Voltage | 4 | 0.00 - 4.50 (V) |
| | Charging | 1 | 0 \| 1 |
| | LED State | 1 | 0 \| 1 |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Last Fix UTC Time | 14 | YYYYMMDDHHMMSS |
| | Pin Mask | 1 | 0 - F |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Digital Input | <=2 | 00 - 03 |
| | Digital Output | <=2 | 00 - 07 |
| | Time Zone Offset | 5 | +\|-HHMM |
| | Daylight Saving | 1 | 0 \| 1 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Motion Status*

The current motion status of the device.

- **16 (Tow)** - The device attached vehicle is ignition off and it is towed.

- **1A (Fake Tow)** - The device attached vehicle is ignition off and it might be towed.
- **11 (Ignition Off Rest)** - The device attached vehicle is ignition off and it is motionless.
- **12 (Ignition Off Motion)** - The device attached vehicle is ignition off and it is moving before it is considered as being towed.
- **21 (Ignition On Rest)** - The device attached vehicle is ignition on and it is motionless.
- **22 (Ignition On Motion)** - The device attached vehicle is ignition on and it is moving.
- **41 (Sensor Rest)** - The device attached vehicle is motionless without ignition signal detected.
- **42 (Sensor Motion)** - The device attached vehicle is moving without ignition signal detected.

✧ *ICCID*

The ICCID of the SIM card.

✧ *CSQ RSSI/RSRP*

The signal strength level.

For 2G/3G network:

| CSQ RSSI | Signal Strength (dBm) |
|---|---|
| 0 | < -113 |
| 1 | -111 |
| 2 - 30 | -109 - -53 |
| 31 | > -51 |
| 99 | Unknown |

For 4G network:

| CSQ RSRP | Signal Strength (dBm) |
|---|---|
| 0 | <-140 |
| 1 | -140 |
| 2 - 96 | -139 - -44 |
| 97 | >= -44 |
| 255 | Unknown |

✧ *CSQ BER*

The quality of the GSM signal. The range is 0-7, and 99 is for unknown strength of signal.

✧ *External Power Supply*

Whether the external power supply is connected.

- **0** - Not connected.
- **1** - Connected.

✧ *External Power Voltage*

The voltage of the external power supply.

✧ *Network Type*

The type of the mobile network the device is currently registered on.

- **0** - Unregistered
- **1** - EGPRS
- **2** - WCDMA
- **3** - LTE

✧ *Backup Battery Voltage*

The voltage of the backup battery. The value of this field is valid only when the external power is not connected.

✧ *Charging*

Whether the backup battery is charging when the main power supply is connected.

- **0** - Not charging.
- **1** - Charging.

✧ *LED State*

A numeral to indicate the working status of all LED lights.

- **0** - All LED lights are turned off.
- **1** - At least one of the LED lights is on.

✧ *Last Fix UTC Time*

The UTC time of the latest successful GNSS fix.

✧ *Pin Mask*

The current working mode of pin.

✧ *Digital Input*

A bitwise hex integer to represent the logic status of the digital input.

✧ *Digital Output*

A bitwise hex integer to represent the logic status of the digital output.

✧ *Time Zone Offset*

The time offset of the local time zone from the UTC time.

✧ *Daylight Saving*

The current setting of the daylight saving.

- **0** - Daylight saving is disabled.
- **1** - Daylight saving is enabled.

## 4.4 RTO Reports

This section describes the feedback messages of runtime operations. Please refer to the details below.

### 4.4.1 GPS(GNSS Information)

After the device receives the command AT+GTRTO to read the GNSS information, it will send the GNSS information to the backend server via the message **+RESP:GTGPS**.

| |
|---|
| Example:<br>+RESP:GTGPS,8020040803,866314060835525,GV58LAU,,,,007F,,,20250312080640,20250312080642,015E$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Message Name | 3 | GPS |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Report Item Mask | 4 | 0000 - FFFF |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Last Fix UTC Time | 14 | YYYYMMDDHHMMSS |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Report Item Mask*

Please refer to <Report Item Mask> of the command AT+GTCFG.

### 4.4.2 ALM(Command Configuration Information)

After the device receives the command AT+GTRTO to read the configurations, it will send corresponding configuration information to the backend server via the message **+RESP:GTALM** according to the configuration mask. This message is only sent via GPRS even if the report mode is forced SMS mode, and it does not support HEX format.

Example:
+RESP:GTALM,8020040803,866314060268081,GV58LAU,,53,40,RMD,1,,,,,46000,46000,46000,46
000,46000,46000,46000,46000,46000,46000,,,46001,46001,46001,46001,46001,46001,46001,46
001,46001,46001,46001,46001,46001,46001,46001,46001,46001,460014,46001,46001,46001,46
001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,460
01,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,4600
1,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,4600
1,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001
4,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,46001,4600
1,46001,46001,46001,46001,46001,46001,46001,46001,,,46002,46002,46002,46002,46002,4600
2,46002,46002,46002,46002,46002,46002,46002,46002,46002,46002,46002,46002,46002,4600
2,,,011804,,,00208A,,,,,2,1,1,100,,,DOS,0,3,1,,,,,,2,,,,,,3,,,,,,7,,,0,,GDO,2,32,3,,,,,,SPD,2,12,24,7,46,
2,1,1,1,,,,,,,,,,,HBM,4,5,0,101,1,3,,61,6,6,,,9,10,,2,1,2,2,26,52,19,52,,,10,10,1,IDL,1,4,3,102,,,,2,1,
3,3,,,,,ASC,,,,,1,,,,,,,,SSR,1,7,7,9,7,1,0,,BZA,3,,,,1,11,11,,,2,11,6,,,2,11,8,,,2,12,8,,,,,,,,,SPA,2,51,,61,

1,,,72,,62,2,,,93,,63,3,,,113,,62,4,,,,,,,,JDC,1,7,2,,3,4,4,3,2,2,2,2,,JBS,1,,12,12,1798,0,32,1,1,3,0,2,1,2,120,60,CRA,1,52,52,52,1,502,502,2,1,2,2,,1,,IOB,0,0001,0001,2,2,1,2,2,,,,,20250312163538,0389$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | ALM |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Format Version | <=3 | 0 - 1 |
| | Total Packets | <=2 | 1 - 25 |
| | Current Packet | <=2 | 1 - 25 |
| | BSI | 3 | BSI |
| | APN | <=64 | |
| | APN User Name | <=30 | |
| | APN Password | <=30 | |
| | Backup APN | <=64 | |
| | Backup APN User Name | <=30 | |
| | Backup APN Password | <=30 | |
| | Network Mode | 1 | 0-6 |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Cutoff Character(Optional) | 2 | \^ |
| | ... | | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Total Packets*

The total number of **+RESP:GTALM**.

✧ *Current Packet*

The sequence number of the current message.

✧ *Cutoff Character(Optional)*

It is controlled by the parameter <Format Version> in AT+GTRTO. If <Format Version> is 1, these two characters are appended to the end of each command configuration, otherwise this field will not exist.

✧ *Network Mode*

- **0** - Auto (LTE & GSM/EGPRS & WCDMA)

- **1** - GSM/EGPRS Only
- **2** - WCDMA
- **3** - LTE Only
- **4** - GSM/EGPRS & WCDMA (GSM/EGPRS First)
- **5** - WCDMA & GSM/EGPRS (WCDMA First)
- **6** - WCDMA & LTE & GSM/EGPRS (WCDMA First)

### 4.4.3 CID(ICCID Information)

After the device receives the command AT+GTRTO to read the ICCID of the SIM card, it will send the ICCID to the backend server via the message **+RESP:GTCID**.

Example:
+RESP:GTCID,8020040803,866314060873492,GV58LAU,89860086127286669760,20250327170510,01AC$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 3 | +RESP:GT |
| | Message Name | 3 | CID |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' - '_' |
| Body | ICCID | 20 | '0' - '9', 'a' - 'z' 'A' - 'Z' |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

### 4.4.4 CSQ(Network Signal Information)

After the device receives the command AT+GTRTO to read the GSM signal level, it will send the GSM signal level to the backend server via the message **+RESP:GTCSQ**.

Example:
+RESP:GTCSQ,8020040803,866314060835525,GV58LAU,70,0,20250312083633,01D2$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | CSQ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | CSQ RSSI\|RSRP | <=3 | 0 - 31 \| 99 \| 0 - 97 \| 255 |
| | CSQ BER | <=2 | 0 - 7 \| 99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *CSQ RSSI/RSRP*

The signal strength level.

For 2G/3G network:

| CSQ RSSI | Signal Strength (dBm) |
|---|---|
| 0 | <-113 |
| 1 | -111 |
| 2 - 30 | -109 - -53 |
| 31 | >-51 |
| 99 | Unknown |

For 4G network:

| CSQ RSRP | Signal Strength (dBm) |
|---|---|
| 0 | <-140 |
| 1 | -140 |
| 2 - 96 | -139 - -44 |
| 97 | >= -44 |
| 255 | Unknown |

✧ *CSQ BER*

The quality of the GSM signal. The range is 0-7, and 99 is for unknown strength of signal.

### 4.4.5 VER(Firmware Version Information)

After the device receives the command AT+GTRTO to get the software version and hardware version, it will send the version information to the backend server via the message **+RESP:GTVER**.

| |
|---|
| Example:<br>+RESP:GTVER,8020040803,866314060835525,GV58LAU,802004,080F,0102,20250312083658,01D8$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | VER |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Device Type | 6 | 802004 |
| | Software Version | 4 | 0000 - FFFF |
| | Hardware Version | 4 | 0000 - FFFF |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

- *Device Type*

  The type of the device.
- *Software Version*

  The software version of the device. The first two characters represent the major version and the last two characters represent the minor version. For example, **010A** means the version **1.10**.
- *Hardware Version*

  The hardware version of the device. The first two characters represent the major version and the last two characters represent the minor version. For example, **010A** means the version **1.10**.

### 4.4.6 BAT(Battery Information)

After the device receives the command AT+GTRTO to read the power supply information, it will send the power supply information to the backend server via the message **+RESP:GTBAT**.

```
Example:
+RESP:GTBAT,8020040803,866314060268081,GV58LAU,1,12008,,0.00,0,1,20250320080726,042A
$
```

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | BAT |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | External Power Supply | 1 | 0 \| 1 |
| | External Power Voltage | <=5 | 0 - 32000(mV) |
| | Reserved | 0 | |
| | Backup Battery Voltage | 4 | 0.00 - 4.50 (V) |
| | Charging | 1 | 0 \| 1 |
| | LED State | 1 | 0 \| 1 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

### 4.4.7 IOS(IO Status Information)

After the device receives the command AT+GTRTO to get the status of all the IO ports, it will send the status to the backend server via the message **+RESP:GTIOS**.

Example:
+RESP:GTIOS,8020040803,866314060835525,GV58LAU,0,,,,01,00,20250312083817,01E3$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | IOS |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Pin Mask | <=2 | 0 - F |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Digital Input | <=2 | 00 - 03 |
| | Digital Output | <=2 | 00 - 07 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

### 4.4.8 TMZ(Time Zone Information)

After the device receives the command AT+GTRTO to get the time zone settings, it will send the time zone information via the message **+RESP:GTTMZ** to the backend server.

| Example: |
|---|
| +RESP:GTTMZ,8020040803,866314060835525,GV58LAU,+0000,0,1,20250312083837,01E8$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | TMZ |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Time Zone Offset | 5 | +\|-HHMM |
| | Daylight Saving | 1 | 0 \| 1 |
| | Network Time Checking | 1 | 0 - 1 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

### 4.4.9 AIF(Basic Device Information)

After the device receives the command AT+GTRTO to get the basic device information, it will send the information via the message **+RESP:GTAIF** to the backend server.

| Example: |
|---|
| +RESP:GTAIF,8020040803,866314060835525,GV58LAU,cmnet,123,456,queclink,7890,01234,898 60015123286906624,73,0,0E9E30A5,10.136.113.117,8.8.8.8,10.10.10.110,2,,,,20250312084211, 01F9$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | AIF |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | APN | <=64 | |
| | APN User Name | <=30 | |
| | APN Password | <=30 | |
| | Backup APN | <=64 | |
| | Backup APN User Name | <=30 | |
| | Backup APN Password | <=30 | |
| | ICCID | 20 | |
| | CSQ RSSI\|RSRP | <=3 | 0 - 31 \| 99 \| 0 - 97 \| 255 |
| | CSQ BER | <=2 | 0 - 7 \| 99 |
| | Cell ID | 4\|8 | XXXX/XXXXXXXX |
| | IP Address | <=15 | 0.0.0.0 |
| | Main DNS | <=15 | 0.0.0.0 |
| | Backup DNS | <=15 | 0.0.0.0 |
| | Auto APN State | 1 | 0-2 |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *CSQ RSSI/RSRP*

The signal strength level.

For 2G/3G network:

| CSQ RSSI | Signal Strength (dBm) |
|---|---|
| 0 | <-113 |
| 1 | -111 |
| 2 - 30 | -109 - -53 |
| 31 | >-51 |
| 99 | Unknown |

For 4G network:

| CSQ RSRP | Signal Strength (dBm) |
|---|---|
| 0 | <-140 |
| 1 | -140 |
| 2 - 96 | -139 - -44 |
| 97 | >= -44 |

| CSQ RSRP | Signal Strength (dBm) |
|---|---|
| 255 | Unknown |

- *CSQ BER*

  The quality of GSM signal. Its range is from 0 to 7, and 99 indicates unknown signal strength.
- *Cell ID*

  The serving cell ID in HEX format.
- *IP Address*

  The IP address of the device.
- *Main DNS*

  The main DNS server.
- *Backup DNS*

  The backup DNS server.
- *Auto APN State*

  The current state of getting APN automatically.
  - **0** - The device cannot obtain the APN or the APN obtained cannot activate PDP context.
  - **1** - The device has obtained the APN and the APN obtained has already activated PDP context.
  - **2** - The device is using the APN set by the command AT+GTBSI.

## 4.4.10 GSV(GPS Satellite Information)

After the device receives the command AT+GTRTO to get the satellite information, it will send the GPS satellite information via the message **+RESP:GTGSV** to the backend server.

Example:
+RESP:GTGSV,8020040803,866314060835525,GV58LAU,10,3,25,4,30,6,21,9,26,11,38,12,40,14,42,17,34,19,37,22,22,20250312084354,0203$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | GSV |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | SV Count | <=2 | 0-24 |
| | SV ID | <=3 | >=0 |
| | SV Power | <=2 | >=0 |
| | … | | |
| | SV ID | <=3 | >=0 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | SV Power | <=2 | >=0 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *SV Count*

The count of satellites found by GPS.

✧ *SV ID*

Satellite ID. If there is no satellite, the field is filled with zero.

✧ *SV Power*

Satellite power. If there is no satellite, the field is filled with zero.

### 4.4.11 RSV(GLONASS Satellite Information)

After the device receives the command AT+GTRTO to get the satellite information, it will send the GLONASS satellite information via the message **+RESP:GTRSV**.

| |
|---|
| Example: |
| +RESP:GTRSV,8020040803,866314060835525,GV58LAU,0,20250312084647,0211$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | RSV |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | SV Count | <=2 | 0 - 24 |
| | SV ID | <=3 | >=0 |
| | SV Power | <=2 | >=0 |
| | ... | | |
| | SV ID | <=3 | >=0 |
| | SV Power | <=2 | >=0 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *SV Count*

The count of satellites found by GLONASS.

✧ *SV ID*

Satellite ID. If there is no satellite, this field is filled with zero.

✧ *SV Power*

Satellite power. If there is no satellite, this field is filled with zero.

### 4.4.12 BSV(Beidou Satellite Information)

After the device receives the command AT+GTRTO to get the satellite information, it will send the Beidou satellite information via the message **+RESP:GTBSV** to the backend server.

| |
|---|
| Example:<br>+RESP:GTBSV,8020040803,866314060835525,GV58LAU,9,7,13,20,28,22,23,26,28,29,16,35,27,38,12,40,16,44,28,20250312084741,0218$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | BSV |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | SV Count | <=2 | 0 - 24 |
| | SV ID | <=3 | > =0 |
| | SV Power | <=2 | > =0 |
| | ... | | |
| | SV ID | <=3 | > =0 |
| | SV Power | <=2 | > =0 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *SV Count*

The count of satellites found by Beidou.

✧ *SV ID*

Satellite ID. If there is no satellite, the field is filled with zero.

✧ *SV Power*

Satellite power. If there is no satellite, the field is filled with zero.

### 4.4.13 CVN(CAN Version Information)

After the device receives the command AT+GTRTO to get the version number of the CAN100, it will send the information to the backend server via the message **+RESP:GTCVN**.

Example:
+RESP:GTCVN,8020040803,866314060000187,GV58LAU-2,2.3.12,,,,,20231208112028,A1BC$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | CVN |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | CAN100 SW Version | <=7 | '0' - '9' 'a' - 'z' |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *CAN100 SW Version*

The software version of the CAN100 device.

## 4.4.14 CSN(CAN Serial Number)

After the device receives the command AT+GTRTO to get the serial number of CAN100, it will send the information to the backend server via the message **+RESP:GTCSN**.

Example:
+RESP:GTCSN,8020040803,866314060000187,GV58LAU-2,2100779,,,,,20231208112146,A1C5$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | CSN |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | CAN100 Serial Number | <=10 | '0' - '9', 'a' - 'z' |
| | Reserved | 0 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *CAN100 Serial Number*

The serial number of the CAN100 device.

## 4.4.15 ATI(Version Information)

After the device receives the command AT+GTRTO to get the basic device information, it will send the information to the backend server via the message **+RESP:GTATI**.

Example:
+RESP:GTATI,8020040803,866314060268081,GV58LAU-A07,001C1031,080F,0101,0402,0104,EFB
A,A4,,20250312165458,0418$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | ATI |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | ATI Mask | <=8 | 00000000 - FFFFFFFF |
| | Firmware Version | 4 | '0' - '9' 'a' - 'z' 'A' - 'Z |
| | Hardware Version | 4 | '0' - '9' 'a' - 'z' 'A' - 'Z |
| | BT Firmware Version | 4 | '0' - '9' 'a' - 'z' 'A' - 'Z |
| | BT Boot Firmware Version | 4 | '0' - '9' 'a' - 'z' 'A' - 'Z |
| | Flash ID | 4 | '0'-'9' 'a'-'z' 'A'-'Z' |
| | Sensor ID | 2 | '0'-'9' 'a'-'z' 'A'-'Z' |
| | Modem IMEI | 15 | IMEI |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *ATI Mask*

This mask is set by the AT+GTRTO command and used to control parameter fields in the **+RESP:GTATI** message.

- *Firmware Version*

  The firmware version of the device. The first two characters represent the branch number, the middle two characters represent the major version and the last two characters represent the minor version. For example, **000101** means version **R00A01V01**.

- *Hardware Version*

  The hardware version of the device. The first two characters represent the major version and the last two characters represent the minor version. For example, **0101** means version **1.01**.

- *BT Firmware Version*

  The Bluetooth firmware version. The first two characters represent the major version and the last two characters represent the minor version. For example, **010A** means the version **1.10**.

- *BT Boot Firmware Version*

  The Bluetooth Boot firmware version. The first two characters represent the major version and the last two characters represent the minor version. For example, **010A** means the version **1.10**.

- *Flash ID*

  It indicates the flash type used by the device.

- *Sensor ID*

  It indicates the sensor type used by the device.

- *Modem IMEI*

  The Modem IMEI of the terminal device.

## 4.4.16 CML(CAN100 Synchronization)

After the device receives the command AT+GTRTO to get the car model ID of the CAN100, it will send the information to the backend server via the message **+RESP:GTCML**.

| |
|---|
| Example:<br>+RESP:GTCML,8020040803,866314060179445,GV58LAU,,,1,,,20231220131924,0021$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | CML |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | CAN100 Car Model ID | <=5 | 0-65535 |
| | CAN100 Car Name | <=50 | ASCII |
| | CAN100 Sync Status | 1 | 1-4 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *CAN100 Car Model ID*

The car model ID of the CAN100 device. If the value is 0, it means that no model has been obtained.

✧ *CAN100 Car Name*

Car Name is human-readable make and model of the car. If this string value is equal to "sync", the queried <CAN100 Car Model ID> is the synchronized class ID.

✧ *CAN100 Sync Status*

- **1** - The synchronization is successful.
- **2** - CAN100 is not properly connected to the CAN bus.
- **3** - The car is not supported by the firmware version of CAN Chipset.
- **4** - CAN100 is not responding.

### 4.4.17 BTI(Bluetooth Peripheral Information)

After the device receives the command AT+GTRTO to get the Bluetooth peripheral information and the list of connected peripherals, it will send the information to the backend server via the message **+RESP:GTBTI**. If the connection between the Primary device and the Ghost device is successful, the battery percentage of the Ghost device will be reported periodically via the message **+RESP:GTBTI** according to the <BTI Report Interval> in AT+GTSVR.

Example:
+RESP:GTBTI,8020040803,866314060873492,GV58LAU,GV58LAU%IMEI,60E93B2073B4,0,0,,,,,,,,,,
20250314083531,008F$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | BTI |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Bluetooth Name | <=20 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Bluetooth Mac Address | 12 | 000000000000 - FFFFFFFFFFFF |
| | Bluetooth State | 1 | 0 \| 1 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Connected Device Number | <=2 | 0 - 11 |
| | Connected Device Name | <=20 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Connected Device Mac | 12 | 000000000000 - FFFFFFFFFFFF |
| | Role | 1 | 0 \| 1 |
| | Real-Time State | 1 | 0 - 1 |
| | Ghost Battery Percentage | 3 | 0 - 100 |
| | Ghost Status | 4 | 0000 - FFFF |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Bluetooth State*

The connection status of the Bluetooth when the device is the Primary device.

- **0** - The Bluetooth is not connected.
- **1** - The Bluetooth is connected.

✧ *Connected Device Number*

The number of the connected peripheral device. If there is more than one device connected, the information for the parameters <Connected Device Name>, <Connected Device Mac> and <Role> will be displayed repeatedly.

✧ *Connected Device Name*

The name of the connected peripheral device.

✧ *Connected Device Mac*

The Mac address of the connected peripheral device.

✧ *Role*

The role type of the peripheral device.

- **0** - Ghost device
- **1** - Primary device

✧ *Real-Time State*

It indicates the data type of <Ghost Battery Percentage> and <Ghost Status> read from the Ghost device.

- **0** - Historical data
- **1** - Real-time data

✧ *Ghost Battery Percentage*

The current capacity of the backup battery in percentage.

Note

The parameter <Connected Device Name> and <Ghost Battery Percentage> will be empty if the terminal device is the Primary device.

✧ *Ghost Status*

Bitwise mask to define status of the Ghost device. 0 indicates "The corresponding abnormal status is not detected", and 1 indicates "The corresponding abnormal status is detected".

- **Bit 0** - Reserved
- **Bit 1** - GNSS fix failure
- **Bit 2** - Reserved
- **Bit 3** - Reserved
- **Bit 4** - Reserved
- **Bit 5** - SIM card error
- **Bit 6** - GSM unavailable
- **Bit 7** - Reserved
- **Bit 8** - Reserved
- **Bit 9 - Bit 14** - Reserved
- **Bit 15** - Communication error between BLE and BB

Note

The parameters <Real-Time State>, <Ghost Battery Percentage> and <Ghost Status> are empty if <Mode> in the command AT+GTSVR is 0 or 2.

## 4.4.18 SCS(Acceleration Calibration Information)

After the device receives the command AT+GTRTO to get the calibration data, it will send the calibration data via the message **+RESP:GTSCS** to the backend server.

Example:
+RESP:GTSCS,8020040803,866314060285200,GV58LAU,2,0.72,0.70,-0.01,0.69,-0.72,-0.09,-0.07,0.06,-1.00,20240508142302,398C$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | SCS |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Self Calibration Status | 1 | 0-2 |
| | X_Forward | <=5 | -1.00 - 1.00 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Y_Forward | <=5 | -1.00 - 1.00 |
| | Z_Forward | <=5 | -1.00 - 1.00 |
| | X_Side | <=5 | -1.00 - 1.00 |
| | Y_Side | <=5 | -1.00 - 1.00 |
| | Z_Side | <=5 | -1.00 - 1.00 |
| | X_Vertical | <=5 | -1.00 - 1.00 |
| | Y_Vertical | <=5 | -1.00 - 1.00 |
| | Z_Vertical | <=5 | -1.00 - 1.00 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Self Calibration Status*

The status of the self-calibration for Acceleration Data.

- **0** - Self-calibration is disabled.
- **1** - Self-calibration is not done.
- **2** - Self-calibration is successful.

✧ *X_Forward, Y_Forward, Z_Forward*

The factors to calculate the new acceleration in forward direction. The formula to calculate the acceleration in Forward direction Xnew is **Xnew = <X_Forward> * X + <Y_Forward> * Y + <Z_Forward> * Z**.

✧ *X_Side, Y_Side, Z_Side*

The factors to calculate the new acceleration in side direction. The formula to calculate the acceleration in Side direction Ynew is **Ynew = <X_Side> * X + <Y_Side> * Y + <Z_Side> * Z**.

✧ *X_Vertical, Y_Vertical, Z_Vertical*

The factors to calculate the new acceleration in vertical direction. The formula to calculate the acceleration in Vertical direction Znew is **Znew = <X_Vertical> * X + <Y_Vertical> * Y + <Z_Vertical> * Z**.

Note

When <Self Calibration Status> is 0 or 1, no calibration factor of the acceleration data will be included in the **+RESP:GTSCS** message. When <Self Calibration Status> is 2, the calibration factors of the acceleration data will be included in the **+RESP:GTSCS** message.

### 4.4.19 ASV(Galileo Satellite Information)

After the device receives the command AT+GTRTO to get the Galileo satellite information, it will send the satellite information via the message **+RESP:GTASV** to the backend server.

Example:
+RESP:GTASV,8020040803,866314060873492,GV58LAU,5,13,34,15,43,23,29,26,42,27,20,20250314083734,00A7$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | ASV |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | SV Count | <=2 | 0 - 24 |
| | SV ID | <=3 | > =0 |
| | SV Power | <=2 | > =0 |
| | ... | | |
| | SV ID | <=3 | > =0 |
| | SV Power | <=2 | > =0 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

- *SV Count*
  The count of satellites found by Galileo.
- *SV ID*
  Satellite ID. If there is no satellite, the field is filled with zero.
- *SV Power*
  Satellite power. If there is no satellite, the field is filled with zero.

## 4.5 Event Reports

This section describes the message related to certain events. Please refer to the details below.

### 4.5.1 Generic Event Report

- PNA: Power-on Report
- PFA: Power-off Report
- PDP: GPRS Connection Establishment Report

Example:
+RESP:GTPNA,8020040803,866314060873492,GV58LAU,20250314083853,00B5$

+RESP:GTPFA,8020040803,866314060873492,GV58LAU,20250314083831,00B3$

+RESP:GTPDP,8020040803,866775051515021,GV58LAU,20221027065853,D524$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | PNA \| PFA \| PDP |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

### 4.5.2 MPN | MPF | BTC

- ✧ MPN(Main Power Connection)
  The report for connecting the main power supply.
- ✧ MPF(Main Power Disconnection)
  The report for disconnecting the main power supply.
- ✧ BTC(Battery Starts Charging)
  Report when backup battery starts charging.

Example:
+RESP:GTMPN,8020040803,866314060873492,GV58LAU,0,0.0,0,69.9,117.129138,31.838847,20250314085039,0460,0000,550B,0E9E30A5,0D,12,210100,0.00,0.00,0.00,20250314085103,0112$

+RESP:GTMPF,8020040803,866314060873492,GV58LAU,1,0.0,0,64.0,117.129299,31.838832,20250314085413,0460,0000,550B,0E9E30A5,0D,12,120000,1.24,1.03,1.61,20250314085414,012B$

+RESP:GTBTC,8020040803,866314060873492,GV58LAU,0,0.0,0,64.0,117.129299,31.838832,20250314085445,0460,0000,550B,0E9E30A5,0D,12,220100,0.00,0.00,0.00,20250314085447,0132$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | MPN \| MPF \| BTC |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX/XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

### 4.5.3 JDR(Network Jamming Indication Notification)

If the <Mode> in the AT+GTJDC command is set to 1, the device will report the **+RESP:GTJDR** message when jamming is detected.

Example:
+RESP:GTJDR,8020040803,866314060254412,GV58LAU,3,0,0.0,0,94.4,117.129868,31.838434,20231227055827,,,,,05,0,210100,20231227140205,04B5$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | JDR |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Jamming Net | 1 | 1-6 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Jamming Net*

The network mode.

- **1** - EGPRS/GSM
- **2** - LTE-Cat4
- **3** - WCDMA, EGPRS/GSM and LTE
- **4** - WCDMA
- **5** - WCDMA and EGPRS
- **6** - WCDMA & LTE & GSM/EGPRS (WCDMA First)

## 4.5.4 JDS(Network Jamming Indication Notification)

If the <Mode> in the AT+GTJDC command is set to 2, the device will report the **+RESP:GTJDS** message when jamming is detected.

Example:
'+RESP:GTJDS,8020040803,866314060000229,GV58LAU,1,3,0,0.0,193,-3.9,117.129708,31.838889,20231220083047,0460,0001,DF5C,027A4F1F,00,20231220083047,03FB$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | JDS |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Jamming Status | 1 | 1 \| 2 |
| | Jamming Net | 1 | 1-6 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Jamming Status*

The current jamming status of the device.

- **1** - Quit the jamming state.
- **2** - Enter the jamming state.

✧ *Jamming Net*

Please refer to the message **+RESP:GTJDR** for details.

Note

1. It may include information of only several neighbor cells or even no neighbor cell. If no neighbor cell is found, all the fields of the neighbor cell will be empty.
2. "ffff" in the fields <LAC(i)> and <Cell ID(i)> means the device does not know the value.
3. This message cannot be sent via SMS.

## 4.5.5 STC(Battery Stops Charging)

Report when backup battery stops charging.

Example:
+RESP:GTSTC,8020040803,866314060873492,GV58LAU,,1,0.0,0,64.0,117.129299,31.838832,20250314085538,0460,0000,550B,0E9E30A5,0D,12,220100,1.28,1.02,1.63,20250314085540,0138$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | STC |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 0 | |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

### 4.5.6 BPL(Backup Battery Low Alarm)

Example:
+RESP:GTBPL,8020040803,866314060268081,GV58LAU,3.65,1,0.0,168,49.4,117.129277,31.839021,20250320034629,0460,0000,550B,0E9E30A5,0D,9,120000,1.21,1.54,1.96,20250320034630,00D3$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | BPL |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Backup Battery Voltage | <=4 | 0.00 - 4.50 (V) |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

## 4.5.7 STT(Device Status Notification)

Report when the device motion status changes.

Example:
+RESP:GTSTT,8020040803,866314060873492,GV58LAU,22,1,0.0,0,64.0,117.129299,31.838832,20250314085415,0460,0000,550B,0E9E30A5,0D,12,220100,1.22,1.00,1.58,20250314085416,012D$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | STT |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Motion Status | 2 | 11 \| 12 \| 21 \| 22 \| 41 \| 42 \| 16 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Motion Status*

The current motion status of the device.

- **16 (Tow)** - The device attached vehicle is ignition off and it is towed.
- **11 (Ignition Off Rest)** - The device attached vehicle is ignition off and it is motionless.
- **12 (Ignition Off Motion)** - The device attached vehicle is ignition off and it is moving before it is considered as being towed.
- **21 (Ignition On Rest)** - The device attached vehicle is ignition on and it is motionless.
- **22 (Ignition On Motion)** - The device attached vehicle is ignition on and it is moving.
- **41 (Sensor Rest)** - The device attached vehicle is motionless without ignition signal detected.
- **42 (Sensor Motion)** - The device attached vehicle is moving without ignition signal detected.

### 4.5.8 IGN(Ignition-on Report)

Example:
+RESP:GTIGN,8020040803,866314060873492,GV58LAU,1,0,0.0,0,64.0,117.129299,31.838832,20250314085617,0460,0000,550B,0E9E30A5,0D,12,220100,0.00,0.00,0.00,,0.0,20250314085618,013F$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | IGN |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Duration of Ignition Off | <=6 | 0 - 999999 (s) |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Hour Meter Count | 13 | 0000000:00:00 - 1193000:00:00 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Duration of Ignition Off*

The duration since last time the ignition is turned off. If it is greater than 999999 seconds, it will be reported as 999999 seconds.

✧ *Hour Meter Count*

If the hour meter count function is enabled by the command AT+GTHMC,total hours the meter counts when the engine is on will be reported in this field. It is formatted with 5 to 7 hour digits, 2 minute digits and 2 second digits. If the function is disabled, this field will be empty.

## 4.5.9 VGN(Virtual Ignition-on Report)

| |
|---|
| Example:<br>+RESP:GTVGN,8020040803,866314060873492,GV58LAU,00,2,73,1,0.0,0,60.5,117.129321,31.839186,20250317025351,0460,0000,550B,0E9E30A5,0D,12,210000,0.83,1.37,1.61,,0.0,20250317025353,0023$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | VGN |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 2 | 00 |
| | Report Type | 1 | 2 \| 4 \| 7 |
| | Duration of Ignition Off | <=6 | 0 - 999999(s) |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Hour Meter Count | 13 | 0000000:00:00 - 1193000:00:00 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Report Type*

This parameter indicates the trigger source of the ignition event.

- **2** - External power voltage mode (virtual ignition detection)
- **4** - Accelerometer mode (virtual ignition detection)
- **7** - Combined detection mode. In this mode, ignition on/off trigger conditions can be selected using parameters <Virtual Ignition On Mask> and <Virtual Ignition Off Mask> in the command AT+GTVMS.

Note

<Virtual Ignition off Mask> must contain <Virtual Ignition On Mask> to prevent logic errors.

✧ *Duration of Ignition Off*

Duration since last time the ignition is off. If it is greater than 999999 seconds, it is reported as 999999 seconds.

✧ *Hour Meter Count*

If the hour meter count function is enabled by the command AT+GTHMC, total hours the meter has counted when the engine is on will be reported in this field. It is formatted with 5 to 7 hour digits, 2 minute digits and 2 second digits. If the function is disabled, this field will be empty.

## 4.5.10 IGF(Ignition-off Report)

| |
|---|
| Example:<br>+RESP:GTIGF,8020040803,866314060873492,GV58LAU,121,1,0.0,0,64.0,117.129299,31.838832,20250314085616,0460,0000,550B,0E9E30A5,0D,12,120000,1.29,1.03,1.65,,0.0,20250314085617,013C$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | IGF |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Duration of Ignition On | <=6 | 0 - 999999 (s) |
| | GNSS Accuracy | <=2 | 0 - 50 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Hour Meter Count | 13 | 0000000:00:00 - 1193000:00:00 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Duration of Ignition On*

The duration since last time the ignition is turned on. If it is greater than 999999 seconds, it will be reported as 999999 seconds.

✧ *Hour Meter Count*

If the hour meter count function is enabled by the command AT+GTHMC, total hours the meter counts when engine is on will be reported in this field. It is formatted with 5 to 7 hour digits, 2 minute digits and 2 second digits. If the function is disabled, this field will be empty.

### 4.5.11 VGF(Virtual Ignition-off Report)

Example:
+RESP:GTVGF,8020040803,866314060873492,GV58LAU,00,2,237,1,0.0,0,60.5,117.129321,31.839186,20250317025749,0460,0000,550B,0E9E30A5,0D,12,110000,0.83,1.05,1.34,,0.0,20250317025750,0035$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | VGF |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 2 | 00 |
| | Report Type | 1 | 2 \| 4 \| 7 |
| | Duration of Ignition On | <=6 | 0 - 999999(s) |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Hour Meter Count | 13 | 0000000:00:00 - 1193000:00:00 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Duration of Ignition On*

Duration since last time the ignition is turned on. If it is greater than 999999 seconds, it is reported as 999999 seconds.

✧ *Hour Meter Count*

If the hour meter count function is enabled by the command AT+GTHMC, total hours the meter has counted when the engine is on will be reported in this field. It is formatted with 5 to 7 hour digits, 2 minute digits and 2 second digits. If the function is disabled, this field will be empty.

## 4.5.12 IDF(Exit Idle State)

Report when the vehicle exits the idle state.

Example:
+RESP:GTIDF,8020040803,866314060873492,GV58LAU,22,64,1,0.0,0,60.5,117.129321,31.839186,20250317031507,0460,0000,550B,0E9E30A5,0D,12,220100,0.85,1.21,1.48,0.0,20250317031509,005F$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | IDF |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Motion Status | 2 | 11 \| 12 \| 16 \| 22 |
| | Duration of Idling Status | <=6 | 0 - 999999 (s) |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

- *Motion Status*

  The current motion status when the vehicle exits the idle state.
- *Duration of Idling Status*

  The period of time during which the vehicle stays in idle state. If it is greater than 999999 seconds, it will be reported as 999999 seconds.

### 4.5.13 GSS(GNSS Signal Status)

Example:
+RESP:GTGSS,8020040803,866314060268081,GV58LAU-A07,1,8,11,,1,0.0,359,61.3,117.128720,31.852358,20250320022748,0460,0000,550B,0E9E30A5,0D,8,110001,1.77,2.11,2.76,20250312163948,03B5$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | GSS |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | GNSS Signal Status | 1 | 0 \| 1 |
| | Satellite Number | <=2 | 0 - 72 |
| | Motion Status | 2 | 11 \| 12 \| 21 \| 22 \| 41 \| 42 \| 16 |
| | Reserved | 0 | |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *GNSS Signal Status*

- **0** - GNSS signal is lost or GNSS fix fails
- **1** - GNSS signal is recovered and GNSS fix is successful

✧ *Satellites in Use*

The number of satellites used for tracking, the high nibble is reserved and the low nibble is valid.

✧ *Motion Status*

The current motion status of the device.

- **16 (Tow)** - The device attached vehicle is ignition off and it is towed.
- **1A (Fake Tow)** - The device attached vehicle is ignition off and it might be towed.
- **11 (Ignition Off Rest)** - The device attached vehicle is ignition off and it is motionless.
- **12 (Ignition Off Motion)** - The device attached vehicle is ignition off and it is moving before it is considered as being towed.
- **21 (Ignition On Rest)** - The device attached vehicle is ignition on and it is motionless.
- **22 (Ignition On Motion)** - The device attached vehicle is ignition on and it is moving.
- **41 (Sensor Rest)** - The device attached vehicle is motionless without ignition signal detected.
- **42 (Sensor Motion)** - The device attached vehicle is moving without ignition signal detected.

## 4.5.14 STR | STP | LSP | IDN

✧ STR: Enter Start State

✧ STP: Enter Stop State

- LSP: Enter Long Stop State
- IDN: Enter Idle State

Example:
+RESP:GTSTR,8020040803,866314060268081,GV58LAU,,,1,53.7,78,60.2,117.202264,31.857749,20250320025739,0460,0000,550B,0E9E30A5,0D,7,220100,1.57,2.25,2.74,0.9,20250320025739,001A$

+RESP:GTSTP,8020040803,866314060268081,GV58LAU,,,1,0.0,79,126.8,117.129264,31.839369,20250320030335,0460,0000,550B,0E9E30A5,0D,6,210100,1.22,1.77,2.14,4.7,20250320030336,0028$

+RESP:GTLSP,8020040803,866314060268081,GV58LAU,,,1,0.0,79,126.8,117.129264,31.839369,20250320030435,0460,0000,550B,0E9E30A5,0D,6,220100,1.22,1.78,2.16,4.7,20250320030436,002C$

+RESP:GTIDN,8020040803,866314060873492,GV58LAU,,,1,0.0,0,60.5,117.129321,31.839186,20250317031404,0460,0000,550B,0E9E30A5,0D,12,210100,0.86,1.37,1.62,0.0,20250317031405,005A$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | STR \| STP \| LSP \| IDN |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 0 | |
| | Reserved | 0 | |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

## 4.5.15 DOS(Wave Shape 1 Output Status)

Example:
+RESP:GTDOS,8020040803,866314060873492,GV58LAU,1,1,1,0.0,0,60.5,117.129321,31.839186,20250317031632,0460,0000,550B,0E9E30A5,0D,12,210101,0.86,1.22,1.50,20250317031633,0067$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | DOS |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Wave1 Output ID | 1 | 1 - 3 \| 7 - 8 |
| | Wave1 Output Active | 1 | 0 - 2 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

- *Wave1 Output ID*

  The ID of the wave shape 1 output.
- *Wave1 Output Active*

  The status of wave shape 1 output. If it is 2, it means that it is currently in the status of gradual wave shape 1 .

### 4.5.16 RMD(Roaming State Report)

If the GSM roaming state of the device changes, the current roaming state will be reported in the **+RESP:GTRMD** message.

Example:
+RESP:GTRMD,8020040900,862170011507322,GV58LAU,1,0,0.0,0,83.9,117.201281,31.833017,20230917071326,0460,0000,5678,2079,05,1,220100,20230917071330,00A4$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | RMD |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Roaming State | 1 | 0 - 3 |
| | GNSS Accuracy | <=2 | 0 - 50 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Roaming State*

A numeral to indicate the roaming state.

- **0** - Home
- **1** - Known roaming
- **2** - Unknown roaming
- **3** - Blocking report

### 4.5.17 CLT(CANBUS Information Alarm)

If the CANBUS Alarm report function is enabled by the command AT+GTCLT, <Alarm Mask 1>, <Alarm Mask 2> and <Alarm Mask 3> meet each trigger condition at the same time, and the trigger event duration time is longer than <Debounce Time>, the **+RESP:GTCLT** alarm message will be sent.

Example:
+RESP:GTCLT,8020040803,866314060873492,GV58LAU,0,00000000,00000000,00000004,,,203FF
FFF,,2,H779133,50.56,20,40,90,,L87.40,3960,32,2.14,1.17,0.97,7436.45,4365,BF,7FFF,3F,1F,3.56,

1.11,FFFFFF,,,,,,,,,,,,,,,,,,,,,0000,621,1324,,,,1,0.0,0,60.5,117.129321,31.839186,20250317032238,0460,0000,550B,0E9E30A5,0D,12,210100,0.97,1.31,1.63,20250317032238,008C$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | CLT |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Group ID | <=2 | 0 - 19 |
| | Alarm Mask 1 | <=8 | 0 - FFFFFFFF |
| | Alarm Mask 2 | <=8 | 0 - FFFFFFFF |
| | Alarm Mask 3 | <=8 | 0 - FFFFFFFF |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | CANBUS Data Mask | <=8 | 0 - FFFFFFFF |
| | VIN | 17 | '0' - '9' 'A' - 'Z' except 'I', 'O', 'Q' |
| | Ignition Key | 1 | 0-2 |
| | Total Distance | <=12 | H(0 - 99999999)/I(I0 - 2147483647) |
| | Total Fuel Used | <=9 | 0.00 - 999999.99(L) |
| | Engine RPM | <=5 | 0 - 16383(rpm) |
| | Vehicle Speed | <=4 | 0 - 6553(km/h) |
| | Engine Coolant Temperature | <=4 | -40 - +215 ( ℃ ) |
| | Fuel Consumption | <=6 | L/100km(0.0 - 999.9)\|L/H(0.0 - 999.9) |
| | Fuel Level | <=8 | L(0.00 - 9999.99)/P(0.00 - 100.00) |
| | Range | <=8 | 0 - 99999999 |
| | Accelerator Pedal Pressure | <=3 | 0 - 100(%) |
| | Total Engine Hours | <=10 | 0.00 - 9999999.99(h) |
| | Total Driving Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Engine Idle Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Idle Fuel Used | <=9 | 0.00 - 999999.99(L) |
| | Axle Weight 2nd | <=5 | 0 - 65535(kg) |
| | Tachograph Information | <=4 | 00 - FFFF |
| | Detailed Information\|Indicators | <=4 | 00 - FFFF |
| | Lights | <=2 | 0 - FF |
| | Doors | <=2 | 0 - FF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Total Vehicle Overspeed Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Vehicle Engine Overspeed Time | <=10 | 0.00 - 9999999.99(h) |
| | CAN Report Expansion Mask | <=8 | 00000000 - FFFFFFFF |
| | Ad-Blue Level | <=8 | L(0.00 - 9999.99)/P(0.00 - 100.00) |
| | Axle Weight 1st | <=5 | 0 - 65535(kg) |
| | Axle Weight 3rd | <=5 | 0 - 65535(kg) |
| | Axle Weight 4th | <=5 | 0 - 65535(kg) |
| | Tachograph Overspeed Signal | 1 | 0 \| 1 |
| | Tachograph Vehicle Motion Signal | 1 | 0 \| 1 |
| | Tachograph Driving Direction | 1 | 0 \| 1 |
| | Analog Input Value | <=5 | 0 - 99999(mV) |
| | Engine Braking Factor | <=10 | 0 - 999999 |
| | Pedal Braking Factor | <=10 | 0 - 999999 |
| | Total Accelerator Kick-downs | <=6 | 0 - 999999 |
| | Total Effective Engine Speed Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Cruise Control Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Accelerator Kick-down Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Brake Applications | <=6 | 0 - 999999 |
| | Tachograph Driver 1 Card Number | <=40 | ASCII |
| | Tachograph Driver 2 Card Number | <=40 | ASCII |
| | Tachograph Driver 1 Name | <=40 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Tachograph Driver 2 Name | <=40 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Registration Number | <=40 | ASCII |
| | Expansion Information | <=4 | 0000 - FFFF |
| | Rapid Brakings | <=8 | 0 - 16711679 |
| | Rapid Accelerations | <=8 | 0 - 16711679 |
| | Engine Torque | <=3 | 0 - 100(%) |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

- *Alarm Mask 1*

  The alarm mask is configured in a bitwise manner. The alarm mask information is based on <Detailed Information / Indicators> of the **+RESP:GTCAN** message.
- *Alarm Mask 2*

  The alarm mask is configured in a bitwise manner. The alarm mask information is based on <Lights> and <Doors> of the **+RESP:GTCAN** message.
- *Alarm Mask 3*

  The alarm mask is configured in a bitwise manner. The alarm mask information is based on <Engine RPM> of the **+RESP:GTCAN** message.

## 4.5.18 CFU(CAN100 FOTA Upgrade Report)

The device will send the message **+RESP:GTCFU** to the backend server during the upgrade process.

| |
|---|
| Example:<br>+RESP:GTCFU,8020040803,866314060873492,GV58LAU,104,,20250317032357,0097$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | CFU |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Code | 3 | |
| | New Version (Optional) | <=10 | '0' - '9' 'a' - 'z' |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Code*

Information code.

- **1x0** - Confirm ok. Start to upgrade.
- **1x1** - Baud rate error. UART does not work for CAN100. Stop upgrade.
- **1x2** - The download process is refused because of low power or an incorrect URL.
- **1x3** - Other upgrade commands are being executed. Stop upgrade.
- **1x4** - CAN chipset response is abnormal.
- **1x5** - The backend server cancels this update process.
- **2x0** - Start to download package.
- **2x1** - Package download succeeds.
- **2x2** - Package download fails.
- **3x0** - Start to upgrade.
- **3x1** - Upgrade succeeds. The reserved parameter is used as follows.

| Fields | Length | Range/Format |
|---|---|---|
| New Version | <=10 | '0' - '9' 'a' - 'z' |

- **3x2** - Upgrade fails.
- **3x3** - Serial Number does not match, firmware upgrade fails.
- **3x4** - The upgrade process is refused because of low power.

Note

**x** here means <Update Type> defined in the command AT+GTCFU.

✧ *New Version*

The version of the new firmware in the CAN100.

### 4.5.19 SVR(Stolen Vehicle Recovery)

Stolen vehicle recovery message reported by the Primary device.

Example:
+RESP:GTSVR,8020040803,866314060268081,GV58LAU,2,780541494A36,300000000000000000
0000,0000,1,0.0,79,42.2,117.129204,31.839002,20250320030700,0460,0000,550B,0E9E30A5,09,
7,1.08,1.50,1.85,20250320030700,0032$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | SVR |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | SVR Working State | 1 | 0-2 |
| | Ghost MAC Address | 12 | 000000000000 - FFFFFFFFFFFF |
| | SVR Appending Information | <=30 | |
| | Target State | <=4 | 0 \| 1 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *SVR Working State*

It indicates the status in which the **+RESP:GTSVR** alert message is generated.

- **0** - Primary loses connection with Ghost.
- **1** - Primary regains connection with Ghost normally.
- **2** - Primary fails to match Ghost during installation phase.

✧ *SVR Appending Information*

It indicates additional Bluetooth information.

✧ *Target State*

A numeral to indicate the status of the target device.

- **0** - Normal.
- **1** - GSM jamming.

Note

If <Mode> is set to 1 in AT+GTSVR, the <Target State> will not be reported.

## 4.5.20 RTP(Remote File Transfer)

Report file transfer information.

Example:
+RESP:GTRTP,8020040803,866314060873492,GV58LAU,0,0,0,301,20250317032807,00B7$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | RTP |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Mode | 1 | 0 |
| | Protocol Type | 1 | 0 |
| | File Type | 1 | 0-2 |
| | Code | 3 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Mode*

The working mode of file transfer.

- **0** - Download file

✧ *Protocol Type*

The type of communication protocol used to obtain data from the backend server.

- **0** - HTTP

✧ *File Type*

It defines the type of file to download from the server.

- **0** - CA certificate
- **1** - Client certificate
- **2** - Client key

✧ *Code*

It indicates the download information.

- **100** - The update command is starting.
- **101** - The update command is refused by the device.
- **200** - The device starts to download the file.
- **201** - The device finishes downloading the file successfully.
- **202** - The device fails to download the file.
- **301** - The device finishes updating the file successfully.
- **302** - The device fails to update the file.

## 4.5.21 PNR(Power-on Reason)

This message indicates the reason for power on.

Example:
+RESP:GTPNR,8020040803,866314060873492,GV58LAU,2,,,,,20250317032942,00C9$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | PNR |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Power On Reason | 1 | 0-3 \| 6 |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Power On Reason*

It indicates the reason for power on.

- **0** - Normal power on
- **1** - FOTA reboot
- **2** - RTO reboot
- **3** - Watchdog reboot
- **6** - Configuration upgrade reboot

### 4.5.22 PFR(Power-off Reason)

This message indicates the reason for power off.

| |
|---|
| Example:<br>+RESP:GTPFR,8020040803,866314060873492,GV58LAU,2,,,,,20250317032921,00C7$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | PFR |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Power Off Reason | 1 | 0-3 |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Power Off Reason*

It indicates the reason for power off.

- **0** - RTO power off
- **1** - Low battery voltage
- **2** - RTO reboot
- **3** - Watchdog reboot

## 4.5.23 CRA(Crash Alarm)

Example:
+RESP:GTCRA,8020040803,866314060873492,GV58LAU,00,1,0.0,0,73.7,117.129100,31.839234,20250317033103,0460,0000,550B,0E9E30A5,0D,12,220101,0.92,1.16,1.48,20250317033104,00D6$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | CRA |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Crash Counter | 2 | 00 - FF |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | HEX |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Crash Counter*

It indicates the crash sequence, combining the report of **+RESP:GTCRA** and **+RESP:GTCRD** into one crash event. It rolls from 0x00 to 0xFF.

### 4.5.24 BCS(Bluetooth Connection Report)

| |
|---|
| Example:<br>+RESP:GTBCS,8020040803,866314060873492,GV58LAU,,1,0.0,0,73.7,117.129100,31.839234,20250317033018,0460,0000,550B,0E9E30A5,0D,12,220101,0.90,0.94,1.30,0D03,GV58LAU%IMEI,60E93B2073B4,1,1,EC543EE65330,,,,,20250317033019,00D2$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | BCS |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 0 | |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Bluetooth Report Mask | 4 | 0000 - FFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Bluetooth Name | <=20 | ASCII |
| | Bluetooth Mac Address | 12 | 000000000000 - FFFFFFFFFFFF |
| | Peer Role | 1 | 0 \| 1 |
| | Peer Address Type | 1 | 0 \| 1 |
| | Peer MAC Address | 12 | 000000000000 - FFFFFFFFFFFF |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Peer Role*

The role type of the peripheral device.

- **0** - Master
- **1** - Slave

✧ *Peer Address Type*

The address type of the peripheral device.

- **0** - Public Device Address or Public Identity Address
- **1** - Random Device Address or Random (static) Identity

## 4.5.25 BDS(Bluetooth Disconnection Report)

| |
|---|
| Example:<br>+RESP:GTBDS,8020040803,866314060873492,GV58LAU,,1,0.0,0,73.7,117.129100,31.839234,20250317033239,0460,0000,550B,0E9E30A5,0D,12,220101,0.86,0.93,1.27,0D03,GV58LAU%IMEI,60E93B2073B4,1,1,EC543EE65330,0,,,,20250317033240,00E4$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | BDS |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 0 | |
| | GNSS Accuracy | <=2 | 0 - 50 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Bluetooth Report Mask | 4 | 0000 - FFFF |
| | Bluetooth Name | <=20 | ASCII |
| | Bluetooth Mac Address | 12 | 000000000000 - FFFFFFFFFFFF |
| | Peer Role | 1 | 0 \| 1 |
| | Peer Address Type | 1 | 0 \| 1 |
| | Peer MAC Address | 12 | 000000000000 - FFFFFFFFFFFF |
| | Reason | 1 | 0 \| 4 |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Peer Role*

The role type of the peripheral device.

- **0** - Master
- **1** - Slave

✧ *Peer Address Type*

The address type of the peripheral device.

- **0** - Public Device Address or Public Identity Address

- **1** - Random Device Address or Random (static) Identity

✧ *Reason*

This parameter indicates the reason of Bluetooth disconnection.

- **0x00** - Normal disconnection.
- **0x04** - Bluetooth peripheral device pairing fails.

### 4.5.26 ASC(Acceleration Calibration Alarm)

The report for calibration data.

Example:
+RESP:GTASC,8020040803,866314060209648,GV58LAU-1,-0.99,0.14,-0.01,0.14,0.99,0.05,0.02,0.05,-1.00,1,26.4,269,51.0,117.078717,31.817927,20250221071806,0460,0000,550B,0F48332C,0D,12,220100,0.91,2.28,2.45,20250221071807,45B9$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | ASC |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | X_Forward | <=5 | -1.00 - 1.00 |
| | Y_Forward | <=5 | -1.00 - 1.00 |
| | Z_Forward | <=5 | -1.00 - 1.00 |
| | X_Side | <=5 | -1.00 - 1.00 |
| | Y_Side | <=5 | -1.00 - 1.00 |
| | Z_Side | <=5 | -1.00 - 1.00 |
| | X_Vertical | <=5 | -1.00 - 1.00 |
| | Y_Vertical | <=5 | -1.00 - 1.00 |
| | Z_Vertical | <=5 | -1.00 - 1.00 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

- *X_Forward, Y_Forward, Z_Forward*
  The factors to calculate the new acceleration in forward direction.
  The formula to calculate the acceleration in Forward direction Xnew is **Xnew = <X_Forward> * X + <Y_Forward> * Y + <Z_Forward> * Z**.
- *X_Side, Y_Side, Z_Side*
  The factors to calculate the new acceleration in side direction.
  The formula to calculate the acceleration in Side direction Ynew is **Ynew = <X_Side> * X + <Y_Side> * Y + <Z_Side> * Z**.
- *X_Vertical, Y_Vertical, Z_Vertical*
  The factors to calculate the new acceleration in vertical direction.
  The formula to calculate the acceleration in Vertical direction Znew is **Znew = <X_Vertical> * X + <Y_Vertical> * Y + <Z_Vertical> * Z**.

## 4.5.27 HBE(Acceleration Information for HBM)

If harsh behavior is over in Mode 5 of the AT+GTHBM command, this message will be sent to the backend server.

Example:
+RESP:GTHBE,8020040803,866314060835525,GV58LAU-2,,2,1,1,20.2,344,43.8,117.078777,31.815984,20250221092032,0460,0000,550B,045E57EE,0D,12,220100,0.87,1.02,1.34,000F000B0056,000AFFFE0050,194,40.3,20250221172033,71E3$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Message Name | 3 | HBE |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 0 | |
| | Self Calibration Status | 1 | 0-2 |
| | Harsh Behavior Type | 1 | 0-4 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | 0000 - FFFF\|00000000 - FFFFFFFF |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Max Acceleration Data | 12 | '0'-'9' 'a'-'f' |
| | Average Acceleration Data | 12 | '0'-'9' 'a'-'f' |
| | Harsh Behavior Duration | <=6 | 0 - 999999(x10ms) |
| | Mileage | <=9 | 0.0 - 4294967.0 (km) |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Max Acceleration Data*

A string made up of 12 characters. It is a set of the maximal values of each axis collected during the occurrence of the harsh driving behavior.

The first 4 characters of these 12 characters represent X axis acceleration data, the middle 4 characters represent Y axis acceleration data and the last 4 characters are for Z axis acceleration data.
The ASCII "0001" indicates HEX value 0x0001, so it means the acceleration is 1. The ASCII "fffd" indicates HEX value 0xFFFD which is the compliment of -3, so it means the acceleration is -3. This value is only valid in Mode 5 of the AT+GTHBM command.

✧ *Average Acceleration Data*

There are 12 characters in a group of acceleration data. It is the average value of acceleration data which triggers this harsh behavior report.
The first 4 characters of these 12 characters represent X axis acceleration data, the middle 4 characters represent Y axis acceleration data and the last 4 characters are for Z axis acceleration data.
The ASCII "0001" indicates HEX value 0x0001, so it means the acceleration is 1. The ASCII "fffd" indicates HEX value 0xFFFD which is the compliment of -3, so it means the acceleration is -3. This value is only valid in Mode 5 of the AT+GTHBM command.

✧ *Harsh Behavior Duration*

The duration of the harsh behavior event. This value is only valid in Mode 5 of the AT+GTHBM command.

✧ *Self Calibration Status*

The status of the self-calibration for Acceleration Data.

- **0** - Self-calibration is disabled.
- **1** - Self-calibration is not done.
- **2** - Self-calibration is successful.

✧ *Harsh Behavior Type*

The type of the harsh behavior.

- **0** - Harsh braking behavior.
- **1** - Harsh acceleration behavior.
- **2** - Harsh cornering behavior.
- **3** - Harsh braking and cornering behavior.
- **4** - Harsh acceleration and cornering behavior.

## 4.5.28 DOM(Waveform Occurs)

The **+RESP:GTDOM** message will be reported when that waveform occurs.

Example:
+RESP:GTDOM,8020040803,866314060268081,GV58LAU,5,1,,1,0.0,258,69.4,117.129069,31.839260,20250320083252,0460,0000,550B,0E9E30A5,0D,12,220100,0.68,0.89,1.12,20250320083254,0486$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Message Name | 3 | DOM |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Wave Shape | 1 | 5 |
| | Wave Output ID | 1 | 1 - 3 |
| | Reserved | 0 | |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Wave Shape*

It represents the type of waveform that occurs.

✧ *Wave Output ID*

The ID of the wave shape output.

## 4.5.29 BAA(Bluetooth Accessory Alarm)

Example:
+RESP:GTBAA,8020040803,866314060268081,GV58LAU,00,1,3,00,011D,DU_100361,1,3500,0,9,0,0,59.9,89,29.4,117.169402,31.853243,20250312073240,0460,0000,550B,0E9E30A5,0D,9,210101,0.00,0.00,0.00,20250312164241,04AA$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | BAA |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Index | <=2 | (HEX) 0 - 9 \| 0xFF \| 0XFE |
| | Accessory Type | <=2 | 0-3 \| 6 \| 11-13 |
| | Accessory Model\|Beacon ID Model | <=2 | 0-5 \| 10 |
| | Alarm Type | 2 | 0-4 \| 7-A \| C-15 |
| | Append Mask | 4\|8 | 0000 - FFFF \| 00000000 - FFFFFFFF |
| | Accessory Name(Optional) | <=20 | '0' - '9', 'a' - 'z', 'A' - 'Z', '-', '_', ' ' |
| | Accessory MAC(Optional) | 12 | '0' - '9' 'A' - 'F' |
| | Accessory Status(Optional) | 1 | 0 \| 1 |
| | Accessory Battery Voltage(Optional) | <=4 | 0 - 5000(mV) |
| | Accessory Temperature(Optional) | <=3 | -40 - 80( ℃ ) |
| | Accessory Humidity(Optional) | <=3 | 0 - 100%(rh) |
| | Accessory Mode(Optional) | <=2 | 0 - 10 |
| | Accessory Event(Optional) | 1 | 0 - 2 |
| | Tire Pressure(Optional) | <=3 | 0 - 500(kPa) |
| | Timestamp(Optional) | 14 | YYYYMMDDHHMMSS |
| | Enhanced Temperature(Optional) | <=5 | -40.00 - 80.00( ℃ ) |
| | Magnet ID(Optional) | 2 | 00 - FF |
| | MAG Event Counter(Optional) | <=5 | 0 - 32767 |
| | Magnet State(Optional) | 1 | 0 - 1 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Accessory Battery Percentage(Optional) | <=3 | 0 - 100(%) |
| | Relay Config Result(Optional) | 1 | 0 - 4 |
| | Relay State(Optional) | 1 | 0 - 1 |
| | CarBLE Conn Status(Optional) | 1 | 0 \| 1 |
| | CarBLE Longitude(Optional) | <=11 | (-)xxx.xxxxxx |
| | CarBLE Latitude(Optional) | <=10 | (-)xx.xxxxxx |
| | CarBLE GNSS UTC Time (Optional) | 14 | YYYYMMDDHHMMSS |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Send Time | 14 | YYYYMMDDHHMMSS |
| Tail | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Index*

The index of the Bluetooth accessory.

- The index of the Bluetooth accessory defined in AT+GTBAS which triggers the **+RESP:GTBAA** message.

For WKF300, it is 0xFF. For other Beacon, it is 0xFE.

✧ *Accessory Type*

The type of the Bluetooth accessory which is defined in the <Index>.

- **0** - No Bluetooth accessory
- **1** - Escort sensor
- **2** - Beacon temperature sensor
- **3** - Bluetooth beacon accessory
- **6** - Beacon Multi-functional sensor
- **11** - Magnet sensor
- **12** - BLE TPMS sensor
- **13** - Relay sensor

✧ *Accessory Model / Beacon ID Model*

The model of the Bluetooth accessory which is defined in AT+GTBAS or the model of the Bluetooth beacon accessory which is defined in AT+GTBID.

✧ *Alarm Type*

The type of the alarm which is generated by the Bluetooth accessory specified by <Accessory Type> and <Accessory Model> in the command AT+GTBAS.

- **0** - The voltage of the Bluetooth accessory is low.
- **1** - Temperature alarm: The current temperature value is lower than <Low Temperature> set in the command **AT+GTBAS**.
- **2** - Temperature alarm: The current temperature value is higher than <High Temperature> set in the command **AT+GTBAS**.
- **3** - Temperature alarm: The current temperature value is within the range defined by <Low Temperature> and <High Temperature> set in the command **AT+GTBAS**.
- **4** - Push button event for WKF300 is detected.
- **7** - Humidity alarm: The current humidity value is lower than <Low Humidity> set in the command **AT+GTBAS**.
- **8** - Humidity alarm: The current humidity value is higher than <High Humidity> set in the command **AT+GTBAS**.
- **9** - Humidity alarm: The current temperature value is within the range defined by <Low Humidity> and <High Humidity> set in the command **AT+GTBAS**.
- **0A** - Angle event notification.
- **0C** - Magnet event notification.
- **0D** - Beacon event notification.
- **0E** - Tire pressure alarm: The current Tire pressure value is lower than <Low Tire pressure> set in the **AT+GTBAS** command.
- **0F** - Tire pressure alarm: The current Tire pressure value is higher than <High Tire pressure> set in the **AT+GTBAS** command.
- **10** - Tire pressure alarm: The current Tire pressure value is within the range defined by <Low Tire pressure> and <High Tire pressure> set in the command **AT+GTBAS**.
- **11** - No available Bluetooth accessory is detected.
- **12** - An available Bluetooth accessory is detected.
- **13** - Door open.
- **14** - Door closed.
- **15** - Relay event notification.

✧ *Append Mask*

Bitwise mask defined in the command AT+GTBAS or AT+GTBID to indicate the reported

Bluetooth accessory data fields.

- **Bit 0** - <Accessory Name>
- **Bit 1** - <Accessory MAC>
- **Bit 2** - <Accessory Status>
- **Bit 3** - <Accessory Battery Voltage>
- **Bit 4** - <Accessory Temperature>
- **Bit 5** - <Accessory Humidity>
- **Bit 7** - <Accessory Input Output Data>, including <Accessory Output Status>, <Accessory Digital Input Status>, <Accessory Analog Input Voltage>
- **Bit 8** - <Accessory Event Notification Data>, including <Accessory Mode>, <Accessory Event>
- **Bit 9** - <Tire Pressure>
- **Bit 10** - <Time stamp>
- **Bit 11** - <Enhanced Temperature>
- **Bit 12** - <MAG Notification Data>, including <Magnet ID>, <MAG Event Counter>, <Magnet State>
- **Bit 13** - <Accessory Battery Percentage>
- **Bit 14** - <Relay Data>, including <Relay Config Result>, <Relay State>
- **Bit 15** - <Expansion Mask>, if Bit 15 is set to 1, Bit 16 - Bit 31 will be valid.
- **Bit 22** - <CarBLE Information>, including <CarBLE Conn status>, <CarBLE Longitude>, <CarBLE Latitude>, <CarBLE GNSS UTC Time>

Note

In the message bit 0 - bit 15 precedes bit 16 - bit 31. Here is an example: <Accessory Append Mask> = 0x081F0007, 0x081F indicates bit 15 - bit 0, 0x0007 indicates bit 31 - bit 16.

✧ *Accessory Name*

The name of the Bluetooth accessory.

✧ *Accessory MAC*

The MAC address of the Bluetooth accessory.

✧ *Accessory Status*

A numeral to indicate whether the accessory is available.

- **0** - The accessory is not available.
- **1** - The accessory is available.

✧ *Accessory Battery Voltage*

The battery voltage of the Bluetooth accessory.

✧ *Accessory Temperature*

Temperature data for the Bluetooth accessory.

✧ *Enhanced Temperature*

High-precision temperature data of Bluetooth accessories.

✧ *Accessory Humidity*

Humidity data for the Bluetooth accessory.

✧ *Accessory Mode*

The operating mode of angle sensor.

✧ *Accessory Event*

The event is generated by the angle sensor.

✧ *Timestamp*

Timestamp of the tire pressure value collection.

- *Magnet ID*

  ID corresponding to different magnet sensors.
- *MAG Event Counter*

  The total number of times detected by the magnet sensor.
- *Magnet State*

  The state of the magnet sensor's two parts.
  - **0** - Separate
  - **1** - Closed
- *Accessory Battery Percentage*

  Percentage of Bluetooth accessory's battery power.
- *Relay Config Result*

  The number representing the response result of the relay, which is controlled and reported by Bit14 of the AT+GTBAS parameter <Accessory Append Mask>.
  - **0** - Configuration updated successfully.
  - **1** - Error in connecting.
  - **2** - The current password is incorrect.
  - **3** - Password update error.
  - **4** - Relay open or close error.
- *Relay State*

  The current state of the WRL300 sensor.
- *CarBLE Conn Status*

  CarBLE connection status.
  - **0** - Lost status.
  - **1** - Connection status.
- *CarBLE Longitude*

  Longitude of CarBLE devices.
- *CarBLE Latitude*

  Latitude of CarBLE devices
- *CarBLE GNSS UTC Time*

  CarBLE device latitude and longitude update time.

## 4.5.30 BID(Bluetooth Beacon ID Report)

| |
|---|
| Example:<br>+RESP:GTBID,8020040803,866314060268081,GV58LAU,6,1,0042,AC233F22863E,-75,1,0042,AC233F228481,-74,1,0042,AC233F22A3BE,-76,1,0042,AC233F22A3B6,-82,1,0042,AC233F228645,-82,1,0042,AC233F229B82,-89,1,0.0,79,114.1,117.129227,31.839377,20250320031242,0460,0000,550B,0E9E30A5,0D,9,220100,1.07,1.33,1.71,20250320031243,004B$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Message Name | 3 | BID |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Number | <=2 | 0 - 3 \| 0 - 15 |
| | Beacon ID Model | <=2 | 0-2 \| 4 \| 5 \| 10 |
| | Accessory Append Mask | 4 | 0000 - FFFF |
| | Accessory MAC (Optional) | 12 | '0' - '9' 'A' - 'F' |
| | Accessory Battery Level (Optional) | <=4 | 0 - 5000(mV) |
| | Accessory Signal Strength (Optional) | <=4 | -120 - 0 |
| | Beacon Type (Optional) | 1 | 0-2 |
| | Beacon Data (Optional) | <=100 | |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Number*

The number of the Bluetooth beacon accessories.

- **WKF300** - The maximum value is 3.
- **iBeacon E6** - The maximum value is 15.
- **IDELA** - The maximum value is 15.
- **WID310** - The maximum value is 15.
- **CarBLE** - The maximum value is 15.
- **WID330** - The maximum value is 15.

✧ *Beacon ID Model*

The model of the Bluetooth beacon accessory which is defined in AT+GTBID.

✧ *Accessory Append Mask*

Bitwise mask defined in the **AT+GTBID** command to configure which data item is reported.

- **Bit 0** - Reserved
- **Bit 1** - <Accessory MAC>
- **Bit 2** - Reserved
- **Bit 3** - <Accessory Battery Level>
- **Bit 4** - Reserved
- **Bit 5** - Reserved
- **Bit 6** - <Accessory Signal Strength>
- **Bit 7** - <Beacon Type> and <Beacon Data>

✧ *Accessory MAC*

The MAC address of the Bluetooth accessory.

✧ *Accessory Battery Level*

The voltage of Bluetooth accessory.

✧ *Accessory Signal Strength*

The signal strength of Bluetooth accessory.

✧ *Beacon Type*

Types of beacons.

- **0** - "ID" Format
- **1** - "iBeacon" Format
- **2** - "Eddystone" Format

✧ *Beacon Data*

Select the data format according to <Beacon Type>:

- If <Beacon Type> is 0, the data format is as follows:

| Fields | Length | Range/Format |
|---|---|---|
| ID_Mfr_Data | 12 | (HEX) |

- If <Beacon Type> is 1, the data format is as follows:

| Fields | Length | Range/Format |
|---|---|---|
| UUID | 32 | (HEX) |
| Major | 4 | (HEX) |
| Minor | 4 | (HEX) |

- If <Beacon Type> is 2, the data format is as follows:

| Fields | Length | Range/Format |
|---|---|---|
| NID | 20 | (HEX) |
| BID | 12 | (HEX) |

## 4.5.31 BAR(Bluetooth Accessory Report)

The device reports the data controlled by <Accessory Type> and <Accessory Model> in AT+GTBAS from the peripheral Bluetooth devices to it via message **+RESP:GTBAR**.

Example:
+RESP:GTBAR,8020040903,866314060000229,GV58LAU,07,7,0,0007,DUT-E S7,0018E9C336E0,1,05,2DF7,58A313002DC27EC87DD48000040000FFFFFFFFFFFF,0,0.0,255,57.3,117.129285,31.839282,20221122062405,0460,0000,550B,0E9E30A5,01,10,20221122142405,12C4$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | BAR |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Index | 2 | 00 - 09 |
| | Accessory Type | 1 | 7 |
| | Accessory Model | 1 | 0 - 4 |
| | Accessory Append Mask | <=4 | 0000 - FFFF |
| | Accessory Name | <=20 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Accessory MAC | 12 | (HEX) |
| | Accessory Status | 1 | 0 - 1 |
| | (Accessory Data) | <=1300 | |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Accessory Type*

The type of the Bluetooth accessory which is defined in the <Index>. The following is supported now:

- **7** - Technoton accessory.

✧ *Accessory Data*

There are accessory data para according <Accessory Type> and <Accessory Model>:

If the <Accessory Type> is 7 and <Accessory Model> is 0 (DUT-E S7), <Accessory Model> is 3 (GNOM DDE S7) or <Accessory Model> is 4(GNOM DP S7):

| Fields | Length | Range/Format |
|---|---|---|
| Version | 2 | (HEX) |
| PGN | 4 | (HEX) |
| PGN Data | <=42 | (HEX) |

If the <Accessory Type> is 7 and <Accessory Model> is 1 (DFM 100 S7):

| Fields | Length | Range/Format |
|---|---|---|
| Version | 2 | (HEX) |
| PGN1 | 4 | (HEX) |
| PGN Data1 | <=42 | (HEX) |
| PGN2 | 4 | (HEX) |
| PGN Data2 | <=42 | (HEX) |
| PGN3 | 4 | (HEX) |
| PGN Data3 | <=42 | (HEX) |

If the <Accessory Type> is 7 and <Accessory Model> is 2 (DFM 250D S7):

| Fields | Length | Range/Format |
|---|---|---|
| Version | 2 | (HEX) |
| PGN1 | 4 | (HEX) |

| Fields | Length | Range/Format |
|---|---|---|
| PGN Data1 | <=42 | (HEX) |
| PGN2 | 4 | (HEX) |
| PGN Data2 | <=42 | (HEX) |
| PGN3 | 4 | (HEX) |
| PGN Data3 | <=42 | (HEX) |
| PGN4 | 4 | (HEX) |
| PGN Data4 | <=42 | (HEX) |

- *Version*
  It indicates the version of accessory software.
- *PGNx*
  It means parameter group number.
- *PGN Datax*
  Different PGN, PGN Data have different data frame formats. It needs to parse according to the TECHNOTON S7 BUS Protocol.

Note

If the total number of characters in a single message is too long, **+RESP:GTBAR** will be divided into multiple messages.

### 4.5.32 UDF(User Defined Function Report)

Report the <Group ID> that triggered the corresponding storage command.

Example:
+RESP:GTUDF,8020040803,866314060268081,GV58LAU,,00000001,0,0.0,79,114.1,117.129227,31.839377,20250320032706,0460,0000,550B,0E9E30A5,0D,8,120000,0.00,0.00,0.00,20250320032707,00A7$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | UDF |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Reserved | 0 | |
| | Group ID MASK | <=8 | 0 - FFFFFFFF |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Device Status (Optional) | 6 | 000000 - FFFFFF |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Group ID Mask*

Bitwise mask which indicates the group ID in the AT+GTUDF command that satisfies all programming conditions.

| ID | Mask Bit | Item |
|---|---|---|
| 1 | Bit 0 | Group ID 0 |
| 2 | Bit 1 | Group ID 1 |
| 3 | Bit 2 | Group ID 2 |
| 4 | Bit 3 | Group ID 3 |
| 5 | Bit 4 | Group ID 4 |
| 6 | Bit 5 | Group ID 5 |
| 7 | Bit 6 | Group ID 6 |
| 8 | Bit 7 | Group ID 7 |
| 9 | Bit 8 | Group ID 8 |
| 10 | Bit 9 | Group ID 9 |
| 11 | Bit 10 | Group ID 10 |
| 12 | Bit 11 | Group ID 11 |
| 13 | Bit 12 | Group ID 12 |

| ID | Mask Bit | Item |
|---|---|---|
| 14 | Bit 13 | Group ID 13 |
| 15 | Bit 14 | Group ID 14 |
| 16 | Bit 15 | Group ID 15 |
| 17 | Bit 16 | Group ID 16 |
| 18 | Bit 17 | Group ID 17 |
| 19 | Bit 18 | Group ID 18 |
| 20 | Bit 19 | Group ID 19 |
| 21 | Bit 20 | Group ID 20 |
| 22 | Bit 21 | Group ID 21 |
| 23 | Bit 22 | Group ID 22 |
| 24 | Bit 23 | Group ID 23 |
| 25 | Bit 24 | Group ID 24 |
| 26 | Bit 25 | Group ID 25 |
| 27 | Bit 26 | Group ID 26 |
| 28 | Bit 27 | Group ID 27 |
| 29 | Bit 28 | Group ID 28 |
| 30 | Bit 29 | Group ID 29 |
| 31 | Bit 30 | Group ID 30 |
| 32 | Bit 31 | Group ID 31 |

For each bit, when it is 1, it indicates that the group ID meets all programming conditions, and when it is 0, it indicates that the group ID does not meet all programming conditions.

## 4.6 Data Flow Reports

This section describes the message related to certain data needs to be sent. Please refer to the details below.

### 4.6.1 CRG(Crash GNSS Data Packet)

The **+RESP:GTCRG** message contains GNSS information before/after the crash. When crash is detected, GNSS information before the crash will be reported to the backend server. And the device will continue to record GNSS information after the crash and report the packed data to the backend server.

Example:
+RESP:GTCRG,8020040803,866314060268081,GV58LAU,1,10,1,1,0.0,79,114.1,117.129227,31.839377,20250320032754,2,1,0.0,79,114.1,117.129227,31.839377,20250320032755,3,1,0.0,79,114.1,117.129227,31.839377,20250320032756,4,1,0.0,79,114.1,117.129227,31.839377,2025032003

2757,5,1,0.0,79,114.1,117.129227,31.839377,20250320032758,6,1,0.0,79,114.1,117.129227,31.839377,20250320032759,7,1,0.0,79,114.1,117.129227,31.839377,20250320032800,8,1,0.0,79,114.1,117.129227,31.839377,20250320032801,9,1,0.0,79,114.1,117.129227,31.839377,20250320032802,10,1,0.0,79,114.1,117.129227,31.839377,20250320032803,20250320032805,00AE$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | CRG |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Data Type | 1 | 0 \| 1 |
| | GNSS Validity Number | <=2 | 0 - 10 |
| | GNSS Point Index | <=2 | 1 |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | ... | | |
| | GNSS Point Index | <=2 | 10 |
| | GNSS Accuracy | <=2 | 0\|1 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Data Type*

The data reported to the backend server is recorded before crash or after crash.

- **0** - Before crash
- **1** - After crash

✧ *GNSS Validity Number*

The number of the successfully fixed GNSS positions included in the report message.

✧ *GNSS Point Index*

The index of total GNSS point before/after the crash.

### 4.6.2 CRD(Crash Data Packet)

The **+RESP:GTCRD** message contains the tri-axial acceleration data before and after the crash, which is determined by the parameters <Samples Before Crash> and <Samples After Crash>. When crash accident is detected, tri-axial acceleration data before crash will be reported to backend server in several frames. And the device will continue to record tri-axial data after crash and report the data to backend server in several frames.

Example:

```
+RESP:GTCRD,8020040803,866314060179445,,00,18,2,1,0002FFFFFFAF0001FFFFFFAF0001FFFFFF
B000010000FFAF0002FFFFFFAF0003FFFFFFAF0001FFFFFFB000010000FFB00002FFFFFFAF0002FF
FFFFAF0002FFFFFFAF0001FFFFFFB00002FFFFFFAF0002FFFFFFAF0002FFFFFFB00002FFFFFFAF000
1FFFFFFAF0001FFFFFFAF0003FFFEFFAF0001FFFFFFAF0003FFFFFFAF0001FFFFFFAF0001FFFFFFAF0
002FFFEFFAF0002FFFFFFAF0002FFFFFFAF0001FFFFFFB00002FFFFFFAF0002FFFFFFB00001FFFFFF
B00002FFFFFFAF0001FFFFFFAF0002FFFFFFAF0002FFFFFFB10002FFFFFFAF0002FFFFFFAF0002FFF
FFFAF0001FFFFFFAF0002FFFFFFAF0002FFFFFFAF0001FFFFFFB00001FFFFFFB00002FFFFFFAF0002
FFFEFFB10002FFFFFFAE0001FFFFFFAF0001FFFFFFAF0002FFFFFFB00003FFFFFFAF0001FFFFFFB00
002FFFFFFAF0003FFFFFFAF0001FFFFFFAF0002FFFFFFAF0002FFFFFFAF0002FFFFFFAF0001FFFFFFA
F0001FFFFFFAF0001FFFFFFAF0002FFFFFFAF0002FFFFFFAF0002FFFFFFAF0002FFFFFFAF0002FFFFF
FAF0001FFFEFFAF0003FFFFFFB00002FFFFFFB00003FFFFFFAF0001FFFFFFB10001FFFFFFB00002FF
FFFFB00002FFFFFFAF0003FFFFFFAF0002FFFFFFAF0002FFFFFFB00001FFFFFFAF0002FFFFFFB1000
2FFFEFFB10001FFFFFFAF0001FFFFFFAF0001FFFFFFB00001FFFFFFAF0002FFFFFFAF0002FFFFFFAF0
002FFFFFFB10001FFFFFFAF0002FFFFFFAF0002FFFFFFAF0002FFFFFFAF00010000FFB00001FFFFFF
AF0002FFFFFFB00002FFFFFFB00002FFFFFFAF0002FFFFFFAF0002FFFFFFAF0002FFFFFFAF0001FFF
FFFAE0002FFFFFFB00001FFFFFFAF,20231208072435,0007$
```

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | CRD |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Crash Counter | 2 | 00 - FF |
| | Data Type | 2 | 00 - 7F |
| | Total frame | <=2 | 1 - 15 |
| | Frame Number | <=2 | 1 - 15 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Data | <=1200 | '0'-'9' 'a'-'f' |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Crash Counter*

It indicates the sequence number of the crash event. The report of **+RESP:GTCRA** and **+RESP:GTCRD** are combined into one crash event. It rolls from 0x00 to 0xFF.

✧ *Data Type*

A hexadecimal parameter to indicate the time of the data (before crash or after crash) and crash direction (+X, -X, +Y, -Y, +Z, -Z or several of them).

| Bits | Description | Range |
|---|---|---|
| Bit 0 | 0: Before crash 1: After crash | 0 - 1 |
| Bit 1 | 0: X-axis crash not detected 1: X-axis crash detected | 0 - 1 |
| Bit 2 | 0: X-axis positive direction 1: X-axis negative direction | 0 - 1 |
| Bit 3 | 0: Y-axis crash not detected 1: Y-axis crash detected | 0 - 1 |
| Bit 4 | 0: Y-axis positive direction 1: Y-axis negative direction | 0 - 1 |
| Bit 5 | 0: Z-axis crash not detected 1: Z-axis crash detected | 0 - 1 |
| Bit 6 | 0: Z-axis positive direction 1: Z-axis negative direction | 0 - 1 |
| Bit 7 | Fixed value | 0 |

✧ *Total Frame*

Total number of messages that are sent to the backend server for the crash event.

✧ *Frame Number*

A numeral to indicate the sequence of the current message.

✧ *Data*

There are 1200 ASCII characters in one message at most which includes acceleration samples in 1 second at most. There are 12 characters in a group. The first 4 characters of these 12 characters represent X axis acceleration data, the middle 4 characters represent Y axis acceleration data and the last 4 characters represent Z axis acceleration data. The ASCII "0001" is equal to 0x0001 in hex format, and the ASCII "afff" is equal to 0xAFFF in hex format. They are two's complement.

✧ *Example*

- +RESP:GTCRD,8020040900,359231038715676,,0,3,1,000100010055…,20120330120443,005C$

  This is the oldest XYZ-axis acceleration data:

  Conversion to hex format: X (axis acceleration data) = 0x0001; Y = 0x0001; Z = 0x0055;

  Decimal format: X (axis acceleration data) = 1; Y = 1; Z = 85;

- +RESP:GTCRD,8020040900,359231038715676,,1,3,3,...fffffff10052,20120330115736,005A$

  This is the last XYZ-axis acceleration data:

  Conversion to hex format: X (axis acceleration data) = 0xFFFF; Y = 0xFFF1; Z = 0x0052;

Decimal format: X (axis acceleration data) = -1; Y = -15; Z = 82;

Note

Acceleration of gravity (+g) is the 82 in decimal format and - g is -82. The linearized acceleration data 1312 represents +16g and -1312 represents -16g.

## 4.7 Data Transmission

This section describes the format of the data transmission messages. Please refer to the details below.

### 4.7.1 DAT(Transparent Data Transmission)

The device supports transparent data transfer between the backend server and the peripheral device connected to its Bluetooth. The device supports bi-directional data transmission. In both directions, the data is transparent to the device.

a) Transfer data from the peripheral device to the backend server:

If the peripheral device supports the AT+GTDAT command, it can transfer data via this command. The peripheral device can send the command **AT+GTDAT** with the data to Bluetooth. According to <Command Type> of **AT+GTDAT**, the device wraps the corresponding data into the backend server with the **+RESP:GTDAT** message either in short format or in long format.

b) Transfer data from the backend server to the peripheral device:

If the backend server needs to send data to the peripheral device, it can send the command **AT+GTDAT** with the data to the device and the device will pick out the raw data and send it to the Bluetooth. The peripheral device can thus get the data from Bluetooth.

#### 4.7.1.1 Short Format

Example:
+RESP:GTDAT,8020040803,866314060179445,GV58LAU,123,20231208074134,0095$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | DAT |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Data to the Backend Server | <=1280 | ASCII Code |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

### 4.7.1.2 Long Format

Example:
+RESP:GTDAT,8020040803,866314060268081,GV58LAU,,,,111111,1,0.0,79,114.1,117.129227,31.839377,20250320032904,0460,0000,550B,0E9E30A5,09,8,1.15,1.49,1.88,,,,,20250320032905,00B1$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | DAT |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Report Type | 1 | 0 \| 2-4 |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Data to the Backend Server | <=1280 | ASCII String |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

- *Report Type*

  It indicates where the data comes from.

  - **0** - Reserved
  - **1** - Reserved
  - **2** - **AT+GTDAT** from Bluetooth
  - **3** - **AT+GTDAT** from over the air
  - **4** - **AT+GTDAT** triggered by events in AT+GTUDF

Note

Data to the Bluetooth starts with a new line and is terminated with \r\n.

## 4.8 CANBUS Information Report

This section describes the message format of CANBUS information. Please refer to the details below.

### 4.8.1 CAN(CANBUS Device Information)

If the CANBUS device information report function is enabled by the command AT+GTCAN, the device will send the CANBUS device information via the message **+RESP:GTCAN** to the backend server periodically.

```
Example:
+RESP:GTCAN,8020040803,866314060268081,GV58LAU,00,1,E03FFFFF,,2,H779133,50.56,20,40,9
0,,L87.40,3960,32,2.14,1.17,0.97,7436.45,4365,BF,7FFF,3F,1F,3.56,1.11,FFFFFF,,,,,,,,,,,,,,,,,,,,,0000,6
21,1324,,,,1,0.0,0,98.3,117.129239,31.839344,20250320092845,0460,0000,550B,0E9E30A5,09,1
2,0.72,0.87,1.13,20250320092846,063C$
```

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | CAN |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Distance Type / Report Type | 2 | X(0 \| 1 )X(0-2) |
| | CANBUS Device State | 1 | 0 \| 1 |
| | CANBUS Report Mask | <=8 | 0 - FFFFFFFF |
| | VIN | 17 | '0' - '9' 'A' - 'Z' except 'I', 'O', 'Q' |
| | Ignition Key | 1 | 0-2 |
| | Total Distance | <=12 | H(0 - 99999999)/I(0 - 2147483647) |
| | Total Fuel Used | <=9 | 0.00 - 999999.99(L) |
| | Engine RPM | <=5 | 0 - 16383 (rpm) |
| | Vehicle Speed | <=4 | 0 - 6553(km/h) |
| | Engine Coolant Temperature | <=4 | -40 - +215 ( ℃ ) |
| | Fuel Consumption | <=6 | L/100km(0.0 - 999.9)\|L/H(0.0 - 999.9) |
| | Fuel Level | <=8 | L(0.00 - 9999.99)/P(0.00 - 100.00) |
| | Range | <=8 | 0 - 99999999(hm) |
| | Accelerator Pedal Pressure | <=3 | 0 - 100(%) |
| | Total Engine Hours | <=10 | 0.00 - 9999999.99(h) |
| | Total Driving Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Engine Idle Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Idle Fuel Used | <=9 | 0.00 - 999999.99(L) |
| | Axle Weight 2nd | <=5 | 0 - 65535(kg) |
| | Tachograph Information | 4 | 00 - FFFF |
| | Detailed Information / Indicators | 4 | 00 - FFFF |
| | Lights | 2 | 0 - FF |
| | Doors | 2 | 0 - FF |
| | Total Vehicle Overspeed Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Vehicle Engine Overspeed Time | <=10 | 0.00 - 9999999.99(h) |
| | CAN Report Expansion Mask | <=8 | 0 - FFFFFFFF |
| | Ad-Blue Level | <=8 | L(0.00 - 9999.99)/P(0.00 - 100.00) |
| | Axle Weight 1st | <=5 | 0 - 65535(kg) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Axle Weight 3rd | <=5 | 0 - 65535(kg) |
| | Axle Weight 4th | <=5 | 0 - 65535(kg) |
| | Tachograph Overspeed Signal | 1 | 0 \| 1 |
| | Tachograph Vehicle Motion Signal | 1 | 0 \| 1 |
| | Tachograph Driving Direction | 1 | 0 \| 1 |
| | Analog Input Value | <=5 | 0 - 99999(mV) |
| | Engine Braking Factor | <=10 | 0 - 4278190079 |
| | Pedal Braking Factor | <=10 | 0 - 4278190079 |
| | Total Accelerator Kick-downs | <=6 | 0 - 999999 |
| | Total Effective Engine Speed Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Cruise Control Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Accelerator Kick-down Time | <=10 | 0.00 - 9999999.99(h) |
| | Total Brake Applications | <=6 | 0 - 999999 |
| | Tachograph Driver 1 Card Number | <=40 | ASCII |
| | Tachograph Driver 2 Card Number | <=40 | ASCII |
| | Tachograph Driver 1 Name | <=40 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Tachograph Driver 2 Name | <=40 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Registration Number | <=40 | ASCII |
| | Expansion Information | <=4 | 0000 - FFFF |
| | Rapid Brakings | <=8 | 0 - 16711679 |
| | Rapid Accelerations | <=8 | 0 - 16711679 |
| | Engine Torque | <=3 | 0 - 100(%) |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | GNSS Accuracy | <=2 | 0 - 50 |
| | Speed | <=5 | 0.0 - 999.9 (km/h) |
| | Azimuth | <=3 | 0 - 359 |
| | Altitude | <=8 | (-)xxxxx.x (m) |
| | Longitude | <=11 | (-)xxx.xxxxxx |
| | Latitude | <=10 | (-)xx.xxxxxx |
| | GNSS UTC Time | 14 | YYYYMMDDHHMMSS |
| | MCC | 4 | 0XXX |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | MNC | 4 | 0XXX |
| | LAC | 4 | XXXX |
| | Cell ID | 4\|8 | XXXX\|XXXXXXXX |
| | Position Append Mask | 2 | 00 - FF |
| | Satellites in Use (Optional) | <=2 | 0 - 72 |
| | Horizontal GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | Vertical GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| | 3D GNSS Accuracy (Optional) | <=5 | 0.00 - 99.99 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Distance Type / Report Type*

It indicates the type of total distance (unit: hm) and the report type.

- Distance Type has the following meanings:
  - **0** - Total distance acquired from CAN Chipset.
  - **1** - Total distance obtained by calculation with GNSS.
- Report Type has the following meanings:
  - **0** - Periodic report.
  - **1** - RTO CAN report.
  - **2** - Ignition event report.

✧ *CANBUS Device State*

A numeral to indicate the communication state with the external CANBUS device.

- **0** - Abnormal.

  It fails to receive data from the external CANBUS device.
- **1** - Normal.

  It is able to receive data from the external CANBUS device.

✧ *CANBUS Report Mask*

Please refer to <CAN Report Mask> in AT+GTCAN.

✧ *VIN*

Vehicle identification number.

✧ *Ignition Key*

A numeral to indicate the ignition status.

- **0** - Ignition off
- **1** - Ignition on
- **2** - Engine on

✧ *Total Distance*

Vehicle total distance. The number is always increasing. The unit is hectometer (H) or distance impulse (I) (if distance from dashboard is not available).

✧ *Total Fuel Used*

The number of liters of fuel used since vehicle manufacture or device installation. The unit is liter.

✧ *Fuel Level*

The level of fuel in vehicle tank. The unit is liter (L) or percentage (P).

✧ *Range*

The number of hectometers to drive on remaining fuel. The unit is hectometer.

✧ *Vehicle Speed*

The vehicle speed based on wheel. The unit is km/h.

✧ *Engine RPM*

The revolutions per minute. The unit is rpm.

✧ *Accelerator Pedal Pressure*

The unit is percentage.

✧ *Engine Coolant Temperature*

The unit is Celsius. Negative value is preceded by negative sign (-), e.g. "-2". If the value is positive, no extra character is inserted, e.g. "20".

✧ *Fuel Consumption*

The fuel consumption is calculated based on values read from vehicle. The unit is L/100Km(M) or L/H(H).

✧ *Total Engine Hours*

Time of engine running since vehicle manufacture or device installation. The unit is hour.

✧ *Total Driving Time*

Time of engine running (non-zero speed) since vehicle manufacture or device installation. The unit is hour.

✧ *Total Engine Idle Time*

Time of engine running during idling status (vehicle at rest) since vehicle manufacture or device installation. The unit is hour.

✧ *Total Idle Fuel Used*

The number of liters of fuel used since vehicle manufacture or device installation. The unit is liter.

✧ *Axle Weight 2nd*

Weight of vehicle's second axle. The unit is kg.

✧ *Tachograph Information*

Two bytes. The high byte describes driver 2, while the low byte describes driver 1.

Each byte format:

| Validity mark | Reserved | Driver working states | | Driver card | Driving time related states | | |
|---|---|---|---|---|---|---|---|
| V | R | W1 | W0 | C | T2 | T1 | T0 |

V: Validity mark (0 - valid driver data, 1 - no valid data)

R: Reserved

C: Driver card (1 - card inserted, 0 - no card inserted)

T2-T0: Driving time related states:

- **0** - Normal / no limits reached.
- **1** - 15min before 4½h.
- **2** - 4½h reached.

- **3** - 15min before 9h.
- **4** - 9h reached.
- **5** - 15minute before 16h (without 8h rest during the last 24h).
- **6** - 16h reached.
- **7** - Other limit.

W1-W0: Driver working states:

- **0** - Rest - sleeping.
- **1** - Driver available - short break.
- **2** - Work - loading, unloading, working in an office.
- **3** - Drive - behind the wheel.

✧ *Detailed Information / Indicators*

A hexadecimal number. Each bit contains information of one indicator.

- **Bit 0** - FL - fuel low indicator (1 - indicator on, 0 - indicator off)
- **Bit 1** - DS - driver seatbelt indicator (1 - indicator on, 0 - indicator off)
- **Bit 2** - AC - air conditioning (1 - on, 0 - off)
- **Bit 3** - CC - cruise control (1 - active, 0 - disabled)
- **Bit 4** - B - brake pedal (1 - pressed, 0 - released)
- **Bit 5** - C - clutch pedal (1 - pressed, 0 - released)
- **Bit 6** - H - handbrake (1 - pulled-up, 0 - released)
- **Bit 7** - CL - central lock (1 - locked, 0 - unlocked)
- **Bit 8** - R - reverse gear (1 - on, 0 - off)
- **Bit 9** - RL - running lights (1 - on, 0 - off)
- **Bit 10** - LB - low beams (1 - on, 0 - off)
- **Bit 11** - HB - high beams (1 - on, 0 - off)
- **Bit 12** - RFL - rear fog lights (1 - on, 0 - off)
- **Bit 13** - FFL - front fog lights (1 - on, 0 - off)
- **Bit 14** - D - doors (1 - any door open, 0 - all doors closed)
- **Bit 15** - T - trunk (1 - open, 0 - closed)

✧ *Lights*

A hexadecimal number. Each bit contains information of one type of light.

- **Bit 0** - Running Lights (1 - on, 0 - off)
- **Bit 1** - Low Beam (1 - on, 0 - off)
- **Bit 2** - High Beam (1 - on, 0 - off)
- **Bit 3** - Front Fog Light (1 - on, 0 - off)
- **Bit 4** - Rear Fog Light (1 - on, 0 - off)
- **Bit 5** - Hazard Lights (1 - on, 0 - off)
- **Bit 6** - Reserved
- **Bit 7** - Reserved

✧ *Doors*

A hexadecimal number. Each bit contains information of one door.

- **Bit 0** - Driver Door (1 - open, 0 - closed)
- **Bit 1** - Passenger Door (1 - open, 0 - closed)
- **Bit 2** - Rear Left Door (1 - open, 0 - closed)
- **Bit 3** - Rear Right Door (1 - open, 0 - closed)

- **Bit 4** - Trunk (1 - open, 0 - closed)
- **Bit 5** - Hood (1 - open, 0 - closed)
- **Bit 6** - Reserved
- **Bit 7** - Reserved

✧ *Total Vehicle Overspeed Time*

The total time when the vehicle speed is greater than the limit defined in CAN100's configuration.

✧ *Total Vehicle Engine Overspeed Time*

The total time when the vehicle engine speed is greater than the limit defined in CAN100's configuration.

✧ *Ad-Blue Level*

The level of Ad-Blue.

✧ *Axle Weight 1st*

Vehicle first axle weight. The unit is Kg.

✧ *Axle Weight 3rd*

Vehicle third axle weight. The unit is Kg.

✧ *Axle Weight 4th*

Vehicle fourth axle weight. The unit is Kg.

✧ *Tachograph Overspeed Signal*

Vehicle overspeed signal from the tachograph.

- **0** - Overspeed is not detected.
- **1** - Overspeed is detected.

✧ *Tachograph Vehicle Motion Signal*

The vehicle motion signal in the tachograph.

- **0** - Motion is not detected.
- **1** - Motion is detected.

✧ *Tachograph Driving Direction*

Vehicle driving direction from the tachograph.

- **0** - Driving forward.
- **1** - Driving backward.

✧ *Analog Input Value*

The value of analog input. The unit is mV.

✧ *Rapid Brakings*

The count of rapid brakings of the vehicle.

✧ *Engine Braking Factor*

It measures how often driver brakes with brake pedal or with engine and stores both counts (always increasing). Decreasing speed with no pedal pressed causes an increase in engine braking factor.

✧ *Pedal Braking Factor*

It measures how often driver brakes with brake pedal or with engine and stores both counts (which are always increasing). Decreasing speed with brake pedal pressed causes an increase in pedal braking factor.

✧ *Total Accelerator Kick-downs*

The count of accelerator pedal kick-downs (with the pedal pressed over 90%).

- *Total Effective Engine Speed Time*
  Total time when the vehicle engine speed is effective. The unit is h.
- *Total Cruise Control Time*
  Total time when vehicle speed is controlled by cruise-control module. The unit is h.
- *Total Accelerator Kick-down Time*
  Total time when accelerator pedal is pressed over 90%. The unit is h.
- *Total Brake Applications*
  The total number of braking processes initiated by brake pedal.
- *Tachograph Driver 1 Card Number*
  The card number of tachograph driver 1.
- *Tachograph Driver 2 Card Number*
  The card number of tachograph driver 2.
- *Tachograph Driver 1 Name*
  The name of tachograph driver 1.
- *Tachograph Driver 2 Name*
  The name of tachograph driver 2.
- *Registration Number*
  The vehicle registration number.
- *Expansion Information*
  A hexadecimal number. Each bit contains information of one indicator.
  - **Bit 0** - Reserved
  - **Bit 1** - BFL - brake fluid low indicator (1 - on, 0 - off or not available)
  - **Bit 2** - CLL - coolant level low indicator (1 - on, 0 - off or not available)
  - **Bit 3** - BAT - battery indicator (1 - on, 0 - off or not available)
  - **Bit 4** - BF - brake system failure indicator (1 - on, 0 - off or not available)
  - **Bit 5** - OP - oil pressure indicator (1 - on, 0 - off or not available)
  - **Bit 6** - EH - engine hot indicator (1 - on, 0 - off or not available)
  - **Bit 7** - ABS - ABS failure indicator (1 - on, 0 - off or not available)
  - **Bit 9** - CHK - "check engine" indicator (1 - on, 0 - off or not available)
  - **Bit 10** - AIR - airbag indicator (1 - on, 0 - off or not available)
  - **Bit 11** - SC - service call indicator (1 - on, 0 - off or not available)
  - **Bit 12** - OLL - oil level low indicator (1 - on, 0 - off or not available)
- *Rapid Brakings*
  The number of total rapid brakings since installation (calculation based on CAN100 settings of speed decrease time and value).
- *Rapid Accelerations*
  The number of total rapid accelerations since installation (calculation based on CAN100 settings of speed increase time and value).
- *Engine Torque*
  The engine torque. Unit: percentage.

## 4.9 Update Configuration

This section describes the message related to firmware and configuration upgrade. Please refer to the details below.

### 4.9.1 UPC(Configuration Update Notification)

The report for over-the-air configuration update.

Example:
+RESP:GTUPC,8020040803,866314060179445,GV58LAU,200,301,http://60.174.225.173:20571/PEO_at.ini,20231208141952,4EF9$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | UPC |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Command ID | <=3 | |
| | Result | 3 | 100-103 \| 200-202 \| 300-302 \| 304-306 |
| | Download URL | <=100 | Complete URL |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Command ID*

The command ID in the update configuration file. It is always 0 before the device starts to update the configuration (<Result> <= 300). It indicates the total number of the commands when <Result> is 301. It indicates the ID of the command which has wrong format when <Result> is 302.It is empty when <Result> is greater than 302.

✧ *Result*

A numeral to indicate whether the configuration is updated successfully.

- **100** - The update command is starting.
- **101** - The update command is confirmed by the device.
- **102** - The update command is refused by the device.
- **103** - The update process is refused because the battery is low.
- **200** - The device starts to download the package.
- **201** - The device finishes downloading the package successfully.

- **202** - The device fails to download the package.
- **300** - The device starts to update the device configuration.
- **301** - The device finishes updating the device configuration successfully.
- **302** - The device fails to update the device configuration.
- **303** - Reserved.
- **304** - <Command Mask>, <GEO ID Mask>, <Stocmd ID Mask> or <Group ID Mask> check fails.
- **305** - The update process is interrupted by abnormal reboot.
- **306** - The update process is interrupted by MD5 verification error.

✧ *Download URL*

The complete URL to download the configuration. It includes the file name.

### 4.9.2 UPD(Firmware Upgrade Report)

The report for over-the-air configuration update.

#### 4.9.2.1 Update Confirmation

The device sends update confirmation information to the backend server if:

✧ The update command is confirmed by the device.
✧ The update command is refused by the device.
✧ The update process is cancelled by the backend server or refused because of an incorrect URL.
✧ The update command is refused because the battery is low.

| Example: |
|---|
| +RESP:GTUPD,8020040803,866775051515021,gv58lau,300,,20221101172249,EEA9$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | UPD |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Code | 3 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *code*

It indicates the confirmation information.

- **1x0** - The update command is confirmed by the device.
- **1x1** - The update command is refused by the device.
- **1x2** - The update process is cancelled by the backend server or refused because of an incorrect URL.
- **1x3** - The update process is refused because the external power supply is not inserted.

Note

x here means <Update Type> defined in the command AT+GTUPD.

### 4.9.2.2 Package Download

The device sends package download information to the backend server if:

- ✧ The device starts to download the package.
- ✧ The device finishes downloading the package successfully.
- ✧ The device fails to download the package.

Example:
+RESP:GTUPD,8020040803,866775051515021,gv58lau,300,,20221101172249,EEA9$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | UPD |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Code | 3 | |
| | Download Times | 1 | 1 \| 2 \| 3 \| 4 |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Code*

It indicates the download information.

- **2x0** - The device starts to download the package.
- **2x1** - The device finishes downloading the package successfully.
- **2x2** - The device fails to download the package.

✧ *Download Times*

The count number of the package download.

Note

x here means <Update Type> defined in the command **AT+GTUPD**.

### 4.9.2.3 Firmware Update

The device sends firmware update information to the backend server if:

- The device starts to update the firmware.
- The device finishes updating the firmware successfully.
- The device fails to update the firmware.
- The update process does not start because the battery is low.

Example:
+RESP:GTUPD,8020040803,866775051515021,gv58lau,300,,20221101172249,EEA9$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | UPD |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Code | 3 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

- *Code*
  It indicates the update information.
  - **3x0** - The device starts to update the firmware.
  - **3x1** - The device finishes updating the firmware successfully.
  - **3x2** - The device fails to update the firmware.
  - **3x3** - The update process does not start because the battery is low.
  - **3x4** - Serial Number does not match, firmware upgrade fails.

Note

x here means <Update Type> defined in the command **AT+GTUPD**.

## 4.9.3 EUC(Extended Configuration Update Report)

Example:
+RESP:GTEUC,8020040803,866314060268081,GV58LAU,0,102,,00000058,,,,,20250320090957,058D$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | EUC |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Command ID | <=3 | |
| | Result | 3 | 100-103 \| 200-202 \| 300-302 \| 304-306 |
| | Download URL | <=100 | Complete URL |
| | Identifier Number | 8 | 00000000 - FFFFFFFF |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Command ID*

The command ID in the update configuration file. It is always 0 before the device starts to update the configuration. It indicates the total number of the commands when the response code is 301 or 306. It indicates wrong format of command ID when the response code is 302. It is empty when the response code is 304 or 305.

✧ *Result*

A numeral to indicate whether the configuration is updated successfully.

- **100** - The update command is starting.
- **101** - The update command is confirmed by the device.
- **102** - The update command is refused by the device.
- **103** - The update process is refused because the battery is low.
- **200** - The device starts to download the package.
- **201** - The device finishes downloading the package successfully.
- **202** - The device fails to download the package.
- **300** - The device starts to update the device configuration.

- **301** - The device finishes updating the device configuration successfully.
- **302** - The device fails to update the device configuration.
- **303** - Reserved.
- **304** - <Command Mask>, <GEO ID Mask>, <Stocmd ID Mask> or <Group ID Mask> check fails.
- **305** - The update process is interrupted by abnormal reboot.
- **306** - The update process is interrupted by MD5 verification error.

✧ *Download URL*

The complete URL to download the configuration. It includes the file name.

✧ *Identifier Number*

Please refer to the parameter <Identifier Number> in the command AT+GTUPC.

## 4.9.4 EUD(Extended Firmware Update Report in ASCII Format)

### 4.9.4.1 Update Confirmation

The device will send update confirmation information to the backend server if:

✧ The device confirms this update command.
✧ The device refuses this update command.
✧ The backend server cancels this update process.
✧ The device refuses this request because the battery is low.

Example:
+RESP:GTEUD,8020040803,866314060268081,GV58LAU,102,,00000000,,,,,20250320091026,0593$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | EUD |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Code | 3 | |
| | Reserved | 0 | |
| | Identifier Number | 8 | 00000000 - FFFFFFFF |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Tail | 1 | $ |

✧ *Code*

It indicates the confirmation information.

- **1x0** - The device confirms this update command.
- **1x1** - The device refuses this update command.
- **1x2** - The backend server cancels this update process.
- **1x3** - The device refuses this request because the battery is low.

Note

x here means <Update Type> defined in the command AT+GTUPD.

#### 4.9.4.2 Package Download

The device will send the package download information to the backend server if:

✧ The device starts to download the package.
✧ The device downloads the package successfully.
✧ The device fails to download the package.

Example:
+RESP:GTEUD,8020040803,866314060268081,GV58LAU,102,,00000000,,,,,20250320091026,059
3$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | EUD |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Code | 3 | |
| | Download Times | 1 | 1 \| 2 \| 3 \| 4 |
| | Identifier Number | 8 | 00000000 - FFFFFFFF |
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Ending Flag | 1 | $ |

✧ *Code*

It indicates the download information.

- **2x0** - The device starts to download the package.
- **2x1** - The device downloads the package successfully.
- **2x2** - The device fails to download the package.

✧ *Download Times*

The count number of the package download.

Note

x here means <Update Type> defined in the command **AT+GTUPD**.

### 4.9.4.3 Firmware Update

The device will send the firmware update information to the backend server if:

✧ The device starts to update firmware.
✧ The device updates the firmware successfully.
✧ The device fails to update the firmware.
✧ The firmware update is cancelled because the battery is low.

Example:
+RESP:GTEUD,8020040803,866314060268081,GV58LAU,102,,00000000,,,,,20250320091026,0593$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 8 | +RESP:GT |
| | Message Name | 3 | EUD |
| | Leading Symbol | 1 | , |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Body | Code | 3 | |
| | Reserved | 0 | |
| | Identifier Number | 8 | 00000000 - FFFFFFFF |
| | Reserved | 0 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Reserved | 0 | |
| | Reserved | 0 | |
| | Reserved | 0 | |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000 - FFFF |
| | Ending Flag | 1 | $ |

✧ *Code*

It indicates the update information.

- **3x0** - The device starts to update the firmware.
- **3x1** - The device updates the firmware successfully.
- **3x2** - The device fails to update the firmware.
- **3x3** - The device cancels the firmware update because the battery is low.
- **3x4** - Serial Number does not match, firmware upgrade fails.

Note

x here means <Update Type> defined in the command **AT+GTUPD**.

## 4.10SACK (Server Acknowledgement)

If server acknowledgement is enabled by the AT+GTSRI command, the backend server should reply to the device whenever it receives a message from the device.

Example:
+SACK:11F0$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 5 | +SACK |
| Tail | Count Number | 4 | 0000-FFFF |
| | Tail | 1 | $ |

✧ *Count Number*

The backend server uses the <Count Number> extracted from the received message as the <Count Number> in the server acknowledgement.

# 4.11HBD (Heartbeat Data)

## 4.11.1 Heartbeat Report

Heartbeat is used to maintain the contact between the device and the backend server while communicating via GPRS. The heartbeat package is sent to the backend server at the interval specified by <Heartbeat Interval> in the AT+GTSRI command.

Example:
+ACK:GTHBD,8020040803,866314060179445,GV58LAU,20231208145533,4F7B$

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +ACK |
| | Leading Symbol | 1 | : |
| | Command Word | 5 | 'A'-'Z', '0'-'9' |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| | Unique ID | 15 | IMEI |
| | Device Name | <=20 | '0' - '9' 'a' - 'z' 'A' - 'Z' '-' '_' |
| Tail | Send Time | 14 | YYYYMMDDHHMMSS |
| | Count Number | 4 | 0000-FFFF |
| | Tail | 1 | $ |

- *Command Word*

  Corresponding to the "Command Word" in the configuration command. For example, it is **"GTHBD"**.
- *Full Protocol Version*

  The protocol version that the device conforms to. It is separated into 3 parts. The first two characters represent the device type. As shown in the example above, 802004 means GV58LAU. The second part includes two characters which indicate the major version number of the protocol and the last part includes two characters which indicate the minor version number of the protocol. Both version numbers are hex digits. For example, 0A01 means version 10.01.
- *Unique ID*

  The International Mobile Equipment Identity of the terminal device.
- *Device Name*

  The name of terminal device.
- *Send Time*

  The local time when the frame was generated, in "YYYYMMDDHHMMSS" format.
  For example, "20191120135807" indicates 13:58:07 on November 20, 2019.
- *Count Number*

  A self-increasing count number in each acknowledgement message. It begins from "0000" and increases by 1 for each acknowledgement message. And it rolls back after "FFFF".

### 4.11.2 Heartbeat Acknowledgement

Whenever the backend server receives a heartbeat package, it should reply with an acknowledgement to the device.

| |
|---|
| Example:<br>+SACK:GTHBD,8020040900,11F0$ |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 5 | +SACK |
| | Leading Symbol | 1 | : |
| | Command Word | 5 | 'A'-'Z', '0'-'9' |
| | Full Protocol Version | 10 | 8000000000 - 80FFFFFFFF |
| Tail | Count Number | 4 | 0000-FFFF |
| | Tail | 1 | $ |

✧ *Command Word*

Corresponding to the "Command Word" in the configuration command. For example, it is **GTHBD**.

✧ *Full Protocol Version*

The protocol version that the device conforms to. It is separated into 3 parts. The first two characters represent the device type. As shown in the example above, 802004 means GV58LAU. The second part includes two characters which indicate the major version number of the protocol and the last part includes two characters which indicate the minor version number of the protocol. Both version numbers are hex digits. For example, 0A01 means version 10.01.

✧ *Count Number*

The backend server uses the <Count Number> extracted from the heartbeat package from the device as the <Count Number> in the server acknowledgement of the heartbeat.

## 4.12Buffer Report

### 4.12.1 Overview

If the buffer report function is enabled by the command AT+GTSRI , the terminal will save the report messages in a local buffer when the following occurs.

✧ GSM/LTE network is not available.

✧ PDP activation for the TCP or UDP connection fails.

✧ Establishment of the TCP connection with the backend server fails.

These messages will be sent to the backend server when connection to the server is recovered again. The buffered reports are saved to the built-in non-volatile memory in case the device is reset. The terminal can buffer up to 10,000 messages (160 bytes per message).

| |
|---|
| Example:<br>+BUFF:GTFRI,8020040900,863286020684354,GV58LAU,,10,1,1,0.0,0,0.5,121.392413,31.164143,20160804044602,0460,0000,1877,03A3,00,104.8,,,,100,210100,,,,20140804044611,2E78$ |

### 4.12.2 Description Information

Detailed information about buffer report is listed below.

- Only +RESP messages excluding **+RESP:GTPDP** and **+RESP:GTALM** are buffered.
- In the buffer report, the original header string "**+RESP**" is replaced by "**+BUFF**" while the other content including the original sending time and count number remains unchanged.
- Buffered messages will be sent only via GPRS by TCP or UDP protocol. They cannot be sent via SMS. If the current report is forced SMS mode, the buffered message will not be sent until the report mode is changed to TCP or UDP.
- The buffered messages will be sent after the real-time messages if <Buffer Mode> in AT+GTSRI is set to 1.
- The buffered messages will be sent before the real-time messages if <Buffer Mode> in AT+GTSRI is set to 2. The SOS message has the highest priority and is sent before buffered messages.

# 5 Report(Hex)

This section defines the HEX formats of the report messages. Please refer to the details below.

## 5.1 ACK (Acknowledgement)

The frame format of **+ACK** is as follows:

Example:
2B41434B016F268020040803080F563F0E060244080100FFFF07E90314052C1F01B9AFD90D0A

2B41434B046F268020040803080F563F0E060244080100FFFF07E90314052D3901C31FB50D0A

2B41434B0B6F268020040803060456 3F0E060849470600FFFF07E8050708053306654D030D0A

2B41434B107F268020040401040C475635384C41552D22008B07E70C1208250ABE9721E30D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +ACK |
| | Message Type | 1 | |
| | Report Mask | 1 | 00 - FF |
| Body | Length | 1 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | ID | 1 | |
| | Serial Number | 2 | 0000 - FFFF |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Message Type*

The ID of the command that the device receives.

| Command | ID |
|---|---|
| AT+GTBSI | 0 |
| AT+GTSRI | 1 |
| AT+GTQSS | 2 |

| Command | ID |
|---|---|
| AT+GTCFG | 4 |
| AT+GTTOW | 5 |
| AT+GTEPS | 6 |
| AT+GTDIS | 7 |
| AT+GTIOB | 9 |
| AT+GTTMA | 10 |
| AT+GTFRI | 11 |
| AT+GTGEO | 12 |
| AT+GTSPD | 13 |
| AT+GTSOS | 14 |
| AT+GTRTO | 16 |
| AT+GTUPD | 21 |
| AT+GTPIN | 22 |
| AT+GTDAT | 23 |
| AT+GTOWH | 24 |
| AT+GTDOG | 25 |
| AT+GTJDC | 27 |
| AT+GTIDL | 28 |
| AT+GTHBM | 29 |
| AT+GTHMC | 30 |
| AT+GTWLT | 34 |
| AT+GTHRM | 35 |
| AT+GTFFC | 36 |
| AT+GTJBS | 37 |
| AT+GTSSR | 38 |
| AT+GTIDA | 43 |
| AT+GTPDS | 45 |
| AT+GTCRA | 46 |
| AT+GTBZA | 47 |
| AT+GTSPA | 48 |
| AT+GTRMD | 53 |
| AT+GTPEO | 54 |
| AT+GTCAN | 57 |
| AT+GTCMD | 61 |
| AT+GTUDF | 62 |
| AT+GTGAM | 65 |

| Command | ID |
|---|---|
| AT+GTOEX | 68 |
| AT+GTIEX | 69 |
| AT+GTUPC | 70 |
| AT+GTCLT | 71 |
| AT+GTCFU | 74 |
| AT+GTASC | 79 |
| AT+GTFVR | 81 |
| AT+GTSVR | 82 |
| AT+GTBTS | 89 |
| AT+GTVVS | 91 |
| AT+GTAVS | 92 |
| AT+GTGDO | 93 |
| AT+GTPEG | 99 |
| AT+GTVMS | 102 |
| AT+GTBAS | 103 |
| AT+GTGNA | 108 |
| AT+GTBID | 109 |
| AT+GTAEX | 116 |
| AT+GTDOS | 119 |
| AT+GTIDS | 131 |
| AT+GTROS | 134 |
| AT+GTMQT | 138 |
| AT+GTTLS | 139 |
| AT+GTRTP | 141 |
| AT+GTSWL | 144 |
| AT+GTBFS | 176 |

- *Report Mask*

  Please refer to <+ACK Mask> in AT+GTHRM.
- *Length*

  The whole length of the acknowledgement message from header to the tail characters.
- *Unique ID*

  If Bit 4 of <+ACK Mask> is 0, the IMEI of the device is used as the unique ID of the device. IMEI is a 15-digit string. In the HEX format message, each 2 digits are encoded into one byte as an integer.

| IMEI | 13 | 57 | 90 | 24 | 68 | 11 | 22 | 0 |
|---|---|---|---|---|---|---|---|---|
| HEX | 0D | 39 | 5A | 18 | 44 | 0B | 16 | 00 |

If Bit 4 of <+ACK Mask> is 1, the device name is used as the unique ID of the device. Device name is an 8-byte string, please refer to <Device Name> in AT+GTCFG. If the length of <Device

Name> is more than 8 bytes, only the first 8 bytes will be acquired. In the HEX format message, each byte is encoded into one byte as an integer. If the device name is less than 8 bytes, the remaining bytes are set to 0.

| Device Name | q | u | e | c | l | i | n | k |
|---|---|---|---|---|---|---|---|---|
| HEX | 71 | 75 | 65 | 63 | 6C | 69 | 6E | 6B |

✧ *ID*

- **AT+GTGEO** - This field represents the GEO command ID.
- **AT+GTPEO** - This field represents the PEO command ID.
- **AT+GTIOB** - This field represents the IOB command ID.
- **AT+GTCLT** - This field represents the CLT group ID.
- **AT+GTCMD** - This field represents the stored command ID.
- **AT+GTUDF** - This field represents the UDF group ID.
- **AT+GTBAS** - This field represents the BAS index.
- **AT+GTIDS** - This field represents the IDS result of issuing **AT+GTIDS**.
- **AT+GTRTO** - This field represents the RTO sub-command ID.
- **AT+GTROS** - This field represents the ROS Output ID.
- **AT+GTXXX** - For other commands, it is 0.

✧ *Serial Number*

The serial number in the configuration command.

✧ *Send Time*

The local time to send the acknowledgement message. 7 bytes in total. The first 2 bytes are for year, and the other 5 bytes are for month, day, hour, minute and second respectively.

| Time | 2011 | 01 | 31 | 06 | 29 | 11 |
|---|---|---|---|---|---|---|
| HEX | 07DB | 01 | 1F | 06 | 1D | 0B |

✧ *Count Number*

A self-increasing count number in each acknowledgement message. It begins from "0000" and increases by 1 for each acknowledgement message. And it rolls back after "FFFF".

✧ *Checksum*

2 bytes. This is an 16-bit CRC checksum, and it is generated by a CRC algorithm with the properties displayed in CRC-16 Calculation. The CRC16 checksum of data between the fields of <Message Header> and <Checksum> (exclude <Message Header> and <Checksum>).

Note

The +ACK frame only indicates that the terminal device has received the command, but does not mean that the command has been successfully executed; the terminal device will send a report to inform the backend server of the execution result of the command when necessary.

## 5.2 Position Related Report

This section describes the format of positioning related messages. Please refer to the details below.

## 5.2.1 Generic Location Report

- **TOW** (Tow Alarm Information)
  If the tow alarm is enabled by the command AT+GTTOW, the device will send the message **+RESP:GTTOW** to the backend server when the motion sensor detects tow.
- **AIS** (Analog Input Alarm)
  The device will send the message **+RESP:GTAIS** to the backend server when analog input voltage enters the alarm range.
- **DIS** (Digital Input Alarm)
  If the status change of digital inputs is detected, the device will send the message **+RESP:GTDIS** to the backend server.
- **IOB** (IO Ports Bind Alarm)
  If the IO combination is set and the corresponding condition is met, the device will report the message **+RESP:GTIOB** to the backend server.
- **SPD** (Speed Alarm Information)
  If the speed alarm is enabled, the device will send the message **+RESP:GTSPD** to the backend server when the speed of the device within the alarm range is detected.
- **RTL** (Realtime Location Information)
  After the device receives the command AT+GTRTO, it will start GNSS to get the current position and then send the message **+RESP:GTRTL** to the backend server.
- **DOG** (Watchdog Reboot Alarm)
  The protocol watchdog reboot message **+RESP:GTDOG**.
- **IGL** (Ignition Location Information)
  The location message **+RESP:GTIGL** for ignition on and ignition off.
- **VGL** (Ignition Location Information)
  The location message **+RESP:GTVGL** for virtual ignition on and ignition off.
- **HBM** (Harsh Behavior Alarm)
  If harsh behavior is detected, the message **+RESP:GTHBM** will be sent to the backend server.
- **EPS** (External Power Alarm)
  If the external power supply monitoring is enabled by the command AT+GTEPS, the device will send the message **+RESP:GTEPS** to the backend server when the voltage of the external power supply enters the alarm range.

Example:

2B5253500182F61FFF0000000200BD8020040803080F475635384C415500002F06000016000C00
0101000000000000006206FB401701E5D47007E903140B162704600000550B0E9E30A5000000000
000001300000000000000000000001000FFFFF000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000FFFF0000000007E903140B162809F2E73E0D0A

2B5253500E00FE1FBF00638020040803060356 3F0E0608494706002EBF0100210C500101000000
0000006A06FB40F401E5D2FF07E80416052816046000001DF5C05FE6667000000000000000000000

0000000000000000007E80416052816034016360D0A

2B5253500582F61FFF0000000200BD8020040803080F475635384C415500002EAB030022000C1
1010100000100000002306FB3FB401E5D62107E903140B1B1B04600000550B0E9E30A500000001
0000000001000000000000000000001000FFFFF0000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000FFFF00000000007E903140B1B1C045FAA490D0A

2B5253500682F61FFF0000000200BD8020040803080F475635384C415500002ED1030022000C0
1010100000000000002306FB3FB401E5D62107E903140B1B3504600000550B0E9E30A500000001
0000000001000000000000000000001000FFFFF0000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000FFFF00000000007E903140B1B36046274A50D0A

2B5253500982F61FFF0000000200BD8020040803080F475635384C415500002EAA030022000C0
1010100000000000006206FB404401E5D45C07E903140B1D0904600000550B0E9E30A500000001
0000000001000000000000000000001000FFFFF0000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000FFFF00000000007E903140B1D0A0469BC0A0D0A

2B5253500982F61FFF0000000200BD8020040803080F475635384C415500002EB1030022000C01
010100000000000006206FB404401E5D45C07E903140B1D2704600000550B0E9E30A5000000010
000000001000000000000000000001000FFFFF00000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000FFFF00000000007E903140B1D28046FF0610D0A

2B5253500C82F61FFF0000000200BD8020040803080F475635384C415500002EE8030022000C13
010100000000000006206FB404401E5D45C07E903140B1E0304600000550B0E9E30A5000000010
000000001000000000000000000001000FFFFF00000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000FFFF00000000007E903140B1E040476B7E30D0A

2B5253501082F61FFF00000002010D8020040803080F475635384C415500002EDB000012000C0
1010100000000000003C06FB401901E5D47907E903140B1F2B04600000550B0E9E30A500000000
0000000001000000000000000000001207FFFFF0000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000FFFF00000000000000000000000000000000000000FFFFFF
0000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000007E903140B1F2B048EB2C30D0A

2B5253501A82F61FFF00000002010D8020040803080F475635384C415500002EDF000022000C2
0010100000000000005B06FB3F1601E5D3AA07E903140B212204600000550B0E9E30A500000000
0000000001000000000000000000001207FFFFF0000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000

000000000000000000000000000000FFFF00000000000000000000000000000000000000FFFFFF
000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000007E903140B212304B14B280D0A

2B5253500F00FE1FBF006380200408030604563F0E06022849040030D20100220C020101001703
004E001606FD7F2A01E6161307E805080608220460001DF5A05A10E1F00000906000000000000
0000000000000000007E805080E08237FB5570F0D0A

2B5253500482F61FFF0000000200BD8020040803080F475635384C415500002EEB010022000C01
010100000000000006206FB401701E5D47007E903140B180D04600000550B0E9E30A5000000000
00000130000000000000000000001000FFFFF00000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000FFFF00000007E903140B180E09FA80B90D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +RSP |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | RSP Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000 |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Report ID \| Report Type | 1 | |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | |
| | Current Mileage | 3 | 0.0 - 65535.0(km) |
| | Total Mileage | 5 | 0.0 - 4294967.0(km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Message Type*

The ID of a specific location report message.

| Message | ID |
|---|---|
| +RESP:GTTOW | 1 |
| +RESP:GTLBC | 3 |
| +RESP:GTEPS | 4 |
| +RESP:GTDIS | 5 |
| +RESP:GTIOB | 6 |
| +RESP:GTFRI | 7 |
| +RESP:GTSPD | 9 |
| +RESP:GTSOS | 10 |
| +RESP:GTRTL | 11 |
| +RESP:GTDOG | 12 |
| +RESP:GTAIS | 14 |
| +RESP:GTHBM | 15 |
| +RESP:GTIGL | 16 |
| +RESP:GTIDA | 17 |
| +RESP:GTERI | 18 |
| +RESP:GTGIN | 20 |

| Message | ID |
|---|---|
| +RESP:GTGOT | 21 |
| +RESP:GTVGL | 26 |
| +RESP:GTSPE | 101 |

✧ *Report Mask*

Please refer to <+RSP Mask> in AT+GTHRM.

✧ *Unique ID*

If Bit 6 of <+RSP Mask> is 0, the IMEI of the device is used as the unique ID of the device. IMEI is a 15-digit string. In the HEX format message, each 2 digits are encoded into one byte as an integer.

| IMEI | 13 | 57 | 90 | 24 | 68 | 11 | 22 | 0 |
|---|---|---|---|---|---|---|---|---|
| HEX | 0D | 39 | 5A | 18 | 44 | 0B | 16 | 00 |

If Bit 6 of <+RSP Mask> is 1, the device name is used as the unique ID of the device. Please refer to <Device Name> in AT+GTCFG for the device name. Device name is an 8-byte string. If the length of <Device Name> is more than 8 bytes, only the first 8 bytes will be acquired. In the Hex format message, each byte is encoded into one byte as an integer. If the device name is less than 8 bytes, the remaining bytes are set to 0.

| Device Name | q | u | e | c | l | i | n | k |
|---|---|---|---|---|---|---|---|---|
| HEX | 71 | 75 | 65 | 63 | 6C | 69 | 6E | 6B |

✧ *External Power Voltage*

The value of external power supply, the unit is mV. The highest bit is used to indicate the length of this value field, 0 means 2 bytes and 1 means 4 bytes. If the highest bit is 1, please use the formula **<External Power Voltage> & 0x7FFFFFFF** to obtain a HEX value. For example, 0x0011 is 17 mV, and 0x80010011 is 65553 mV.

✧ *Digital Input Status*

| Input Status Mask | ID |
|---|---|
| Ignition Detection | 0x01 |
| Digital Input1 | 0x02 |

✧ *Digital Output Status*

| Output Status Mask | ID |
|---|---|
| Digital Output1 | 0x01 |
| Digital Output2 | 0x02 |
| Digital Output3 | 0x04 |

✧ *Motion Status*

The current motion status of the device.

✧ *Satellites in Use*

Number of satellites being used for tracking. The high nibble is reserved and the low nibble is valid. But if Bit 1 of <RSP Expansion Mask> is set to 1, this parameter would be expanded to two bytes, the high byte is reserved and the low byte is valid.

✧ *Report ID / Report Type*

The high nibble is for <Report ID> and the low nibble is for <Report Type>.

- *Speed*

  3 bytes in total. The first 2 bytes are for the integer part of the speed and the last byte is for the fractional part. The fractional part has 1 digit.

- *Longitude*

  The longitude of the current position. 4 bytes in total. The device converts the longitude to an integer with 6 implicit decimals and reports this integer in HEX format. If the value of the longitude is negative, it is represented in Two's Complement format.

| **Longitude (121.390847)** | **121390847** | | | |
|---|---|---|---|---|
| HEX | 07 | 3C | 46 | FF |

- *Latitude*

  The latitude of the current position. 4 bytes in total. The device converts the latitude to an integer with 6 implicit decimals and reports this integer in HEX format. If the value of the latitude is negative, it is represented in Two's Complement format.

| **Latitude (31.164503)** | **31164503** | | | |
|---|---|---|---|---|
| HEX | 01 | DB | 88 | 57 |

- *Altitude*

  The altitude from GNSS. If the altitude is negative, it is represented in Two's Complement format. Unit: meter.

- *GNSS UTC Time*

  UTC time obtained from the GNSS chip. 7 bytes in total. The first 2 bytes are for year, and the other 5 bytes are for month, day, hour, minute and second respectively.

| **GNSS UTC Time** | **2011** | | **07** | **14** | **08** | **24** | **13** |
|---|---|---|---|---|---|---|---|
| HEX | 07 | DB | 07 | 0E | 08 | 18 | 0D |

- *Current Mileage*

  3 bytes in total. The first 2 bytes are for the integer part of the current mileage and the last byte is for the fractional part. The fractional part has 1 digit.

| **Current Mileage** | **0** | | **0** |
|---|---|---|---|
| HEX | 00 | 00 | 00 |

- *Total Mileage*

  5 bytes in total. The first 4 bytes are for the integer part of the total mileage and the last byte is for the fractional part. The fractional part has 1 digit.

| **Total Mileage** | **0** | | | | **0** |
|---|---|---|---|---|---|
| HEX | 00 | 00 | 00 | 00 | 00 |

- *Total Hour Meter Count*

  6 bytes in total. The first 4 bytes represent the hour part, the fifth byte represents the minute part, and the sixth byte represents the second part.

| **Total Hour Meter Count** | **0** | | | | **0** | **0** |
|---|---|---|---|---|---|---|
| HEX | 00 | 00 | 00 | 00 | 00 | 00 |

- *CAN Data*

  Please refer to the **+RESP:GTCAN** report in hex format. The <CAN Data> contains the following

fields:
From <CANBUS Device State> up to <GNSS Accuracy>, but excluding <GNSS Accuracy>. This field can be analyzed as per the **+RESP:GTCAN** report.

### 5.2.2 FRI(Fixed Report Information)

The location report message **+RESP:GTFRI** in HEX format is as follow.

Example:
2B5253500782F60FFF0000000200BB8020040803080F475635384C41550000030022000C100101
0000000000006206FB404401E5D45C07E903140B1C2004600000550B0E9E30A50000000100000
000010000000000000000000001000FFFFF000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000FFFF0000000007E903140B1C2104641A590D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +RSP |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | RSP Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Report ID \| Report Type | 1 | |
| | Number | 1 | 1 - 15 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved\|GNSS Trigger Type(Optional) | 1 | 00\|00 - 04 |
| | Current Mileage | 3 | 0.0 - 65535.0(km) |
| | Total Mileage | 5 | 0.0 - 4294967.0(km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *GNSS Trigger Type*

The trigger type of GNSS point has the following meanings.

- **00** - Time point
- **01** - Corner point
- **02** - Distance point
- **03** - Mileage point
- **04** - Optimum point (time & mileage)

### 5.2.3 LBC(Location By Call Alarm)

If the parameter <Location Request Mask> is enabled by the command AT+GTCFG, the device will get and send the current position to the backend server via the message **+RESP:GTLBC** when there is an SMS. The message **+RESP:GTLBC** in HEX format is as below.

Example:
2B5253500382F61FFF0000000201148020040803080F475635384C415500002EB8000022000C00
7017855503680F0101000000000000007306FB412E01E5D46507E90315013827046000000550B0E9E
30A5000000000000000000000000000000000000001207FFFFF00000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000FFFF00000000000000000000000000000000

000000FFFFFF0000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000007E9031501382713F9FD
650D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +RSP |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | RSP Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Report ID \| Report Type | 1 | |
| | Number Length \| Number Type | 1 | |
| | Phone Number | <=10 | |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0(km) |
| | Total Mileage | 5 | 0.0 - 4294967.0(km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Number Length / Number Type*

The high nibble is for <Number Length> and the low nibble is for <Number Type>.
<Number Length> is the total number of bytes which is equal to the length of <Phone Number> in bytes plus the length of the parameter <Number Length / Number Type>.
<Number Type> indicates if there is a "+" sign before the phone number. 1 means "with the sign", and 0 means "without the sign".

| Message | Number Length | Number Type |
|---|---|---|
| HEX | 7 | 0 |

✧ *Phone Number*

No more than 10 bytes. In each byte, use the high nibble and low nibble to represent one digit of the phone number respectively.
If there is no digit for the last low nibble to represent, fill it with 0xF.

| Phone Number (02154450293) | 02 | 15 | 44 | 50 | 29 | 3 |
|---|---|---|---|---|---|---|
| HEX | 02 | 15 | 44 | 50 | 29 | 3F |

### 5.2.4 SOS(SOS Alarm Notification)

If the SOS function is enabled, the device will send the message **+RESP:GTSOS** to the backend server when the corresponding digital input port triggers SOS. The location report message **+RESP:GTSOS** in HEX format is as below.

Example:

2B5253500A82F61FFF00000002010E8020040803080F475635384C415500002EB3030022000C10
0001010000000000000007306FB412E01E5D46507E9031501392404600000550B0E9E30A50000000
0000000000000000000000000000001207FFFFF000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000FFFF0000000000000000000000000000000000000FFFFF

F000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000007E9031501392414001E840D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +RSP |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | RSP Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Report ID \| Report Type | 1 | |
| | Reserved | 1 | 00 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | |
| | Current Mileage | 3 | 0.0 - 65535.0(km) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Total Mileage | 5 | 0.0 - 4294967.0(km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

### 5.2.5 IDA(ID Authentication Alarm)

According to <Mode> of the command AT+GTIDA, the reporting mode of **+RESP:GTIDA** varies.

If <Mode> is set to 1 or 2, **+RESP:GTIDA** will be reported according to the <Report Mode> setting. If <Mode> is set to 0, **+RESP:GTIDA** will always be reported without checking the status of ID authorization.
The location report message **+RESP:GTIDA** in HEX format is as below.

Example:
2B5253501182F61FFF0000000201118020040803080F475635384C415500002EB8010022000C00
021203010101000000000000007306FB412E01E5D46507E903150200290460000055 0B0E9E30A50
0000000000000000000000000000000000001207FFFFF000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000FFFF0000000000000000000000000000000000000
0FFFFFF000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000007E9031502002A14103B060D
0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +RSP |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | RSP Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Report ID \| Report Type | 1 | 00 |
| | ID Length | 1 | 6 |
| | ID | <=0E | ASCII (not including '=' and ',') |
| | ID Report Type | 1 | 0-3 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *ID*

The ID that is currently read.

✧ *ID Report Type*

The type of reported ID.

- **0** - IDA function is disabled.
- **1** - The ID is authorized.
- **2** - The ID has logged out.
- **3** - The ID is unauthorized.

### 5.2.6 ERI(Expand Fixed Report Information)

If **+RESP:GTERI** is enabled, the device will send the message **+RESP:GTERI** to the backend server instead of **+RESP:GTFRI**.

The message **+RESP:GTERI** in HEX format is as below.

Example:
2B5253501202FE0FFF00000100008D80200406000604475635384C415500000101210C10000100
0103000400000063011F44555F31303033363100F022A2143F36010DAC0009000101000000000
0006C06FB403101E5D3F707E8050801271104600000550B0E9E30A50000000000000000000000
0000000000000000000000007E8050801271206E7AE8D0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +RSP |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | RSP Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | ERI Mask | 4 | 00000000 - FFFFFFFF |
| | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Report ID \| Report Type | 1 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Reserved | 1 | 00 |
| | Bluetooth Accessory Data(Optional) | | |
| | Number | 1 | 1 - 15 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved\|GNSS Trigger Type(Optional) | 1 | 00\|00 - 04 |
| | Current Mileage | 3 | 0.0 - 65535.0(km) |
| | Total Mileage | 5 | 0.0 - 4294967.0(km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ **Bluetooth Accessory Data(Optional)**

| Fields | Length | Range/Format |
|---|---|---|
| Bluetooth Accessory Number | 1 | 0 - 10 |
| Index | 1 | 0 - 9 |
| Accessory Type | 1 | 0-2 \| 6-8 \| 10-13 |
| Accessory Model | 1 | 0-5 |
| Raw Data Length | 2 | 0000 - FFFF |
| Raw Data(Optional) | <=4 | |
| Accessory Append Mask | 2 | 0 - FFFF |
| Accessory Name(Optional) | <=21 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |

| Fields | Length | Range/Format |
|---|---|---|
| Accessory MAC(Optional) | 6 | 000000000000 - FFFFFFFFFFFF |
| Accessory Status(Optional) | 1 | 0 - 1 |
| Accessory Battery Voltage(Optional) | 2 | 0 - 5000(mV) |
| Accessory Temperature(Optional) | 1 | -40 - 80 ( ℃ ) \| 0x7F |
| Accessory Humidity(Optional) | 1 | 0 - 100%(rh) |
| Accessory Output Status(Optional) | 1 | 00 - 03 |
| Accessory Digital Input Status(Optional) | 1 | 00 - 01 |
| Accessory Analog Input Voltage(Optional) | 2 | 0 - 32000(mV) |
| Accessory Mode(Optional) | 1 | 0 - 10 |
| Accessory Event(Optional) | 1 | 0 - 2 |
| Tire Pressure(Optional) | 2 | 0 - 500(kPa) |
| Timestamp(Optional) | 7 | YYYYMMDDHHMMSS |
| Enhanced Temperature(Optional) | 2 | -40.00 - 80.00( ℃ ) \| 0x7FFF |
| Magnet ID(Optional) | 1 | 00 - FF |
| MAG Event Counter(Optional) | 2 | 0 - 32767 |
| Magnet State(Optional) | 1 | 0 - 1 |
| Accessory Battery Percentage(Optional) | 1 | 0 - 100(%) |
| Relay State(Optional) | 1 | 0 - 1 |

- *Bluetooth Accessory Number*

  It indicates the number of accessories connected with the device .
- *Index*

  The index of the Bluetooth accessory.
  - The index of Bluetooth accessory defined in AT+GTBAS which triggers the **+RESP:GTBAA** message.
- *Accessory Type*

  The type of the Bluetooth accessory which is defined in the <Index>. The following is supported now:
  - **0** - No Bluetooth accessory
  - **1** - Escort sensor
  - **2** - Beacon temperature sensor
  - **6** - Beacon multi-functional sensor
  - **7** - Techonoton accessory
  - **8** - External Input Output Bluetooth accessory

- **10** - Mechatronics Bluetooth accessory
- **11** - Magnet sensor
- **12** - BLE TPMS sensor
- **13** - Relay sensor

- *Accessory Model*

  The model of the Bluetooth accessory which is defined in AT+GTBAS.
- *Raw Data Length*

  It indicates the length of <Raw Data>.
- *Raw Data*

  The raw data is read from Bluetooth accessory. It varies depending on <Accessory Type> and <Accessory Model>.
- *Accessory Name*

  The name of the Bluetooth accessory. It ends with 0x00.
- *Accessory MAC*

  The MAC address of the Bluetooth accessory.
- *Accessory Status*

  A numeral to indicate whether the accessory is available.
  - **0** - The accessory is not available.
  - **1** - The accessory is available.
- *Accessory Battery Voltage*

  It indicates the voltage of the battery in Bluetooth accessory.
- *Accessory Temperature*

  Temperature data of Bluetooth accessory. If the temperature is not obtained, the parameter is reported as 0x7F.
- *Accessory Humidity*

  Humidity data for the Bluetooth accessory.
- *Enhanced Temperature*

  It instructs Bluetooth accessories to measure temperature with high precision. If the temperature is not obtained, the parameter is reported as 0x7FFF.

| Temperature (16.66) | 1666 | |
|---|---|---|
| HEX | 06 | 82 |

Note

The current temperature value. A total of 2 bytes. The temperature value is converted to an integer with 2 implicit decimals, and the integer is reported in HEX format. If the temperature value is negative, it is expressed in 2's complement format.

- *Magnet ID*

  ID corresponding to different magnet sensors.
- *MAG Event Counter*

  The total number of times detected by the magnet sensor.
- *Magnet State*

  The state of the two parts of the magnet sensor.
  - **0** - Separate
  - **1** - Closed

✧ *Accessory Battery Percentage*

Percentage of Bluetooth accessory's battery power.

✧ *Relay State*

The current state of the relay sensor.

✧ *GNSS Trigger Type*

The trigger type of GNSS point has the following meanings.

- **00** - Time point
- **01** - Corner point
- **02** - Distance point
- **03** - Mileage point
- **04** - Optimum point (time & mileage)

Note

The word "Optional" means the item is controlled by the parameter <ERI Mask>.

### 5.2.7 GIN|GOT(Geo-Fence Information)

The messages **+RESP:GTGIN** and **+RESP:GTGOT** in HEX format are as below.

Example:

2B5253501482F61FFF0000000201168020040803080F475635384C415500002EB5010022000C00
01000000000000000101010000000000007306FB412E01E5D46507E9031502043404600000550
B0E9E30A50000000000000000000000000000000000000001207FFFFF00000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000FFFF000000000000000000000000000
00000000000FFFFFF0000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000007E903150204341
4273BB00D0A

2B5253501582F61FFF0000000201168020040803080F475635384C415500002EAD010021000C00
01000000000000000101010000000000007306FB412E01E5D46507E9031502051604600000550
B0E9E30A50000000000000000000000000000000000000001207FFFFF00000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000FFFF000000000000000000000000000
00000000000FFFFFF0000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000007E903150205161
42B9F9D0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +RSP |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | RSP Expansion Mask | 4 | 00000000 - FFFFFFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Supply Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Area Type | 1 | 0 - 1 |
| | Mask Group | 1 | 00 - 1F |
| | Area Mask Group 1 | 8 | 0000000000000001 - FFFFFFFFFFFFFFFF |
| | Area Mask Group 2 | 8 | 0000000000000001 - FFFFFFFFFFFFFFFF |
| | Area Mask Group 3 | 8 | 0000000000000001 - FFFFFFFFFFFFFFFF |
| | Area Mask Group 4 | 8 | 0000000000000001 - FFFFFFFFFFFFFFFF |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Mask Group*

Bitwise mask to determine whether to report <Area Mask>. Bit 0 is for Area Mask Group 1 and Bit 1 is for Area Mask Group 2. 1 means "Report the information", and 0 means "Do not report the information".

✧ *Area Mask Group*

Bitwise mask for trigger condition composition of the corresponding PEO or GEO ID. Each bit, from the lowest bit to the highest bit, represents the logic status of the corresponding ID to trigger the entering or exiting event. 1 means that the event of the ID set is triggered, and 0 means the event of the ID set is not triggered. In a group, if no event of ID is triggered, the bitwise mask will be null.

- *Area Mask Group 1*
  From bit 0 to bit 63 represents ID 0 to 63.
- *Area Mask Group 2*
  From bit 0 to bit 34 represents ID 64 to 127.
- *Area Mask Group 3*
  From bit 0 to bit 63 represents ID 128 to 191.
- *Area Mask Group 4*
  From bit 0 to bit 63 represents ID 192 to 199.

## 5.2.8 SPE(Speed Alarm inside the Geo-Fence)

Example:

2B5253506500FE1FBF00668020040803060456 3F0E0602284904003041010022 0C010000000101
001202010D001F06FAF56F01E5A52607E80508080626046000 01DF5C05A88B0B0000090600000
000000000000000000000000007E80508100628 82314B2F0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +RSP |
| | Message Type | 1 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | RSP Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Report ID \| Report Type | 1 | |
| | Group ID | 1 | 0 - 19 |
| | Reserved | 2 | 0000 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | |
| | Current Mileage | 3 | 0.0 - 65535.0(km) |
| | Total Mileage | 5 | 0.0 - 4294967.0(km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Group ID*

A numeral to indicate in which group the speed alarm is triggered.

## 5.3 Device Information Reports

This section describes the message related to device generic information. Please refer to the details below.

### 5.3.1 INF(Device Information)

Information report messages include the messages shown in the default value of the following table. However, only **+RESP:GTINF** includes all the items while other messages only include information of items related to themselves.

Example:

2B494E4601FFFFD24D003400FD475635384C4155008020040803080F010200000000000000000000
0000000000000000000001000022000C0007E903150206090000007F00001E000258000000902E
DF0000008986001512328690662 44F0004000000010460000055 0B0E9E30A5004F0A0128020F0
41B07270824091D101F1A161B271E2500100120061A071E0B200C1C101D171E19171C1C201522
232518271C28192B213B1E0006332E302E31370733323031363132000000020147563 5384C415
525494D45490060E93B2073B4000143414E2D4C6F6769737469635F33323031363132 00F60C94
0CE03301000000000307E9031502060A1431FD900D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +INF |
| | Message Type | 1 | |
| | Report Mask | 2 | 0000 - FFFF |
| Body | INF Expansion Mask | 2 | 0000 - FFFF |
| | INF Expansion2 Mask | 2 | 0000 - FFFF |
| | Length | 2 | |
| | Unique ID | 8 | IMEI/Device Name |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Firmware Version | 2 | 0000 - FFFF |
| | +RESP:GTVER | | |
| | +RESP:GTIOS | | |
| | +RESP:GTGPS | | |
| | +RESP:GTBAT | | |
| | +RESP:GTCID | | |
| | +RESP:GTCSQ | | |
| | +RESP:GTTMZ | | |
| | +RESP:GTGSM | | |
| | +RESP:GTGSV<br>+RESP:GTRSV<br>+RESP:GTBSV<br>+RESP:GTASV | | |
| | +RESP:GTCVN | | |
| | +RESP:GTCSN | | |
| | +RESP:GTCML | | |
| | +RESP:GTSCS | | |
| | +RESP:GTBTI | | |
| | Network Type | 1 | 00 - 03 |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Message Type*

The ID of a specific information report message.

| Message | ID |
|---|---|
| +RESP:GTINF | 1 |
| +RESP:GTGPS | 2 |
| +RESP:GTCID | 4 |
| +RESP:GTCSQ | 5 |
| +RESP:GTVER | 6 |
| +RESP:GTBAT | 7 |
| +RESP:GTIOS | 8 |
| +RESP:GTTMZ | 9 |
| +RESP:GTGSM | 10 |
| +RESP:GTGSV | 11 |

| Message | ID |
|---|---|
| +RESP:GTCVN | 13 |
| Reserved | ... |
| +RESP:GTCSN | 20 |
| +RESP:GTRSV | 21 |
| +RESP:GTBSV | 22 |
| +RESP:GTBTI | 24 |
| +RESP:GTCML | 26 |
| Reserved | ... |
| +RESP:GTSCS | 28 |
| Reserved | ... |
| +RESP:GTASV | 37 |

- *Report Mask*

  Please refer to <+INF Mask> in AT+GTHRM.

- *Unique ID*

  If Bit 1 of <+INF Mask> is 0, the IMEI of the device is used as the unique ID of the device. IMEI is a 15-digit string. In the HEX format message, each 2 digits are encoded into one byte as an integer.

| IMEI | 13 | 57 | 90 | 24 | 68 | 11 | 22 | 0 |
|---|---|---|---|---|---|---|---|---|
| HEX | 0D | 39 | 5A | 18 | 44 | 0B | 16 | 00 |

If Bit 1 of <+INF Mask> is 1, the device name is used as the unique ID of the device.
Device name is an 8-byte string, please refer to <Device Name> in AT+GTCFG.
If the length of <Device Name> is more than 8 bytes, the device will only acquire the first 8 bytes.
In the HEX format message, each byte is encoded into one byte as an integer. If the device name is less than 8 bytes, the remaining bytes are set to 0.

| Device Name | q | u | e | c | l | i | n | k |
|---|---|---|---|---|---|---|---|---|
| HEX | 71 | 75 | 65 | 63 | 6C | 69 | 6E | 6B |

- *Device Type, Protocol Version, Firmware Version*

  If <Message Type> is 6 (**+RESP:GTVER**) in the message, Bit 2 (<Device Type>)/Bit 3 (<Protocol Version>)/Bit 4 (<Firmware Version>) in <+INF Mask> will be forced to 1, and these fields will be always present in the HEX report of **+RESP:GTVER**.

**+RESP:GTVER**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTVER | Hardware Version | 2 | 0000 - FFFF |
| | Reserved | 2 | 0000 |
| | Reserved | 2 | 0000 |

**IOS**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTIOS | Reserved | 1 | 00 |
| | Reserved | 2 | 0000 |
| | Reserved | 2 | 0000 |
| | Reserved | 2 | 0000 |
| | Reserved | 1 | 00 |
| | Reserved | 2 | 0000 |
| | Reserved | 2 | 0000 |
| | Reserved | 2 | 0000 |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Pin Mask | 1 | 00 - 0F |

**GPS**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTGPS | Motion Status | 1 | 0x11 \| 0x12 \| 0x21 \| 0x22 \| 0x41 \| 0x42 \| 0x16 \| 0x1A |
| | Reserved | 1 | 00 |
| | Satellites in Use | 1 | 0 - 72 |
| | Enable Power Saving \| OWH Mode \| Outside Working Hours \| AGPS | 1 | 00 - FF |
| | Last Fix UTC Time | 7 | YYYYMMDDHHMMSS |
| | Reserved | 1 | 00 |
| | FRI Discard No Fix | 1 | 0\|1 |
| | Report Item Mask | 2 | |
| | IGN Interval | 3 | |
| | IGF Interval | 3 | |
| | Reserved | 2 | 0000 |
| | Reserved | 1 | 00 |

✧ *Enable Power Saving, OWH Mode, Outside Working Hours, AGPS*

The highest bit, or Bit 7, is reserved; Bit 5 and Bit 6 are for <Enable Power Saving>; Bit 4 and Bit 3 are for <OWH Mode>; Bit 2 is for <Outside Working Hours>, and Bit 0 and Bit 1 are for <AGPS>. <Outside Working Hours> is used to indicate whether the device is currently outside working hours. 1 means "Outside working hours".

- **Bit 0 - 1** - AGPS
- **Bit 2** - Outside Working Hours
- **Bit 3 - 4** - OWH Mode

- **Bit 5 - 6** - Enable Power Saving

**BAT**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTBAT | Power Supply \| Backup Battery On \| Charging \| LED State \| Power Mode | 1 | 00 - FF |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Backup Battery Voltage | 2 | 0 - 4500(mV) |
| | Backup Battery Percentage | 1 | 0 - 100 |

✧ *Power Supply, Backup Battery On, Charging, LED State, Power Mode*

The highest bit, or Bit 7, is for <Power Supply> which indicates whether the external power supply is connected to the device. Bit 6 is for <Backup Battery On> which indicates whether the backup battery is working. Bit 5 is for <Charging> which indicates whether the backup battery is currently charging. Bit 4 is for <LED State>, indicating at least one LED indicator is blinking or steady on when it is 1, and all LEDs are off when it is 0. Bit 0 is for <Power Mode>.

- **Bit 0** - Power Mode
- **Bit 1 - 3** - Reserved
- **Bit 4** - LED State
- **Bit 5** - Charging
- **Bit 6** - Backup Battery On
- **Bit 7** - Power Supply

**CID**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTCID | ICCID | 10 | ICCID |

✧ *ICCID*

ICCID is a 20-digit string. In the HEX format message, every 4 bits are used to represent one digit of the 20 digits of the ICCID.

| ICCID | 89 | 86 | 00 | 00 | 09 | 09 | 17 | 21 | 49 | 53 |
|---|---|---|---|---|---|---|---|---|---|---|
| HEX | 89 | 86 | 00 | 00 | 09 | 09 | 17 | 21 | 49 | 53 |

**CSQ**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTCSQ | CSQ RSSI\|RSRP | 1 | 0-31 \| 99 \| 0-97 \| 255 |
| | CSQ BER | 1 | 0 - 7 \| 99 |

**TMZ**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTTMZ | Time Zone Offset Sign \| Enable Daylight Saving \| Network Time Checking | 1 | 00 - FF |
| | Time Zone Offset | 2 | HHMM |

✧ *Time Zone Offset Sign, Enable Daylight Saving, Network Time Checking*

Bit 2 is for <Network Time Checking> which indicates whether to check GNSS UTC time with network time. Bit 1 is for <Enable Daylight Saving> which indicates whether the daylight saving function is currently enabled. Bit 0 is for <Time Zone Offset Sign> which indicates the positive or negative offset of the local time from UTC time. 1 means "negative offset".

**GSM**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTGSM | GIR Trigger Type | 1 | |
| | Cell Number | 1 | |
| | MCC | 2 | |
| | MNC | 2 | |
| | LAC | 2 | |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | RX Level | 1 | |

✧ *GIR Trigger Type*

The ID of fix type. It indicates what kind of GNSS fix this cell information is for.

- **INF** - This cell information is for INF request.
- **SOS** - This cell information is for SOS request.
- **RTL** - This cell information is for RTL request.
- **LBC** - This cell information is for LBC request.
- **TOW** - This cell information is for TOW request.
- **FRI** - This cell information is for FRI request.
- **GIR** - This cell information is for sub command **C** in the **AT+GTRTO** command.
- **ERI** - This cell information is for ERI request.

| Fix Type | ID |
|---|---|
| INF | 0 |
| SOS | 1 |
| RTL | 2 |
| LBC | 3 |
| TOW | 4 |
| FRI | 5 |
| GIR | 6 |

| Fix Type | ID |
|---|---|
| ERI | 7 |

✧ *Cell Number*

The number of cells. It also indicates the number of cell information groups. One cell information group consists of MCC, MNC, LAC, and Cell ID.

**GSV**

**RSV**
**BSV**
**ASV**

| Messages | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTGSV<br>+RESP:GTRSV<br>+RESP:GTBSV<br>+RESP:GTASV | SV Count | 1 | |
| | SV ID | 1 | |
| | SV Power | 1 | |
| | … | | |
| | SV ID | 1 | |
| | SV Power | 1 | |

**CVN**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTCVN | CAN100 SW Version Length | 1 | 0 - 10 |
| | CAN100 SW Version | <=10 | |

**CSN**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTCSN | CAN100 Serial Number length | 1 | 0 - 10 |
| | CAN100 Serial Number | <=10 | |

**CML**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTCML | CAN100 Car Model ID | 2 | 0x0000 - 0xFFFF |
| | CAN100 Car Name Length | 1 | 0 - 50 |
| | CAN100 Car Name | <=50 | |
| | CAN100 Sync Status | 1 | 1 - 4 |

**SCS**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTSCS | Self Calibration Status | 1 | 0-2 |
| | X_Forward | 1 | -100 - 100 |
| | Y_Forward | 1 | -100 - 100 |
| | Z_Forward | 1 | -100 - 100 |
| | X_Side | 1 | -100 - 100 |
| | Y_Side | 1 | -100 - 100 |
| | Z_Side | 1 | -100 - 100 |
| | X_Vertical | 1 | -100 - 100 |
| | Y_Vertical | 1 | -100 - 100 |
| | Z_Vertical | 1 | -100 - 100 |

**BTI**

| Message | Fields | Length | Range/Format |
|---|---|---|---|
| +RESP:GTBTI | Bluetooth Name | <=20 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Bluetooth MAC Address | 6 | 000000000000 - FFFFFFFFFFFF |
| | Bluetooth State | 1 | 0\|1 |
| | Connected Device Number | 1 | 0 - 11 |
| | Connected Device Name | <=20 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Connected Device MAC | 6 | 000000000000 - FFFFFFFFFFFF |
| | Role | 1 | 0 \| 1 |
| | Real-Time State | 1 | 0 - 1 |
| | Ghost Battery Percentage | 1 | 0 - 100 |
| | Ghost Status | 2 | 0000 - FFFF |

Note

<Connected Device Name> ends with 0x00.

## 5.3.2 ATI(Version Information)

After the device receives the command AT+GTRTO to get the basic device information, it will send the information to the backend server via the message **+RESP:GTATI**.

Example:
2B41544938802004080308 0F563F0E0608493102001C1031080F010204020104EFBAA400000000
00000000007E9031502351D158B191E0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Message Header | 4 | +ATI |
| Body | Length | 1 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI |
| | ATI Mask | 4 | 00000000 - FFFFFFFF |
| | Firmware Version | 2 | '0' - '9' 'A' - 'F' |
| | Hardware Version | 2 | '0' - '9' 'A' - 'F' |
| | BT Firmware Version | 2 | 0000 - FFFF |
| | BT Boot Firmware Version | 2 | 0000 - FFFF |
| | Flash ID | 2 | 0000 - FFFF |
| | Sensor ID | 1 | 00 - FF |
| | Modem IMEI | 8 | IMEI |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

- *ATI Mask*

  This mask is set by the AT+GTRTO command and used to control parameter fields in **+ATI** message.

## 5.4 Event Reports

This section describes the message related to certain events. Please refer to the details below.

### 5.4.1 Generic Event Report

- **PNA** (Power-on Report)
- **PFA** (Power-off Report)
- **MPN** (Main Power Connection)
- **MPF** (Main Power Disconnection)
- **BTC** (Battery Starts Charging)
- **STC** (Stop Charging Notification)
- **STT** (Device Status Notification)
- **PDP** (GPRS Connection Establishment Report)
- **IDN** (Enter Idle State)
- **STR** (Enter Start State)

- ✧ **STP** (Enter Start State)
- ✧ **LSP** (Enter Start State)

Example:

2B4556540102FE1FFF00678020040803080F475635384C41550000000000100410001000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000007E90315020A38145A7B5D0D0A

2B4556540202FE1FFF01078020040803080F475635384C415500002EBD0100220C010100000000
00007006FB3FBF01E5D45407E90315020A1F04600000550B0E9E30A5000000000000000000000
00000000000000001207FFFFF00000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000FFFF00000000000000000000000000000000000000FFFFFF0000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000007E90315020A21145850800D0A

2B4556540302FE1FFF01078020040803080F475635384C415500002EE201002100010000000000
00000000000000000000000000000000000004600000550B0E9E30A5000000000000000000000
00000000000000000207FFFFF00000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000FFFFFF0000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000007E90315020A3A145D05150D0A

2B4556540402FE1FFF01078020040803080F475635384C4155000000000000120C010100000000
0000A706FB413201E5D48107E90315020F0404600000550B0E9E30A5000000000000000000000
00000000000000001207FFFFF00000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000FFFF00000000000000000000000000000000000000FFFFFF0000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000007E90315020F0514969C420D0A

2B4556540702FE1FFF01078020040803080F475635384C415500002EC20100220C010100000000
0000A706FB413201E5D48107E90315020F2504600000550B0E9E30A5000000000000000000000
00000000000000001207FFFFF00000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000FFFF00000000000000000000000000000000000000FFFFFF0000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000007E90315020F27149C145C0D0A

2B4556540802FE1FFF01078020040803080F475635384C415500002F110100220C010100000000
0000A706FB413201E5D48107E90315020F3704600000550B0E9E30A5000000000000000000000
00000000000000001207FFFFF00000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000

00000000000000000FFFF00000000000000000000000000000000000000FFFFFF0000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000007E90315020F39149E66E00D0A

22B4556540902FE1FFF01078020040803080F475635384C415500002EC80100220C01010000000
000007306FB412E01E5D46507E9031502073004600000550B0E9E30A50000000000000000000
000000000000000001207FFFFF0000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000FFFF00000000000000000000000000000000000000FFFFFF000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000007E90315020730143CDAB70D0A

2B4556540C02FE1FFF006780200408030604475635384C415500002F010100220C010100000000
00006206FB3FF201E5D58707E80508032916000000000000000000000000000000000000000000
000000000000000000000000007E80508032917077964940D0A

2B4556541002FE1FFF01078020040803080F475635384C415500002EF60100210C010100000000
0000A706FB413201E5D48107E9031502121D04600000550B0E9E30A500000000000000000000
00000000000000001207FFFFF00000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000FFFF00000000000000000000000000000000000000FFFFFF0000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000007E9031502121E14AA02040D0A

2B4556541700FE1FBF00628020040803040D563F0E06014F2C0500300E0100220C0101002E0601
0D002B06FA975901E5809B07E70C1A053B0404600000550B0E9E30A5000004080000007F09000
00000000000000007E70C1A0D3B0500B9FA960D0A

2B4556541800FE1FBF00628020040803040D563F0E06014F2C05002FFC0100210C010100000001
0E002706FA8BDD01E57EE007E70C1A060020046000000550B0E9E30A5000005010000008002000
00000000000000007E70C1A0E002100BFE1D70D0A

2B4556541D00FE1FBF00628020040803040D563F0E06014F2C050030160100210C01010000000
10E002706FA8BDD01E57EE007E70C1A06021F04600000550B0E9E30A500000501000000800200
000000000000000007E70C1A0E022000C4FD7D0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Message Type*

The ID of a specific event report message.

| Message | ID |
|---|---|
| +RESP:GTPNA | 1 |
| +RESP:GTPFA | 2 |
| +RESP:GTMPN | 3 |
| +RESP:GTMPF | 4 |
| +RESP:GTBPL | 6 |
| +RESP:GTBTC | 7 |
| +RESP:GTSTC | 8 |
| +RESP:GTSTT | 9 |
| +RESP:GTPDP | 12 |
| +RESP:GTIGN | 13 |
| +RESP:GTIGF | 14 |
| +RESP:GTUPD | 15 |
| +RESP:GTIDN | 16 |
| +RESP:GTIDF | 17 |
| +RESP:GTDAT | 18 |
| +RESP:GTJDR | 20 |
| +RESP:GTGSS | 21 |
| +RESP:GTSTR | 23 |
| +RESP:GTSTP | 24 |
| +RESP:GTCRA | 25 |
| +RESP:GTDOS | 27 |
| +RESP:GTGES | 28 |
| +RESP:GTLSP | 29 |
| +RESP:GTJDS | 32 |
| +RESP:GTRMD | 33 |
| +RESP:GTUPC | 40 |
| +RESP:GTCLT | 41 |
| +RESP:GTCFU | 42 |
| +RESP:GTVGN | 45 |
| +RESP:GTVGF | 46 |
| +RESP:GTASC | 47 |
| +RESP:GTPNR | 48 |
| +RESP:GTPFR | 49 |
| +RESP:GTHBE | 51 |
| +RESP:GTBCS | 52 |
| +RESP:GTBDS | 53 |

| Message | ID |
|---|---|
| +RESP:GTBAA | 65 |
| +RESP:GTBID | 67 |
| +RESP:GTBAR | 70 |
| +RESP:GTSVR | 73 |
| +RESP:GTDOM | 75 |
| +RESP:GTEUC | 84 |
| +RESP:GTRTP | 86 |
| +RESP:GTUDF | 104 |

- *Report Mask*

  Please refer to <+EVT Mask> in AT+GTHRM.

- *Unique ID*

  If Bit 6 of <+EVT Mask> is 0, the IMEI of the device is used as the unique ID of the device. IMEI is a 15-digit string. In the HEX format message, each 2 digits are encoded into one byte as an integer.

| IMEI | 13 | 57 | 90 | 24 | 68 | 11 | 22 | 0 |
|---|---|---|---|---|---|---|---|---|
| HEX | 0D | 39 | 5A | 18 | 44 | 0B | 16 | 00 |

  If Bit 6 of <+EVT Mask> is 1, the device name is used as the unique ID of the device. Device name is an 8-byte string, please refer to <Device Name> in AT+GTCFG. If the length of <Device Name> is more than 8 bytes, the device will only acquire the first 8 bytes. In the Hex format message, each byte is encoded into one byte as an integer. If the device name is less than 8 bytes, the remaining bytes are set to 0.

| Device Name | q | u | e | c | l | i | n | k |
|---|---|---|---|---|---|---|---|---|
| HEX | 71 | 75 | 65 | 63 | 6C | 69 | 6E | 6B |

- *Satellites in Use*

  Number of satellites being used for tracking. The high nibble is reserved and the low nibble is valid. But if Bit 1 of <EVT Expansion Mask> is set to 1, this parameter would be expanded to two bytes, the high byte is reserved and the low byte is valid.

### 5.4.2 BPL(Backup Battery Low Alarm)

The event report message **+RESP:GTBPL** in HEX format is as below.

Example:

2B4556540600FE1FBF00648020040803040C563F0E06023629020A00000000110C0E7401010000
000000006506FB400D01E5D44D07E70C1602120B04600000550B0E9E30A500000000000000000
000000000000000000007E70C160A120C08A2D6E00D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Backup Battery Voltage | 2 | 0 - 4500(mV) |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.4.3 JDR(Jamming Indication Notification)

Jamming indication. The event report message **+RESP:GTJDR** in HEX format is as below.

Example:
2B4556541400FE1FBF006380200408030604563F0E06084947060030FE01002100030100000000
000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000007E8050808042A08399E820D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Jamming Net | 1 | 1-5 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

### 5.4.4 JDS(Jamming Indication Notification)

Jamming indication. The event report message **+RESP:GTJDS** in HEX format is as below.

Example:

2B4556542000FE1FBF0064802004080306045 63F0E060849470600312401002100020301000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000007E8050808002C0834D14B0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Jamming Status | 1 | 1 \| 2 |
| | Jamming Net | 1 | 1-5 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Jamming Status*

The current jamming status of the device.

- **1** - Quit the jamming state.
- **2** - Enter the jamming state.

## 5.4.5 IGN | IGF

✧ IGN - Ignition-on Report
✧ IGF - Ignition-off Report

The HEX format of the event report messages **+RESP:GTIGN** and **+RESP:GTIGF** is as follows. For these two types of messages, the <Current Mileage> and <Total Mileage> fields will always be present regardless of the <+EVT Mask> setting.

Example:

```
2B4556540D02FE1FFF010B8020040803080F475635384C41550000000001002100000000020100
00000000000000000000000000000000000000000000004600000550B0E9E30A50000000000000
0000000000000000000000000207FFFFF0000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000FFFFFF00000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000007E90315020A39145B78D30D0A

2B4556540E02FE1FFF010B8020040803080F475635384C415500002EB90000120C000000D40101
0000000000008D06FB3F7501E5D40B07E90315020D0A04600000550B0E9E30A500000000000000
0000000000000000000000001207FFFFF0000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000FFFF000000000000000000000000000000000000000FFFFFF00000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000007E90315020D0B1476B9DA0D0A
```

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Duration of Ignition On or Ignition Off | 4 | 0 - 999999 (s) |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.4.6 VGN | VGF

- ✧ VGN - Virtual Ignition-on Report
- ✧ VGF - Virtual Ignition-off Report

The HEX format of the event report messages **+RESP:GTVGN** and **+RESP:GTVGF** is as follows. For these two messages, the <Current Mileage> and <Total Mileage> fields will always be present regardless of the <+EVT Mask> setting.

Example:
2B4556542D02FE1FFF010D8020040803080F475635384C4155003C2EE40000220C00020000003
50101000000000000AC06FB417301E5D46507E903150225330460000550B0E9E30A500000000
000000000000000000000000000001207FFFFF0000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000FFFF00000000000000000000000000000000000000FFFFFF
0000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000007E90315022535151F17000D0A

2B4556542E02FE1FFF010D8020040803080F475635384C4155003D2EE40000120C000200000019
0101000000000000AC06FB417301E5D46507E903150226100460000550B0E9E30A5000000000
000000000000000000000000000001207FFFFF000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000FFFF000000000000000000000000000000000000000FFFFFF0
0000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000007E9031502261215258C4B0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Reserved | 1 | 00 |
| | Report Type | 1 | 0-4 |
| | Duration of Ignition On or Ignition Off | 4 | 0 - 999999(s) |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

### 5.4.7 IDF(Exit Idle State)

The HEX format of the event report message **+RESP:GTIDF** is as follows.

Example:
2B4556541102FE1FFF010B8020040803080F475635384C415500002EB00000120C0000003A0101
000000000000A706FB413201E5D48107E9031502131B04600000550B0E9E30A50000000000000
0000000000000000000000001207FFFFF000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000FFFF00000000000000000000000000000000000000FFFFFF00000

00000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000007E9031502131C14B3725E0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Duration of Idling | 4 | |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.4.8 GSS(GNSS Signal Status)

The HEX format of the event report message **+RESP:GTGSS** is as follows.

Example:
2B4556541500FE1FBF00678020040803060456 3F0E0602284904003075010021000000000000001
00000A050154003706FB429901E5D4C207E8050807151C04600001DF5C027A4F1F00000906000
0000000000000000000000000007E805080F1621812779EA0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | GNSS Signal Status | 1 | 0\|1 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Reserved | 4 | 00000000 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Satellites in Use*

Number of satellites being used for tracking. The high nibble is reserved and the low nibble is valid. But If Bit 1 of <EVT Expansion Mask> is 1 this parameter would be expanded to two bytes, the high byte is reserved and the low byte is valid.

✧ *GNSS Signal Status*

0 means "GNSS signal lost or no successful GNSS fix", and 1 means "GNSS signal recovered and successful GNSS fix".

### 5.4.9 DOS(Wave Shape 1 Output Status)

The HEX format of the event report message **+RESP:GTDOS** is as follows.

Example:
2B4556541B02FE1FFF01098020040803080F475635384C4155003F2EF600011A0C010101010000
000000006806FB40B401E5D44B07E90315022713046000005 50B0E9E30A5000000000000000000

```
000000000000000000001207FFFFF0000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000FFFF00000000000000000000000000000000000000FFFFFF000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000007E90315022714152E750C0D0A
```

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Wave1 Output ID | 1 | 1 - 3 \| 7 - 8 |
| | Wave1 Output Active | 1 | 0 - 1 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.4.10 GES(Parking-Fence Information)

The HEX format of the event report message **+RESP:GTGES** is as follows.

Example:
2B4556541C02FE1FFF0113802004080308 0F475635384C415500002ED000001208000001150000
01F40000000501010000000000004606FB3E6601E5DDF907E9031506200B04600000550B0E9E3
0A500000000000000000000000000000000000001207FFFFF0000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000FFFF000000000000000000000000000000000
00000FFFFFF00000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000007E9031506200C191FBE2
D0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Trigger GEO ID | 2 | 0 - 99 |
| | Enable Trigger GEO | 1 | 0 \| 1 |
| | Trigger Mode | 1 | 0 \| 21 \| 22 |
| | Radius | 4 | 50 - 6000000(m) |
| | Check Interval | 4 | 0 \|5 - 86400(s) |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Trigger GEO ID*

The ID of Geo-Fence.

✧ *Enable Trigger GEO*

The current Parking-Fence is active or inactive.

- **0** - The current Parking-Fence is inactive.
- **1** - The current Parking-Fence is active.

## 5.4.11 RMD(Roaming State Report)

The HEX format of the event report message **+RESP:GTRMD** is as follows.

Example:

2B4556542102FE1FFF0108802004080308 0F475635384C41550015000000001A0C020101000000
0000008A06FB40DA01E5D41D07E903150218330460000055 0B0E9E30A50000000000000000000
000000000000000000001207FFFFF0000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000FFFF0000000000000000000000000000000000000000FFFFFF000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000007E9031502183414D78F120D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Roaming State | 1 | 0 - 3 |
| | Number | 1 | 1 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.4.12 CLT(CANBUS Information Alarm)

The HEX format of the event report message **+RESP:GTCLT** is as follows.

Example:
2B4556542902FE1FFF010B8020040803080F475635384C415500422EBB0001110C000000000000
0000000000000401010000000000006806FB40B401E5D44B07E9031502283804600000550B0E9
E30A500000000000000000000000000000000000001207FFFFF000000000000000000000000000
000000002000BE37D000000323800140028005A0000000000005728000000431400000F780020
000000020E00000001110000000006100001D0C2D110D00BF7FFF3F1F00000003380000000010B0
0000000001FFFFF00000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000007E9031502283915371679 0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Group ID | 1 | 00 - 19 |
| | Alarm Mask 1 | 4 | 00000000 - FFFFFFFF |
| | Alarm Mask 2 | 4 | 00000000 - FFFFFFFF |
| | Alarm Mask 3 | 4 | 00000000 - FFFFFFFF |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Group ID*

The ID of CANBUS alarm group. The CANBUS alarm function supports settings of a total of 20 groups.

✧ *Alarm Mask 1*

The alarm mask is configured in a bitwise manner. The alarm mask information is based on <Detailed Information / Indicators> of the **CAN** (ASCII)/(HEX) message.

✧ *Alarm Mask 2*

The alarm mask is configured in a bitwise manner. The alarm mask information is based on <Lights> and <Doors> of the **CAN** (ASCII)/(HEX) message.

✧ *Alarm Mask 3*

The alarm mask is configured in a bitwise manner. The alarm mask information is based on <Engine RPM> of the **CAN** (ASCII)/(HEX) message.

## 5.4.13 PNR(Power-on Reason)

Example:

2B4556543002FE1FFF00688020040803080F475635384C4155003900000100410000010000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000007E9031502143B14BA3A200D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |

| Parts | Fields | Length | Range/Format |
| --- | --- | --- | --- |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Power On Reason | 1 | 0-3 \| 6 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 000000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.4.14 PFR(Power-off Reason)

Example:
2B4556543102FE1FFF0108802004080308 0F475635384C4155000000000000120C010101000000

000000A706FB413201E5D48107E9031502133004600000550B0E9E30A500000000000000000000
00000000000000000001207FFFFF0000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000FFFF000000000000000000000000000000000000000FFFFFF00000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000007E9031502133114B88BC20D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Power Off Reason | 1 | 0-3 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.4.15 ASC(Acceleration Calibration Alarm)

The HEX format of the event report message **+RESP:GTASC** is as follows.

Example:
2B4556542F00FE1FBF006B8020040803060456 3F0E06022849040030AD0100220C630002009D0
702F99D01010012080052000906FE4FD301E5956A07E805080625140460 0001DF3A05FE5A1F00
00090600000000000000000000000000007E805080E25158057F83D0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | | | 0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | X_Forward | 1 | -100 - 100 |
| | Y_Forward | 1 | -100 - 100 |
| | Z_Forward | 1 | -100 - 100 |
| | X_Side | 1 | -100 - 100 |
| | Y_Side | 1 | -100 - 100 |
| | Z_Side | 1 | -100 - 100 |
| | X_Vertical | 1 | -100 - 100 |
| | Y_Vertical | 1 | -100 - 100 |
| | Z_Vertical | 1 | -100 - 100 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 0000 - FFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *X_Forward, Y_Forward, Z_Forward*

The factors to calculate the new acceleration in forward direction. The formula to calculate the acceleration in Forward direction Xnew is Xnew = <X_Forward> * X + <Y_Forward> * Y +

<Z_Forward> * Z.

✧ *X_Side, Y_Side, Z_Side*

The factors to calculate the new acceleration in side direction. The formula to calculate the acceleration in Side direction Ynew is Ynew = <X_Side> * X + <Y_Side> * Y + <Z_Side> * Z.

✧ *X_Vertical, Y_Vertical, Z_Vertical*

The factors to calculate the new acceleration in vertical direction. The formula to calculate the acceleration in Vertical direction Znew is Znew = <X_Vertical> * X + <Y_Vertical> * Y + <Z_Vertical> * Z.

## 5.4.16 HBE(Acceleration Information for HBM)

XYZ-axis acceleration data in one harsh behavior. The HEX format of the event report message **+RESP:GTHBE** is as follows.

Example:

2B4556543300FE1FBF0074802004080306 04563F0E0602284904 0030CC0100220C000202000800
11004C0004FFF30046000005010100220200530 01706FD743901E614D807E80508060709046000
01DE1505A1F90B000009060000000000000000000000000000 7E805080E070A7FA482EE0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Reserved | 1 | 00 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Self Calibration Status | 1 | 0-2 |
| | Harsh Behavior Type | 1 | 0 - 4 |
| | Max Acceleration Data | 6 | '0'-'9' 'a'-'f' |
| | Average Acceleration Data | 6 | '0'-'9' 'a'-'f' |
| | Harsh Behavior Duration | 3 | 0 - 999999(x10ms) |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Max Acceleration Data*

There are 12 characters in a group of acceleration data. It is the maximal value of acceleration data which triggers this harsh behavior report. The XYZ-axis of the <Max Acceleration Data> in the **+RESP:GTHBE** report correspond to those of the device's coordinate system.

The first 4 characters of these 12 characters represent X axis acceleration data, the middle 4 characters represent Y axis acceleration data and the last 4 characters are for Z axis acceleration data.

The ASCII "0001" indicates HEX value 0x0001, so it means the acceleration is 1. The ASCII "fffd" indicates HEX value 0xFFFD which is the compliment of -3, so it means the acceleration is -3. This value is only valid in Mode 5 of the AT+GTHBM command.

✧ *Average Acceleration Data*

There are 12 characters in a group of acceleration data. It is the average value of acceleration data which triggers this harsh behavior report.

The first 4 characters of these 12 characters represent X axis acceleration data, the middle 4 characters represent Y axis acceleration data and the last 4 characters are for Z axis acceleration data.

The ASCII "0001" indicates HEX value 0x0001, so it means the acceleration is 1. The ASCII "fffd" means HEX value 0xFFFD which is the compliment of -3, so it means the acceleration is -3. This value is only valid in Mode 5 of the AT+GTHBM command.

✧ *Harsh Behavior Duration*

The duration of the harsh behavior event. This value is only valid in Mode 5 of the AT+GTHBM command.

✧ *Self Calibration Status*

The status of the self-calibration for Acceleration Data.

- **0** - Self-calibration is disabled.
- **1** - Self-calibration is not done.
- **2** - Self-calibration is successful.

✧ *Harsh Behavior Type*

The type of the harsh behavior.

- **0** - Harsh braking behavior.
- **1** - Harsh acceleration behavior.
- **2** - Harsh cornering behavior.
- **3** - Harsh braking and cornering behavior.
- **4** - Harsh acceleration and cornering behavior.

### 5.4.17 CRA(Crash Alarm)

The HEX format of the event report message **+RESP:GTCRA** is as follows.

Example:

2B4556541902FE1FFF01088020040803080F475635384C415500432EC000001A0C040101000000
0000006806FB40B401E5D44B07E90315022A2004600000550B0E9E30A500000000000000000000
0000000000000000001207FFFFF000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000FFFF00000000000000000000000000000000000000FFFFFF00000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000007E90315022A221541EEEE0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Crash Counter | 1 | 00 - FF |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Tail | 2 | 0x0D 0x0A |

✧ *Crash Counter*

A parameter to indicate the crash sequence. It combines the report **+RESP:GTCRA** and **+RESP:GTCRD** into one crash event. It rolls from 0x00 to 0xFF.

## 5.4.18 BCS(Bluetooth Connection Alarm)

The report for Bluetooth connection.

Example:

2B4556543402FE1FFF01248020040803080F475635384C415500002EC00100210C0D0347563538
4C415525494D45490060E93B2073B40101F60C940CE0330101000000000000A706FB413201E5D
48107E90315020E100460000550B0E9E30A5000000000000000000000000000000000000000020
7FFFFF000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000FFFFFF00000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000007E90315020E11148C469A0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Bluetooth Report Mask | 2 | 0000 - FFFF |
| | Bluetooth Name | <=21 | ASCII |
| | Bluetooth MAC Address | 6 | 000000000000 - FFFFFFFFFFFF |
| | Peer Role | 1 | 0 \| 1 |
| | Peer Address Type | 1 | 0 \| 1 |
| | Peer MAC Address | 6 | 000000000000 - FFFFFFFFFFFF |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Bluetooth Name*

The name of the device for Bluetooth identification. It ends with 0x00.

### 5.4.19 BDS(Bluetooth Disconnection Alarm)

The report for Bluetooth disconnection.

Example:

2B4556543502FE1FFF0125802004080308OF475635384C415500002ED20100210C0D034756353

84C415525494D45490060E93B2073B40101F60C940CE0330001010000000000008D06FB3F7501
E5D40B07E90315020B0804600000550B0E9E30A5000000000000000000000000000000000000
0207FFFFF000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000FFFFFF00000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000007E90315020B0A14639F9C0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Bluetooth Report Mask | 2 | 00000 - FFFF |
| | Bluetooth Name | <=21 | ASCII |
| | Bluetooth MAC Address | 6 | 000000000000 - FFFFFFFFFFFF |
| | Peer Role | 1 | 0 \| 1 |
| | Peer Address Type | 1 | 0 \| 1 |
| | Peer MAC Address | 6 | 000000000000 - FFFFFFFFFFFF |
| | Reason | 1 | 0 \| 4 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Bluetooth Name*

The name of the device for Bluetooth identification. It ends with 0x00.

### 5.4.20 BAA(Bluetooth Accessory Alarm)

The HEX format of the event report message **+RESP:GTBAA** is as follows.

Example:
2B4556544102FE1FFF007580200408030604475635384C415500002EB30100220CFE030212000A
EC619EA10C0A000001010000000000006206FB3FF201E5D58707E805080611020460000055 0B0
E9E30A500000000000000000000000000000000000000000000000007E8050806110307D7F13B0D
0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Index | 1 | 0-9 \| FF \| FE |
| | Accessory Type | 1 | 0-3 \| 6 \| 11-13 |
| | Accessory Model / Beacon ID Model | 1 | 0-5 \| 10 |
| | Alarm Type | 1 | 0-4 \| 7-A \| C \| E-15 |
| | Append Mask | 2\|4 | 0000 - FFFF \| 00000000 - FFFFFFFF |
| | Accessory Name(Optional) | <=21 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Accessory MAC(Optional) | 6 | '0'-'9' 'A'-'F' |
| | Accessory Status(Optional) | 1 | 0 - 1 |
| | Accessory Battery Voltage(Optional) | 2 | 0 - 5000(mV) |
| | Accessory Temperature(Optional) | 1 | -40 - 80 ( ℃ ) \| 0x7F |
| | Accessory Humidity(Optional) | 1 | 0 - 100%(rh) |
| | Accessory Mode(Optional) | 1 | 0 - 10 |
| | Accessory Event(Optional) | 1 | 0 - 2 |
| | Tire Pressure(Optional) | 2 | 0 - 500(kPa) |
| | Timestamp(Optional) | 7 | YYYYMMDDHHMMSS |
| | Enhanced Temperature(Optional) | 2 | -40 - 80 ( ℃ ) \| 0x7FFF |
| | Magnet ID(Optional) | 1 | 00 - FF |
| | MAG Event Counter(Optional) | 2 | 0 - 32767 |
| | Magnet State(Optional) | 1 | 0 - 1 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Accessory Battery Percentage(Optional) | 1 | 0 - 100(%) |
| | Relay Config Result(Optional) | 1 | 0 - 4 |
| | Relay State(Optional) | 1 | 0 - 1 |
| | CarBLE Conn Status(Optional) | 1 | 0 \| 1 |
| | CarBLE Longitude(Optional) | 4 | (-)xxx.xxxxxx |
| | CarBLE Latitude(Optional) | 4 | (-)xx.xxxxxx |
| | CarBLE GNSS UTC Time (Optional) | 7 | YYYYMMDDHHMMSS |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Index*

The index of the Bluetooth accessory.

- The index of Bluetooth accessory defined in AT+GTBAS which triggers the message **+RESP:GTBAA**.
- **0xFF** - For WKF300.
- **0xFE** - For other Beacon.

✧ *Accessory Type*

The type of the Bluetooth accessory which is defined in the <Index>.

- **0** - No Bluetooth accessory.
- **1** - Escort sensor.
- **2** - Beacon temperature sensor.
- **3** - Bluetooth beacon accessory.
- **6** - Beacon Multi-functional sensor.
- **11** - Magnet sensor.
- **12** - BLE TPMS sensor.
- **13** - Relay sensor.

✧ *Accessory Model / Beacon ID Model*

The model of the Bluetooth accessory which is defined in AT+GTBAS or the model of the Bluetooth beacon accessory which is defined in AT+GTBID.

✧ *Alarm Type*

The type of the alarm which is generated by the Bluetooth accessory specified by <Accessory Type> and <Accessory Model> in the command AT+GTBAS.

- **0** - The voltage of the Bluetooth accessory is low.
- **1** - Temperature alarm: The current temperature value is below <Low Temperature> set in the command AT+GTBAS.
- **2** - Temperature alarm: The current temperature value is above <High Temperature> set in the command AT+GTBAS.
- **3** - Temperature alarm: The current temperature value is within the range defined by <Low Temperature> and <High Temperature> set in the command AT+GTBAS.
- **4** - Push button event for WKF300 is detected.
- **7** - Humidity alarm: The current humidity value is below <Low Humidity> set in the command AT+GTBAS.
- **8** - Humidity alarm: The current humidity value is above <High Humidity> set in the command AT+GTBAS.
- **9** - Humidity alarm: The current temperature value is within the range defined by <Low Humidity> and <High Humidity>, which are set in the command AT+GTBAS.
- **0A** - Angle event notification.
- **0C** - Magnet event notification.
- **0E** - Tire pressure alarm: The current Tire pressure value is lower than <Low Tire pressure> set in the AT+GTBAS command.
- **0F** - Tire pressure alarm: The current Tire pressure value is higher than <High Tire pressure> set in the AT+GTBAS command.
- **10** - Tire pressure alarm: The current Tire pressure value is within the range defined by <Low Tire pressure> and <High Tire pressure> set in the command AT+GTBAS.
- **11** - No available Bluetooth accessory is detected.
- **12** - An available Bluetooth accessory is detected.
- **13** - Door open.
- **14** - Door closed.
- **15** - Relay event notification.

✧ *Append Mask*

Bitwise mask defined in the command AT+GTBAS or AT+GTBID to indicate the reported

Bluetooth accessory data fields.

- **Bit 0** - <Accessory Name>
- **Bit 1** - <Accessory MAC>
- **Bit 2** - <Accessory Status>
- **Bit 3** - <Accessory Battery Voltage>
- **Bit 4** - <Accessory Temperature>
- **Bit 5** - <Accessory Humidity>
- **Bit 7** - <Accessory Input Output Data>, including <Accessory Output Status>, <Accessory Digital Input Status>, <Accessory Analog Input Voltage>
- **Bit 8** - <Accessory Event Notification Data>, including <Accessory Mode>, <Accessory Event>
- **Bit 9** - <Tire Pressure>
- **Bit 10** - <Time Stamp>
- **Bit 11** - <Enhanced Temperature>
- **Bit 12** - <MAG Notification Data>, including <Magnet ID>, <MAG Event Counter>, <Magnet State>
- **Bit 13** - <Accessory Battery Percentage>
- **Bit 14** - <Relay Data>, including <Relay Config Result>, <Relay State>.
- **Bit 15** - <Expansion Mask>, if Bit 15 is set to 1, Bit 16 - Bit 31 will be valid.
- **Bit 22** - <CarBLE Information>, including <CarBLE Conn Status>, <CarBLE Longitude>, <CarBLE Latitude>, <CarBLE GNSS UTC Time>

Note

In the message bit 0 - bit 15 precedes bit 16 - bit 31. Here is an example: <Accessory Append Mask> = 0x081F0007, 0x081F indicates bit 15 - bit 0, 0x0007 indicates bit 31 - bit 16.

✧ *Accessory Name*

The name of the Bluetooth accessory which ends with '0'(0x00). If the accessory name is not found, this filed will be filled with 0x00.

✧ *Accessory MAC*

The MAC address of the Bluetooth accessory.

✧ *Accessory Status*

A numeral to indicate whether the accessory is available.

- **0** - The accessory is not available.
- **1** - The accessory is available.

✧ *Accessory Battery Voltage*

The battery voltage of the Bluetooth accessory.

✧ *Accessory Temperature*

Temperature data of Bluetooth accessory. If the temperature is not obtained, the parameter is reported as 0x7F.

✧ *Enhanced Temperature*

It instructs Bluetooth accessories to measure temperature with high precision. If the temperature is not obtained, the parameter is reported as 0x7FFF.

Note

The current temperature value. A total of 2 bytes. The temperature value is converted to an integer with 2 implicit decimals, and the integer is reported in HEX format. If the temperature value is negative, it is expressed in 2's complement format.

| Temperature (16.66) | 1666 | |
|---|---|---|
| HEX | 06 | 82 |

- *Accessory Humidity*

  Humidity data of the Bluetooth accessory.
- *Accessory Mode*

  The operating mode of angle sensor.
- *Accessory Event*

  The event is generated by the angle sensor.
- *Timestamp*

  Timestamp of the tire pressure value collection.
- *Magnet ID*

  ID corresponding to different magnet sensors.
- *MAG Event Counter*

  The total number of times detected by the magnet sensor.
- *Magnet State*

  The state of the magnet sensor's two parts.
  - **0** - Separate
  - **1** - Closed
- *Accessory Battery Percentage*

  Percentage of Bluetooth accessory's battery power.
- *Relay Config Result*

  The number representing the response result of the relay, which is controlled and reported by Bit14 of the parameter <Accessory Append Mask> in AT+GTBAS.
  - **0** - Configuration updated successfully.
  - **1** - Error in connecting.
  - **2** - The current password is incorrect.
  - **3** - Password update error.
  - **4** - Relay open or close error.
- *Relay State*

  The current state of the relay sensor.
- *CarBLE Conn Status*

  CarBLE connection status.
  - **0** - Lost status.
  - **1** - Connection status.
- *CarBLE Longitude*

  Longitude of CarBLE devices.
- *CarBLE Latitude*

  Latitude of CarBLE devices.
- *CarBLE GNSS UTC Time*

  CarBLE device latitude and longitude update time.

## 5.4.21 BID(Bluetooth Beacon ID Report)

The HEX format of the event report message **+RESP:GTBID** is as follows.

Example:
2B4556544302FE1FFF007C80200406000604475635384C415500002EF00100220C02010042AC23
3F2297F6AD010042AC233F228481A70101000000000000006206FB3FF201E5D58707E8050806113
904600000550B0E9E30A5000000000000000000000000000000000000000000000007E8050806
113907DB4B9D0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Number | 1 | 0 - 3 \| 0 - 15 |
| | Beacon ID Model | 1 | 0-2 \| 4 \| 5 \| 10 |
| | Accessory Append Mask | 2 | 0000 - FFFF |
| | Accessory MAC (Optional) | 6 | '0'-'9' 'A'-'F' |
| | Accessory Battery Level (Optional) | 2 | 0 - 5000(mV) |
| | Accessory Signal Strength (Optional) | 1 | -120 - 0 |
| | Beacon Type (Optional) | 1 | 0-2 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Beacon Data (Optional) | <=100 | |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Number*

The number of Bluetooth beacon accessories.

- **WKF300**. The maximum value is 3.
- **iBeacon E6**. The maximum value is 15.
- **ID ELA**. The maximum value is 15.
- **WID310**. The maximum value is 15.
- **CarBLE**. The maximum value is 15.
- **WID330** - The maximum value is 15.

✧ *Beacon ID Model*

The model of the Bluetooth beacon accessory which is defined in AT+GTBID.

✧ *Accessory Append Mask*

Bitwise mask defined in the AT+GTBID command to indicate the reported Bluetooth beacon accessory data fields.

- **Bit 0** - Reserved
- **Bit 1** - <Accessory Mac>
- **Bit 2** - Reserved

- **Bit 3** - <Accessory Battery Level>
- **Bit 4** - Reserved
- **Bit 5** - Reserved
- **Bit 6** - <Accessory Signal Strength>
- **Bit 7** - <Beacon Type> and <Beacon Data>

✧ *Accessory MAC*

Bluetooth accessory MAC address.

✧ *Accessory Battery Level*

The voltage of Bluetooth accessory.

✧ *Accessory Signal Strength*

The signal strength of Bluetooth accessory.

✧ *Beacon Type*

Types of beacons.

- **0** - "ID" Format
- **1** - "iBeacon" Format
- **2** - "Eddy stone" Format

✧ *Beacon Data*

Select the data format according to <Beacon Type>:

- If <Beacon Type> is 0, the data format is as follows:

| Fields | Length | Range/Format |
|---|---|---|
| ID_Mfr_Data | 6 | (HEX) |

- If <Beacon Type> is 1, the data format is as follows:

| Fields | Length | Range/Format |
|---|---|---|
| UUID | 16 | (HEX) |
| Major | 2 | (HEX) |
| Minor | 2 | (HEX) |

- If <Beacon Type> is 2, the data format is as follows:

| Fields | Length | Range/Format |
|---|---|---|
| NID | 10 | (HEX) |
| BID | 6 | (HEX) |

### 5.4.22 BAR(Bluetooth Accessory Report)

The HEX format of the event report message **+RESP:GTBAR** is as follows.

Example:

```
2B4556544602FE1FFF0136802004080306044756353844C415500002EEE0101210C000704000747
4E4F4D20445020533700010182CD7CE0010148F76D014600000000FFFFFFFFFFFFFFFFFFFFFFFF
FFF0101000000000000007206FB409601E5D47707E805090A361004600000550B0E9E30A5000000
0000000000000000000000000000001207FFFFF00000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000004700
```

000000200000000027000000001000000000000000000000000000000000000000000000FFFF
FF0000000000000000000000000000000000158000003210000000000000000200000000000000
0000000000000005000000000000000000000000000000000007E805090A36100D30806D0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Index | 1 | 0 - 9 |
| | Accessory Type | 1 | 7 |
| | Accessory Model | 1 | 0 - 4 |
| | Append Mask | 2 | 0000 - FFFF |
| | Accessory Name | <=20 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Accessory MAC | 6 | '0'-'9' 'A'-'F' |
| | Accessory Status | 1 | 0 - 1 |
| | (Accessory Data) | <=650 | |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Accessory Data*

There are accessory data para according <Accessory Type> and <Accessory Model>.

- If the <Accessory Type> is 7 and <Accessory Model> is 0 (DUT-E S7) or <Accessory Model> is 4 (GNOM DP S7):

| Fields | Length | Range/Format |
|---|---|---|
| Version | 1 | 00 - FF |
| PGN | 2 | 0000 - FFFF |
| PGN Data | 21 | 000000000000000000000000000000000000000000<br>--<br>FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF |

- If the <Accessory Type> is 7 and <Accessory Model> is 3(GNOM DDE S7):

| Fields | Length | Range/Format |
|---|---|---|
| Version | 1 | 00 - FF |
| PGN | 2 | 0000 - FFFF |
| PGN Data | 20 | 0000000000000000000000000000000000000000<br>--<br>FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF |

- If the <Accessory Type> is 7 and <Accessory Model> is 1 (DFM 100 S7):

| Fields | Length | Range/Format |
|---|---|---|
| Version | 1 | 00 - FF |
| PGN1 | 2 | 0000 - FFFF |

| Fields | Length | Range/Format |
|---|---|---|
| PGN Data1 | 20 | 0000000000000000000000000000000000000000<br>--<br>FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF |
| PGN2 | 2 | 0000 - FFFF |
| PGN Data2 | 20 | 0000000000000000000000000000000000000000<br>--<br>FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF |
| PGN3 | 2 | 0000 - FFFF |
| PGN Data3 | 20 | 0000000000000000000000000000000000000000<br>--<br>FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF |

- If the <Accessory Type> is 7 and <Accessory Model> is 2 (DFM 250DS7):

| Fields | Length | Range/Format |
|---|---|---|
| Version | 1 | 00 - FF |
| PGN1 | 2 | 0000 - FFFF |
| PGN Data1 | 20 | 0000000000000000000000000000000000000000<br>--<br>FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF |
| PGN2 | 2 | 0000 - FFFF |
| PGN Data2 | 20 | 0000000000000000000000000000000000000000<br>--<br>FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF |
| PGN3 | 2 | 0000 - FFFF |
| PGN Data3 | 20 | 0000000000000000000000000000000000000000<br>--<br>FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF |
| PGN4 | 2 | 0000 - FFFF |
| PGN Data4 | 20 | 0000000000000000000000000000000000000000<br>--<br>FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF |

## 5.4.23 RTP(Remote File Transfer)

The HEX format of the event report message **+RESP:GTRTP** is as follows.

Example:
2B4556545602FE1FFF012F8020040803080F475635384C415500002EBE01002208000000006468
7474703A2F2F36302E3137342E3232352E3137333A33303039392F63612E637274000101003B05
004E003106FC53D101E61A9307E90315062D2D04600000550B0E9E30A50000070700000007070
0000000000000000001207FFFFF000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000FFFF0000000000000000000000000000000000000000FFFFFF000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000007E90315062D2D194D4E3D0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Analog Input Mode | 2 | |
| | Analog Input1 Voltage | 2 | |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Mode | 1 | 0 |
| | Protocol Type | 1 | 0 |
| | File Type | 1 | 0 - 2 |
| | Code | 2 | 100 \| 101 \| 200 \| 201 \| 202 \| 301 \| 302 |
| | Download URL | <=101 | Complete URL |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Mode*

The working mode of file transfer.

- **0** - Download file

✧ *Protocol Type*

The type of communication protocol used to obtain data from the background server.

- **0** - HTTP

✧ *File Type*

It defines the type of file to download from the server.

- **0** - CA certificate
- **1** - Client certificate
- **2** - Client key

✧ *Code*

It indicates the download information.

- **100** - The update command is starting.
- **101** - The update command is refused by the device.
- **200** - The device starts to download the file.
- **201** - The device finishes downloading the file successfully.
- **202** - The device fails to download the file.
- **301** - The device finishes updating the file successfully.

- **302** - The device fails to update the file.

✧ *Download URL*

The complete URL to download the configuration. It includes the file name and ends by 0x00.

## 5.4.24 SVR(Stolen Vehicle Recovery)

Stolen vehicle recovery message reported by the Primary device.

Example:
2B4556544902FE1FFF011B8020040803080F475635384C415500002F19010022080060E93B2073
B40B300000000000000000000000101003100004E003006FC420D01E6178607E903150706140
4600000550B0E9E30A500001E090000001E09000000000000000000001207FFFFF0000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000FFFF00000000000000000
000000000000000000000FFFFFF000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000007E90
315070614197ECEE00D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | SVR Working State | 1 | 0-2 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Ghost/Primary Device MAC Broadcast | 6 | 000000000000 - FFFFFFFFFFFF |
| | SVR Appending Length | 1 | |
| | SVR Appending Information | <=15 | |
| | Target State | 1 | 0 \| 1 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.4.25 CFU(CAN100 FOTA Upgrade Report)

The HEX format of the event report message **+RESP:GTCFU** is as follows.

Example:
2B4556542A02FE1FFF010A8020040803040C475635384C4155000031900101210C00C80001010
000000000005F06FB415401E5D46907E70C13080B3704600001DF5C027A4F1F0000000000000
000000000000000000000001207FFFFF0000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000

```
000000000000000000000000000000000000000000000000000000000000000000000000FFFFFF000000
000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000007E70C13080B380056F70A0D0A
```

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Code | 2 | |
| | Reserved | 1 | 00 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.4.26 DOM(Waveform Occurs)

The HEX format of the event report message **+RESP:GTDOM** is as follows.

Example:

2B4556544B02FE1FFF010A8020040803080F475635384C415500002EC8010022080501FF010100
3104005D002B06FCD6C001E60A2C07E90315070A0604600000550B0E9E30A5000022060000002
20600000000000000000001207FFFFF00000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000FFFF00000000000000000000000000000000000000FFFFFF0000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000007E90315070A071982208D0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Wave Shape | 1 | 5 |
| | Wave Output ID | 1 | 1 - 3 |
| | Wave Output Active | 1 | FF |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.4.27 UDF(User Defined Function Report)

The device will send the message **+RESP:GTUDF** to the backend server during the upgrade process. The message in HEX format is as follows.

Example:
2B4556546802FE1FFF010B8020040803080F475635384C415500502EE80000120C000000010101
0000000000006806FB40B401E5D44B07E9031502331804600000550B0E9E30A50000000000000
00000000000000000000000001207FFFFF000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000FFFF00000000000000000000000000000000000000FFFFFF00000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000007E903150233191575CCCB0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Group ID Mask | <=4 | 0 - FFFFFFFF |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.5 Data Flow Reports

This section describes the message related to certain data needs to be sent. Please refer to the details below.

### 5.5.1 Generic Data Flow Report

CRG(Crash GNSS Data Packet)

Example:

2B444154040000007F0119802004080308 0F475635384C415500010A0101000000000000006006F
B40A201E5D43D07E90315061426020100000000000006006FB40A201E5D43D07E903150614270
30100000000000006006FB40A201E5D43D07E903150614280401000000000000006006FB40A201E5
D43D07E90315061429050100000000000006006FB40A201E5D43D07E9031506142A06010000000
000006006FB40A201E5D43D07E9031506142B070100000000000006006FB40A201E5D43D07E903
1506142C08010000000000006006FB40A201E5D43D07E9031506142D090100000000000006006F
B40A201E5D43D07E9031506142E0A0100000000000006006FB40A201E5D43D07E9031506142F0
7E9031506143118F6EEFA0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +DAT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | +RESP:GTCRG | | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Message Type*

The ID of a specific data report message.

| Message | ID |
|---|---|
| +RESP:GTCRG | 4 |

**+RESP:GTCRG**

| Fields | | Length | Range/Format |
|---|---|---|---|
| +RESP:GTCRG | Data Type | 1 | 0\|1 |
| | GNSS Validity Number | 1 | 0 - 10 |
| | GNSS Point Index | 1 | 1 |
| | GNSS Accuracy | 1 | 0 \| 1 - 50 |
| | Speed | 3 | 0.0 - 999.9(km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | (-)xxxxx.x (m) |
| | Longitude | 4 | (-)xxx.xxxxxx |
| | Latitude | 4 | (-)xx.xxxxxx |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | ... | | |
| | GNSS Point Index | 1 | 10 |
| | GNSS Accuracy | 1 | 0 \| 1 - 50 |
| | Speed | 3 | 0.0 - 999.9(km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | (-)xxxxx.x (m) |
| | Longitude | 4 | (-)xxx.xxxxxx |
| | Latitude | 4 | (-)xx.xxxxxx |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |

- *Report Mask*

  Please refer to <+DAT Mask> in AT+GTHRM.
- *Data Type*

  The data reported to the backend server is recorded before crash or after crash.
  - **0** - Before crash.
  - **1** - After crash.
- *GNSS Validity Number*

  The number of the successfully fixed GNSS positions included in the report message.
- *GNSS Point Index*

  The index of GNSS point.

## 5.5.2 CRD(Crash Data Packet)

The HEX format of the report message **+RESP:GTCRD** is as follows.

Example:

2B435244007D02828020040803080F563F0E0608493102056901010258000DFFFEFFC5000CFFFE
FFB4FFFA0018FFA6FFF40011FFAE0001000CFFB0FFFF0004FFB10000FFFAFFAFFFFE0009FFAF0000
0002FFAF0000FFFEFFB10001FFF9FFAF00010001FFB100000000FFB10001FFFEFFB100010000FFB0
00000000FFB10001FFFFFFAF00010001FFB000000000FFB100010000FFB000000000FFB10001000
0FFAF00010000FFB100010000FFAF00010000FFB000010000FFB100000001FFAF00010000FFB000
010000FFAE00010000FFB100010000FFAF00010001FFB100000001FFB0FFFFFFFFFFB000030004F
FB1FFFEFFFEFFB00000FFFEFFAFFFFFFFFFFFAEFFFFFFFEFFB000010002FFB100020002FFB1000300
01FFB200020004FFB100010004FFB100020000FFAE00010001FFB100030001FFAF00030002FFAF0
0010005FFAFFFFDFFFFFFB100010000FFB10001FFFDFFB0FFFF0000FFB100010002FFAF00040006
FFAF0004FFFFFFAF00030002FFAF00040000FFB0FFFCFFFEFFAFFFFA0000FFAF00000002FFAF0001
0001FFAF00010000FFAF00030000FFB000020001FFAFFFFF0002FFB100000000FFAF0001FFFEFFAF
0001FFFEFFAF00020002FFB10000FFF9FFB1FFFD0006FFB0FFFDFFFDFFAFFFFEFFFFFFAF0001FFF9
FFB100030009FFB100040004FFB000060000FFB000010001FFB000010000FFB200010000FFB000
000000FFB000000000FFAF00010000FFAF00010000FFAF00010000FFB000010000FFAF00010000F
FAF00010000FFAF00010000FFAF00000000FFB100010001FFB000010000FFB000010000FFB00001
0000FFB000010000FFAF00010000FFAF00000002FFB000010002FFAF00010000FFAF07E90315023
4061582FAFD0D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Message Header | 4 | +CRD |
| | Report Mask | 2 | 0000 - FFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Crash Counter | 1 | 00 - FF |
| | Data Type | 1 | 0x00 - 0x7F |
| | Total Frame | 1 | 1 - 8 |
| | Frame Number | 1 | 1 - 8 |
| | Data Length | 2 | 0 - 1200 |
| | Data | <=1200 | HEX |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

- *Report Mask*

  Please refer to <+CRD Mask> in AT+GTHRM.

- *Unique ID*

  If Bit 1 of <+CRD Mask> is 0, the IMEI of the device is used as the unique ID of the device. IMEI is a 15-digit string. In the HEX format message, each 2 digits are encoded into one byte as an integer.

| IMEI | 13 | 57 | 90 | 24 | 68 | 11 | 22 | 0 |
|---|---|---|---|---|---|---|---|---|
| HEX | 0D | 39 | 5A | 18 | 44 | 0B | 16 | 00 |

  If Bit 4 of <+CRD Mask> is 1, the device name is used as the unique ID of the device. Device name is an 8-byte string, please refer to <Device Name> in AT+GTCFG. If the length of <Device Name> is more than 8 bytes, only the first 8 bytes will be acquired. In the HEX format message, each byte is encoded into one byte as an integer. If the device name is less than 8 bytes, the remaining bytes are set to 0.

| Device Name | q | u | e | c | l | i | n | k |
|---|---|---|---|---|---|---|---|---|
| HEX | 71 | 75 | 65 | 63 | 6C | 69 | 6E | 6B |

- *Crash Counter*

  It indicates the sequence number of the crash event. The reports of **+RESP:GTCRA** and **+RESP:GTCRD** are combined into one crash event. It rolls from 0x00 to 0xFF.

- *Data Type*

  A hexadecimal parameter to indicate the time of the data (before crash or after crash) and crash direction (+X, -X, +Y, -Y, +Z, -Z or several of them). Please refer to the following table for the detailed syntax.

| Bits | Description | Range |
|---|---|---|
| Bit 0 | 0: Before crash 1: After crash | 0 - 1 |
| Bit 1 | 0: X-axis crash not detected 1: X-axis crash detected | 0 - 1 |
| Bit 2 | 0: X-axis positive direction 1: X-axis negative direction | 0 - 1 |
| Bit 3 | 0: Y-axis crash not detected 1: Y-axis crash detected | 0 - 1 |
| Bit 4 | 0: Y-axis positive direction 1: Y-axis negative direction | 0 - 1 |

| Bits | Description | Range |
|---|---|---|
| Bit 5 | 0: Z-axis crash not detected 1: Z-axis crash detected | 0 - 1 |
| Bit 6 | 0: Z-axis positive direction 1: Z-axis negative direction | 0 - 1 |
| Bit 7 | Fixed value | 0 |

- *Total Frame*

  Total number of the messages that are sent to the backend server for the crash event.
- *Frame Number*

  A numeral to indicate the sequence of the current message.
- *Data*

  There is a maximum of 1200 bytes in one frame which contains acceleration samples in 2 seconds at most. There are 6 bytes in a group: the first 2 bytes represent X axis acceleration data, the middle 2 bytes represent Y axis acceleration data and the last 2 bytes are Z axis acceleration data. And they are two's complement. Regarding how to convert the two's complement to the original value, please refer to the Appendix.

## 5.6 Data Transmission

This section describes the message related to certain data needs to be sent. Please refer to the details below.

### 5.6.1 DAT(Transparent Data Transmission)

The HEX format of the event report message **+RESP:GTDAT** is as follows.

Example:
2B4556541202FE1FFF0113802004080308 0F475635384C4155004C2EC80000120C000A61616161
61616161616101010000000000006806FB40B401E5D44B07E90315023421046000000550B0E9E3
0A500000000000000000000000000000000000001207FFFFF0000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000FFFF000000000000000000000000000000000
00000FFFFFF000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000007E9031502342315 86A9
400D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Data Length | 2 | |
| | Data | | |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Tail | 2 | 0x0D 0x0A |

✧ *Data Length*

It indicates the length of the parameter <Data>.

## 5.7 CANBUS Information Report

This section describes the message format of CANBUS information. Please refer to the details below.

### 5.7.1 CAN(CANBUS Device Information)

The CANBUS device information report message **+RESP:GTCAN** uses the format below.

Example:
2B43414E01000007FF00E68020040803080F475635384C4155001001207FFFFF00000000000000
000000000000000000002000BE37D000000323800140028005A000000000000572800000431
400000F780020000000020E000000011100000000006100001D0C2D110D00BF7FFF3F1F00000003
38000000010B000000000001FFFFF000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000100000000000006806FB40
B401E5D44B07E9031502291904600000550B0E9E30A50007E9031502291B153ACDF80D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +CAN |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI |
| | Distance Type\|Report Type | 1 | X(0 \| 1 )X(0-2) |
| | CANBUS Device State | 1 | 0 \| 1 |
| | CANBUS Report Mask | 4 | 0x00000000 - 0xFFFFFFFF |
| | VIN | 17 | |
| | Ignition Key | 1 | 0-2 |
| | Total Distance | 4 | H(0 - 99999999)/I(0 - 2147483647) |
| | Total Fuel Used | 5 | 0.00 - 999999.99(L) |
| | Engine RPM | 2 | 0 - 16383(rpm) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Vehicle Speed | 2 | 0 - 6553(km/h) |
| | Engine Coolant Temperature | 2 | -50 - +215( ℃ ) |
| | Fuel Consumption | 3 | L/100km(0.0 - 999.9)\|L/H(0.0 - 999.9) |
| | Fuel Level (Liter) | 5 | L(0.00 - 9999.99) |
| | Fuel Level (Percentage) | 5 | P(0.00 - 100.00) |
| | Range | 4 | 0 - 99999999(hm) |
| | Accelerator Pedal Pressure | 2 | 0 - 100(%) |
| | Total Engine Hours | 5 | 0.00 - 9999999.99(h) |
| | Total Driving Time | 5 | 0.00 - 9999999.99(h) |
| | Total Engine Idle Time | 5 | 0.00 - 9999999.99(h) |
| | Total Idle Fuel Used | 5 | 0.00 - 999999.99(L) |
| | Axle Weight 2nd | 2 | 0 - 65535(kg) |
| | Tachograph Information | 2 | 00 - FFFF |
| | Detailed Information/Indicators | 2 | 00 - FFFF |
| | Lights | 1 | 0x00 - 0xFF |
| | Doors | 1 | 0x00 - 0xFF |
| | Total Vehicle Overspeed Time | 5 | 0.00 - 9999999.99(h) |
| | Total Vehicle Engine Overspeed Time | 5 | 0.00 - 9999999.99(h) |
| | Total Distance Impulses | 4 | 0 - 21474836 |
| | CANBUS Report Expansion Mask | 4 | 0x00000000 - 0xFFFFFFFF |
| | Ad-Blue Level | 2 | 0 - 100(%) |
| | Axle Weight 1st | 2 | 0 - 65535(kg) |
| | Axle Weight 3rd | 2 | 0 - 65535(kg) |
| | Axle Weight 4th | 2 | 0 - 65535(kg) |
| | Tachograph Overspeed Signal | 1 | 0\|1 |
| | Tachograph Vehicle Motion Signal | 1 | 0\|1 |
| | Tachograph Driving Direction | 1 | 0\|1 |
| | Analog Input Value | 4 | 0 - 99999(mV) |
| | Engine Braking Factor | 4 | 0 - 4278190079 |
| | Pedal Braking Factor | 4 | 0 - 4278190079 |
| | Total Accelerator Kick-downs | 4 | 0-999999 |
| | Total Effective Engine Speed Time | 5 | 0.00 - 9999999.99(h) |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Total Cruise Control Time | 5 | 0.00 - 9999999.99(h) |
| | Total Accelerator Kick-down Time | 5 | 0.00 - 9999999.99(h) |
| | Total Brake Applications | 4 | 0 - 999999 |
| | Tachograph Driver 1 Card Number | <=40 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Tachograph Driver 2 Card Number | <=40 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Tachograph Driver 1 Name | < =40 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Tachograph Driver 2 Name | < =40 | '0'-'9' 'a'-'z' 'A'-'Z' '-' '_' |
| | Registration Number | <=40 | ASCII |
| | Expansion Information | 2 | 0x0000 - 0xFFFF |
| | Rapid Brakings | 4 | 0 - 16711679 |
| | Rapid Accelerations | 4 | 0 - 16711679 |
| | Engine Torque | 1 | 0 - 100(%) |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Message Type*

The ID of the CAN report message.

| Message | ID |
|---|---|
| +RESP:GTCAN | 1 |

✧ *Report Mask*

Please refer to <+CAN Mask> in AT+GTHRM.

✧ *Length*

The length of the whole message from header to the tail characters.

✧ *Unique ID*

If Bit 6 of <+CAN Mask> is 0, the IMEI of the device is used as the unique ID of the device. IMEI is a 15-digit string. In the HEX format message, each 2 digits are encoded into one byte as an integer.

| IMEI | 13 | 57 | 90 | 24 | 68 | 11 | 22 | 0 |
|---|---|---|---|---|---|---|---|---|
| HEX | 0D | 39 | 5A | 18 | 44 | 0B | 16 | 00 |

If Bit 6 of <+CAN Mask> is 1, the device name is used as the unique ID of the device. Device name is an 8-byte string, please refer to <Device Name> in AT+GTCFG for the device name. If the length of <Device Name> is more than 8 bytes, the device will only acquire the first 8 bytes. In the Hex format message, each byte is encoded into one byte as an integer. If the device name is less than 8 bytes, the remaining bytes are set to 0.

| Device Name | q | u | e | c | l | i | n | k |
|---|---|---|---|---|---|---|---|---|
| HEX | 71 | 75 | 65 | 63 | 6C | 69 | 6E | 6B |

✧ *Distance Type / Report Type*

The high nibble is for <Distance type> and the low nibble is for <Report Type>.

Distance type has the following meanings.

- **0** - Total distance acquired from CAN Chipset.
- **1** - Total distance obtained by calculation with GNSS.

Report type has the following meanings.

- **0** - Periodic report.
- **1** - RTO CAN report.
- **2** - Ignition event report.

✧ *CANBUS Device State*

A numeral to indicate the state of communication with the external CANBUS device.

- **0** - Abnormal. It fails to receive data from the external CANBUS device.
- **1** - Normal. It is able to receive data from the external CANBUS device.

✧ *CANBUS Report Mask*

Please refer to <CAN Report Mask> in AT+GTCAN.

✧ *VIN*

Vehicle identification number.

✧ *Total Distance*

Vehicle distance. The number is always increasing. The unit is hectometer. If it is set to 0, the distance is not available.

✧ *Total Fuel Used*

A total of 5 bytes. The first 4 bytes are for the integer part of the total fuel used and the last byte is for the decimal part. The decimal part has 2 digits.

✧ *Engine Coolant Temperature*

The engine coolant temperature of the vehicle. 2 bytes in total. If this value is negative, it is represented in Two's complement format.

✧ *Fuel Level (Liter)*

5 bytes in total. The first 4 bytes are for the integer part of the fuel level (liters) and the last

byte is for the fractional part. The fractional part has 2 digits. This field is controlled by Bit 8 in <CANBUS Report Mask>.

✧ *Fuel Level (Percentage)*

5 bytes in total. The first 4 bytes are for the integer part of the fuel level (percentage) and the last byte is for the fractional part. The fractional part has 2 digits. This field is controlled by Bit 8 in <CANBUS Report Mask>.

✧ *Fuel Consumption*

3 bytes in total. The first byte indicates the unit. The unit L/100km is represented as FE, and the unit L/H is represented as FF. The other two bytes indicate the value. The fuel consumption value is converted to an integer with 1 implicit decimal digit by multiplying it by 10 and the integer is reported in HEX format.

| Fuel Consumption Value 12.1 | 121 | |
|---|---|---|
| HEX | 00 | 79 |

✧ *Total Engine Hours*

5 bytes in total. The first 4 bytes are for the integer part of the total engine hours and the last byte is for the fractional part. The fractional part has 2 digits.

✧ *Total Driving Time*

5 bytes in total. The first 4 bytes are for the integer part of the total driving time and the last byte is for the fractional part. The fractional part has 2 digits.

✧ *Total Engine Idle Time*

5 bytes in total. The first 4 bytes are for the integer part of the total engine idle time and the last byte is for the fractional part. The fractional part has 2 digits.

✧ *Total Idle Fuel Used*

5 bytes in total. The first 4 bytes are for the integer part of the total idle fuel used and the last byte is for the decimal part. The decimal part has 2 digits.

✧ *Tachograph Information*

Two bytes. The high byte describes driver 2, and the low byte describes driver 1.
Each byte format:

| Validity mark | Reserved | Driver working states | | Driver card | Driving time related states | | |
|---|---|---|---|---|---|---|---|
| V | R | W1 | W0 | C | T2 | T1 | T0 |

- **V** - Validity mark (0 - valid driver data, 1 - no valid data)
- **R** - Reserved
- **C** - Driver card (1 - card inserted, 0 - no card inserted)
- **T2-T0** - Driving time related states:
  - **0** - Normal / no limits reached
  - **1** - 15min before 4½h
  - **2** - 4½h reached
  - **3** - 15min before 9h
  - **4** - 9h reached
  - **5** - 15 minutes before 16h (without 8h rest during the last 24h)
  - **6** - 16h reached
  - **7** - Other limit

- **W1-W0** - Driver working states:
  - **0** - Rest - sleeping
  - **1** - Driver available - short break
  - **2** - Work - loading, unloading, working in an office
  - **3** - Driver - behind the wheel

✧ *Detailed Information / Indicators*

2 bytes in total. Each bit contains information of one indicator.

- **Bit 0** - FL - fuel low indicator (1 - indicator on, 0 - off)
- **Bit 1** - DS - driver seatbelt indicator (1 - indicator on, 0 - indicator off)
- **Bit 2** - AC - air conditioning (1 - on, 0 - off)
- **Bit 3** - CC - cruise control (1 - active, 0 - inactive)
- **Bit 4** - B - brake pedal (1 - pressed, 0 - released)
- **Bit 5** - C - clutch pedal (1 - pressed, 0 - released)
- **Bit 6** - H - handbrake (1 - pulled-up, 0 - released)
- **Bit 7** - CL - central lock (1 - locked, 0 - unlocked)
- **Bit 8** - R - reverse gear (1 - on, 0 - off)
- **Bit 9** - RL - running lights (1 - on, 0 - off)
- **Bit 10** - LB - low beams (1 - on, 0 - off)
- **Bit 11** - HB - high beams (1 - on, 0 - off)
- **Bit 12** - RFL - rear fog lights (1 - on, 0 - off)
- **Bit 13** - FFL - front fog lights (1 - on, 0 - off)
- **Bit 14** - D - doors (1 - any door open, 0 - all doors closed)
- **Bit 15** - T - trunk (1 - open, 0 - closed)

✧ *Lights*

A hexadecimal number. Each bit contains information of one light.

- **Bit 0** - Running Lights (1 - on, 0 - off)
- **Bit 1** - Low Beam (1 - on, 0 - off)
- **Bit 2** - High Beam (1 - on, 0 - off)
- **Bit 3** - Front Fog Light (1 - on, 0 - off)
- **Bit 4** - Rear Fog Light (1 - on, 0 - off)
- **Bit 5** - Hazard Lights (1 - on, 0 - off)
- **Bit 6** - Reserved
- **Bit 7** - Reserved

✧ *Doors*

A hexadecimal number. Each bit contains information of one door.

- **Bit 0** - Driver Door (1 - open, 0 - closed)
- **Bit 1** - Passenger Door (1 - open, 0 - closed)
- **Bit 2** - Rear Left Door (1 - open, 0 - closed)
- **Bit 3** - Rear Right Door (1 - open, 0 - closed)
- **Bit 4** - Trunk (1 - open, 0 - closed)
- **Bit 5** - Hood (1 - open, 0 - closed)
- **Bit 6** - Reserved
- **Bit 7** - Reserved

✧ *Total Vehicle Overspeed Time*

5 bytes in total. The first 4 bytes are for the integer part of the total vehicle overspeed time and the last byte is for the fractional part. The fractional part has 2 digits.

- *Total Vehicle Engine Overspeed Time*

  5 bytes in total. The first 4 bytes are for the integer part of the total vehicle engine overspeed time and the last byte is for the fractional part. The fractional part has 2 digits.

- *Total Distance Impulses*

  Vehicle distance in impulses. The number is always increasing. The unit is imp. If it is set to 0, the distance in imp is not available.

- *Ad-Blue Level*

  The level of Ad-Blue. 2 bytes in total.

- *Axle Weight 1st*

  Vehicle first axle weight. The unit is kg.

- *Axle Weight 3rd*

  Vehicle third axle weight. The unit is kg.

- *Axle Weight 4th*

  Vehicle fourth axle weight. The unit is kg.

- *Tachograph Overspeed Signal*

  Vehicle overspeed signal from the tachograph.

  - **0** - Overspeed is not detected.
  - **1** - Overspeed is detected.

- *Tachograph Vehicle Motion Signal*

  The vehicle motion signal in the tachograph.

  - **0** - Motion is not detected.
  - **1** - Motion is detected.

- *Tachograph Driving Direction*

  Vehicle driving direction from the tachograph.

  - **0** - Driving forward.
  - **1** - Driving backward.

- *Analog Input Value*

  The value of analog input. The unit is mV.

- *Engine Braking Factor*

  It measures how often driver brakes with brake pedal or with engine and stores both counts (which are always increasing). Decreasing speed with no pedal pressed causes an increase of the engine braking factor.

- *Pedal Braking Factor*

  It measures how often driver brakes with brake pedal or with engine and stores both counts (which are always increasing). Decreasing speed with brake pedal pressed causes an increase of pedal braking factor.

- *Total Accelerator Kick-downs*

  The count of accelerator pedal kick-downs (with the pedal pressed over 90%).

- *Total Effective Engine Speed Time*

  Total time when the vehicle engine speed is effective. The unit is hour. The first 4 bytes are for the integer part of the total engine idle time and the last byte is for the fractional part. The fractional part has 2 digits.

✧ *Total Cruise Control Time*

Total time when vehicle speed is controlled by cruise-control module. The unit is hour. The first 4 bytes are for the integer part of the total engine idle time and the last byte is for the fractional part. The fractional part has 2 digits.

✧ *Total Accelerator Kick-down Time*

Total time when accelerator pedal is pressed over 90%. The unit is hour. The first 4 bytes are for the integer part of the total engine idle time and the last byte is for the fractional part. The fractional part has 2 digits.

✧ *Total Brake Applications*

The count of braking processes initiated by brake pedal.

✧ *Tachograph Driver 1 Card Number*

The card number of tachograph driver 1. The value is a numeric string and ends by 0x00.

✧ *Tachograph Driver 2 Card Number*

The card number of tachograph driver 2. The value is a numeric string and ends by 0x00.

✧ *Tachograph Driver 1 Name*

The name of tachograph driver 1. The value is a name string and ends by 0x00.

✧ *Tachograph Driver 2 Name*

The name of tachograph driver 2. The value is a name string and ends by 0x00.

✧ *Registration Number*

The vehicle registration number. The value is a numeric string and ends by 0x00.

✧ *Expansion Information*

A hexadecimal number. Each bit contains information of one indicator.

- **Bit 0** - W - webasto (1 - on, 0 - off or not available)
- **Bit 1** - BFL - brake fluid low indicator (1 - on, 0 - off or not available)
- **Bit 2** - CLL - coolant level low indicator (1 - on, 0 - off or not available)
- **Bit 3** - BAT - battery indicator (1 - on, 0 - off or not available)
- **Bit 4** - BF - brake system failure indicator (1 - on, 0 - off or not available)
- **Bit 5** - OP - oil pressure indicator (1 - on, 0 - off or not available)
- **Bit 6** - EH - engine hot indicator (1 - on, 0 - off or not available)
- **Bit 7** - ABS - ABS failure indicator (1 - on, 0 - off or not available)
- **Bit 8** - Reserved
- **Bit 9** - CHK - "check engine" indicator (1 - on, 0 - off or not available)
- **Bit 10** - AIR - airbags indicator (1 - on, 0 - off or not available)
- **Bit 11** - SC - service call indicator (1 - on, 0 - off or not available)
- **Bit 12** - OLL - oil level low indicator (1 - on, 0 - off or not available)

✧ *Rapid Brakings*

The number of total rapid brakings since installation (calculation based on CAN100 settings of speed decrease time and value).

✧ *Rapid Accelerations*

The number of total rapid accelerations since installation (calculation based on CAN100 settings of speed increase time and value).

✧ *Send Time*

The local time to send the acknowledgement message. 7 bytes in total. The first 2 bytes are for year, and the other 5 bytes are for month, day, hour, minute and second respectively.

| Send Time | 2011 | | 01 | 31 | 06 | 29 | 11 |
|---|---|---|---|---|---|---|---|
| HEX | 07 | DB | 01 | 1F | 06 | 1D | 0B |

## 5.8 Update Configuration

This section describes the message related to firmware and configuration upgrade. Please refer to the details below.

### 5.8.1 UPC(Configuration Update Notification)

The HEX format of the event report message **+RESP:GTUPC** is as follows.

Example:

2B4556542802FE1FFF01318020040803040C475635384C4155000031840101210C00012C687474
703A2F2F36302E3137342E3232352E3137333A32303537312F50454F5F61742E696E6900010100
00000000007306FB40E501E5D4F307E70C1307202104600001DF5C027A4F1F000000000000000
00000000000000000000001207FFFFF0000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000FFFFFF0000000
000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000007E70C13072022001F6AA30D0A


| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | | | 0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Command ID | 1 | |
| | Result | 2 | 100-103 \| 200-202 \| 300-302 \| 304-306 |
| | Download URL | <=101 | Complete URL |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

✧ *Command ID*

The command ID in the update configuration file. It is always 0 before the device starts to update the configuration (<Result><=300). It indicates the total number of the commands when <Result> is 301. It indicates the ID of the command which has wrong format when <Result> is 302.It is empty when <Result> is greater than 302.

✧ *Result*

A numeral to indicate whether the configuration is updated successfully.

- **100** - The update command is starting.
- **101** - The update command is confirmed by the device.
- **102** - The update command is refused by the device.

- **103** - The update process is refused because the battery is low.
- **200** - The device starts to download the package.
- **201** - The device finishes downloading the package successfully.
- **202** - The device fails to download the package.
- **300** - The device starts to update the device configuration.
- **301** - The device finishes updating the device configuration successfully.
- **302** - The device fails to update the device configuration.
- **303** - Reserved.
- **304** - <Command Mask>, <GEO ID Mask>, <Stocmd ID Mask> or <Group ID Mask> check fails.
- **305** - The update process is interrupted by abnormal reboot.
- **306** - The update process is interrupted by MD5 verification error.

✧ *Download URL*

The complete URL to download the configuration. It includes the file name and ends by 0x00.

### 5.8.2 UPD(Firmware Upgrade Report)

The HEX format of the event report message **+RESP:GTUPD** is as follows. For this message, <Protocol Version> and <Firmware Version> will always be present regardless of the <+EVT Mask> setting.

Example:
2B4556540F02FE1FFF006A802004060006044756353 84C415500002F0F0100210C006601010100
00000000006206FB3FF201E5D58707E8050805282D04600000550B0E9E30A5000000000000000
0000000000000000000000000000007E8050805282E07948DA90D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Code | 2 | |
| | Retry | 1 | |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

### 5.8.3 EUC(Extended Configuration Update Report)

The HEX format of the event report message **+RESP:GTEUC** is as follows, which is similar to **+RESP:GTUPC**.

Example:
2B4556545402FE1FFF013680200406000604475635384C415500002EB30101210C000000CA6874

74703A2F2F3139322E3136382E31302E37353A32303537312F4130355F61742E696E6900000000
58000100000000000000007206FB409601E5D47707E805090B162504600000550B0E9E30A500000
0000000000000000000000000000000001207FFFFF00000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000470
000000020000000002700000000100000000000000000000000000000000000000000000FFF
FFF000000000000000000000000000000000015800000321000000000000000020000000000000
0000000000000005000000000000000000000000000000000007E805090B16250D9DF5F00D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +EVT |
| | Message Type | 1 | |
| | Report Mask | 4 | 00000000 - FFFFFFFF |
| | EVT Expansion Mask | 4 | 00000000 - FFFFFFFF |
| Body | Length | 2 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |
| | Unique ID | 8 | IMEI/Device Name |
| | Battery Level | 1 | 0 - 100 |
| | External Power Voltage | 2 | 0 - 32000(mV) |
| | Digital Input Status | 1 | 00 - 03 |
| | Digital Output Status | 1 | 00 - 07 |
| | Motion Status | 1 | 0x11 \| 0x12<br>0x21 \| 0x22<br>0x41 \| 0x42<br>0x16 \| 0x1A |
| | Satellites in Use | 1\|2 | 0 - 72 |
| | Command ID | 2 | |
| | Result | 2 | 100-103 \| 200-202 \| 300-302 \| 304-306 |
| | Download URL | <=101 | Complete URL |
| | Identifier Number | 4 | |
| | Reserved | 1 | 00 |
| | Number | 1 | 1 |
| | GNSS Accuracy | 1 | 0 - 50 |
| | Speed | 3 | 0.0 - 999.9 (km/h) |
| | Azimuth | 2 | 0 - 359 |
| | Altitude | 2 | |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Longitude | 4 | |
| | Latitude | 4 | |
| | GNSS UTC Time | 7 | YYYYMMDDHHMMSS |
| | MCC | 2 | 0000 - FFFF |
| | MNC | 2 | 0000 - FFFF |
| | LAC | 2 | 0000 - FFFF |
| | Cell ID | 4 | 00000000 - FFFFFFFF |
| | Reserved | 1 | 00 |
| | Current Mileage | 3 | 0.0 - 65535.0 (km) |
| | Total Mileage | 5 | 0.0 - 4294967.0 (km) |
| | Current Hour Meter Count | 3 | HHMMSS |
| | Total Hour Meter Count | 6 | HHHHHHHHMMSS |
| | CAN Data | <=400 | |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

## 5.9 HBD (Heartbeat Data)

### 5.9.1 Heartbeat Report

Heartbeat is used to maintain the contact between the device and the backend server while communicating via GPRS. The heartbeat package is sent to the backend server at the interval specified by <Heartbeat Interval> in the AT+GTSRI command.

Example:
2B484244FF228020040803080F475635384C41550007E90314062B230295F2470D0A

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| Head | Header | 4 | +HBD |
| | Report Mask | 1 | 00 - FF |
| Body | Length | 1 | |
| | Device Type | 3 | 802004 |
| | Protocol Version | 2 | 0000 - FFFF |
| | Firmware Version | 2 | 0000 - FFFF |

| Parts | Fields | Length | Range/Format |
|---|---|---|---|
| | Unique ID | 8 | IMEI/Device Name |
| Tail | Send Time | 7 | YYYYMMDDHHMMSS |
| | Count Number | 2 | 0000 - FFFF |
| | Checksum | 2 | 0000 - FFFF |
| | Tail | 2 | 0x0D 0x0A |

- *Report Mask*

  Please refer to <+HBD Mask> in AT+GTHRM.

- *Unique ID*

  If Bit 4 of <+HBD Mask> is 0, the IMEI of the device is used as the unique ID of the device. IMEI is a 15-digit string. In the HEX format message, each 2 digits are encoded into one byte as an integer.

| IMEI | 13 | 57 | 90 | 24 | 68 | 11 | 22 | 0 |
|---|---|---|---|---|---|---|---|---|
| HEX | 0D | 39 | 5A | 18 | 44 | 0B | 16 | 00 |

  If Bit 4 of <+HBD Mask> is 1, the device name is used as the unique ID of the device. Device name is an 8-byte string, please refer to <Device Name> in AT+GTCFG. If the length of <Device Name> is more than 8 bytes, the device will only acquire the first 8 bytes.

  In the Hex format message, each byte is encoded into one byte as an integer. If the device name is less than 8 bytes, the remaining bytes are set to 0.

| Device Name | q | u | e | c | l | i | n | k |
|---|---|---|---|---|---|---|---|---|
| HEX | 71 | 75 | 65 | 63 | 6C | 69 | 6E | 6B |

  If the mask of <UID> in <+HBD Mask> of AT+GTHRM is set to 0, the heartbeat message will not report device name or IMEI information. If the mask of <UID> is set to 1, then the heartbeat message will report device name or IMEI information according to the mask of <Device Name>.

## 5.10Buffer Report in Hex Format

When HEX format messages go into the local buffer, the device will replace the second byte of the report messages with "B". Thus, **+BSP** is buffered report for **+RSP**, **+BNF** is buffered report for **+INF**, **+BRD** is buffered report for **+CRD** and **+BVT** is buffered report for **+EVT**. The remaining part of the report messages is kept unchanged.

# 6 Appendices

Here are the appendices of the @Track protocol, including the modification history of this protocol, copyright notice, etc., Please refer to the details below.

## 6.1 Two's Complement

For a positive value, the two's complement is itself. Take 17 as an example. Its hex format is 0x11 and the two's complement for it is 0x11. For a negative value, the following gives detailed calculations.

-X is a negative value.
Firstly, get to know the number of bits for the negative value N, then the two's complement for it is:

2^N - X

For example, to use 16 bits to represent -100, the two's complements for it should be:

2^16 - 100 = 65436 = 0xFF9C

Above is two's complement for -100 in hex format.

On the contrary, the two's complement can also be converted to the hex value that it represents in a similar way.

1. Get to know the number of bits for the two's complement.
2. Get the sign of the value, positive or negative. If the highest bit is 1, it is a negative value. If the highest bit is 0, it is a positive value.
3. If it is a positive value, there is no need for conversion. It is the value.
4. If it is a negative value, get the real value through the following calculation:

-(2^N - X)

Where:

N is the number of bits for the two's complement.
X is the value that is converted from the two's complement directly.

For example, if the number of bits for the two's complement is 16 and the two's complement is 0xFF9C, then it is a negative value as the highest bit is 1, and the detailed calculation for it is:

-(2^16 - 0XFF9C) = -100

## 6.2 Accessory Index

| Accessory Model Name | Accessory Type | Accessory Model | Alarm Type | Append Mask |
|---|---|---|---|---|
| Escort_Fuel_BLE | 1 | 0 | 0 | 001F |
| Escort_Angle_BLE | | 3 | 0 \| A | 011F |
| WTS300 | 2 | 0 | 0 - 3 | 001F |
| Temperature ELA | | 1 | 0 - 3 | 081F |
| BLE CAN100 | 4 | 0 | | |
| WTH300 | 6 | 2 | 0-3 \| 7-9 | 083F |
| ELA sensor | | 3 | 0-3 \| 7-9 | 0837 |
| DTH100(WMS301) | | 4 | 0-3 \| 7-9 \| 13-14 | 283F |
| WTH301 | | 5 | 0-3 \| 7-9 | 283F |
| DUT-E S7 | 7 | 0 | | 0007 |
| DFM 100S7 | | 1 | | 0007 |
| DFM 25ODS7 | | 2 | | 0007 |
| GN0M DDE S7 | | 3 | | 0007 |
| GNOM DP S7 | | 4 | | 0007 |
| WBC300 | 8 | 0 | | 0087 |
| Fuel Sensor | 10 | 0 | | 2017 |
| Angle Sensor | | 1 | | 0017 |
| MAG ELA | 11 | 0 | C | 100F |
| ATP100/ATP102 | 12 | 0 | 0 \| E-10 | 0617 |
| WRL300 | 13 | 0 | 15 | 4007 |

## 6.3 CRC-16 Calculation

The 16-bit CRC checksum in the report should be calculated according to the properties in the table below:

Example:
CRC (00 7F 24 1E 01 00 01 11 47 56 37 36 4D 47 2D 34 00 FF FF 07 E4 06 1C 09 1B 24 01 58) = 60D9H (0x60D9)

| Property | Value |
|---|---|
| Name | CRC-16 |
| Width | 16 bits |

| Property | Value |
|---|---|
| Polynomial | 0x1021 ($X^{16} + X^{12} + X^5 + 1$) |
| Initialization | FFFFH (0xFFFF) |
| Reflect input | False |
| Reflect output | False |
| Final XOR | 0000H (0x0000) |

Here is a corresponding CRC-16 algorithm routine written in C language:

```
static const uint16_t crc16_table[256] =
{
        0x0000,0x1021,0x2042,0x3063,0x4084,0x50A5,0x60C6,0x70E7,
        0x8108,0x9129,0xA14A,0xB16B,0xC18C,0xD1AD,0xE1CE,0xF1EF,
        0x1231,0x0210,0x3273,0x2252,0x52B5,0x4294,0x72F7,0x62D6,
        0x9339,0x8318,0xB37B,0xA35A,0xD3BD,0xC39C,0xF3FF,0xE3DE,
        0x2462,0x3443,0x0420,0x1401,0x64E6,0x74C7,0x44A4,0x5485,
        0xA56A,0xB54B,0x8528,0x9509,0xE5EE,0xF5CF,0xC5AC,0xD58D,
        0x3653,0x2672,0x1611,0x0630,0x76D7,0x66F6,0x5695,0x46B4,
        0xB75B,0xA77A,0x9719,0x8738,0xF7DF,0xE7FE,0xD79D,0xC7BC,
        0x48C4,0x58E5,0x6886,0x78A7,0x0840,0x1861,0x2802,0x3823,
        0xC9CC,0xD9ED,0xE98E,0xF9AF,0x8948,0x9969,0xA90A,0xB92B,
        0x5AF5,0x4AD4,0x7AB7,0x6A96,0x1A71,0x0A50,0x3A33,0x2A12,
        0xDBFD,0xCBDC,0xFBBF,0xEB9E,0x9B79,0x8B58,0xBB3B,0xAB1A,
        0x6CA6,0x7C87,0x4CE4,0x5CC5,0x2C22,0x3C03,0x0C60,0x1C41,
        0xEDAE,0xFD8F,0xCDEC,0xDDCD,0xAD2A,0xBD0B,0x8D68,0x9D49,
        0x7E97,0x6EB6,0x5ED5,0x4EF4,0x3E13,0x2E32,0x1E51,0x0E70,
        0xFF9F,0xEFBE,0xDFDD,0xCFFC,0xBF1B,0xAF3A,0x9F59,0x8F78,
        0x9188,0x81A9,0xB1CA,0xA1EB,0xD10C,0xC12D,0xF14E,0xE16F,
        0x1080,0x00A1,0x30C2,0x20E3,0x5004,0x4025,0x7046,0x6067,
        0x83B9,0x9398,0xA3FB,0xB3DA,0xC33D,0xD31C,0xE37F,0xF35E,
        0x02B1,0x1290,0x22F3,0x32D2,0x4235,0x5214,0x6277,0x7256,
        0xB5EA,0xA5CB,0x95A8,0x8589,0xF56E,0xE54F,0xD52C,0xC50D,
        0x34E2,0x24C3,0x14A0,0x0481,0x7466,0x6447,0x5424,0x4405,
        0xA7DB,0xB7FA,0x8799,0x97B8,0xE75F,0xF77E,0xC71D,0xD73C,
        0x26D3,0x36F2,0x0691,0x16B0,0x6657,0x7676,0x4615,0x5634,
        0xD94C,0xC96D,0xF90E,0xE92F,0x99C8,0x89E9,0xB98A,0xA9AB,
        0x5844,0x4865,0x7806,0x6827,0x18C0,0x08E1,0x3882,0x28A3,
        0xCB7D,0xDB5C,0xEB3F,0xFB1E,0x8BF9,0x9BD8,0xABBB,0xBB9A,
        0x4A75,0x5A54,0x6A37,0x7A16,0x0AF1,0x1AD0,0x2AB3,0x3A92,
        0xFD2E,0xED0F,0xDD6C,0xCD4D,0xBDAA,0xAD8B,0x9DE8,0x8DC9,
        0x7C26,0x6C07,0x5C64,0x4C45,0x3CA2,0x2C83,0x1CE0,0x0CC1,
        0xEF1F,0xFF3E,0xCF5D,0xDF7C,0xAF9B,0xBFBA,0x8FD9,0x9FF8,
        0x6E17,0x7E36,0x4E55,0x5E74,0x2E93,0x3EB2,0x0ED1,0x1EF0,
};
```

```
uint16_t getCrc16(uint8_t* pdata, uint16_t len)
{
    uint16_t crc16 = 0xffff;

    while(len > 0) {
        crc16 = crc16_table[*pdata ^ (uint8_t)(crc16 >> 8)] ^ (crc16 << 8);
        pdata++;
        len--;
    }

    return crc16;
}
```